# Golden Reasoning

Soft Horizons, Survivorship, and the Two-Channel Muffled Hazard

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**Abstract**

A biosphere that survives indefinitely must satisfy two conditions, not one. We model total hazard as a two-channel process, an external background muffled by accumulated capacity plus an internal term that the same capacity feeds, and we place the noise on the negative logarithmic derivative of the hazard, one integration further out than is standard. Survival then requires a *sign* condition on how hazard responds to capacity, which we call *technoanatomy*: where the sign is adverse, viability is identically zero for every parameter value and every driver, so that no amount of capacity helps, while where it is favourable, viability is the product of that indicator with the driver's escape probability, and the two conditions are independent. In the critical case the survival tail is $t^{-1/4}$, the persistence exponent of integrated Brownian motion transferred to the soft-killing functional by an explicitly flagged Ansatz whose pre-registered falsifiers are tested numerically. Conditioning on survival converges to a Doob $h$-transform, locally absolutely continuous with respect to the unconditioned law but mutually singular from it at infinity, and from the singularity follows an epistemic ceiling: no finite observation certifies which measure one is under. The ceiling is proved once and then turned on this work's own conclusions as firmly as on its targets.

Two familiar positions fall to the same structure. Programmes that measure civilisational progress by capacity alone are indexing the multiplicand and not the indicator, and programmes that would withdraw capacity leave the background unmuffled, reaching viability zero by the other route. Both conclusions are derived rather than argued separately. The model is one of three movements. The second is a conditioning argument: that the record we hold reads as the past-light-cone restriction of a conditioning on an event in our *future* light cone, the indefinite survival of the biosphere, with the companion program's observer-conditioning recovered as its pre-technological restriction. The future-conditioned tradition of Barrow and Tipler, the Omega Point, and axiarchism is placed against the argument delta by delta, and the strong forward reading is developed with a phenomenology and a discriminating quantity but, by the ceiling, no decisive test.

The third movement is a set of positions, each conditional on a single adoptable premise and each fenced: a grounding for the normative vocabulary, a decision rule whose limit form is a ratio of persistence-harmonic values, an ethics, and, for the reader the abstract's first sentence does not reach, a third route to meaning, *inferred* rather than revealed or constructed. The route is stated at two tiers, an unconditional claim that the constructed-versus-discovered dichotomy is broken and a Strong-conditional claim that the settlement's inference inverts; the post-Nietzschean crisis is classified as a formal invalidity and reversed at its root; and the framework's reception-scale predictions are entered scoreable, with the conditions of their failure attached. The dependency between the movements runs one way, so that the results and the argument consume nothing from the positions, and a reader

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unpersuaded by the positions holds the first two intact. The whole is organised in five parts, with the chain from evidence to conduct stated once, link by link, at its grades.

# Contents

## Notation

Symbols are listed by the object they name. Where a letter does double duty the two uses are separated by section, and the collisions that remain are flagged. One is flagged here rather than in a table note: $\varkappa$, the schedule channel's mean offspring (§5.7), is not $\kappa$, the exceedance factor of the imported Accordance bound (§3.2, used again in §7); the glyphs are close and the objects are not.

**The hazard model** (§5).

| | |
|---|---|
| $h_t^{\mathrm{tot}}$ | total hazard rate, (5.7) |
| $B_t$ | external background; $B_t = \sum_i e^{\epsilon W_t^{(i)}}$, random |
| $\varsigma_b$ | muffling factor, logistic, $\varsigma_b(0) = 1$ exactly |
| $\Gamma$ | internal coefficient (the trough at zero capacity) |
| $\rho$ | *technoanatomy*: hazard's response to capacity, Definition 5.32 |
| $x_t$ | muffling exponent (*technostrength*); $x = \int \mu$, (5.5) |
| $\mu_t$ | driver, $\mu = -\frac{d}{dt} \log h$; Brownian post-onset, (5.4) |
| $\sigma$ | driver volatility    $b$ — muffling offset    $k$ — $\log(B/\Gamma)$ |
| $\Lambda_t$ | cumulative hazard $\int_0^t h\, ds$ |
| $T$ | departure time (Cox construction, Definition 5.4) |
| $\tau$ | hard-wall functional    $\mathcal{I} = \{\Lambda_\infty < \infty\}$ — viability, Definition 5.5 |
| $h_*$ | hazard floor under adverse anatomy    $x_*$ — its argument |

**Measures and the survivorship limit** (§§5.8, 6).

| | |
|---|---|
| $\mathbb{P}$ | unconditioned law — the *General* biosphere |
| $Q$ | survivorship limit, a Doob $\mathfrak{h}$-transform — the *Golden* biosphere, Definition 6.1 |
| $\mathbb{P}^{(t)}$ | $\mathbb{P}(\cdot \mid T > t)$, the *family* interpolating them; not a measure, Remark 6.2 |
| $\mathfrak{h}$ | persistence-harmonic function; $dQ/d\mathbb{P}\vert_{\mathcal{F}_t} = \mathfrak{h}(X_t)/\mathfrak{h}(X_0)$, (5.47) |
| $s(\cdot)$ | scale function    $\mathcal{F}_t$ — filtration |

**The schedule channel** (§5.7).

| | |
|---|---|
| $n$ | gates in the schedule    $m$ — options per gate |
| $\lambda_{k,j}$ | rate of option $j$ at gate $k$; $\lambda_k = \sum_j$, $\pi_k = \lambda_{k,1}/\lambda_k$ |
| $Z_t$ | outstanding debt    $r$ — gates owed per wrong gate |
| $\varkappa$ | mean offspring $r(1-\pi)$; criticality at $\mathbb{E}[\log \varkappa] = 0$ |
| $u$ | completion weight, least root of $u = s\, f(u)$ |
| $c_{\mathrm{eff}}$ | effective contrast, $r \log_{10}(1 + (B+\Gamma)/\lambda)$ |
| $\gamma_k, \nu_k$ | staged leak on credit and driver; both zero at $k = n$ |

**Decision, ethics, and the Route World** (§§9, 8).

| | |
|---|---|
| $A, B$ | options (§9); $P(A)$ — Aethic weight, read as attainability |
| Aur | Aurutance, log of the decision ratio |
| AM, RM | automorality (act-indexed), relative morality (maxim-indexed) |
| $\mathcal{G}$ | the Golden Point as a conditioning event |
| $\mathcal{L}_t, \mathcal{L}_*$ | accumulated and total log-evidence, Definition 8.12 |
| $\kappa$ | exceedance parameter of the Accordance Principle, (7.2) |

**Collisions the reader should know about.** $s$ is the scale function in §5 and the per-gate survival factor in §5.7; $T$ is the departure time throughout except in §6.3.1, where $T = 4$ is a conditioning horizon; $n$ is the background count in (5.7), the gate count in §5.7, and the institution count in §11; $b$ is the muffling offset and, in §11, a Heider weight; $\kappa$ is the exceedance parameter and, in Appendix A, a gain minimum; $\varkappa$ (curly) is the schedule's mean offspring and is a distinct symbol. $\Phi$ with a numeral indexes premises; $\Phi$ alone is the normal CDF.

## Glossary of coined terms

Terms introduced by this work or by its companions, with the place each is defined. Terms marked † are imported from the companion program and are stated at §3.2.

**Technostrength** The muffling exponent $x$: accumulated capacity to lower hazard. (5.5)

**Technoanatomy** $\rho$: the *sign* of hazard's response to capacity. Definition 5.32

**Crest / trough channel** The external and internal terms of (5.7).

**Soft wall** The killing functional replacing a hard boundary; Ansatz 5.18 relates the two.

**Golden Point** The greatest fixed point $\nu F$ of the safety operator: a state admitting no foreclosing continuation.

**Departure Point** Its dual $\mu G$: a state all of whose continuations foreclose — the biosphere's losing condition, which precedes physical death and is separable from it in both directions. Remark 2.19

**Departure-sufficient event** The referent of the hazard: an occurrence whose realization forecloses, with death-events its typical occupants and nothing biological presupposed. Remark 5.6

**Golden-categorical robustness** The definitional-root premise: feature-terms are well-defined to the degree their reference is robust relative to the asymptotic survivorship structure; the companion's Aethic-categorical criterion at full depth. Remark 5.7

**Task-object** An object individuated by its contribution to a designated root Aethus; the constructive form of Golden-categorical robustness, with the Golden Point its generative root and the referent of contribution. Definition 4.1, Remark 4.2

**Golden Aethus** The Aethus stating exactly the Golden attribute, all else blank; the index of the Golden Nexus, and, under the Strong reading, the substrate on which every thing is a weight-carrying fluctuation. Definition 4.4

**Commitment primitive (node)** An undefended normative placement from which practical argument propagates; judged non-arbitrary to the degree it traces to the Golden root. §9.7

**Inferred meaning (the third route)** Meaning discovered in the record and its conditioning structure — neither constructed by the observer nor revealed through a non-natural channel; unconditional at the dichotomy-breaking tier, Strong-conditional at the inversion tier. §9.8

**Nietzschean settlement** The modern operating frame holding that constructed meaning is the only honest option; engaged and superseded at §9.8, its diagnosis retained.

**Species-level self-awareness** The criterion: a species is self-aware when it possesses the concept of its own modal poles ($\mu G$, $\nu F$) as such; on this test the discontinuity is conceptual and, as of writing, ahead. Remark 10.1

**Inference flip** The vantage-indexed best-inference operator's change of output at a named evidential arrival; the meaning-question's flip condition is the coupling of Miracles, on the Tricky-Principle precedent. Remark 9.10

**Load path** Part IV's ten links stated as one auditable chain, coupling through owned conduct, graded prefix-to-suffix. §9.11

**Miracleism** The meaning tradition of Part IV, named for its source: the Miracles fund the meaning; the Golden Point actualizes it, preventing vanity. Remark 9.14

**General / Golden Biosphere** A biosphere under $\mathbb{P}$ / under $Q$. Definition 6.1

**Goldening** The interpolation $\mathbb{P}^{(t)}$ between them.

**Golden Filter** The reweighting survivorship performs on an ensemble. Proposition 2.9

**Golden Tendency** The ensemble consequence of the Filter. Principle 1

**Golden Question / Answer** *Should our biosphere live forever?*, and what the Filter does to frozen answers. §2.4

**Golden Imperative** Passage through the Golden Point as the only route avoiding $\mu G$. Principle 2

**Golden Extrapolation** Self-placement at finite $t$ on the goldening. Definition 2.18

**Golden Initiative** Persistence-directed practice, named from inside. Definition 9.1

**Golden Score / Aurutance** The decision ratio $\mathfrak{h}(A)/\mathfrak{h}(B)$ and its logarithm. §9.3

**Departric** Prefix: automorality negative. **Biotic**: positive. **Neffective**: zero. Table 1

**Schedule / gate / debt / leak** The third hazard channel and its parts. §5.7

**Route World** The world of the Strong extrapolation hypothesis. Definition 8.1

**Golden Optimation** Optimation under a future-cone conditioning event. Definition 8.3

**Golden causality** The pattern in which Golden Optimation accounts for a present event. Definition 8.6

**Investment / Invested Golden causality** The weight-versus-necessity bound: a realized route's improbability lower-bounds how much more it accomplishes, at exceedance confidence. Proposition 7.6; phenomenological face at Remark 8.7

**Departric invalidity / Mattering residue** That past $\mu G$ the adopted ordering is undefined rather than small, a claim about an index, and the pruning whose residue the dominance argument runs on. §9.6.5, §8.6

**Retrocausal (conditioning)** Conditioned on an attribute in one's future light cone; nothing further. Definition 7.1

**Retrocausal Accordance** The Accordance Principle run forward: one's present cannot be arbitrary to one's future. Principle 7, and (7.3) for the bound form.

**Finale paradox** Why one occupies the present rather than deep time, on a reading with no resource to discharge the fluke; the relay point at which the companion's "when does Optimation stop" frontier is dissolved rather than answered. §7.4, Remark 7.2

**Adversivity** Net antagonism in an institutional balance. §11

**Aethus / Aethae** $^{\dagger}$ An information state and its refinements.

**Nexus / Cosmic Nexus** $^{\dagger}$ A weighting scheme over Aethae; the unconditioned one.

**Optimation** $^{\dagger}$ The fluke content a conditioned record displays.

**Accordance Principle** $^{\dagger}$ The typicality constraint binding attribute by attribute.

**Imperfection Magnitude Drop** $^{\dagger}$ A weight-suppressing perturbation short of foreclosure.

# 1 Introduction

Will the biosphere continue indefinitely? This work treats that question as one with enough structure to carry theorems at one end and an account of meaning at the other, and it makes the crossing in one direction only, so that the mathematics never borrows from the philosophy and every claim along the way is carried at a stated grade. Concretely, the document contains a stochastic model of survivorship together with its theorems (Part II); a conditioning argument over the historical record (Parts I and III); and a set of normative, ethical, and existential positions that consume the first two while being consumed by nothing (Parts IV and V). A reader who wants only the probability may therefore read Part II and stop, holding everything this work proves, while a reader who came for the older question underneath will find it answered, at its grade, in Part IV.

*One must imagine Sisyphus happy. A pale blue dot. You make your own meaning.* The reader knows these consolations, because they are the standing settlement of the educated present: the shared conclusion that the universe supplies no meaning, that construction is what remains, and that maturity consists in bearing this well. This paper engages the settlement on its own terms, steelmanning it in §9.8 as the uniquely reasonable position given what its authors could see, and then argues that its inventory was incomplete. Meaning has been offered as *revealed*, and after the revelation failed it has been offered as *constructed*; but there is a third cell in that table, meaning *inferred*, read off an object under stated inference discipline with published falsifiers and a proven ceiling on certainty, and that cell has stood empty less because anyone examined and rejected it than because nothing had ever seemed to occupy it. This document is an attempt to occupy it at a stated price, and the price is displayed before anything is purchased.

### 1.1 The thesis

Capacity to act on the world is usually treated as the quantity that decides whether a civilisation lasts, and this work argues that it is the wrong quantity, because the right one is a sign rather than a magnitude. Consider a system facing hazard from two directions, an external background it can learn to muffle and an internal channel that its own capacity feeds. Accumulating capacity lowers the first and does something to the second, and what it does to the second is a structural property of the system rather than a matter of how much capacity it holds. Call that property technoanatomy and write it $\rho$. The central result is that its sign is decisive: adverse $\rho$ makes indefinite survival impossible outright, not unlikely but impossible, for every parameter value and every history, while favourable $\rho$ makes survival possible without making it likely, since a second and independent condition must also hold.

Two consequences follow immediately, and both cut against widely held positions. A programme that measures progress by energy captured or capability acquired is measuring a quantity that cannot distinguish the two cases, while a programme that would withdraw capacity leaves the background unmuffled and arrives at the same zero from the other side. These are not two objections but one, and the paper derives them together.

### 1.2 The model in three lines

A reader who wants the mathematics before the vocabulary needs only the following, which §5 develops properly. *The hazard.* $h_t^{\mathrm{tot}} = B_t \varsigma_b(x_t) + \Gamma e^{-\rho x_t}$, (5.7): an external background $B_t$ attenuated by a muffling factor $\varsigma_b$ that capacity $x$ improves, plus an internal term that capacity feeds or starves according to the sign of $\rho$. The normalisation is $\varsigma_b(0) = 1$, so at zero capacity the crest is the bare background.

*The driver.* $x_t = \int_0^t \mu_s\, ds$ with $\mu$ Brownian: the noise sits on $\mu = -\frac{d}{dt} \log h$, one integration beyond where standard practice puts it. That placement is the paper's principal modelling claim, and it is what moves the persistence exponent from $\frac{1}{2}$ to $\frac{1}{4}$. *Departure.* $T$ is the first time the cumulative hazard exhausts an independent exponential (Definition 5.4), and viability is the event $\{\Lambda_\infty < \infty\}$ — that the total hazard ever accrued stays finite. Departure with probability one and infinite life expectancy are compatible, and both hold in the critical case.

### 1.3 What this work does not claim

One reconciliation is owed at the outset, because it is the objection a reader will arrive holding. This paper builds a decision rule and an ethics on a conditional distribution, and a conditional distribution entails no obligation, so the two must be squared before anything else is said. The reconciliation is a division of labour between two premises. $\Phi2$ (Premise $\Phi2$) governs what the apparatus outputs, which is distributions, from which no duty follows. Φ6 (Premise Φ6) governs what an agent does with a choice the apparatus never makes, namely which grounding to occupy, given that some grounding is occupied whether or not anything is said. The two premises operate at different levels and do not meet, with the consequence that everything normative in this work is conditional on Φ6, which is offered and priced and may be declined, and an agent who declines it contradicts nothing proved here. §2.2 states the split in four checkable steps, and §9.5 restates it where the decision rule might otherwise be over-read.

One structural fact should be stated alongside it, because it is the protection the mathematics needs from the rest of the document. This work contains three kinds of thing: results, meaning the hazard model of §5 with its theorems and the survivorship transform of §6; a graded argument, meaning the conditioning principle of §3, carried at [**S**] with its extrapolation hypotheses at [**O**]; and positions, meaning the grounding of §2, the decision rule of §9, the ethics of §9.6, and the placement of §10 against extinctionism and techno-optimism, all carried at [Φ] and all conditional on the one adoptable premise. The dependency between the three runs one way. The results consume nothing from the positions, the graded argument consumes nothing from the positions, and the positions consume both, so a reader unpersuaded by the positions is left holding the results and the argument intact, with nothing in the positions usable against them. A reader who finds the positions objectionable, reading a paper on survivorship conditioning that takes a stance on longtermism as advocacy, should read the results and the argument first and the positions never, and will have read everything this work proves.

Beyond that, three standing limits. The epistemic ceiling (Theorem 6.18) forbids certifying any system's position under survivorship from finite observation — including one's own, and including this work's. No actor, institution, or programme is scored anywhere in this paper. And the model's practical relevance rests on technoanatomy being a process rather than a fixed parameter, which §5.6.3 does not yet supply; three sections record that as their inherited dependency, and it is the paper's principal exposure.

### 1.4 The situation: two announcements and an empty cell

Two sentences, ninety-five years apart, frame the condition this work addresses. The first is Nietzsche's, from 1882: "God is dead. God remains dead. And we have killed him." [1] Read structurally, what the sentence announced is that the *revealed* cell of the meaning inventory had emptied, and with it, by an inference that went unexamined because no alternative was visible, that meaning must henceforth be made rather than found. The second is Weinberg's, from 1977, delivered from inside the enterprise that had done the emptying: "The more the universe seems comprehensible, the more it also seems pointless." [2] His successor sentence offered comprehension itself as the consolation, research as one of the few things lifting human life above the level of farce, and that is the settlement in its noblest form: lucidity as the remaining good, held by a mind that saw no further inventory to check.

Where the revealed cell's contents came from deserves one disciplined paragraph, because it fixes what any successor must not be. The premodern supply of meaning ran on a mechanism that this program's companion writings name *the premodern dragon*: the coupling of ancestral fear, the inheritance of a lineage that spent its formative epochs as prey, with the human appetite for control over other humans. The gods that domesticated the terror also collected the rent, and the arrangement priced meaning in obedience. What the scientific enlightenment did, on this reading, was not so much refute a doctrine as remove a mechanism, and the whiplash of the modern condition follows from the fact that the operation which removed a false floor removed the only floor then in inventory. The settlement is the sound of a civilisation deciding, reasonably enough on its evidence, that floors were never load-bearing to begin with. On the essential point the present work sides with the settlement without reservation: the dragon must stay dead, and §9.10 prices in six checkable deltas the claim that nothing here restores it. But the open question was never whether to go back. It is whether a dead dragon and an empty cell exhaust the options.

The diagnosis this framework offers of Nietzsche's announcement is therefore epistemic inexperience rather than categorical limitation: humanity had never yet held an object from which meaning could be inferred, and mistook never having held one for the impossibility of its existing. Nobody in 1882 was positioned to detect the mistake, which is why the settlement was, and is steelmanned here as, the uniquely reasonable reading of its evidence. The deadlock that followed, described most exactly by MacIntyre as interminable and incommensurable assertion [3], is classified in this work as a crisis on the inference side: every tradition rightly discounts every other tradition's root commitments, and its own, because commitments admit no inference in either direction, a condition to which §9.7 gives its criterion. MacIntyre closed by waiting for another, doubtless very different, Benedict. The fallback proposed here is not a person and not a community but an *object*: a conditioning target with a mathematics, a ladder of grades, published falsifiers, and a proven ceiling on what anyone may claim to know about it. Part IV states what inferring from it costs, in the near-unconditional tier and the Strong tier separately, and enters its reception-scale predictions scoreable.

None of the mathematics depends on this subsection, since the dependency runs the other way, as §1.3 has already priced, and a reader who wants the theorems alone loses nothing by proceeding directly to Part II. A reader for whom this subsection named their situation should know that the paper was written in the conviction that such readers exist at scale, and that §9.11 states, link by link and at its grades, what this work believes it can offer them.

## 1.5 How to read this

The document is long and its parts are separable. Four entry points. *For the mathematics*, start at §5 and read to §6. That stretch is self-contained: a hazard process, its persistence asymptotics, the anatomy trichotomy, the two-condition theorem, and the survivorship transform. It consumes nothing from the companion program and can be assessed on its own terms. *For the argument about capacity and risk*, read §5.8 for the two-condition theorem and then §10, which applies it to extinctionism and to techno-optimism in four pages. *For the philosophical grounding*, read §2. It is the most demanding section and depends on results stated later; a reader who wants it first should take §1.2 as the model and return.

*For the anthropic material*, read §3, beginning with §3.2, which states the companion apparatus this section consumes; its human-history argument (§3.7) is the one to read if reading only one. *For the argument that the Golden Point is the right target*, read §7, which develops the retrocausal accordance principle, runs it on a worked case at human scale, and derives from it the *shape* the target must have — and then identifies the Golden Point as what has that shape. It is the primary argument for the target; §3's reductio clears the ground for it.

*For the framework's most distinctive prediction*, read §8, the Route World, which develops the strong forward reading of the conditioning argument and states what would, and by theorem what would not, distinguish it from its rivals. It is carried at **[O]** throughout, save the conduct argument of §8.6, at [Φ] on an adopted valuation, and can be read after §3 without the intervening sections.

Two conventions run throughout. Every substantive claim carries a grade, [**P**] proved, [**C**] cited, [**A**] conditional on a stated Ansatz, [**S**] argued at scaling or schematic level, [Φ] premise carrying its own falsifier, [**O**] open, and the grades are load-bearing: §3's central claims are [Φ] and [**S**], and what it proves is proved conditional on them, which it says. And each section closes with a ledger stating what it costs, what it buys, and what it does not claim. A reader in a hurry can read the ledgers.

As of the structural pass the paper is organized in five parts, and the part map is the fastest orientation: Part I poses the Question and fixes its objects; Part II is the survivorship mathematics, self-contained; Part III is the inference discipline, Golden reasoning, accordance, the Route World, with the ceiling as its governing limit; Part IV is the normative and existential engine, the Score through the third route; Part V is the human position. A reader wanting only the probability reads Part II; only the philosophy of meaning, Part IV after Part I's vocabulary; the whole argument, in order.

# Part I: The Question and Its Target

Part I fixes the question and its objects, and proves nothing quantitative. Agency's curse is posed and the Golden Question is stated at its price (§2); the conditioning target is then chosen, with the Departure Point built as a least fixed point together with its cascade, continuity defended, and the rival periodization engaged (§3); and task-objects supply the individuation that the perturbation arguments consume, an arc that closes on the Golden-categorical premise and, under the Strong reading only, on the Golden Aethus (§4). A reader leaves this part holding a target, an ontology of what is conditioned on, and the vocabulary in which every later part speaks.

## 2 Agency's curse and the Golden Question

An agent cannot stop acting, and cannot see far enough to know what its acting does. This section names the resulting problem, *agency's curse*, argues that no agent escapes it, and proposes the one countermeasure the predicament leaves available. Since some question must be held ungrounded, prefer one whose answer the agent's own existence constrains. That is Φ6, this work's single doorway through which obligation can enter.

The candidate offered is the Golden Question, *should our biosphere live forever?*, and what recommends it is not that its answer can be proved but that survivorship selects it: a biosphere organizing its practice around the negative does less to persist, and the Filter thins it accordingly. The affirmative is therefore selected rather than deduced, which is a positional claim and not a truth claim, and the difference is kept in view throughout.

Two things scope the whole. Whether a biosphere *can* reach what the answer affirms is settled by $\rho$ and known to nobody (§6.8); and the section's practical relevance rests on technoanatomy being a process rather than a fixed parameter — which Remark 10.8 argues the framework itself requires, and which §5.6.3 has yet to supply.

### 2.1 The problem

Aethic reasoning demotes several classical metaphysical problems — some to irrelevance, some to attributions of constructs [4]. The problem of this subsection is comparatively undemoted:

the demotions pass over it, and what they leave standing is the business of the present section. We call it *agency's curse*. It is the conjunction of two conditions on any agent.

*Condition 1. Forced action.* An agent is at every moment committed to acts of influence on the universe; inaction, sleep, and self-destruction are themselves selections among alternatives with divergent consequences. The position is that of someone crossing a burning building toward a child: the tempo is set by the fire, not by the crosser, and every hesitation is itself a move. But the comparison flatters. The ordinary case ends; the house has walls; the parent knows what they are trying to do. Here the stakes of each selection are unbounded relative to the time allotted for it, and the situation admits neither exit nor completion — declining to decide is itself a decision, so the condition cannot be suspended, and it attaches to agency as such rather than to human psychology.

*Condition 2. No accessible grounding.* The agent has no access to an objective morality, meaning, or interpretation of the universe against which these forced selections could be weighed — if any such thing exists at all. Picture a spectrum from inert matter to omniscience. A rock is not cursed, because it does not act; an omniscient agent is not cursed, because it can see; agents occupy the one altitude at which the curse can attach, high enough to be bound by Condition 1 and too low to weigh what it binds them to. The altitude is not a human peculiarity: it is a side effect of the kind of thing an agent is.[1]

The curse is the conjunction. Bound to selections of unbounded consequence and given nothing to weigh them against, an agent is driving blind — and blind not at the periphery, where instruments can be added, but in the most intimate of its outlooks, at the point where any instrument would have to be read. Stating this exactly requires three terms.

**Definition 2.1** (Questions, answers, and frozen questions)**.** A *question* is a state an agent takes to be held or not held; an *answer* is the state the agent assigns to it. An agent typically assigns an answer by borrowing from the answers to other questions, recursively — *where are the keys* borrows from *where were they last held*, and so on, until the borrowing terminates. A *frozen question* is a question whose answer the agent firmly holds but which borrows from no question, answer, or other information the agent can access; its answer is a *frozen answer*.

The ostension is the child's why-game: each answer is met with another *why* until the adult can only say *I don't know*. The question immediately preceding that admission is frozen — it is the foundation of every question that traced to it, while itself resting on nothing the agent can produce.

*Remark* 2.2 (What the definition excludes)*.* A frozen question must be neither paradoxical nor inconsequential to the agent's other commitments; a question whose answer changed nothing downstream, or which admitted no coherent answer, would be frozen only vacuously. The restriction is substantive: a later candidate for the role will have to be shown to satisfy it, and the demonstration will be owed, not assumed.

*Remark* 2.3 (Three properties, and which one carries the section)*.* The frozen questions the curse concerns have three further properties. They are held by an agent under forced action (Condition 1); they concern what the agent is to *do*, not merely what is the case; and they are

[1] The refusal of human privilege here is a standing commitment of the paper, made again in the choice of conditioning target, at recursive anchoring, and, at its full width, at continuity (3.20, 3.21, §3.4).

*selectable* — the predicament fixes that some frozen question is held, not which one. The third property carries the remainder of the section: everything built after this point turns on the difference between an unavoidable set and an unavoidable member. What would refute it, in its own vicinity: a demonstration that an agent's frozen questions are fixed by their circumstances rather than chosen within them — Proposition 2.4 would stand, and everything built after it would fall. Nearby notions in the literature share some of these properties and not others; the placement is deferred to §2.7, which will trade on exactly these three.

With the vocabulary in place, the second condition sharpens into a claim.

**Proposition 2.4** (Non-emptiness; [Φ]). *An agent under Conditions 1 and 2 holds at least one frozen question at every moment.*

The proposition is argued, not proved, and the argument is a dilemma over an exhaustive disjunction, there is an objective morality or there is not, so that the two horns below are all there is. What would refute it is either an agent under forced action all of whose answers terminate in something the agent can access that is itself a reason, a regress with an accessible floor, or a transmission protocol of the kind ruled out on the second horn below. Neither has been exhibited; the claim is graded accordingly.

*First horn: there is no objective morality.* Then for any forced selection $x$, the chain descending from *why is $x$ to be done* terminates — not in nothing, but in something that is not itself a reason: a preference, a convention, a disposition. A constructivist is free to call the terminus a fact, and it is one. What it is not is a reason of the kind the chain was descending through, and the question why *that* should bear the weight of everything above it borrows from nothing the agent can access. The frozen question arrives at once.

*Second horn: there is an objective morality.* This is the horn that makes the curse survive the existence of the very thing whose absence seemed to generate it, and it turns on a communication problem. Suppose the objective moral truth exists, and suppose, granting the most favorable case, that some agent is in full possession of it and undertakes to transmit it, by any channel whatsoever: written across the sky, planted in dreams, implanted beneath consciousness.

Consider the recipient's information state after the transmission. It is compatible with exactly three hypotheses: the content is true; the content is a deception; the content is irrelevant, applying in some circumstances but not the recipient's. The claim is not that deception is probable, or that the recipient will in fact feel doubt — it is that *no evidence available to the recipient separates the three hypotheses.* Any certificate of the transmission's veracity would itself have to be an objective moral truth, and the only source of such truths is the channel already under assessment. The request for the discharging evidence re-invokes the very source it was meant to check; the regress is not long, it is a loop of length one. The trichotomy, true, deceptive, or irrelevant, is therefore itself a question whose answer the recipient can hold but cannot derive from anything accessible. It is a frozen question, and the horn closes.[2]

**Why the second horn is structural, and where it could fail.** The argument is about the recipient's information state, not the recipient's psychology. A recipient of perfect equanimity, untroubled by any felt doubt, is in the same position: the three hypotheses remain undistinguished by anything in their possession, and their equanimity is a frozen answer to the

[2] Aethically restated: the recipient holds the transmitted content as an attribute whose warrant is blank, and the blankness cannot be filled from within — any filling would itself be an attribute with a blank warrant from the same source. The vocabulary is §3.1's, and the horn is a special case of it.

trichotomy, not an escape from it. The horn would fail if a certificate of veracity could be produced whose warrant did not draw on the transmitting source — an independent channel to the same objective truth. But an independent channel is just a second transmission, and the trichotomy attaches to it identically. What the horn cannot rule out, and does not need to, is the truth of the content; it rules out only the recipient's ever standing in an accessible derivational relation to it.

Both horns close. On either, the agent holds at least one frozen question; and Condition 1 makes the holding load-bearing, since every forced selection is weighed, when it is weighed at all, against a frozen answer or against nothing.

**What the curse removes, and what it does not.** The curse does not imply that any frozen answer is probably wrong. A frozen answer may be excellently calibrated; nothing above bears on that, and no claim about the accuracy of anyone's morality has been made. What the curse removes is authority of a specific kind: any appeal available *between* agents whose frozen answers conflict. Each can trace their chain to where it stops; neither chain reaches anything the other is obliged to accept; and there the exchange ends — not because either party is unreasonable, but because the resource an adjudication would spend does not exist. Collectives inherit the position with the stakes multiplied, collectively held frozen answers have historically been catastrophic, and the removal binds even that sentence: the condemnation is itself issued from a frozen answer and inherits its standing. It is an instance of the structure under discussion, not an exception to it.

Four neighbors resemble the curse; none is it. The simulation problem is defanged by Aethic reasoning directly: an Aethus is not demoted in its reality by the scenario, and the felt metaphysical blow is an artifact [4]. The problem of induction concerns whether what sits behind the whole of human inference operates as extrapolation from within would have it. Epistemological skepticism, and its Aethic counterpart, concern whether the information held by the Aethus can be attained by an agent with any surety; Aethic accidentalism carries the same burden into metaphysics [4].

*Remark* 2.5 (The differentia, and why quietism is unavailable). Each of the neighbors admits a quietist response: an agent may register the problem, shrug, reinterpret their place in reality, and continue, at no cost the problem itself can name. This is not a defect in the neighbors; it is what they are like, because each concerns the metaphysical *outside* — what stands behind induction, whether the world is as presented, whether the Aethus can be reached. Agency's curse concerns the metaphysical *us*, and it binds at the point of forced action: the shrug is itself a selection, the continuation is itself a selection, and each is weighed against a frozen answer or against nothing. Quietism toward the curse is therefore not a response to it but an instance of it. The difference is in kind, not degree, and no ranking of the problems is claimed or needed.

One thing has gone unexamined. The argument fixes that some frozen question is held; it is silent on which. Whether the silence is an opening is the question of §2.2.

## 2.2 The countermeasure

§2.1 closed on a silence: the argument fixes that some frozen question is held, and says nothing about which. The third property of Remark 2.3, selectability, is where the whole of §2's positive content enters, and it enters as one premise. An agent cannot decline to stand somewhere. What can differ is the footing.

*Premise* Φ6 (Minimal arbitrariness; *meta-normative selection rule*). Among the frozen questions an agent cannot avoid holding, prefer one whose answer is constrained by the agent's own existence over one whose answer is not.

*Remark* 2.6 (The falsifier). What would refute it, in its own vicinity, is one of two exhibits. Exhibit a frozen question whose answer is compelled by the agent's existence to a degree comparable with the candidate the section goes on to supply, and which delivers a *different* answer to the Golden Question — then Φ6 selects a family rather than a point, and its use downstream fails even though the premise stands. Or show that "constrained by the agent's own existence" admits no ordering at all — then the premise selects nothing and is empty rather than false. The first exhibit attacks the use; the second attacks the premise. Neither has been produced, and the section's downstream claims are priced against both.

*Remark* 2.7 (What the constraint is, and what it is not). The premise's whole content sits in one phrase, and the phrase should be said rather than leant on. The constraint is positional, not truth-conferring; nothing here reaches truth. Some frozen questions are such that the answer an agent holds is bound up with the agent's being in a position to hold it — the holding is not a free selection among alternatives but something the position has already narrowed. Others are not: an agent could hold either answer without anything about its situation differing. The first kind carries information about the agent's circumstances; the second is a coin already flipped, where remembering how it landed adds nothing. In the paper's own terms: Φ6 asks for the frozen question that behaves most like a conditioning event. The whole apparatus is built on conditioning, and the premise is the demand that one's grounding be a conditioned quantity rather than a free one — which is what makes Φ6 a resident of this framework rather than an import to it.

*Remark* 2.8 (Why the selection is scarce). Constraint alone is cheap, and degenerately so: *am I asking something?* is fully constrained by the asking and entirely inconsequential. What Φ6 asks for is a conjunction — an answer heavily constrained by position *and* consequential enough to carry what Condition 1 puts on frozen answers. The conjunction is what makes the premise non-trivial: candidates are not everywhere. And it is where a promise is cashed. Remark 2.2 held that a later candidate for the role would owe a demonstration of admissibility rather than an assumption of it; the conjunction is that debt restated as a criterion, and the demonstration follows once the machinery it consumes is in place.

Two further things about Φ6, and then the reconciliation a reader of §6 will arrive already holding. *First, it is a selection rule over an unavoidable set, not a derivation of a value.* Φ6 ranges over groundings, not over actions. It does not say the agent ought to have a grounding; Proposition 2.4 established that the agent has one whether or not anything is said, and that its content is underdetermined by everything the agent can access. What Φ6 adds is only this: arbitrariness admits of degrees, and among foundations the agent must occupy anyway, less is preferable to more. The ordering is ordinal and nothing stronger — some frozen questions are more constrained by the agent's existence than others, and no figure will be put on the compulsion, here or later. And a preference for the least arbitrary footing must not be inflated into the claim that the least arbitrary footing is *correct*. It is less arbitrary. Nothing in the paper can make it correct, and §6.8 is the standing reason: correctness of a grounding is exactly the kind of thing no finite observation certifies.

*Second, no ought is derived from an is.* The objection every reader will have is that Φ6 smuggles a norm out of a fact — that "your existence constrains this answer" is an is, and "therefore hold it" is an ought, and the gap between them is the oldest gap there is. The answer is that the ought is not derived because it is not new. Condition 1 of §2.1 already places the agent under forced action; the practical necessity of standing on *some* foundation is a structural fact about agency, in place before Φ6 says anything, supplied by the predicament and not by the premise. Φ6 operates entirely downstream of that necessity, on the one question the predicament leaves open — which arbitrary foundation to occupy. A rule for choosing among footings an agent will occupy regardless derives no obligation to stand; the standing was never optional. That is what *meta-normative* means in the premise's kind, and it is the whole of the premise's ambition.

**What Φ2 denies, and what Φ6 claims.** Premise Φ2 declares Golden reasoning inferential: it licenses conditional expectations, not duties. $Q$ is a conditional distribution; a distribution entails no obligation; Remark 6.7 leans on exactly this, and the fence of §9.5 will lean on it again. If Φ6 contradicted that, §2 would be dismantling the paper's own load-bearing wall. The split: Φ2 is a claim about what the mathematics *outputs*, and the mathematics outputs distributions, from which no duty follows, exactly as Φ2 says. Φ6 is a claim about what an agent does with a choice the mathematics never made for them, the choice of grounding, which Proposition 2.4 shows is forced on the agent by their situation and left open by everything the apparatus proves.

That the two premises do not meet can be checked by enumeration rather than taken from prose. The section's normative chain has four steps:

(i) an agent must hold some frozen answer (Proposition 2.4);

(ii) among the frozen questions the agent cannot avoid holding, Φ6 prefers the constrained (Premise Φ6);

(iii) one candidate is maximally constrained — discharged once the machinery it consumes is built;

(iv) if Φ6 is adopted and the candidate answered affirmatively, certain comparisons follow (§9.2).

At no step does a proposition of the form "$Q$ implies that one ought to $x$" appear. Step (i) is structural, step (ii) is adopted, step (iii) is a constraint claim, and step (iv) is conditional on both adoptions. The apparatus outputs distributions at every step and duties at none.

One further protection, stronger than the level-distinction. The grounding runs through a fact about the situation of *whoever is asking*, not through the interests of anyone who will exist in the futures $Q$ describes. Remark 6.7 answers an objection that requires such beings — beneficiaries whose weight the apparatus stands accused of smuggling. Here there are none in the argument at all: no future being's interests appear at any of the four steps, and an agent alone in the last hour of a biosphere could run them unchanged.

An agent who declines Φ6, then, is in conflict with nothing the paper proves, and the paper is, in exactly Φ2's sense, neutral about the declining. But declining has a shape, and §2.1 already named it. Declining Φ6 in favor of a competing selection criterion is coherent, and the section holds no argument against it. Declining it and thereby standing outside the problem is not a position: indifference among frozen questions is not abstention from the choice but the choice made without a criterion — a frozen answer of the least examined kind. The structure is Remark 2.5's, met one level up: quietism toward the selection is an instance of the selection.

What the paper is not neutral about is the price of adopting. An agent who adopts Φ6 *and* answers the Golden Question affirmatively does thereby acquire commitments — and the decision rule of §9.2 is those commitments made computable, which is a sentence that should be read as a cost rather than an advertisement. The accounting is this: Φ2 keeps the apparatus obligation-free; Φ6 is the single doorway through which obligation can enter; and everything on the far side of the doorway is the agent's adopted grounding, carried at [Φ], revocable by the agent and certified by nothing. Whether the doorway leads anywhere, whether there exists a frozen question whose answer an agent's existence does compel, is the question §2.4 exists to answer.

### 2.3 The Golden Tendency

§2.2 closed on whether the doorway leads anywhere; the candidate that answers is named in §2.4, and its credentials will consume machinery this subsection must build first. A late observer does not see what was made; they see what is left. Survivorship is not a force — it selects nothing, prefers nothing, and does not act; it is what remains when a population is observed at a later time than it was generated. This subsection states what remains, in two parts, and then says how much of it survives contact with the scoping the paper already imposes on itself.

The first part is arithmetic.

**Proposition 2.9** (The Golden Filter; **[P]**)**.** *Let a cohort of biospheres originate together with types $\theta \sim \nu$, each type carrying hazard rate $\lambda_\theta(\cdot)$ and survival function $S_\theta(t) = \exp\big(-\int_0^t \lambda_\theta(\tau)\, d\tau\big)$. Order types by efficiency so that $\lambda_\theta$ is pointwise nonincreasing in $\theta$. Then the composition of the cohort still standing at $t$ is the originating composition reweighted by survival,*

$$dN_t(\theta) \;\propto\; S_\theta(t)\, d\nu(\theta),$$

*and it stochastically dominates $\nu$, with the dominance nondecreasing in $t$.*

The proof is the statement: the number of type-$\theta$ members still present is the number that started times the probability one lasted, and normalizing gives the reweighting; the dominance is monotone in $t$ because the likelihood ratio between a more and a less efficient type, $S_\theta(t)/S_{\theta'}(t) = \exp\int_0^t (\lambda_{\theta'} - \lambda_\theta)\, d\tau$, is nondecreasing in $t$ exactly when the hazards are ordered — which is why the hypothesis orders hazards rather than survival functions. Nothing is hypothesized beyond the ordering. All of the content of the Filter is in what it gets applied to, and the deflation is worth insisting on, because the Filter is routinely mistaken for a substantive posit when it is a bookkeeping identity about two different distributions that were never the same object.

The galaxy is what the identity looks like when it is allowed to run. Relax the Rare-Earth hypothesis to galactic scale, let biospheres undergo abiogenesis independently and far apart, each drawing a hazard rate at random. The fragile fizzle; the durable persist and fizzle later. Watch the populations rather than the biospheres: the durable accumulate, because arrivals outpace departures, while the fragile turn over in waves and hold roughly constant. Then the observation recurses, among the durable, the most durable stand in exactly this relation to the rest, and the skew propagates into the tail as far as the tail extends.[3]

The Filter is best named by what it sits between. On one side is the distribution of *generation* — what abiogenesis produces, the Cosmic Nexus. On the other is the distribution of *abundance*

[3] Strictly, the galaxy is the renewal rather than the cohort version: origination is ongoing, so the standing population is a convolution of the origination rate with survival rather than a simple reweighting of a single $\nu$. Proposition 2.9 is exact for the cohort; the galaxy inherits the same concentration conclusion asymptotically by the same mechanism, and nothing below turns on the difference.

— what a late observer finds. The Filter is the map from the first to the second, and the map is nothing but conditioning on having lasted. In the paper's vocabulary this is the pair $\mathbb{P}$ and $Q$ of Definition 6.1: General and Golden, generation and abundance, the two objects the galaxy separates by hand.

*Remark* 2.10 (The Filter accumulates; it does not aim). The galaxy contains no striving. No biosphere in it is trying to be durable, no selection pressure is applied to any individual, and no biosphere's prospects improve by so much as an instant on account of the Filter. The durable accumulate because they were durable, and that is the whole mechanism. The paper's anti-teleological commitment (Remark 3.40, and the calibration attached to Optimation's name in [5]) is not a caution appended to this passage but a description of it: *is populated by*, *accumulates*, *is selected* are exact; *aims*, *seeks*, *progresses toward* are not, and every teleological reading of what follows is an error introduced by the reader rather than a claim inherited from the model.

The apparatus that makes this precise is already in the paper.[4] Three identifications carry it. The Filter *is* the survivorship interpolation $\mathbb{P}^{(t)} \to Q$ of §5.8, whose limit Theorem 5.75 establishes. That high efficiency accumulates *is* Lemma 5.74: every bias field is a nondecreasing function of the state, so conditioning on survival to any horizon is an upward tilt, at every horizon, with no further assumption. And the rate at which abundance departs from generation is the goldening rate $s/4t$ of (6.1), with the persistence exponent entering as the linear-response coefficient.

So far the ensemble has been a population of biospheres; the Tendency reweights a different one — the continuations of a single history, held in disagreeing superposition. The substitution is free, because survivorship reweighting does not consult what indexes its ensemble: Proposition 2.9 multiplies each member's weight by that member's survival function and renormalizes, indifferent to whether the members are separated in space or in superposition. The one assumption the substitution would cost, that continuations carry no term coupling them, is discharged by construction at §5.8, where the object is a conditioning of a single path measure, so no such term arises to be assumed away.

**Principle 1** (The Golden Tendency; **[S]**)**.** Let a biosphere have favorable technoanatomy, $\rho > 0$. Then the state space contains a region on which the total hazard tends to zero (Proposition 5.35(i)), and as $t$ grows, the futures in which the biosphere has descendants at horizon $t$ are drawn increasingly from that region.

*Remark* 2.11 (The claim is about an ensemble, not a path). This is the section's central fence, and it is stated here in the form a reader arriving from §6.4.2 will already expect. The Tendency does not say that a biosphere improves. Nothing here moves any system along any path; $\rho$ does not change in the fixed model, and what conditioning does is reweight an ensemble. The selection

[4] Provenance. A 2024 draft derived the Filter by hand. It defined survival efficiency as a quantile $x$ of the survival probability $s(t, x)$ at horizon $t$, introduced a "Survival Nexus" $N_S \equiv \{S > t\}$, and computed the efficiency expected under it as $\mathbb{E}[X_t] = \int_0^1 s\,x\,dx / \int_0^1 s\,dx$, exhibiting toy hazard families under which $\mathbb{E}[X_\infty]$ rose to 3/4 and 5/6 from the unconditioned 1/2. What that construction anticipated: the object (a survival-conditioned reweighting of efficiency), the direction (the conditioned expectation exceeds the unconditioned one, and increases with the horizon), and the two-Nexus framing (generation against abundance), which survives here intact. What it did not anticipate, and could not have: that the limit exists at all rather than merely trending; that the limiting object is a Doob transform (Theorem 5.75) rather than a family of reweightings indexed by $t$; that the approach is clockless; and that the result is scoped by anatomy in the way Remark 2.12 records. Its measured descendant is the anatomy selection of §6.3.2, where the crude 3/4 becomes $0.75 \to 0.84 \to 0.91$ across horizons. The 2024 equations are superseded and are not reproduced.

is Darwinian in form rather than developmental — which is exactly what the folk version of the technological-adolescence argument gets wrong, and what Cooper [6] is right to object to in objecting that no evidence supports a trajectory of moral improvement. The ensemble form is not a weaker version of the trajectory form. It is the version that survives that rebuttal, and it is the reason the section can afford the claim at all.

Where, then, would a trajectory reading live? Not nowhere, and not forbidden — unbuilt. It becomes available only where technoanatomy carries its own dynamics, so that anatomy may escape upward and fail to return (§5.6.3, Corollary 5.43); those dynamics are **[O]**. A reader who wants improvement is requesting a piece of the framework that does not yet exist — and which Remark 10.8 argues is required for independent reasons anyway.

*Remark* 2.12 (What the antecedent costs, and who is not entitled to it). The antecedent is anatomy, not regime, because the Tendency holds in two of the three cells that §6.2.1 distinguishes and they are exactly the cells with $\rho > 0$ — the cells in which the vanishing-hazard region the statement names exists at all. In the *critical* regime it holds in the form stated, and the machinery above is the machinery of it: the tail is polynomial, the limit exists, and the reweighting is the whole content. In the *viable* regime it holds and stops being interesting, because survival has positive probability outright and conditioning on it is an ordinary conditional rather than a limit statement. In the *trapped* regime the antecedent fails outright: $\rho \leq 0$ by Theorem 5.36, and by Proposition 5.35 no region of vanishing hazard exists to be drawn from — survivors concentrate on a single path, and the principle does not misfire there; it does not fire. So the Tendency is not a theorem, and calling it one would be the section's most attractive error. What is proved is that conditioning tilts upward (Lemma 5.74) and that the limit exists where the tail is polynomial (Theorem 5.75). That descendants' futures are *in fact* so drawn requires in addition that this biosphere has $\rho > 0$ — and by §6.8 no one is entitled to that antecedent from finite observation. The Tendency is therefore stated by an author who cannot certify its condition. That is the normal case rather than a defect: the conditional is what the mathematics supports, and the ceiling is what forbids upgrading it.

Two things remain before the subsection closes.

*The Lindy relation is prior work, and the priority is not the paper's.* That civilizational longevity follows a Lindy relation is established in [7], [8], [9], and [10]; §6.7(c) states this and states that the paper's only addition is the derived exponent. The Tendency inherits that concession without amendment. The 2024 material reached the Lindy relation by a different route, aggregate many independent hazards and the aggregate flattens, so near-constant hazard is what many hazards become rather than an assumption imposed on them, and that argument is sound about the object it addresses. The model, moreover, contains that object: the background channel of (5.6) is exactly a sum of many independent random exponentials, treated in Appendix B as a partition function with freezing. Where the 2024 material erred is that a component was described and taken for the whole. This paper does not diffuse $h$; it diffuses $-\frac{d}{dt}\log h$, one integration further out, which is what makes $\log h$ a $C^1$ object, the state degenerate and 3/2-self-similar, and the survival tail polynomial rather than exponential. The noise placement is the locating novelty flagged in the relation-to-literature passage of §5. The tail is a property of the evolution, and the aggregation argument, correct about the background channel, never reaches it.

*Contingency is not a competitor.* The standing reply to the Tendency is that it makes descendants sound determined — that a claim of this shape cannot hold of beings who choose. The gas law is the familiar answer: a statement about a volume is not falsified by the fact that no molecule obeys it, and the order at the aggregate is not a constraint imposed on any constituent from outside. But the framework's own answer is sharper, and it is not an analogy. The Tendency is a statement about a measure over futures, not a prediction about any one of them. Contingency is not what the Tendency competes with; contingency is what the measure is defined over. An objector who establishes that the future is radically open has established the existence of the object the Tendency reweights, and has said nothing whatever about the reweighting. The two claims cannot conflict, because they are not claims about the same thing.

What the Tendency does not supply is an agent-side counterpart. The mathematics describes a reweighting; whether a biosphere's conscious effort has an analogous structure, and what a decision rule built on it would compute, is taken up in §9.1 and not here. A related caution: the observation that a biosphere might generate a sub-component targeting its own extinction pressures, and that we are, as it happens, such a sub-component, reads as a coincidence too large to be an accident. It is not an exclamation but an inference, and the form it takes is the contrapositive of the Theorem of Aimless Optimation [5]: unequal weights under two Nexae entail non-independence from $H_N$. Stated that way it licenses a recorded correlation and nothing more.

## 2.4 The Golden Question

### 2.4.1 The statement

$\Phi 6$ selects nothing unless a candidate exists. The candidate:

> *Should our biosphere live forever?*

The contrast that locates it is with the cogito, and one sentence carries it: the cogito secures an existence claim and grounds nothing further, while the Golden Question is chosen for its downstream reach — its interest is entirely in what it grounds. (The historical placement beside Descartes is a separate and stronger claim, and it is §2.7's to make at its price.)

Two debts fall due here, and they are the two halves of Remark 2.8's conjunction. The *consequence* half is the admissibility demonstration that Remark 2.2 put on account: the question is not paradoxical, since either answer is coherent and neither dissolves the asking; and it is not inconsequential, since its answer bears on every forced selection Condition 1 names — a biosphere that should persist forever and one that need not are different backdrops for every decision an agent under forced action makes. That reach is the whole of the not-Descartes contrast restated as a property. The *constraint* half is the forcing argument.

Begin with what the asking itself supplies, which is less than it appears. By some future point the biosphere persists or it does not; the asking is an act of a persisting biosphere's sub-component, so wherever the asking occurs, persistence to date has already been conditioned on. The asker does not survey the dichotomy from above it: into every system in which the asking is placed, the asker is placed as well. This establishes what the question is about, the asker's own situation rather than a detached one, and it establishes no more. The bare fact of asking does not narrow the answer: an asker holding the negative sits on the same branch, with nothing about their situation differing, and by Remark 2.7's own test a position of that kind is a coin already flipped. The narrowing needs a mechanism, and the mechanism is the one the section already owns.

**Proposition 2.13** (Forcing; [$\Phi$])**.** *In the ensemble of biospheres in which a frozen answer to the Golden Question is held and acted upon at biosphere scale, survivorship reweights toward* *the affirmative: the weight carried by continuations in which the negative is held falls with the persistence of its host, and the affirmative incurs no corresponding penalty.*

(One placement for later: in the vocabulary of §9.7, the frozen answer is the paradigm *universe-sourced commitment node* — the existence proof consumed there.) The argument is three steps, each already established elsewhere. *Consequence.* By the admissibility just discharged, the answer bears on practice: an answer with no downstream difference would be inconsequential and inadmissible, so a biosphere-scale frozen answer organizes what its holder does. *The middle step.* A biosphere whose practices are organized around *this biosphere should not persist* does less to persist. The step is near-analytic, persistence-directed practice is what the affirmative organizes, and its absence is what the negative licenses, but near-analytic is not analytic, and the fence below states what would break it. *Reweighting.* Proposition 2.9 does the rest: the standing ensemble thins in hosts of the acted-on negative and does not thin in hosts of the affirmative, so the answer found held in persisting biospheres skews affirmative, increasingly in $t$. The asymmetry is real because it turns on what happens when the answer is *acted on*, not on the bare fact of asking.

**A stronger sense in which the question is the asker's.** One consideration bears on *which question* $\Phi 6$ selects rather than on which answer survivorship leaves, and it raises the price of the first falsifier of Remark 2.6: a rival frozen question, comparably constrained, delivering a different answer. The companion argues that a reasoner cannot detach from its biosphere even notionally. The dependence runs past life support to contextual-epistemic grounding, so the attempted separation dissolves the Aethic truth conditions the reasoner's own statements rested on, in the way imagining masslessness dissolves the frame under Mach's principle [5, 11]; the chain of inferences by which anything is fixed terminates in the reasoner's being a component of the biosphere.

If that holds, the Golden Question's subject is not merely correlated with the asker's existence — it is what the asker's capacity to frame *any* question is constituted by. That is a stronger reading of "constrained by the agent's own existence" than the positional one argued above, and a rival question would have to match it. Two limits, since the argument is easy to over-draw. It establishes a dependence and not an asymmetry: the negative answer is framed with the same constituted apparatus, so nothing here discriminates between *answers*, and the forcing argument's reliance on survivorship is not relieved. And it is the companion's argument at the companion's grade, imported here only to make the falsifier harder to satisfy rather than to close it.

**What agency is, in this framework, and why it is not a primitive.** One background commitment should be surfaced here, since it governs how every reference to an agent in this section is to be read. Agency is not a primitive of the framework: the companion's act-robustness doctrine forbids it [4]. What stands in for it is *contingency that continues to move* — an attribute whose weight in the Aethic progression keeps shifting across successive Aethae without settling to stated. That is the difference between a choice still open and a fact already in the books: the object in the sky, once its distance is resolved, is stated and one-and-done; whether to attempt the difficult thing is a weight that the progression keeps revising until it does not. An agent, on this reading, is a system some of whose relevant attributes are of the second kind, and "the agent chooses" is shorthand for the weight settling. Nothing in this section requires more than that, and Remark 2.14 below is what keeps the shorthand from being read as causal purchase.

*Remark* 2.14 (Position, not causal purchase). Φ6 is a rule for an agent; Proposition 2.13 is about an ensemble, and the bridge must be stated rather than implied. An individual's holding does not move their biosphere's hazard — one agent among a biosphere's many is not a lever on it. What the individual has is position: they are asking from inside a biosphere that has persisted, and by Proposition 2.9 that is evidence about which answer the biospheres one can be asking from tend to hold. The relation is evidential, not causal — the relation the record bears to the Accordance Principle throughout, and the one Remark 3.40 insists on when it refuses to let a conditioning event act. The constraint Φ6 asks after is positional in exactly this sense: what the asker's existence narrows is not which answer they can hold, the nihilist holds the negative unimpeded, but which answer their existence is evidence for at the scale the question itself concerns. No situational feature is invoked beyond existence within a persister, and the narrowing runs through the same survivorship that structures the whole paper.

*Objection* 2.15 (Why call the Golden Point good?). Grant everything. Survivorship selects the affirmative; biospheres organized around the negative thin out. Why does that make the Golden Point *good*? And if this work will not supply an objective morality, why is the conclusion not simply that persisting forever is bad and a biosphere ought to stop?

The objection is met by conceding it, and the concession costs nothing that was ever claimed. *Nothing here says the Golden Point is good.* Remark 2.16 states the limit already: what is established is positional stability, not truth, and reading a selection effect as a demonstration is the error Remark 3.40 guards against. Φ6 asks for the *least arbitrary* grounding among those an agent must occupy anyway, not the correct one — and §6.8 is the standing reason no argument could supply the latter, since correctness of a grounding is exactly what no finite observation certifies. A reader who concludes that the Golden Point is not good has contradicted nothing in this work.

*The second half has an answer that asks the objector to concede nothing about value.* One may hold that a biosphere ought to stop. Proposition 2.13 does not refute the position; it locates it. Held and acted on at biosphere scale, that answer removes its own holder, so the ensemble still holding it thins with time. The position is not false — it is *unstable*, and the difference is the whole of what this subsection claims. An answer may be perfectly coherent and still not be one the biospheres one can be asking from tend to hold.

*Remark* 2.16 (What the forcing argument shows, and what it cannot). The proposition is a conditioning claim of exactly the paper's usual shape, and it inherits the paper's usual discipline. It fixes what survivorship selects among persisting biospheres, and hence what an asker is positioned to find; it does not fix what is the case. Selection effects do not act (Remark 2.10), and being positioned inside the selection is not evidence that the selected answer is true — the constraint is positional, not truth-conferring, which is Φ6's own scoping (Remark 2.7) applied to Φ6's best candidate. What would refute the argument, in its own vicinity: exhibit a biosphere organizing its practice around the negative while doing no less to persist — the middle step is near-analytic, not analytic, and this is its exhibit; or exhibit a coupling by which affirmative-organized practice lowers persistence comparably, restoring the symmetry the argument denies. And the standing rival exhibit of Remark 2.6 remains open: a frozen question comparably compelled that delivers a different answer would leave this argument intact and its use void.

One consequence is worth its own paragraph. The selection just argued leans on Proposition 2.9, which reweights in every cell of the trichotomy, and not on Principle 1, which needs $\rho > 0$ to name a destination. The Golden Answer's selection therefore does not inherit the Tendency's antecedent: the answer is selected wherever there is survivorship at all, and $\rho > 0$

governs only whether what it affirms is *available*. That is the separation of §2.4.2 arriving from the mechanism rather than by stipulation.

### 2.4.2 Should and can

The question asks whether the biosphere *should* live forever, and the affirmative is what survivorship selects wherever an asker can stand. Whether it *can* is a separate matter, and Theorem 5.38 is the separation — a separation the mechanism has already exhibited, since the selection ran through Proposition 2.9, which operates in every cell, while availability is $\rho$'s business alone. In the trapped regime the purchase is exactly nothing: $P(\text{viable}) = 0$ identically (Theorem 5.36), so an agent there is positioned inside the selection of the Golden Answer and inhabits a biosphere that cannot act on it. The answer is yes and the thing is unavailable, and both halves of that sentence stand at full strength.

This is the paragraph that keeps the section from reading as optimism, and it should be taken at the calibration the paper has already set: we are not practically Golden to any interesting tolerance, and Golden reasoning does not claim we are (Remark 6.6). Nor does the forced answer sort the options it condemns: Remark 10.4 shows the extinctionist future and the settled-at-$x_*$ future carrying viability zero alike, so affirming the Golden Answer while settling is not a middle path but the same zero by a longer route. What follows from a forced *should* and an uncertified *can* is not despair and not reassurance; it is Remark 10.8's conclusion that anatomy must be a process — the one response that is neither.

### 2.4.3 Meaning, at its price

The meaning claim, stated at exactly its strength: for an agent who adopts Φ6, the slot that philosophy labels *meaning* is filled, and filled non-arbitrarily. That is the whole of it. The claim is downstream of an adopted premise, carried at [Φ], revocable with the adoption, and certified by nothing — the far side of §2.2's doorway, described rather than promoted.

Stating the claim's compatibility with Remark 10.9 is what earns the paragraph. That remark disclaims a verdict on human meaning-making issuing *from a source outside human self-interpretation* — and resists, at the paper's grade, the temptation to read the framework as supplying one. Nothing here supplies one. The grounding runs through Φ6, which is adopted rather than output; the answer is positional rather than true; and an agent who declines the premise receives no verdict of any kind, favorable or otherwise. What the disclaimed reading wanted was meaning underwritten from outside; what §2 delivers is an agent's unavoidable slot filled from inside, by a criterion the agent holds. Those are different kinds of claim, and only the first is disclaimed. The unconditional form, that this is what meaning *is*, search concluded, would collide with Remark 10.9 and Φ2 simultaneously, and it is not made.

This subsection was written before the apparatus that pays it existed; the reader should know the debt is now discharged. The meaning here purchased at its stated price is developed in full as the third route, inferred meaning, with its two-tier claim, its literature, and its defenses, at §9.8, and this early statement should be read as that development's reservation, not its substitute.

### 2.4.4 An alternative phrasing

One rephrasing is worth naming and not running: that Golden reasoning stands to agency's curse as the scientific method stands to skepticism, with *least circular* and *minimally arbitrary* as the same selection in two vocabularies. The argument that earns the parallel, and the burden the parallel carries, are §2.7's in full.

*Remark* 2.17 (The Question's neighbors: an ally and two opponents). The Golden Question has one close published ally and two standing opponents, and naming all three fixes what the affirmative does and does not claim.

*The ally.* Jonas's first imperative, "that there be a mankind", is the affirmative answer held as the ground of an ethics for the technological age [12], and it is the nearest published neighbor of $\Phi6$ read through the Question. The delta is the direction of support: Jonas *derives* the imperative, from an ontology of purposes this framework neither needs nor asserts, whereas here the affirmative is priced as a premise with its falsifier stated and its adoption left to the reader (§2.2) — and the forcing proposition supplies what Jonas's derivation cannot, a survivorship mechanism by which biospheres that persist come to hold the affirmative without anyone's having proved it.

*The opponents.* Williams's tedium argument and Hägglund's finitude argument hold that endless life would be, respectively, unlivable and valueless — meaning requiring mortal stakes [13, 14]. Both run at the individual scale, and the category shift to the biosphere already blunts them; but the sharper reply is supplied by the hazard base itself, and it is worth stating because it dissolves rather than resists the objection. The Golden Point is not the removal of danger: under the Cox construction the hazard is strictly positive at every instant forever, and unbounded lifespan means only that the accumulated hazard converges — immortality as a convergent series of mortal moments, never as invulnerability. So the condition both opponents claim meaning requires, live stakes, real fragility, something genuinely at risk, is satisfied at *every* time under the affirmative, and what the objections target is a picture of immortality (stakes ended, danger over) that the formalism does not contain. The finitude tradition's quarrel is with a Golden Point no one here has defined.

## 2.5 Golden Extrapolation

The conditioning of §5.8 may be anchored at any Aethus in a lineage; Proposition 3.20 establishes that the anchoring is recursive, and Remark 3.21 draws the two consequences — no spacetime event is privileged, no point in a lineage is privileged, and the appearance of privilege is an artifact of which cone one occupies. Golden Extrapolation is what those results look like read from the inside.

**Definition 2.18** (Golden Extrapolation; [$\Phi$])**.** To place oneself by Golden Extrapolation is to take one's position as an output of the conditioning rather than as its anchor: a point on the goldening interpolation $\mathbb{P}^{(t)} \to Q$ of Definition 6.1, indexed by one's own age, occupied for the same reason any point occupies its own.

The reading is available because the mathematics already refuses the alternative. An anchor is a stipulation, and Proposition 3.20 shows the conditioning does not require one — so a lineage point that took itself for the anchor would be claiming a privilege the construction declines to grant anywhere. What replaces the privilege is not diminishment: every point on the interpolation is an output in the same sense, and a position's being unprivileged says nothing about its stakes.

Nor is the gradient a figure. It is the family of measures the paper already owns, and placement on it carries a number. Definition 6.3 indexes the interpolation by age, a biosphere of age $t$ is a Practical Golden Biosphere to tolerance $\varepsilon$ on windows of length $s$ where $\mathbb{P}^{(t)}$ and

$Q$ agree to within $\varepsilon$ on $\mathcal{F}_s$, and by Proposition 6.4 with Remark 6.5, a position at age $t$ confers $\varepsilon$-Goldenness over windows of length $4\varepsilon t$: a constant fraction of one's own history, at every age. Self-placement by Extrapolation is therefore measured, not merely unprivileged. And $\mathbb{P}^{(t)}$ is a family of measures, not a path, so the reading sits on the right side of Remark 2.11 by construction rather than by phrasing.

One comparative remains, and the result that governs it is Remark 10.5: stakes rise monotonically along the chain, because a step late in it carries the improbability of every step behind it. A monotone sequence has no interior maximum. So the present may well be the most consequential moment in human history, and cannot be the most consequential in the biosphere's — under monotonicity that superlative is always ahead. The claim ranks no positions; what rises is what is at issue at them, and that is §10.1.4's business.

## 2.6 The Golden Imperative

The mathematics first, because it is already done. By §3.4.1, $G = \neg F \neg$, so $\nu F$ and $\mu G$ are complements outright: the dichotomy between reaching the Golden Point and reaching the Departure Point is inherited rather than argued, holds for arbitrary branching, and is exhaustive on definite states. **[P]**, and nothing in this subsection adds to it. What the subsection adds is a modal reading and a reframing, both carried at [$\Phi$].

**Principle 2** (The Golden Imperative; [$\Phi$])**.** Reaching the Golden Point is the only route by which the Departure Point is avoided.

The modal form is exactly as strong as the complementarity beneath it and no stronger: with exactly one of the two points given to occur, avoiding one *is* reaching the other, and the principle is a near-immediate consequence rather than a separate result — which is the accurate presentation. Calling it an *imperative* is the overlay, and the overlay imports $\Phi$6 and the Golden Answer: an agent who has adopted neither reads the principle as a true disjunction with no instruction in it, and the paper gives them no argument that they are wrong to. The expectancy form of the same claim, that every alternative to the Golden Point carries a finite expectancy, is §10.3's, stated once there; the two forms point at each other and neither is repeated.

*Remark* 2.19 (Departure is the losing condition, not death). One consequence of the modal definition should be stated before the outlook it licenses, because it is where this framework's primitives part company with those of the fields it borders. Astrobiology and futurism take *extinction* as the quantity to estimate. This work does not, and the reason is that departure and death come apart in both directions.

*A biosphere can be departed and alive.* Past $\mu G$ every continuation fails to advance, so the Golden Point is unreachable, modally, not merely improbably, and viability is zero. Nothing has yet happened to it. Its hazard has fired no more than before; its members are unaware; the physical extinction lies ahead and may lie far ahead. What has been lost is the *possibility* of the thing worth reaching, and that loss is complete at the moment of departure and not at the moment of death. In the framework's own terms, a biosphere departs when its fate is sealed and not when the sealing is executed, and the interval between the two can be long. The framework's name for the first event is the *phenomenological death* of the biosphere: the death this work prices arrives exclusively with the Departure Point, even where the biological end is yet to come, and the gap between the two is exactly the bounded remainder the trapped tail measures. This is the reading §10.1 consumes when it finds the undisturbed world already departed — entry into $\mu G$ by configuration is the death, and the hazards that later fire are its consequences.

*And a biosphere can be undeparted and certain to die.* The critical regime is the case: departure occurs with probability one, and life expectancy is nonetheless infinite (§5.4), because advancing continuations exist and merely carry measure zero. Such a biosphere has not departed, $\mu G$ is a modal condition and this one fails it, and the difference is not a technicality, since everything the survivorship transform computes is computed there.

So neither implication holds. Extinction probability is therefore not the quantity this framework estimates, and a reader who translates $\mathbb{P}(\mathcal{G})$ into "probability of not going extinct" has substituted a measure claim for a modal one and will get the trapped and critical cases exactly backwards. *What replaces it.* The existential dichotomy for a biosphere is Departure Point against Golden Point, not death against life. Physical death is downstream of the first and uninformative about it: where anatomy is adverse it is certain and its expectancy is finite (Theorem 5.36(ii)); where anatomy is favourable but recurrence holds it is certain and its expectancy is infinite; and in neither case does the timing of death report the state that decided the outcome. The losing condition is the closure of possibility, and it is the only thing in this framework that a biosphere can lose *once*.

**The outlook, in one sentence.** The dichotomy admits a plain statement of what it makes of life, and the statement should be given once at full width before the reframing below. Life, from its earliest origins on a planet, undergoes a race to reach its Golden Point before its Departure Point claims it. That is not a metaphor laid over the mathematics but the mathematics read in prose: the two points are complementary fixed points, every actual biosphere carries weight on both, and the weight moves. The reading corrects the character the Departure Point is usually given. It is not a dark finish line recently come into view, to be crossed or avoided by human ingenuity alone — as if it appeared with us. It has been the other pole of every biosphere's superposition since that biosphere began, and what is recent is only the capacity to move the weight deliberately.

That correction has a consequence for how the record should be read, and it is a continuity argument rather than a rhetorical one. The naive picture places the Departure Point ahead of us as one possible destination among several — a crossroads at the modern day, at which humanity may or may not turn the wrong way. The framework's picture is that at *every* moment of a biosphere's history there stood the same possibility of diverging onto it, and that our ancestors, in surviving improbably, were not preparing for a contest we would later fight but fighting the same one. By Principle 4 the ancestral struggle and the agency-mediated one are one activity under different mediation; what Remark 2.19 adds is that they have one *losing condition* as well, and it was available throughout. The Miracles of §3.8 are accordingly not lucky preliminaries to a modern crisis. They are earlier rounds of it, and the record we hold is the record of the rounds that were won.

Two things follow for reading the record. The stakes are higher now, by Remark 10.5, and only by that — the difference is magnitude, not kind. And no moment in the record was ever safe: a biosphere's not having departed at any given point is a fact about that point and confers nothing on the next, which is why §2.5 places one on a gradient rather than past a threshold.

*Remark* 2.20 (The losing condition's neighbors, and the deltas). Three bodies of writing stand near Remark 2.19, and each should be located exactly, because each contains part of the correction and none contains the object.

*Parfit's asymmetry* is the source intuition of the field: that the difference between near-total and total destruction exceeds the difference between none and near-total, because the second difference destroys the future rather than more of the present [15]. That is the modal claim in embryo, what total destruction uniquely removes is not more people but the *possibility*, and this framework reads its own fixed point as the formalization Parfit's intuition was waiting for: $\mu G$ is what "destroying the future" denotes, the asymmetry becomes a theorem about the poles rather than a judgment about magnitudes, and the separation results say where the intuition's usual event-probability rendering fails it.

*Schell's second death* is the nearest literary precursor: the death of humankind as an object distinct from, and worse than, the sum of first deaths — "the death of death," the cancellation of the unborn [16]. The kinship with phenomenological death is close enough that the delta must be said plainly: Schell names the object and mourns it; the fixed point defines it, separates it from biological death in both directions, and prices what follows. A named object supports rhetoric; a defined one supports theorems.

*The s-risk literature* has, under the banner "fates worse than extinction," been circling the departed-and-alive quadrant — futures in which a biosphere persists at scale inside a configuration from which nothing advances [17]. The framework supplies that literature the category it has lacked: such futures are not *worse than* the losing condition, they are the losing condition, occupied at length — departure with the trapped tail still running. And the exchange runs the other way at a stated price: a suffering-focused reader is precisely the mixed-currency reader of §8.6, for whom the residue argument honestly breaks; the framework locates their exit rather than disputing their values.

The reframing is what Definition 3.5 already carries, read at full width: the Departure Point is not the destruction of life, it is the inevitability of the destruction — so a biosphere past its Departure Point differs from the abiotic in tense, not in outcome. Two consequences. *The fundamental dichotomy moves.* The division that matters is not extant against extinct, and not abiotic against biotic, the latter boundary is not an extremum by any fundamental measure, but abiotic-or-Departric against Golden, since the complementarity above is exactly the statement that everything ends up on one side of that line or the other. *Life as it stands is an intermediate.* A biosphere still inside the disagreeing superposition between the two points (Remark 3.15) is neither of the things the dichotomy names; its standing is not a category but a pendency, resolved in one direction or the other by the same complementarity. Neither consequence is a demotion, pendency is where all the section's stakes live, but both are $\Phi 6$-conditional readings of a **[P]** result, and the tags do not mix.

## 2.7 Placement

The problem and its countermeasure now stand, and four neighbors sit close enough that silence about them would read as either ignorance or identity. This subsection states each delta once, at one property each, and then makes the historical claim the section has so far declined to make — at its price.

*Hinge propositions.* The closest neighbor by a distance is Wittgenstein's hinges [18], and a reader who knows them will have assumed identity since Definition 2.1. The resemblance is real and should be conceded in full: a hinge is exempt from grounding, functions as the foundation of a practice, and answers the child's *why* with *it just is* — all of which is the frozen question's phenomenology exactly. The delta is Remark 2.3, property for property. Hinges are world-directed: *here is a hand*, *the world existed before my birth* — claims about what is the case, held by believers whether or not any action is forced. Frozen questions of the curse's kind are action-directed and bind at forced action. And, the property that carries everything, hinges are not selectable: they are inherited with a form of life, held involuntarily, and the hinge framework contains no operation by which an agent chooses among candidate hinges, because on that framework the choosing is not a thing an agent does. Φ6 *is* such an operation. The premise the countermeasure turns on is thus not merely absent from the hinge framework but inexpressible in it, and that is the difference doing work. Neither framework is thereby refuted; they answer different questions, and the frozen question is not a hinge renamed.

*The Agrippan trilemma.* Regress, circle, or dogma [19]: every justification ends in one of the three, and the frozen question is the dogmatic horn — encountered from inside, by an agent under forced action, rather than catalogued from outside by an epistemologist. That much is placement, not addition. The addition is one sentence: the trilemma is symmetric over stopping points, it has no resources to prefer one dogma to another, since its whole content is that justification ends, and Φ6 is precisely a preference over stopping points. The section's claim against the trilemma is therefore exactly as strong as Φ6 and no stronger: a meta-normative selection among the dogmatic horn's occupants, priced at [Φ], falsified where Φ6 is (Remark 2.6).

*Error theory.* Mackie's argument from queerness [20] shares the first horn's terminus, no objective values, and draws from it a semantic thesis: moral claims are systematically false. The curse is indifferent to the semantics; it operates identically whether moral claims are false, truth-valueless, or expressive, because its engine is the agent's information state rather than the status of moral language. The sharper difference is that the curse survives the negation of Mackie's conclusion. Error theory assumes the absence of objective values; the second horn of §2.1 is what happens in their *presence*, and the communication argument has no counterpart in Mackie because his framework never needs one. The curse is not error theory with different vocabulary; it is what remains standing on both sides of the question error theory answers.

*Absurdism.* The nearest neighbor in temperament rather than structure [21]. The absurd is a confrontation, the demand for meaning against a silent universe, and the canonical response is a stance: revolt, lucidity, the refusal of both hope and suicide. The curse is narrower and colder: a structural fact about an information state, with no confrontation in it and no stance demanded. And the tradition sits inside the problem rather than beside it: the absurdist's assurance of unsureness is itself a frozen answer, held with the same groundlessness as any conviction it replaces. What the curse's framework adds is not a better stance but a different kind of object — Φ6 is a criterion, and a criterion is answerable to falsifiers in a way a stance is not.

*The historical claim.* What remains is the analogy the section has been holding back since its first sentence, and it is stated here in full so that its price can be attached in the same breath. Skepticism was never dissolved, and could not be; it was defanged, by a procedure that created a demarcation, the scientific against the ascientific, and the demarcation proved fecund. The procedure's own defense is circular and knows it: to deny the method's standing is to assert a framework in which knowledge and empirical observation are uncorrelated, which defeats the asserting, so the defense is circular and, by its own measure, the least circular available — no rival argument even bounds its own circularity. The parallel now claimed: agency's curse stands to philosophy as skepticism stood to inquiry; Φ6 creates a demarcation among groundings, constrained by the agent's existence, or not, as the method created one among claims; and *least circular* and *minimally arbitrary* are the same selection expressed in the two vocabularies, which is why the analogy is structural rather than incidental.

The price. Everything in the analogy up to fecundity is a restatement of results already graded — the demarcation exists if Φ6 selects, and Remark 2.6 says what would show it selects nothing. The placement of the present moment beside Bacon and Descartes is a different kind of claim: it asserts that the demarcation will produce work, and that is a forecast, evidentially empty today, carried at **[O]** and at nothing less. The scientific demarcation earned its history in retrospect, by centuries of what it enabled; this one has enabled §2.4 and §9, and whether that is the beginning of a comparable retrospect or the whole of a small one is exactly what no one now writing can know. The analogy is offered with its falsifier and its forecast separated, and only the first is asserted.

**The register of this section, stated for the reader who noticed.** This section is philosophical and does not pretend otherwise. It is not armchair, in the specific sense that it links to the mathematics at every load-bearing step, the two-condition theorem, the Filter, the ceiling, and the relation runs one way: *the mathematics limits the possibility space, and the philosophy is what stands out in what remains.* Where the section crosses from one register to the other it does so by naming a premise, carried at [Φ], rather than by letting a philosophical claim inherit a mathematical grade. So the section merges the practices of two disciplines deliberately, and the merger has a discipline of its own: nothing here revokes the grounding of §5 or §6, and everything here that is not a theorem says so.

## 2.8 Ledger

**Costs.**

1. *Non-emptiness* (Proposition 2.4), [Φ]. Falsifier: an agent under forced action all of whose answers terminate in something accessible that is itself a reason; or, on the second horn, a veracity certificate whose warrant does not draw on the transmitting source.

2. *Selectability* (Remark 2.3, third property), [Φ]. Falsifier: a demonstration that an agent's frozen questions are fixed by their circumstances rather than chosen within them — Proposition 2.4 would stand, and everything built after it would fall.

3. *Minimal arbitrariness* (Φ6, Premise Φ6), [Φ], kind: meta-normative selection rule. Falsifiers: a frozen question comparably compelled by existence that delivers a different Golden Answer (kills the use); no ordering on "constrained by the agent's own existence" (kills the premise).

4. *Forcing* (Proposition 2.13), [Φ]. Falsifiers: a biosphere organizing its practice around the negative answer while doing no less to persist (the middle step is near-analytic, not analytic, and this is its exhibit); a coupling by which affirmative-organized practice lowers persistence comparably. The setup separately voided by an asking that does not condition on persistence.

5. *The Golden Filter* (Proposition 2.9), **[P]**, under the hazard-ordering hypothesis; exact for a cohort, asymptotic for the renewal case.

6. *The Golden Tendency* (Principle 1), **[S]**, antecedent $\rho > 0$ inside the statement, the conditioned-from region existing by Proposition 5.35(i); regime scoping inherited from §6.2.1 unchanged; ensemble form only.

7. *The meaning claim* (§2.4.3), [Φ], Φ6-conditional throughout; falls with Φ6's falsifiers.

8. *Golden Extrapolation* (Definition 2.18), [Φ], a reading of cited results (Proposition 3.20, Remark 3.21, Proposition 6.4); the generation comparative governed by Remark 10.5 alone — possibly maximal in human history, never in the biosphere's, the superlative always ahead.

9. *The Golden Imperative* (Principle 2), [Φ] overlay on the inherited **[P]** complementarity of §3.4.1; instructive force Φ6-conditional.

10. *Placement's four deltas*, [Φ], each priced at Φ6's strength where they consume it; *the fecundity forecast*, **[O]**, asserted as forecast only, its falsifier and forecast explicitly separated.

**Buys.** A grounding for the paper's normative vocabulary that is stated rather than assumed — one premise, one doorway, and a four-step chain a hostile reader can check for the absence of any "$Q$ implies one ought" step. A meaning claim explicitly conditional on an adopted premise and therefore compatible with Remark 10.9. A should/can separation that arrives from the mechanism, the Golden Answer's selection runs through the Filter, which reweights in every cell, while availability is $\rho$'s alone, rather than by stipulation; in consequence, the selection does not inherit the Tendency's antecedent. A self-placement that is measured, not merely unprivileged. A provenance record that prices the 2024 material honestly, including where it took a component for the whole. And a placement that locates the section's one premise as inexpressible in its nearest neighbor's framework rather than merely absent from it.

**Not claimed.** That the Golden Answer is true. That any agent ought to adopt Φ6. That the biosphere will reach the Golden Point, or that $\rho > 0$ obtains — no one is entitled to the antecedent (§6.8). That agency's curse is dissolved; it is countered, and the counter is adopted. That anything in the section generates a duty — Φ2 stands, and the enumeration of §2.2 exhibits it standing. That any biosphere improves — every claim is ensemble-form, and the trajectory reading remains unbuilt at **[O]**. That an individual's holding moves a biosphere's hazard — position, not causal purchase. That constraint confers truth, that compulsion admits a number, or that constraint and consequence trade off by any stated relation. That any generation's contribution equals any other's — contributions rise with the stakes. That the historical placement beside Bacon and Descartes is anything but a forecast.

**Inherited dependency.** §2 inherits the exposure that §3.11 records, and more acutely. Everything after the countermeasure consumes survivorship over futures, and under a fixed adverse $\rho$ there is nothing of the kind to consume: §2.4.2 has nothing to separate, Principle 2 is satisfied vacuously, and the section's claims are then true and useless. The exposure is reduced only where technoanatomy is a process rather than a parameter, and those dynamics are **[O]** (§5.6.3); Remark 10.8 is the standing argument that the process is required regardless. The exposure is real and is not reduced by the arguments' internal soundness.

# 3 Continuing from Nexic reasoning

The companion development derives a typicality constraint, the Accordance Principle, and applies it to the natural-historical record under a weighting it calls Optimation: the cosmic weighting conditioned on observer-existence [5, 22, 23]. That weighting is there a construction, adopted because observer-existence is the obvious thing to condition on. This section argues that it is the wrong thing, that the right thing is the *Golden Point*, and that Optimation is recovered exactly as a consequence rather than assumed as a premise.

Two claims carry the argument. The *Past-Alignment Principle* (§3.5) states that conditioning the cosmic weighting on the Golden Point and then discarding every attribute outside one's own past light cone returns precisely the Nexus of Optimation — so that the record we observe is, attribute for attribute, the past-light-cone restriction of the Golden Nexus. The *Golden-Conditioning Hypothesis* (§3.6) argues by contradiction that this is no coincidence: observer-existence is too weak a condition to have produced the record, since it survives perturbations that the record does not. What conditioning on observers cannot explain, conditioning on the Golden Point does.

The consequence is a reversal of order. Optimation ceases to be the framework's starting point and becomes its first derived result; the Golden Biosphere ceases to be an object defined by survival asymptotics and becomes the conditioning target from which the rest follows. What the argument does *not* settle is the future: whether the alignment that holds behind us extends ahead of us is left to the three hypotheses of §3.9, which are stated and not adjudicated.

## 3.1 Vocabulary and standing conventions

**Convention 3.1** (Terms carried from the companion)**.** The argument runs in the companion's vocabulary [5, 22], and the six terms it uses are fixed here. An *Aethus* (plural *Aethae*) is a partial information state: some attributes stated, all others genuinely blank, individuated by its stated content and closed under entailment from it. The *Cosmic Nexus* is the occurrence-weighted measure over Aethae, with no conditioning absorbed into it. *Optimation* is that measure conditioned on the biosphere-level observer attribute, and the *Nexus of Optimation* the resulting weighting. The *Aetherian Nexus* is the Aethus of our own natural history — the one whose stated attributes we read off the record. *Centric unfolding* is the accumulation of stated attributes along a particular reasoner's history. And the *Golden Point* is the event of a biosphere's passage into the Golden regime, in the sense §6 makes precise; the *Golden Nexus* is the cosmic weighting conditioned on that event. The *Golden Aethus* is the Aethus stating exactly that event as a future-referring attribute, all else blank — the index of that conditioning (§4.6). Two usage notes are carried from the companion's fuller statement. *Optimation* names the conditioned measure; the fluke content a record so conditioned displays against the Cosmic Nexus's own rates is what the measure *shows*, and this work uses the word for the measure and speaks of its fluke content when the display is meant. And the Aetherian Nexus is not a weighting but an Aethus, the one of which our own Aethus is a proper child, and is used in that sense throughout. To *complement* a stated attribute is to replace it by a strictly mutually exclusive alternative: the perturbed Aethus states something incompatible with what ours states at that place, rather than lacking what ours has or carrying it alongside further content. Aethae being individuated by stated content, every perturbation in this framework is such a substitution, and perturbed Aethae are accordingly disjoint from ours rather than related to it by inclusion. An *Imperfection Magnitude Drop* is what results from complementing a stated attribute correlated with the remainder of an Aethus: the success-intersection of the perturbed history is displaced by many orders of magnitude, becoming drastically minute without reaching zero [5, §IMD]. The *High-Contrast Earth Hypothesis* asserts the size of that displacement and not its sign, the sign being supplied separately by the Accordance Principle. The Drop is the graded counterpart of departure and the object Definition 3.14 refines, and the *Strong Accordance Principle* supplies the direction as the statement $c > k$, with $k$ the prior-odds toll [23].

**Convention 3.2** (Grades)**.** As elsewhere: **[P]** proved, **[C]** cited, **[S]** argued at scaling or schematic level, **[O]** open, and [Φ] a philosophical premise carrying its own falsifier. This section's two central claims, the conditioning principle and the Golden Tendency's antecedent, are [Φ] and **[S]** respectively, and the extrapolation hypotheses are **[O]** by construction. What is **[P]** here is arithmetic downstream of a premise: the weight-ratio bound, the disjointness lemma it rests on, and the credence and Lindy laws of §8, each proved conditional on an input the section states and does not prove. Nothing here is asserted at a grade the argument does not reach.

### 3.2 Imported apparatus

This section consumes results from the companion program as premises, and a reader without those texts cannot check the arguments that follow. The apparatus is therefore stated here, in the companion's own terms and at its own grades, so that what is borrowed is visible and what is argued here is separable from it. Nothing in this subsection is claimed as new; nothing outside it is assumed.

**Principle 3** (Accordance; imported at [Φ])**.** For a natural-historical attribute $A$ with canonical alternative $A'$ its complement, and $H$ the production of an observer,

$$\frac{\mathbb{P}(H \mid A')}{\mathbb{P}(H \mid A)} \lesssim \frac{\mathbb{P}(A)}{\mathbb{P}(A')}, \tag{3.1}$$

read attribute by attribute over a partial state and never as a global optimisation over complete histories: *improbability on the path to an observer must be bought by observer-relevance* [5, 24].

The relation $\lesssim$ is not a tolerance. It is *defined* by an exceedance form: for any $\kappa \geq 1$,

$$\Pr\left[\frac{\mathbb{P}(H \mid A')}{\mathbb{P}(H \mid A)} > \kappa \, \frac{\mathbb{P}(A)}{\mathbb{P}(A')}\right] \leq \frac{1}{1+\kappa}, \tag{3.2}$$

the probability taken over the realisation, so violations are priced rather than forbidden and the whole tail profile is fixed with no free parameter. (3.1) is the $\kappa = 1$ reading. Two consequences travel with it and are used in this work: the alternative is *canonical*, the complement, so nothing is analyst-supplied, and the exceedance form is its own multiplicity guard, bounding the expected number of $\kappa$-fold violations among $n$ interrogated pairs by $n/(1+\kappa)$, with the standing caution that $n$ must count attributes *scanned* and not merely those interrogated.

**Convention 3.3** (The sum-collapse)**.** Over a finite product of $n$ interrogable slots of $b$ values each, alternatives at perturbation depth $\Delta$ are counted by $\binom{n}{\Delta}(b-1)^{\Delta}$ and suppressed by $10^{-c_{\mathrm{eff}}\Delta}$, so the total closes by the binomial theorem at $\big(1 + (b-1)10^{-c_{\mathrm{eff}}}\big)^n$. The realised configuration carries a share of order unity exactly when $c_{\mathrm{eff}} > \log_{10}\big(n(b-1)\big)$ — contrast per slot outrunning the logarithm of combinatorial supply [23].

*Remark* 3.4 (What is imported, and what this section owes)*.* Three notes on the standing of the above.

*They are premises here, not results.* Each is argued in the companion and none is re-argued in this work, so every claim in §3 that consumes one inherits its grade. A reader who rejects Principle 3 should read this section's conclusions as conditionals on it. *The companion's own limits transfer with them.* In particular the Accordance Principle is explicitly *not* a licence for global position inferences: a position in a completed roster has no attribute-wise translation, and the retort that the position could be made an attribute fails because its answer space presupposes the roster it was meant to replace [24]. Where this work makes a position inference, Ansatz 8.13, it does so on its own account and says so.

*The vocabulary this section runs on.* §3.7 and §3.8 state their arguments in the 2022 Laws, Second Law, Third Law, Golden Linkage, the $L$-threshold, while this subsection states the companion's constraint in the companion's current notation. The two are the same apparatus under older and newer letters; the companion's own notational-provenance section supplies the translation and records that the Laws are its constraint as first written, and a reader moving between the papers should use it rather than translate by hand [5]. Where this work refines a companion operation, the topological criterion on Imperfection Magnitude Drops refines the companion's counterfactual operation, and the coupling analysis of §3.4.1 consumes its closure formalisation, the companion now carries canonical anchors for both, and the dependency is checkable there.

*And some companion machinery is used more lightly than the above.* Where a further companion result is consumed at a single site, it is stated where it is used rather than here; the Accordance Principle and the sum-collapse are the two this section leans on repeatedly, and the vocabulary they run in is fixed at §3.1.

## 3.3 The problem: what does Optimation condition on?

Optimation is introduced in the companion as the cosmic weighting conditioned on observer-existence, and the choice is natural: an observer is what is doing the reasoning, so an observer is what the reasoning conditions on. The choice is nevertheless a choice, and it is load-bearing — every rarity verdict the companion issues is relative to it. The difficulty is that observer-existence is *robust* in a way the record is not. Strip from our history the coal that powered industrialization, or the metallurgical sequence that made rocketry possible, or any of a dozen similar contingencies, and what remains is still a world with observers in it. Observerhood survives these perturbations; the specific character of our record does not. A conditioning target that is indifferent to the difference cannot explain why we find the difference realized.

That gap is what the rest of this section fills. The proposal is that the operative condition was never observer-existence but the Golden Point, and that Optimation is what Golden-conditioning looks like when restricted to the region of spacetime one can actually see. Two questions follow, and the rest of this section answers them in order. *Which* target should replace observer-existence, and on what grounds — §3.4. And what reason is there to think the replacement is not merely better motivated but actually the condition under which the record was produced — §§3.5–3.7, in two independent legs.

## 3.4 Choosing the target: continuity and non-privilege

Before the argument for Golden-conditioning, a prior question: why the Golden Point rather than some more familiar target? "Technologically advanced civilization" is the obvious candidate and is the wrong one, for two reasons that compound. It is not well-defined. Any threshold one names, writing, metallurgy, industry, spaceflight, is a stipulation, and the arbitrariness is not cosmetic: every rarity verdict issued relative to such a target inherits it.

And it is *discontinuous with natural history.* A conditioning target must be applicable across the whole record if the record is to be read as one object. "Civilization" applies to the last few millennia and to nothing before, so a framework built on it cannot run a single conditioning across both the end-Cretaceous impact and the Industrial Revolution. It must instead splice two analyses at a joint it has no principled way to locate.

**Definition 3.5** (Departure Point)**.** The *Departure Point* is the event past which *every* continuation fails to advance — the first moment at which all scenarios lead to a failure to advance past that point. The modal quantifier is the definition and not an embellishment of it: what is being named is the closure of possibility, not the observation of non-progress. *This is what makes it the dual of the Golden Point.* That was defined as the point past which a biosphere is too capable at self preservation to regress past it — which, stated with the quantifiers exposed, is the point past which *some* available course has *every* continuation avoiding foreclosure. The asymmetry between the two is deliberate: departure is $\forall\exists$ — every course admits a foreclosing continuation — and Goldenness is $\exists\forall$ — some course admits none.

Two $\forall$-statements could not be complements of one another; $\exists\forall$ and $\forall\exists$ can be, and §3.4.1 shows that these two are, by exhibiting them as the complementary fixed points of De Morgan-dual operators. The quantifier asymmetry is therefore not a wrinkle in the duality but its mechanism. The pair is exhaustive on definite states, and an Aethus carries weight across both. §3.4.4 makes that precise, since it is best stated once the grading is in hand. The reading this forces bears stating, since a looser gloss would lose it. Goldenness is not the guarantee that nothing can go wrong; it is the possession of a *course* along which nothing does, whatever the environment answers. It is a claim about available strategy, not about the absence of hazard — which is why a Golden biosphere still faces a universe that would destroy it and is Golden anyway.

Both are *points* for the same reason: irreversibility. A state one may still leave is not a point but a phase, and neither concept is about phases. Two consequences of stating it this way. *Biological death is a mechanism, not the definition.* Death is one way a trajectory is foreclosed, and it happens to be the most visible one; irreversible loss of the capacity to advance is another, and the definition is indifferent between them. What is extracted is the *phenomenology of foreclosure*, taken apart from the particular mechanism of biosphere-death in which it is most legible — and once extracted it applies wherever there is a trajectory to foreclose, which is why the notion carries across the scales Principle 4 requires.

*Non-advancement alone does not suffice.* A biosphere that fails to progress for an age has not thereby reached its Departure Point; it has had a lull, and a lull is reversible. Only when no continuation remains that advances has the point been passed. This is the distinction that the phrase "failure to advance" elides and that the quantifier restores. Two consequences for how the model of §5 should be read. The hazard $h_t^{\mathrm{tot}}$ of (5.7) is the hazard *of foreclosure*, not of death; the muffling exponent $x$ is accumulated against that; and every Imperfection Magnitude Drop is a movement toward it. And the analytic form of the modal claim is *pathwise* divergence, $\Lambda_\infty = \infty$ along every continuation, which the trapped regime satisfies outright (Theorem 5.36(i)), of which $\mathbb{P}(\Lambda_\infty < \infty) = 0$ is the measure shadow; the two come apart exactly where Remark 2.19 separates them, since the critical regime carries the shadow while advancing continuations survive at measure zero. This is why the trapped regime of Proposition 5.35(iii) has departed: not because a biosphere halting at $x_*$ has stopped, but because with $\rho < 0$ the viability probability is identically zero, so no continuation available to it advances. Halting is not the departure; having nowhere left to go is.

The term is used rather than "extinction" deliberately, and the reason is the same one that governs the choice of the Golden Point itself. "Extinction" is a biological category, applying to taxa and carrying the vocabulary of one discipline; the Departure Point must apply at every scale the record contains, to a Mesozoic lineage as readily as to a biosphere tomorrow, and must name an event in the Aethic block universe rather than a fact about organisms. It is therefore defined at the same primitive level as the Golden Point, which is what allows Principle 4 to be stated at all.

*Internal and external.* The Departure Point divides by the locus of its cause. An *internal* Departure Point is brought about by factors within the biosphere's own homeworld; an *external* one is inflicted from outside, and is roughly constant in magnitude over long timescales because the cosmos is roughly constant in makeup. That division is the ontological origin of the two channels of (5.7): the crest channel carries the external hazard and is bounded, the trough channel carries the internal and is not. Technoanatomy (Definition 5.32) is then the statement of how the second responds to capacity acquired against the first.[5]

### 3.4.1 The implication cascade: departure as a least fixed point

Definition 3.5 carries a recursion that is worth making explicit, because it is already available in the companion framework in finished form and because it changes what the model's hazard is a hazard *of.* The internal and external hazards of (5.7) *imply* departure; they are not exhaustive of it. A biosphere may reach a configuration from which no route avoids those hazards, and by the modal definition that configuration *is* a Departure Point, attained before either hazard has fired. The statement "no matter what you do, you reach a Departure Point" is itself a Departure Point — and since that statement can again be true one step earlier, the implication propagates.

*Remark* 3.6 (This is the third postulate's operator, and it is proved). The companion's third postulate supplies exactly this recursion [4]. Over states ordered by descent, with Base collecting the non-recursive modes, it defines a safety operator and its dual

$$
\begin{aligned}
F(Y) &= \big\{A \notin \mathrm{Base} :\ \exists B \sqsubseteq A,\ \forall C \sqsubseteq B,\ C \in Y\big\},\\
G(X) &= \mathrm{Base} \cup \big\{A :\ \forall B \sqsubseteq A,\ \exists C \sqsubseteq B,\ C \in X\big\},
\end{aligned}
$$

both monotone, so that by Knaster–Tarski the greatest fixed point $\nu F$ and least fixed point $\mu G$ exist for arbitrary branching and are complements of one another; for finitely-branching trees the closure ordinal is at most $\omega$, so the cascade terminates.

Read Base as departure by internal or external hazard — departure-sufficient occurrences in the sense of Remark 5.6, with nothing about death presupposed. Then $G$ is the doom operator in the sense above, every route has a doomed continuation, and the Departure Point is

$$\mathcal{D} \;=\; \mu G \;\supsetneq\; \mathrm{Base}. \tag{3.3}$$

Three consequences, each already carried by the companion's machinery rather than assumed here.

*The implication is one-directional by design.* The companion states the third postulate as an implication and not a biconditional, which is precisely the containment in (3.3): the hazards

[5] *Provenance.* The Departure Point, its internal/external division, the constancy of the external threat, and the observation that a biosphere may come to draw energy out of itself are all set out in our *Golden Science* of August 2022 [25], some years before the present formalism and before the drafting collaboration. That document also states the Golden Point as the point past which a biosphere cannot regress, derives a positive immortality probability from an exponentially decaying hazard, argues for "biosphere" over "civilization" as the unit, and names the Trace Connection as an open problem — which §3.5 is the attempt to close. The formalism here should be read as supplying machinery for commitments already made there, not as generating them.

suffice for departure and are not necessary to it. *The two-generation form is the companion's, and was not imported.* The quantifiers above are the third postulate's own: $F$ asks for a child all of whose grandchildren are safe, $G$ for a state every child of which has a doomed grandchild, and it is the *pair* of generations that makes the operators monotone and the fixed points complementary. Read that way the Golden Point is an event horizon in a precise sense, the state past which the capacity to trigger departure is closed off, which is $\nu F$'s $\exists\forall$ said in the other direction, and the Departure Point is its dual, the state at which every remaining possibility retains that capacity. This parallel is recorded rather than used, and its provenance is worth a line: the companion's derivation was conducted under a discipline of *not* reaching for parallels to the Nexic and Golden material, on the ground that the surprise of agreeing superpositions gave no warrant to expect every construction to transfer. The parallel therefore holds without having been arranged, which is a weaker thing than a proof and a stronger thing than a resemblance.

*The Golden/Departure duality is a theorem, not a construction.* The Golden Point is $\nu F$, there exists a route all of whose continuations remain safe, and the Departure Point is $\mu G$. Since $G = \neg F \neg$, the two fixed points are complements outright, so the duality asserted in Definition 3.5 need not be stipulated: it is inherited, and the pair is exhaustive. *The complementarity is a fact about definite states.* The fixed-point analysis lives on the Boolean lattice of definite states, where the result above is exact. An Aethus is a weighted superposition of such states, and what that implies for the categories is taken up in §3.4.4.

*Human extinction sits in the difference* $\mu G \setminus \mathrm{Base}$. It is not itself a biosphere-level departure by internal or external hazard, since the biosphere persists; but with no muffling agent remaining, every continuation reaches Base (Proposition 10.2). It therefore enters at the first step of the cascade. The classical reading, that human extinction is a Departure Point because it is only a matter of time, is the $n = 1$ case of the recursion, and (3.3) is what licenses it.

*Remark* 3.7 (The model's hazard is a lower bound). The consequence for §5 should be stated plainly, since it qualifies every rate in this work. The hazard $h_t^{\mathrm{tot}}$ of (5.7) is the hazard of Base: it prices the internal and external mechanisms and nothing else. The hazard of *departure* is the hazard of $\mu G$, which by (3.3) is at least as large and in general strictly larger — because a biosphere departs when its fate is sealed, not when the sealing is executed, and the interval between those can be long.

So the model's timescales are conservative in a specific direction: where it reports a departure time $T$, the Departure Point proper may have been passed considerably earlier, and the survival curves of §5.4 are upper bounds on the probability of not having departed. Nothing in the asymptotic analysis is disturbed by this, a cascade that fires earlier by a bounded lead time leaves exponents untouched, but the quantitative readings inherit it, and the anatomy urgency of §5.6.3 is sharpened by it: the window in which one may still act closes at the cascade's rate, not at the hazard's. Computing that rate is open **[O]**; §5.7 is a first attempt at it, exhibiting the excess as a third hazard channel whose completion weight is a least fixed point (5.36).

A second approach, complementary rather than competing, is available in the companion and is worth recording here because it constrains what the rate can be. A sealing-hazard is not a hazard on a trajectory but a *settlement* rate on the books: what it prices is the acquisition of statedness by the attribute that the fate is sealed, and that is an event of the present index whatever its referent's date — statedness being a property of the indexing Aethus and never of a clock reading, so that blanks lack tense [4, 26]. Three consequences follow, and the first is restrictive. The weight the current books place on a fixed future-referring attribute is, by the framework's law of total probability, a *bounded martingale* along the progression, the current state prices its own next resolutions, so its expected rate of increase vanishes identically and *the sealing-hazard cannot be a drift.* What survives as a per-unit-time object is the settlement flux, the weight mass absorbed into statedness per unit proper time, together with the direction-blind quadratic-variation rate, which a refined account could consult alongside it. And since a bounded martingale converges, the current weight audits its own hazard: it decomposes into the price of eventual settlement plus the expected never-settled residue, so the number available at an instant is already an integral over the flux.

That last observation has a consequence for how the open problem should be posed, and it is the one worth carrying. The flux and the level are not rival candidates for *the* sealing-hazard; they are the two ends of a one-parameter family, obtained by pricing settlement at a discount,

$$h_\gamma(B) \;=\; \mathbb{E}_B\big[e^{-\gamma(\rho-t)}W_\rho\big], \qquad \gamma \geq 0, \tag{3.4}$$

with $\rho$ the earlier of settlement and the path's terminus. At $\gamma \to 0$ optional stopping returns the level itself; at $\gamma \to \infty$ the exponential concentrates and $\gamma h_\gamma$ returns the instantaneous rate. And the kernel's shape is *forced* rather than chosen: requiring the pricing to commute with refinement of the progression, the price of the children's prices returning the parent's, makes the delay factor multiplicative across concatenated intervals, whose bounded monotone solutions are exactly the exponentials [4].

Two things follow for this work. The exponential of the Cox construction (Definition 5.4) is therefore derivable from partition consistency rather than adopted with it. And this paper already uses both ends of the family without having named them as ends: (5.7)'s $h^{\text{tot}}$ is a rate, the $\gamma \to \infty$ face; the schedule channel's completion weight (5.36) is a level, the $\gamma \to 0$ face. Reading them as one family says what the open rate of this remark actually is — not a single number awaiting computation, but a curve whose urgency parameter is itself a modelling commitment, and whose mixtures over urgencies are, by Bernstein's theorem, exactly the completely monotone kernels.

The construction is developed at individual scale in [4], where the settling attribute is lineage termination; what transfers here is the form, by Principle 4, and what does not transfer is any of its quantities.

**Proposition 3.8** (One permanent floor suffices; **[P]** given the model)**.** *Suppose some channel $h_j$ of* (5.7) *acquires a* permanent floor*: there are $c > 0$ and $t_0$ with $h_j(t) \geq c$ for all $t \geq t_0$, no route remaining by which it falls arbitrarily again. Then $\Lambda_\infty = \infty$, so $\mathbb{P}(\Lambda_\infty < \infty) = 0$ and the state lies in $\mu G$.*

*Proof.* The channels are non-negative, so $h^{\text{tot}} \geq h_j$ pointwise, and

$$\Lambda_\infty = \int_0^\infty h_t^{\text{tot}}\,dt \;\geq\; \int_{t_0}^\infty h_j(t)\,dt \;\geq\; \int_{t_0}^\infty c\,dt = \infty. \qquad \square$$

Two things the proof does *not* use are worth marking, since both are tempting and one is false. It uses no property of the other channels: in particular the internal channel need not participate, and a permanent floor in the external channel alone places the state in $\mu G$ — which is the claim's whole point, that one does not need the base hazard entire. And it does not pass through "$\Lambda_\infty < \infty$ requires $h_t^{\text{tot}} \to 0$", which would be the natural route and is not available: a non-negative function may have finite integral without tending to zero, as tall narrow spikes of shrinking width show. The floor hypothesis makes the detour unnecessary, since it bounds the integrand below on a set of infinite measure directly.

*Example* 3.9 (Human extinction as an Imperfection Magnitude Drop). Human extinction is an Imperfection Magnitude Drop, and for reasons that are structural rather than sentimental. The grades differ across the parts and are marked where they fall: (A) and (B) below are claims about historical individuation [$\Phi$], their combination with Proposition 3.8 is [**P**] given the model, and the smallness of $\varepsilon$ is imported at the companion's grade (Remark 3.11). Two reasons, kept separate because they are referred to separately below:

(A) *The lineage is spent.* The cascade to human evolution ran through the human–chimpanzee last common ancestor, a particular historical individual which no longer exists to be run through again.

(B) *The setting is spent.* The specific sequence of climatic and geographical events that brought that lineage down to the grasslands has been cashed out, the world now standing at a different stage.

The pathway is therefore not merely improbable on repetition — it is *unrepeatable*, its preconditions spent. Two claims must be kept apart here, since the argument moves between them and they carry different values. That *the human route recurs* is not improbable but unavailable: (A) and (B) name stated attributes that are gone, so this contributes nothing at all to what follows. That *the Golden Point is reached without* (A) and (B) is a different proposition, and it is minute rather than zero — an Imperfection Magnitude Drop, in the exact sense the next paragraphs derive. Everything positive that remains after human extinction belongs to the second; the first is simply closed.

*The derivation, run on (A) and (B).* An Imperfection Magnitude Drop is a targeted perturbation away from one's own Aethus, yielding a natural history whose success-intersection is drastically minute but never quite zero [5, §IMD]. Human extinction complements (A) and (B): the last common ancestor is no longer a stated attribute available to be run through, and the climatic-geographical setting has been cashed out. Any Aethus continuing from that state therefore carries those two attributes complemented.

Two established results then apply, in an order that matters. The High-Contrast Earth Hypothesis fixes the *magnitude*: complementing an attribute correlated with the remainder of the Aethus displaces $P(H \mid \cdot)$ by many orders of magnitude, to an effectively random order — the hypothesis asserting the size of the gap and not its sign [5]. The Accordance Principle then fixes the *direction*, placing ours on the greater side. The perturbed Aethae are accordingly drastically minute, which is what it is to be an Imperfection Magnitude Drop.

*Why the deep-time reply does not reach it.* "Re-evolving humans is unlikely" invites the answer that deep time makes unlikely things happen, and a post-extinction biosphere has deep time in abundance. That answer would land against a claim about rates. It does not land here, because the displacement is a property of the *perturbation* rather than of a waiting period: complementing a correlated attribute re-randomizes the success-intersection wherever in time the continuation is read, and elapsed time supplies no mechanism returning it to the unperturbed value. The claim is thus made within probability and not outside it, as the Disaster-Difficulty Unification Principle requires — the two sit on the severity spectrum wherever their probability puts them, and what distinguishes them is not their kind but that their probability does not accumulate.

### 3.4.2 Alternative lineages, by disjointness

The objection that remains is the one about successors: humans will not return, but some other lineage, another primate, or the corvid or cetacean lineages in which complex cognition has arisen independently, might find its own way to the Golden Point. It is answered by the derivation already given, and by a single observation about what such a lineage is.

*Proposition* 3.10 (Every alternative lineage is an Imperfection Magnitude Drop relative to (A) and (B)). *Let $L$ be any extant lineage other than our own, and let $A_L$ be the Aethus of a natural history in which $L$ reaches the Golden Point. The attributes of $L$ mediating its route are* disjoint *from those mediating (A) and (B): $L$ did not descend through the human–chimpanzee last common ancestor, and was not the lineage driven into the African grassland setting. That disjointness is not an added premise — it is what makes $L$ an alternative lineage rather than ours. Hence $A_L$ carries (A) and (B) complemented, and by the displacement above $A_L$ is an Imperfection Magnitude Drop relative to our own Aethus.*

Three features make the proposition economical. The disjointness is definitional, so it need not be established lineage by lineage. The displacement is already in hand, so no second mechanism is introduced. And the conclusion attaches to *each* alternative individually, so nothing turns on how many there are. Each is a drop, for the same reason, by the same result. Two clarifications, since the proposition is easy to over-read.

*It does not say a successor is impossible.* An Imperfection Magnitude Drop is minute and never zero, so $A_L$ retains positive weight for every $L$. What is claimed is that the weight is displaced by the contrast, not that it is annihilated — and the residual is precisely the $\varepsilon$ of §3.4.4. *It does not depend on which lineages are named.* The primates, corvids and cetaceans are referents rather than premises: they say what the disjoint attributes belong to. The proposition would hold for a lineage nobody has thought to nominate, since disjointness from (A) and (B) is a consequence of being a different lineage and not a fact discovered about any particular one.

### 3.4.3 The conclusion

The pieces now assemble, and the last step is worth running in full rather than by reference, since it is where a claim about *lineages* becomes a claim about the *Departure Point* and the two are not obviously the same kind of thing. (A) and (B) remove the human route; Proposition 3.10 makes every alternative route an Imperfection Magnitude Drop. Consider then any branch carrying no new muffling agent. On such a branch no route remains by which one re-arises, that is what the two previous results jointly establish, and a channel whose reduction depends on an agent that no longer exists and cannot re-arise is a channel that will never again fall. Fix any $c$ below its prevailing value and any $t_0$ after the loss: $h^{\text{ext}}(t) \geq c$ for all $t \geq t_0$, which is precisely a *permanent floor* in the sense of Proposition 3.8.

That proposition then closes the argument in two lines. Channels are non-negative, so $h^{\text{tot}} \geq h^{\text{ext}}$ pointwise, and

$$\Lambda_\infty \;\geq\; \int_{t_0}^{\infty} h^{\text{ext}}(t)\,dt \;\geq\; \int_{t_0}^{\infty} c\,dt \;=\; \infty,$$

so $\mathbb{P}(\Lambda_\infty < \infty) = 0$. By Definition 3.5 that is the analytic form of the modal claim, no continuation advances, and the state therefore lies in $\mu G$. One such channel suffices, and no property of the other channel, of $\rho$, or of the driver is used anywhere in the step: a single floor anywhere in the total hazard is enough, whatever the rest of the model is doing.

It is worth marking what this does *not* require. It does not require that the floor be high, only that it be positive and permanent; a biosphere left at a very low but irreducible external hazard departs just as surely as one left at a high one, merely later. And it does not require extinction to be the mechanism — what carries the step is the loss of the *agent*, and any event removing the capacity to muffle without removing the biosphere would carry it identically. On every such branch, therefore, human extinction is a Departure Point *outright* — in the modal sense of Definition 3.5, and not merely a step toward one.

Across the full Aethic weight the statement is graded rather than outright. Positive weight survives on the branches Proposition 3.10 prices, those on which some alternative lineage reaches the Golden Point, and $\varepsilon$ is that weight. It is positive because an Imperfection Magnitude Drop is minute and never zero, and small because each such branch takes the displacement of the High-Contrast Earth Hypothesis (Remark 3.11). Its formal statement is §3.4.4.

*Remark* 3.11 (What carries the argument, and what would break it). The conclusion closes an objection that would otherwise stand against §10, that a successor lineage might re-evolve the muffling capacity, so the accumulation is inherited rather than forfeited, and it closes it with the same machinery that establishes the drop, not with an additional argument. What is contributed here is only the identification of (A) and (B) as the attributes perturbed, together with the observation that any alternative lineage is disjoint from them (Proposition 3.10); the displacement is then supplied by the High-Contrast Earth Hypothesis and its direction by the Accordance Principle, both imported at the companion's grade [5, 23].

Three things would break it, and they attack different components. That (A) or (B) is misidentified, that the last common ancestor or the climatic sequence is not in fact a stated attribute of the Aethus in the relevant sense, is an attack on this section's own contribution. That the contrast is not vast, or that the interrogated attributes fall outside the correlated class the Hypothesis is scoped to, is an attack on the imported magnitude. And that the direction runs the other way is an attack on the imported Accordance step. The first is where a critic should aim if the target is the present argument rather than the framework it runs on.

### 3.4.4 Grading, and why two states suffice

The cascade settles what a Departure Point *is*. What remains is that no realistic state satisfies it exactly, since some weight always survives on branches that advance — so the operative statements are graded, and the grading attaches to weight rather than to kind.

*Remark* 3.12 (The recovery branch: a genuine Aethic superposition, and the grading it supports). An Imperfection Magnitude Drop does not drive the probability of later recovery to zero, and this is definitional rather than incidental: a drop is a perturbation whose success-intersection is drastically minute *but never quite zero* [5]. The weight remaining after human extinction is therefore exactly the drop's residual weight — that of the perturbed Aethae in which the Golden Point is attained regardless. Nothing further is needed to say why it is positive, and nothing further is needed to say why it is small: the first is what an Imperfection Magnitude Drop is, and the second is the displacement of Example 3.9.

That small positive weight sits in apparent tension with a Departure Point defined by a strict modal quantifier: if *some* continuation advances, however faintly, then not every continuation fails, and the label should not apply. *The post-drop state is an Aethic superposition, and that* *is the whole of the resolution.* An Imperfection Magnitude Drop does not replace one state by another; it produces a state carrying both a departed component and a recovery component, each genuinely stated, in whatever proportion the physics dictates [27]. Neither component is an artifact of description, and neither is deleted by the other being large. The recovery branch is therefore *present in the Aethus*, not absent from it and merely improbable — which is precisely the condition a strict universal quantifier is sensitive to.

*So the label does not attach, and grading is forced rather than chosen.* If a component that advances is genuinely stated, then it is false that every continuation fails, and Definition 3.5 does not apply however small that component's weight. What the situation calls for is not a verdict about whether the state is departed but a measurement of how much of it is: the superposition is graded, so the description of it must be graded too. Nothing is rounded and nothing is discarded to reach that conclusion — the grading is read off the state rather than imposed on it.

*Remark* 3.13 (The graded object at individual scale, and a result that transfers)*.* The definition below has an exact individual-scale instance in the companion, and the companion proves something about it this work does not: for a child $B$ of a state with survival weight $s_A$, define its *death grade* $g(B) := 1 - s(B)/s_A$, the fraction of survival weight the child destroys. Then the level-price is *affine* in the grade, and a graded slide of grade $g$ is equivalent at zero urgency to a mixture placing weight $q = g$ on total settlement and $1 - q$ on no change [4]. Read across, an $\varepsilon$-Departure Point is a child of death grade $1 - \varepsilon$ at biosphere scale, and the affinity and the mixture equivalence transfer unchanged — which is worth recording because it says the grading is linear in exactly the coordinate the survivor's own accounting uses (Remark 5.3), and not in its logarithm.

*Definition* 3.14 ($\varepsilon$-Departure Point)*.* A state is an *$\varepsilon$-Departure Point* when the total Aethic weight of continuations that advance is at most $\varepsilon$. A Departure Point in the strict sense of Definition 3.5 is the case $\varepsilon = 0$.

Human extinction is then an $\varepsilon$-Departure Point with $\varepsilon$ the weight of the recovery branch — which is to say, in the phrase that states the situation exactly, *almost precisely a Departure Point, probabilistically.* That is a stronger claim than the binary label would have been, not a weaker one: it carries a number where the label carried only a verdict, and it is true where the label was not.

This settles the question the definition deferred.

*Remark* 3.15 (Where the variation lies)*.* Write $w(A)$ for the total Aethic weight that state $A$ places on continuations which advance. The categories are $\nu F$ and $\mu G$, complementary and exhaustive on definite states (§3.4.1). What varies across biospheres is therefore not *which* category one occupies but *how weight is distributed* across the two: a state at $0 < w < 1$ is a superposition of the pair in the proportion $w : 1 - w$, exactly as Remark 3.12 describes the post-drop state.

Three consequences.

*The weights carry what a finer taxonomy would.* The framework supplies Aethic weights already, and $w$ is a number where a category would be a verdict; the finer information is in the weight and is available without further vocabulary. *The grading attaches to the state, not to the kind.* Definition 3.14 grades by weight, which is the correct place for grading to sit: an $\varepsilon$-Departure Point is a state whose Departure-tending weight is $1 - \varepsilon$.

*Every actual biosphere is superposed, and this is the subject matter.* None is secured and none has $w = 0$ while advancing continuations remain, so every case of interest carries weight on both poles. That is also why §6's interpolation $\mathbb{P}^{(t)}$ is the right object: it tracks the redistribution of weight between the two, rather than arrival at either.

*Remark* 3.16 (Where the strictness survives, and the two layers). Two things should be kept apart, because the grading does not reach everywhere.

*The model's verdicts remain strict.* Within §5, the configuration $x \equiv 0$ gives $\mathbb{P}(\Lambda_\infty < \infty) = 0$ *exactly*, and adverse anatomy gives it exactly for every $\rho \leq 0$: these are identities, not small numbers. The $\varepsilon$ of Definition 3.14 does not live inside the model at all. It lives at the Aethic level, where the weighting distributes over *which configuration obtains* — and the recovery branch is precisely the branch on which a new muffling agent arises, which the model of a fixed biosphere does not represent. So the model says what happens given a configuration, strictly; the Aethic weight says how much weight each configuration carries.

*This is what the two-layer algebra is for.* The companion's construction is an unweighted layer carrying the structural facts and a weighted layer laid over it without disturbing them [27]. The cascade of §3.4.1 is a Boolean recursion and belongs to the first: $\mu G$ is a set, computed from implication alone. The grading of Definition 3.14 belongs to the second. Neither displaces the other. Making the cascade itself graded would require a fixed point of an $M$-valued doom operator, which is open **[O]** and is the natural next question this remark raises. Remark 5.62 exhibits one concrete instance of such a fixed point — single-type and carrying one number rather than ranging over the lattice, but computed rather than assigned.

**Principle 4** (Continuity; [$\Phi$])**.** The struggle our ancestors underwent against the Departure Point in natural history is continuous, philosophically and ontologically, with a society's agency-mediated struggle against the Departure Point. They are not analogs but the same activity under different mediation.

The Golden Point is the target that respects this, and it does so because of what kind of definition it has: *survival conditioning of its type.* That is primitive enough to apply without alteration to a Mesozoic mammal evading theropod predation and to a modern society avoiding its own weapons. Both lower a hazard; both prevent an Imperfection Magnitude Drop; neither requires the vocabulary of the other. The substance is continuous where the societal construct is not.

*Objection* 3.17 (Agency). A Mesozoic mammal does not know it is fighting the Departure Point. There is no agency, no goal, no strategy — only an animal avoiding a predator. Calling that continuous with deliberate societal risk-management equivocates on the word "fighting".

The objection assumes agency is what individuates the activity. It is not. By §4, what individuates a task-object is its *contribution to the root*, and agency is nowhere in that criterion — which is precisely why task-objects were required before this principle could be stated. What replaces agency is the mediating mechanism, and Optimation conditions on the mechanism rather than around it. Where a society lowers its hazard through deliberation, a lineage lowers its through evolutionary selection; the conditioning does not care which, because what it fixes is that the hazard be lowered. It therefore reaches *through* the mechanism to whatever contingencies the mechanism requires in order to operate.

*Example* 3.18 (Conditioning through the medium: the $\alpha$-gal case; **[S]**). Our $\alpha$-gal case exhibits the structure exactly, and its details matter because they show the conditioning having no alternative route [5].

Some thirty million years ago the Catarrhine lineage came under pathogen pressure of a particular kind. The pathogen presented $\alpha$-gal, a carbohydrate our own cells also produced, so that no immune response was available which did not attack the host. What survived carried a mutation disabling *GGTA1*, the gene responsible for producing it — a gene retained by every other mammalian lineage, and lost in ours alone.[6]

Now read it under Optimation. The gene was ancient and useful, which is why it is universal among mammals; and evolutionary selection removes a gene if and only if there is pressure to remove it. That is the mechanism's own condition of operation, and it is not negotiable. So under ordinary conditions the loss was not merely improbable but close to unavailable: selection had nothing to act on, because the gene was not costing anything.

A conditioning requiring the loss therefore cannot act on the gene. It must act on whatever *supplies the pressure* — and what supplied it was the contingent existence of a pathogen presenting the very carbohydrate in question, which converts the gene from an asset into the one thing standing between the lineage and extinction. Optimation reweights the pathogen's occurrence to prominence, and the loss follows *through* selection rather than around it.

This is the general shape. Optimation does not reach past evolutionary selection to install outcomes; it reweights the contingencies that make selection deliver them. Agency, where it exists, is one such mechanism among others, the deliberative rather than the Darwinian medium, and the conditioning is indifferent between them.

*Remark* 3.19 (What the principle buys, and its cost). It buys the object the rest of this work analyzes. Without continuity, §5's hazard process would model two different things spliced at an arbitrary joint; with it, one process runs from abiogenesis to the present under a single conditioning, and the technological present is a *position* along it rather than a change of subject.

The cost is that the principle is a philosophical premise and not a result, and its scope should be stated at full width rather than left to be inferred, because it is larger than the conditioning question that motivated it. Continuity licenses treating a Mesozoic mammal's evasion of predation, a body's suppression of its own cancers, and an institution's not turning its capacity against itself as values of *one variable*. That is what permits §11's institutional machinery to bear on §5's hazard dynamics at all: without it, the adversivity model would describe the politics of societies and the survival model would describe the fate of lineages, and no argument would carry between them.

So the premise is not merely buying a conditioning target. *It is what makes this one work rather than several.* Should it fail, the sections do not degrade gracefully — they separate, and each retains whatever it had independently while the connections lapse. *Falsifier:* a demonstration that deliberative and evolutionary hazard-reduction differ in a respect the conditioning must track — that some feature of the record is explicable under one and not the other. The agency objection is the natural candidate and is answered above; a better one would consist in exhibiting such a feature rather than in restating the intuition. A second and more targeted falsifier follows from the width just claimed: exhibit a hazard-lowering mechanism at one scale whose governing variable provably fails to correspond to anything at another — a property of institutional self-restraint with no analog in organismal or lineage-level hazard control, or the reverse.

[6]The inactivation of *GGTA1* in Old World primates is established, as is pathogen-mediated selection as its leading explanation. The *severity* of the episode, whether it approached extinction of the lineage, is an inference and not a finding, and the argument does not rest on it: pressure sufficient to remove a gene universal among mammals is all that the reasoning requires, and a milder episode delivering that pressure would serve the example unchanged.

**The principle transfers form, and not only vocabulary.** One further thing the principle does is easily missed, because the statement above is defensive in shape — it says what continuity licenses one to treat as a single variable, which sounds like permission to use a word. It also licenses *inference*, and an instance is now available.

§5.7 represents pre-onset hazard as a sequence of gates. Continuity says the pre-onset struggle and the post-onset one are the same activity under different mediation. It therefore predicts that post-onset internal hazard is representable the same way — that the trough channel of (5.7) should decompose into gates, its smoothness being a coarse-graining rather than a fact. Remark 5.67 carries out the decomposition and finds two things the prediction did not have to deliver: the exponential form falls out of geometric compounding rather than being assumed, and the anatomy trichotomy turns out to be the schedule's own criticality test with capacity substituted for gate count. The construction also agrees with the boundary-scaling derivation of §6.6.1, which was arrived at independently.

That is the principle doing deductive work. It did not merely permit the two hazards to be discussed in one vocabulary; it said what form one should take given the other, and the form checked out against a derivation that did not consume it. A reader who regards continuity as a philosophical convenience should notice that a convenience does not generate a correct structural prediction about a channel it was not formulated to describe.

The falsifier stated above is unchanged and now has a sharper instance available: exhibit an internal hazard whose dependence on capacity provably admits no gate decomposition, and the transfer fails at exactly the place this passage claims it succeeds.

### 3.4.5 Why continuity is wanted: the non-privilege of humans

The principle has now been stated and defended, but not motivated, and its motivation is not tidiness. Continuity is wanted because it is what makes the framework non-anthropocentric, and the non-anthropocentrism is essential rather than incidental. Begin from what the target is. *The Golden Point is a property of a biosphere* — not of a civilization, not of a species. That is why it applies to an alien biosphere as readily as to ours, and applies without translation: nothing in "survival conditioning of its type" mentions humans, technology, or society, so nothing must be reinterpreted when the biosphere in question is not ours.

**Proposition 3.20** (Recursive anchoring; **[S]**). *Because the target is biosphere-level and the principle is continuous across mediation, the conditioning may be anchored at* any *Aethus in the lineage consistent with ours. Take a Mesozoic mammal, its Aethus a parent of ours rather than an Imperfection Magnitude Drop, and ask what the continuation of Aethic attributes within* its *future light cone must look like for its biosphere to reach the Golden Point. The answer runs through the remainder of the lineage, and so through us, without us having been named anywhere in the question.*

Our position is therefore an *output* of the conditioning rather than a stipulated anchor for it. Humans are not a privileged split point; the technological present is where the recursion happens to pass, and its role is emergent from a primitive that mentions no species. The same question asked of a mammal in the Maastrichtian, of a lineage that has not yet speciated, or of a biosphere elsewhere entirely, is one question.

**Three things a human-centered target would cost.** The alternative, conditioning on "technological civilization" or on observers, is not merely less elegant. It forfeits three things, in increasing order of seriousness.

*Robustness across biospheres.* An alien case would require separate treatment, since the target would be defined by features of our own history. A framework whose central conditioning cannot be applied to the systems it is meant to reason about has not achieved generality; it has assumed its instance. *Robustness across perturbations.* §3.5.2 evaluates histories in which humans do not arise. Under a human-centered target those histories cannot be evaluated at all, the target simply fails to apply, and precisely the perturbations that carry the argument become unaskable.

*Non-circularity.* A human-centered target takes *us* as primitive, so the framework would build its base unit out of the thing it is supposed to explain. The Continuity Principle is what prevents this, and that is the deepest reason to want it.

**A distinction to keep: definition versus conditioning.** Two claims are easily run together and must not be, since the argument's validity depends on their separation.

The *definition* of the Golden Point, a survival property of a biosphere, in the sense §5 makes precise, is stipulative.[7] Someone who rejects everything else in this section may adopt it, and the hazard analysis of §5 stands under it regardless.

The *conditioning claim*, that this property is what the Accordance Principle has been conditioning on, is not stipulative and is what §§3.6–3.7 argue for, by reductio on natural history and again on human history. Running the two together would make the argument circular, defining the conditioning target into its role and then observing that it occupies it. Keeping them apart is what leaves the argument something it could have failed at.

*Remark* 3.21 (Two non-privilege claims, and their symmetry). The section now makes two claims of this shape, and they are the same claim along different axes. Principle 5 indexes its

[7]A dated public record fixes the concept's provenance. The author's grade-school idea journal [28] records the coinage in mid-January 2021, in the eleventh grade: "the point after which our society will be too capable in self preservation to regress past that point," with the surrounding reasoning already survivorship-shaped — constant danger decaying survival exponentially, hence extinction eventually certain without the threshold, which is Theorem 5.36(ii) in embryo. The same record carries three further anticipations worth a sentence each. "Important can be defined as that which tends to it" is the Score of §9 as a germ. An August 2021 entry defines "Golden Achievement" as the natural logarithm of the likelihood of reaching the Golden Point, which is the additive log-likelihood structure of the credence method (§8.2) before any of its machinery existed. And a July 2021 notation distinguishes the Miracles elapsed by the modern day from the total "if there are more Miracles needed in the future" — the live-third-term question of Remark 5.70, posed as a subscript four years before it was taken up as a hypothesis with an evidence route. The record shows the conception arising from the survivorship reasoning alone, with no contact with the future-conditioned cosmological tradition placed at Remark 8.8. A second dated document, the author's *Golden Science* of August 2022 [25], is the link between that coinage and the present formalism, and several of this paper's core objects appear in it by name. Its "Outlook Probability" and "Golden Probability" are the unconditioned and conditioned laws, the $\mathbb{P}/Q$ pair, defined and used; its internal/external division of the Departure Point is the two-channel decomposition of §5, with the external channel argued roughly constant and the internal one fed by the biosphere itself; its survival analysis runs on the integrated hazard, computing a positive immortality probability $e^{a/b}$ for exponentially decaying danger, which is the $\Lambda_\infty < \infty$ criterion in closed form; its Third Law is the inequality $b/a < p/(1-p)$ with the L-threshold named — the vocabulary Proposition 3.25 still uses; its "Doomsday Pacing" is the ancestor of the invested-timing role claim of Remark 9.12; Optimation and the task-object both appear as sections; and its "Trace Connection," a placeholder named for "some unknown form of causal (or causal mimicking) connection" between our position and the Golden Point, "until it is officially derived," is the promissory note that retrocausal accordance (§7) later paid. The same document also contains reasoning the mature discipline now fences, the every-attribute optimality conjectures among it, so the record documents the corrections as well as the anticipations, which is what a provenance record is for.

light-cone restriction to one's own present *because that is where the observable cone is anchored, not because the event is distinguished*; the construction at any other event returns that event's cone. Proposition 3.20 says the same of position in a lineage: the conditioning may be anchored anywhere consistent, and returns that anchor's continuation. No privileged spacetime event, no privileged lineage point — and in both cases the appearance of privilege is an artifact of which cone the reasoner happens to occupy.

*Remark* 3.22 (A third axis: no privileged tense). The two axes above concern *where* a reasoner stands. A third concerns *when*, and it follows from the same two results rather than needing anything new. We read the ancestral record as a settled chain in which, at juncture after juncture, the beneficial thing happened — which is what conditioning predicts of any cone read from its own end. By Proposition 3.20 the construction is available at any anchor, so an observer anchored far enough downstream reads *our* actions the same way: a settled chain, cropped to their past cone, and by Principle 5 aligned with their own Nexus of Optimation. We are, to that anchor, what the Cretaceous is to us. The distinction between the static record one reads and the dynamic outcomes one produces is therefore not a distinction in the objects. It is an artifact of which cone the reasoner occupies — the sentence the previous remark ends on, now applied to tense.

Two guards, both standing elsewhere in this work and both binding here. *Ensemble, not trajectory.* Nothing in this reweights anyone's actions. Conditioning reweights the ensemble a reasoner is drawn from, and the downstream anchor's reading says which ensemble rather than what any biosphere will do; Remark 2.11 is disqualifying about the other reading. *And determinacy is indexed to information, not to date.* The downstream cone holds our actions as determinate entries because that anchor has read them; ours does not, and nothing about the reading reaches back. That is exactly the footing on which the companion treats Newcomb's problem, determinacy indexed to an information state rather than to a position in time, so that an unread outcome is a blank entry and blanks lack tense [26], and Remark 3.40's refusal to let a conditioning event *act* is unamended by it.

The relation to Principle 4 is close and is not identity, and the two are easily merged. Continuity is a claim about the *object*: the ancestral struggle and the agency-mediated one are one activity under different mediation. This is a claim about the *reading*: the epistemic relation to that object is the same from any anchor. Neither entails the other, a single activity could be read asymmetrically across anchors, and readings could coincide for activities that were not one, so they are neighbors, and the argument here consumes only the second.

### 3.4.6 The rival periodization: the Precipice, engaged

The continuity principle has a standing rival, and it is the most serious one the field possesses: the periodization of Ord's *Precipice* [29]. This work owes it a direct engagement — not because the rivalry is total, but for the opposite reason. Of everything in the existential-risk literature, the Precipice's central definition is the nearest approach to the losing condition as this framework draws it, and the methodology companion's discipline applies with full force to nearest neighbors [30]: where a position is close, the deltas must be named exactly, because a dense neighborhood is precisely where a distinct position gets misfiled as a variant of what each reader already holds. The deltas here are three, the quantity, the shape of time, and the endpoint, and each is stated below after the position itself is stated at its strongest.

**The position, at its strongest.** The Precipice defines an existential catastrophe not as extinction but as *the destruction of humanity's long-term potential* — explicitly including unrecoverable collapse and unrecoverable dystopia alongside extinction — and argues that humanity entered, in 1945, a period different in kind from all prior history: capability has outrun wisdom, anthropogenic risk now exceeds the natural background by orders of magnitude (the book's considered figures: something like one in six per century against roughly one in ten thousand), the period is short by civilizational standards, and the task it sets is *existential security* — risk brought low and locked low — after which a Long Reflection can settle what the future is for. It is the most careful and most quantified statement of its picture in the literature, its treatment of its own uncertainties is honest, and its definition is modal in spirit: "destruction of potential" is closure-of-possibility language, and the inclusion of non-extinction catastrophes is exactly the concession that death and loss come apart. The definition descends from Parfit's asymmetry [15] (Remark 2.20), and the descent matters below: what the Precipice inherits from Parfit is the intuition, and what it does not inherit, because nothing in the lineage supplies it, is a formal object for the intuition to be *about*. All of that should be said before any of what follows, because what follows is not a dismissal. It is the claim that the Precipice's definition is right, that its own operationalization does not honor it, and that the periodization built on that operationalization inverts in the cases that decide.

**Delta one: the quantity. The ledger reverts to events.** The Precipice's definition points at membership in $\mu G$, a configuration property, the closure of possibility (Definition 3.5, Remark 2.19), but its arithmetic is risk-per-century of catastrophe *events*. The two come apart in both directions, and the separations are not corner cases; they are this framework's central theorems. A biosphere can be departed with no catastrophe on any ledger: past $\mu G$ nothing has yet happened, hazards fire later, members are unaware (Remark 10.6), and the sharpest instance is the undisturbed world of §10.1, which minimizes per-century catastrophe probability on the event bookkeeping and is already departed on the modal one: the safest world the ledger can describe is a lost world. And a biosphere can be undeparted while the event bookkeeping condemns it: in the critical regime departure is certain, expectancy is nonetheless infinite, advancing continuations exist, and everything the survivorship theory computes is computed there — a world an event metric writes off with probability one is the world in which the entire structure of hope resides. So a ranking of worlds by existential-catastrophe probability misorders them relative to the quantity the Precipice's own definition names, and this framework's position is not that Ord's definition is wrong but that it deserves a formalization his ledger cannot give it: the framework sides with the definition against the arithmetic.

**Delta two: the shape of time. Gradient, not age.** The Precipice's picture is a threshold crossed — one short age, different in kind, bracketed by Trinity on one side and by security or ruin on the other. Against this the framework sets three results it did not build for the purpose. First, continuity (Principle 4, and the cascade of §3.4.1): the Departure Point has been available at every moment of the biosphere's history, the record's Miracles are earlier rounds of the same engagement rather than preliminaries to a modern one, and what 1945 changed is mediation and magnitude — never the arrival of a losing condition that was absent before. Second, and this is the part that preserves everything the Precipice is right to insist on: the stakes claim is not disputed but *re-derived*, and strengthened. Remark 10.5 obtains by combinatorial construction what the Precipice obtains by periodization, each moment carries the highest stakes the lineage has yet faced, because a failure late in the chain forfeits the whole accumulated product, and a monotone sequence distinguishes every later point over every earlier one without distinguishing any point categorically (Remark 8.23, §2.5). The gradient picture therefore keeps the urgency and discards only the step. Third, the quantitative heart of the kind-claim, the ratio of anthropogenic to natural risk, divides a numerator estimated by judgment by a denominator estimated from the track record, and the track record is precisely what survivorship conditioning prices: a record read under conditioning on its reader shows rare natural catastrophe whether or not the underlying rates were kind. The literature knows this as anthropic shadow [31], and the Precipice acknowledges it; the delta is that this framework supplies the machinery (§5.8) in which the caveat stops being a footnote and becomes the subject — the bias field is explicit, its magnitude is computable in-model, and "natural risk is small" and "we are among survivors" are hypotheses the finite record cannot separate. A periodization whose different-in-kind claim rests on that ratio rests on the one number the framework's own theorems say a survivor cannot read off their record.

**Delta three: the endpoint. Security decomposes.** The Precipice's terminal state, existential security, is risk brought low *and known to be low*: a far side, reached and certified, on which the Long Reflection can convene. The framework decomposes this into one component it preserves and one it proves unavailable. Risk brought low is anatomy work, $\rho$ held positive, patching capacity maintained against acquired capability, and it is continuous, never completed, with the losing condition permanently adjacent; on this the two pictures can trade freely. Known to be low is a certification, and the ceiling (Theorem 6.18, Remark 6.19) forbids it to any finite observer: no bounded record places a biosphere on the favorable side, forfeiture itself is silent (Remark 10.6), and the unavailability points in exactly the direction that would have been consoling. So "the Precipice, safely passed" is not a state anyone could know themselves to occupy, and a research program whose success condition is unobservable in principle has misdrawn its own finish line — the metaphor is doing load-bearing work the theory cannot support, since a precipice is the kind of thing that has a far side, and a superposition one inhabits is not. What replaces the finish line is not despair but the correct object: maintenance of the anatomy condition, graded and continuous, with the Golden Point as $\nu F$, a modal fixed point one may be inside without ever certifying it, rather than security as a date.

**A corollary: the hinge dissolves rather than resolves.** The debate downstream of the Precipice, whether the present is the "hinge of history" [32, 33], is conducted on a shared assumption this framework denies: that hingeness is categorical, so that the present either is or is not the special moment. The framework's answer is neither side's. Every present is the hingiest yet, by the monotone-stakes construction — so the influence-skeptic's base-rate argument attacks a claim the framework never makes — and no present is categorically special, by continuity and by the tense-relativity of the conditioning's labels (Remark 8.4) — so the hinge-proponent's conclusion is an artifact of reading a label that moves with the observer as a property of the moment. A graded hinge is had for free; a categorical one is unavailable; and the binary framing, not either answer to it, is what the formalism removes.

**What survives, and the fences.** Much survives, and it should be listed rather than implied: the urgency (re-derived, by construction rather than by period); the dominance of the anthropogenic channel (the trough channel is the unbounded one, and the framework gives that asymmetry a functional form rather than an estimate); the priority of risk-work over expansion (the anti-Kardashev results agree with the Precipice against the accelerationists); and most of the book's concrete policy content, which re-grounds without loss on the gradient picture. What does not survive is the quantity, the age, and the finish line. Three fences bound the engagement. Nothing here consumes the Route World — the counter runs from the losing-condition analysis, the monotone-stakes construction, and the ceiling, so a reader who rejects every conditioning argument in this section keeps deltas one and three whole and loses only the third prong of delta two. Nothing here disputes the Precipice's numbers by offering rivals — the framework's claim is that the decisive ratio is unreadable from a survivor's record, which is a reason to compute less, not differently. And nothing here diminishes the practical convergence: a reader persuaded by the Precipice to work on existential risk has been persuaded of the right conduct, on this framework's own showing (§8.6), by an argument whose bookkeeping this subsection corrects and whose urgency it keeps.

**The provenance pattern, recorded once.** The engagement instantiates the methodology companion's criterion exactly [30]: the nearest neighbor's *definition* already contains the correction, destruction of potential *is* closure of possibility, while the neighborhood's density is what kept the modal formalization unoccupied, each camp filing the vacancy as territory already held. The prediction that discipline makes about reception is stated here so it can be scored: this engagement will be misfiled, confidently and in both directions, as "a variant of Ord with extra mathematics" and as "a rejection of existential-risk studies." It is neither, and the deltas above are the scoring key.

With the target fixed and its motivation stated, the question becomes evidential: is there reason to believe the record was in fact produced under it? The remainder of the section argues that there is, first by saying what alignment with such a target would look like (§3.5), then by two independent reductios, one on natural history (§3.6), one on human history (§3.7), of which the second does not require the premise the first leans on.

## 3.5 What alignment with the target would mean

### 3.5.1 Statement

**Principle 5** (Past-Alignment; [Φ])**.** Construct the *Golden Nexus* by conditioning the Cosmic Nexus on the Golden Point. From it, discard every stated attribute whose referential event lies outside one's own past light cone — in the stricter reading, discarding the spacelike attributes as well as the future-directed ones. What remains is the Nexus of Optimation.

Equivalently: the past light cone of Aethic attributes we actually observe is the same one as the past light cone through our position in the Aethic block universe of the Golden Nexus.[8]

Two features of the statement are deliberate and should not be read past. *The restriction is to the past cone only.* The principle takes no position on whether attributes in one's future light cone agree with the near-optimal part of the Golden Nexus. That silence is not modesty but scope: one cannot reliably infer future-cone attributes prior to engaging in Golden reasoning, so a principle that assumed agreement there would be assuming its own conclusion. §3.9 treats the forward question separately and does not settle it.

[8] More precisely than "our actual light cone": owing to random contingency in centric unfolding, our own Aethus is a proper child of the Aetherian Nexus rather than the Aetherian Nexus entire, so the object being compared is the past light cone of the Aetherian Nexus's own Aethic block universe.

*The spacetime point is not privileged.* The principle is indexed to one's own present because that is where the observable cone is anchored, not because that event has any distinguished status. The same construction at any other event returns that event's cone.

**What the principle claims in the companion's own terms.** Read against the Optimal-Earth Conjecture [23], the principle sharpens what "optimal" is optimal *for*. The conjecture there holds our record observer-optimal on effectively any interrogable attribute. Past-Alignment says the optimand is not vaguely observer-production but specifically the Golden Point: one asks for the optimal Aethic history *to the Golden Point*, and then surgically reopens every attribute whose occurrence lies in the future cone of the indexing event. If the principle holds, cropping the Golden Nexus to one's past light cone reproduces Optimation exactly — and it is the exactness, not the plausibility, that would make this a first-principles derivation of Optimation rather than a restatement.

### 3.5.2 The argument from prerequisite attributes

The supporting argument has two steps.

*Step one: identify the attributes that position us for the Golden Point.* Our record contains features which, whatever else they do, make this biosphere unusually able to reach the Golden regime. The capacity to build rocket vehicles is the cleanest example, since it is what would allow the biosphere to persist past the eventual death of its star. *Step two: perturb the record and watch them fail.* Consider Aethic perturbations in which the dinosaurs never go extinct, in which animal life never evolves at all, or in which Africa does not exist or develops very differently. Each comparison presupposes that the perturbed history and ours share a common term against which a missing capacity registers as missing, which ordinary object-individuation cannot supply and §4 does: what is compared across these histories is a Task-Earth and a task-lineage (Definition 4.1), not an Earth and a lineage. With that in place, each perturbation removes the capacity:

- A world in which the dinosaurs persist is one in which, by the Accordance Principle's own verdict, the lineage does not reach our level of intelligence, and so does not build rocket vehicles.
- A world of fungi lacks the animal capacities the construction requires.
- A world without the specific African climatic sequence leaves great apes arboreal, without the bipedal orientation and strategic cognition the construction requires.[9]

The examples are illustrative rather than exhaustive, and the generalization is what matters. By the substitution property of the Accordance Principle [5], the argument runs at *any* rung: a trait of ours may itself be placed in the conditioning seat, and alternative natural histories are then deduced to fail its occurrence almost surely. The maneuver requires the trait to be identifiable in histories where it does not occur, which is a task-object identification rather than an object one (§4.7). So for any Golden-relevant trait one names, perturbations of natural history systematically undermine its occurrence probability.

[9] The climatic sequence in Africa is what drove our great-ape ancestors to bipedality.

**Corollary 3.23** (Local optimality of the record; [**S**]). *If for every Golden-relevant trait — individuated as a task-object, per §4 — the perturbed histories systematically depress its probability, then our Aethus locally maximizes the probability of the Golden Point among its neighbors — and our natural history is accordingly the past-light-cone restriction of the optimal Aethus for the Golden Nexus.*

*Remark* 3.24 (The grade, stated honestly). Corollary 3.23 is [**S**] and not [**P**]. It inherits the substitution property's own license, rung-screening, flagged as such at the source [23], and it establishes *local* optimality among interrogated perturbations, not global optimality over the space of histories. What would strengthen it is a perturbation exhibited that leaves a Golden-relevant trait's probability *undiminished*; what would break it is a systematic class of such perturbations.

Past-Alignment as stated records an agreement. It does not yet show the agreement is more than coincidence, and the two arguments that follow are the argument that it is not.

### 3.6 The conditioning principle, and the two arguments for it

Past-Alignment as stated is an observation about agreement. The following argues that the agreement is not coincidental — that Nexic natural history was never conditioned on Optimation in the first place, but on the Golden Nexus throughout.

**Principle 6** (Golden-Conditioning; [**S**], by reductio). Suppose, for contradiction, that the Accordance Principle conditions on personal observer-existence rather than on the Golden Point.

Observerhood is robust under Aethic perturbation. One may strip probabilistic prerequisites of the Golden Point, rocket capability, the presence of industrialization-driving coal, and replace them by their complements without disturbing one's standing as an observer. A world in which Earth simply lacked the resources to begin an industrial revolution is a world whose inhabitants are no less observers. Now select a handful of such attributes: necessary for the Golden Point, unlikely to survive reopening in the Aetherian Nexus, and admissible in the sense of Remark 3.34 — reopening them must leave the dependent structure intact, so that the resulting Aethae remain individuated by the same task-objects as ours (§4.7). The totality of the counterfactual Aethae so obtained carries *substantially more* Aethic weight than the Aetherian Nexus, while carrying a comparatively minute probability of the Golden Point.

If Accordance were conditioned on observer-existence alone, it would therefore place us in that high-weight counterfactual sum rather than in the Aetherian Nexus. But the Aetherian Nexus is in fact our parent Aethus. Contradiction. Hence the Accordance Principle is not conditioned on observers, so long as "observer" is robust enough to survive these perturbations.

### 3.7 The undeniable case: human history

Principle 6 admits two arguments, and they differ in one respect that decides their order. Both need observerhood to be robust under the perturbations considered — that one can strip prerequisites of the Golden Point without disturbing anyone's standing as an observer. On natural history a sufficiently demanding notion of "observer" can contest that premise (Remark 3.41). On human history it cannot: no one holds that human observers required Alexander's survival, or Newcastle's coal, or Newton. Observers existed before each and would have continued after. So the human-history argument is the one on which the robustness premise is simply not deniable, and it is put first for that reason; the natural-history argument follows as the same reasoning in a domain where the premise is contestable and the conclusion correspondingly weaker.

The argument also does something the natural-history one cannot. What these contingencies deliver is not observers but specifically the technological condition — and observer-conditioning is indifferent to every one of them while Golden-conditioning is not. That isolates what the conditioning must be sensitive to, and it says something about the companion that the companion cannot say about itself: its conditioning target, an eventual observer in one's biosphere, is exactly right for the pre-technological record and provably insufficient beyond it. Not a criticism, the companion reads the pre-human record and stops, by design, but the reason for stopping becomes visible only here.

### 3.7.1 Flukes establish their own significance

**Proposition 3.25** (Significance from route-deletion, at the self-sampling price; **[S]**). *Let $F$ be an event established as a* fluke*, of low outlook probability $p$, with no claim yet made about its significance. Suppose $F$ were statistically independent of the Golden Point. By the Second Law its Golden Probability equals its outlook probability, $g = p$, and we should expect with probability $1 - p$ to occupy the counterfactual $\neg F$. We do not. Hence $F$ is not independent of the Golden Point, and by the Third Law its Golden Linkage exceeds the L-threshold [5, 25] — in the notation of §3.2, the Accordance bound* (3.1) *with $H$ the Golden Point, $A := F$, $A' := \neg F$, so that $\mathbb{P}(G \mid \neg F)/\mathbb{P}(G \mid F) \lesssim p/(1-p)$ with $\kappa$-fold violations priced by* (3.2) *(Remark 3.4 records the translation).*

*Two premises deserve their names before the conclusion is stated at its licensed strength. The step "we should expect with probability $1 - p$ to occupy $\neg F$" is a* self-sampling *premise, location weighted within a reference class spanning the $F$- and $\neg F$-worlds, of the lineage this corpus polices elsewhere [34]; it is adopted exactly once, at the root, and Remark 7.5 governs its normalizer. Granting it, what the proposition delivers is a* conditional-likelihood *claim:* for route-deleting *$F$ (Remark 3.26), observing $F$ raises the Golden linkage above the L-threshold. Significance is deduced given the counterfactual's route-deleting character — and establishing that character is case work, as the scope remark says, not a free rider on the flukiness. Aggregating over a set of route-deleting flukes, the union of their counterfactual Aethae carries a far greater Aethic weight than our own, and the principle refuses to book our location within the lesser one as brute luck.*

*Remark* 3.26 (What the proposition quantifies over, and the counterfactuals it excludes). The template above, $F$ is improbable, we observe it, therefore $F$ is linked to the Golden Point, runs on *any* observed improbability if nothing restricts the range of $F$, and would then license absurdities: the precise configuration of a Cretaceous raindrop is astronomically improbable and is not thereby significant. The restriction that blocks this is already implicit in the proposition's use and is stated here, because a reader is entitled to it before the stockpile rather than after.

*The counterfactual must delete a route, not permute one.* Write $\neg F$ for the alternative. The inference is licensed when $\mathbb{P}(G \mid \neg F)$ is genuinely lowered by the substitution, when the counterfactual removes a prerequisite of the Golden Point, and is void when $\neg F$ merely reshuffles which particular events fill the same structural role. The raindrop case is a permutation: some configuration obtains, and every one of them leaves the downstream dependency structure untouched, so $\mathbb{P}(G \mid \neg F) = \mathbb{P}(G \mid F)$ and the bound (3.1) is satisfied with room to spare rather than violated. Improbability alone therefore establishes nothing; improbability *of a route-deleting attribute* establishes the linkage, and the proposition should be read with that restriction throughout.

*The restriction is not free, and it is what the stockpile is selected for.* Deciding whether a counterfactual deletes or permutes is a claim about dependency structure and is not delivered by the attribute's weight — which is Remark 8.10's point arriving early: a marginal test on weights recovers the hubs of the dependency graph, and the route-deleting attributes are exactly the hubs. So the criterion is principled and its application is [**S**], case by case. This is also the honest register in which the stockpile's items differ. The coal has the route-deleting character on the economic historians' own analysis, Allen's price ratios make mechanization pay in one place and nowhere else, so the counterfactual removes the route rather than relocating it [35], while the Granicus is weaker on exactly this axis, since the claim that no comparable transmission survives Alexander's death at the Granicus is a strong fragility premise about the Hellenistic world and not a computed dependency. That ordering is the same one Remark 3.29 reaches from the evidential side, and the agreement of two independent criteria on which items are strongest is worth more than either.

*Remark* 3.27 (The retrospective-target null, faced at the site). The strongest rival to the proposition is not the determinist's substitutability but the *retrospective-target* null: any realized history, read backward, contains improbable events that were prerequisites of whatever that history's distinctive features happen to be, so a stockpile of flukes pointing at the realized outcome is expected under no conditioning whatever — the sharpshooter who paints the target around the shots. The paper's answer exists but lived two sections away; it belongs here. First, the target is *ex ante*: the Golden Point is picked out by the fixed-point structure of §3.4.1, a future event defined before and independently of which record happens to obtain, so the target is drawn before the shots in the only sense the null requires. Second, the evidential force claimed is *comparative*, and the comparison is stated where the historical cases are heard (§8.5): consistency with a hypothesis is not evidence for it over a rival the same data also fit. Third, what survives that discipline is the human-history leg's genuinely distinctive content: the stockpile's flukes are prerequisites of the *technological* condition specifically, to which observer-conditioning is provably indifferent — and that asymmetry between the rivals, not raw improbability, is what the proposition's linkage tracks.

**The dinosauroid, answered (worked case; [S], one import at [Φ]).** The question whether the dinosaurs could have evolved into humans, posed in its classic form as the dinosauroid [36] and generalized in the debate between contingency and convergence [37, 38], receives a direct answer once its vague target is replaced. "Evolving to intelligence" is not a well-posed endpoint; "acquiring the capacity to trigger the Golden Point" is, and with that substitution the framework's answer is no, by deduction rather than by paleontological intuition. The classic rejoinder, that some birds are strikingly intelligent, is met on its own terms: a lineage may become smart along one metric or another without ever rising to high capacity for high Golden-Point probability, and the delicate Golden-leading activities, rocketry as the emblem, would never have entered its realm of possibility. The companion's Imperfection-Magnitude-Drop argument [5], an IMD being a targeted perturbation of one's own natural history whose observer-production probability is drastically minute, sharpens "less" to "vastly less": such a lineage's Golden-Point probability would sit far below our own, so that with the vast majority of the likelihood it would have become smart, if at all, along a tangential metric. The substitution also dissolves the standoff it was borrowed from. Convergence may well hold for cognition along some metric, as its defenders argue, while contingency holds at the level that matters here, because Golden capacity is a delicate, Miracle-coupled competence and not a convergent target;

the two camps were arguing about different endpoints under one word.

The case is the flukes proposition's canonical instance, and it shows the route-deletion criterion doing its case work rather than being asserted. The end-Cretaceous impact is the fluke $F$, the world in which the dinosaurs continue is $\neg F$, and the claim that $\neg F$ carries vastly lower Golden-Point probability is the claim that $F$ was route-deleting rather than a permutation, which is exactly what Remark 3.26 requires to be established case by case. The "birds" objection is the substitutability reading of Remark 3.28, answered here by the metric distinction. One pitfall must be avoided, and the companion names it: the naive counterfactual "no humans" removes the conditioning on us, and with it Optimation, so that the counterfactual Earth destabilizes quickly and the comparison becomes trivial. The comparison is therefore run already conditioning on Earth maintaining its habitability, and the answer survives that concession, which is what gives it force. The structure is the natural-historical mirror of the extinction proposition (§10.1): remove the muffling lineage and the background stands unmuffled, in the deep past as in the counterfactual present.

*Remark* 3.28 (This settles the contingency-determinism debate, in one direction). Historians divide over whether the path to modernity was contingent or overdetermined, and the dispute is usually conducted by weighing narratives. Proposition 3.25 makes it an inference. The determinist holds that the apparent flukes were substitutable: no Alexander, then someone else; no British coal, then some other route. Substitutability is exactly high $b$, a good chance of the Golden Point without $F$, hence low Golden Linkage, hence $g \approx p$, hence the expectation that we occupy $\neg F$. The Third Law, the Accordance bound (3.1) with $H$ the Golden Point, bounds it quantitatively: $b/a < p/(1-p)$, so the flukier the event, the more severely substitutability is constrained by our observing it. The determinist is not refuted by a better narrative; they are constrained by an inequality whose inputs are the event's improbability and its route-deleting character (Remark 3.26) — nothing else.

### 3.7.2 The stockpile

Two cases, deliberately ordered so that the weaker rhetorically comes first. *A single sword-stroke.* At the Granicus in 334 BCE, two months into an eleven-year campaign, the Persian noble Spithridates brought a battle-axe down on Alexander's helmet and split its crest; Cleitus the Black severed his arm before the second blow [39, 40]. Had the stroke landed, the expedition would have ended within days of beginning, and with it the Hellenistic world, its Alexandrias, and the transmission through which much of what followed ran.

*Where the coal happened to be.* Pomeranz argues that England was a "fortunate freak" whose coal lay close to water and ports, making steam economically feasible, while China's coal sat in the northwest, far from the Jiangnan textile economy that might have used it — a geological contingency placing coal and the Americas nearer the western than the eastern end of Eurasia [41]. Allen adds the labor side: Newcastle carried the highest ratio of labor to energy costs in the world in the early eighteenth century, and only under that ratio did mechanization pay [35]. His summary judgment is an Imperfection Magnitude Drop stated by an economic historian who has never encountered one: there was a single path to the twentieth century, and it ran through northern Britain.

A third case is available but requires reframing. Diamond's account of Eurasian advantage [42] is *determinist* at the level of human history, geography made the outcome near-inevitable, and is criticized on that ground. What the present argument takes from it lies one level down: the distribution of domesticable species and the continent's east–west axis are contingent *across* *possible Earths* while determining *within* ours. That is precisely the Accordance profile, low outlook probability with high Golden Linkage, and Diamond's determinism is at the wrong level rather than mistaken.

### 3.7.3 Why this leg is the stronger one

Two reasons, and the second is the more important. *The robustness premise becomes uncontroversial.* The natural-history reductio needs it granted that observers survive the removal of the end-Cretaceous impact, which can be resisted. Nothing comparable is available here. No one holds that human observers required Alexander's survival, or Newcastle's coal seams, or Newton. Humans existed for millennia before each and would have continued after their removal. The premise that carried the weight and could be denied is here simply not deniable.

*The conclusion sharpens.* What these contingencies deliver is not observers but specifically the *technological* condition — skyscrapers, the internet, the capacity to leave a dying star. Observer-conditioning is indifferent to every one of them; Golden-conditioning is not. So the human-history argument does not merely repeat the first at lower risk; it isolates what the conditioning must be sensitive to.

*Remark* 3.29 (The narrative-bias objection, and the test that answers it)*.* The serious objection is selection in the record rather than in the world. Dramatic near-misses make better stories, are more often written down, and are more often repeated; a history dense with them may report a fact about historiography. The framework predicts an asymmetry that narrative bias does not, and the prediction is already on record. Attributes with high Golden Linkage magnitude but *low connectivity* to other attributes should have low Golden Probability — lucky avoidance of adversity outranks lucky survival of it, so isolated close calls should be *rare* in the record while densely connected contingencies should be *numerous* [25]. Narrative bias inflates both alike; the framework inflates one and suppresses the other.

The discrimination favors the framework for a plain reason: *the strongest cases are the dull ones.* Nobody narrates the Newcastle labor-to-energy price ratio, and no popular account turns on Jiangnan's distance from northwestern coal seams. Yet that is where the peer-reviewed economic history converges [41, 35]. Were narrative selection generating the pattern, the undramatic structural contingencies would be missing — and they are the best-attested items in the file. The Granicus is the vivid illustration; the coal is the evidence, and the ordering of §3.7.2 inverts the rhetorical one deliberately.

Both legs concern the past cone, and neither speaks to the future. That silence is deliberate, §3.5.1 recorded it as scope rather than modesty, and the three positions available on the forward question are set out next without adjudication.

## 3.8 The contestable case: natural history

The same reductio, run on the natural-historical record. Its premise, that observerhood survives the perturbations considered, is contestable here in a way it was not above, and the argument is correspondingly the weaker of the two; it is included because it reaches further back, and because it is where the conditioning claim was first posed.

**The Trace Connection, named in 2022.** The claim this subsection advances was posed as an open problem, under its own name, four years before the present argument. Our *Golden Science* of August 2022 [25] sets out the Golden Principle in four numbered points, of which the third states that our observations are far closer to representing those of an average Golden-Point biosphere than those of an average position in the universe, Past-Alignment in prose, and the fourth states that some connection, causal or causal-mimicking, holds between our existence at this time and place and the Golden Point itself. That paper then names the connection and declines to supply it: it is to be called *the Trace Connection* until it is officially derived.

§3.6 is offered as that derivation. What the 2022 statement supplies, and what no later argument could manufacture for itself, is that the gap was identified, named, and dated before any candidate filling existed — so the reductio cannot be read as having been reverse-engineered to fit a conclusion already in hand. The same document also carries, in prose and without the stochastic apparatus of §5: the hazard-function framing of biosphere survival; the observation that an exponentially decaying hazard yields a strictly positive limiting survival probability; the division of the failure mode into internal and external channels, together with the insistence that a Golden biosphere must lessen *both* rather than either; the thesis that domination is inefficient because it feeds the internal channel, which is Φ3; the Kardashev critique including the self-consumption argument; and the definition of the Golden Point as the point past which a biosphere cannot regress, drawn there against the Hubble horizon. The formal content of §§5–11 is later; the apparatus it formalizes is not.

**Corollary 3.30** (What remains; **[S]**)**.** *What the reductio leaves standing is the empirical artifact of Past-Alignment itself: conditioning on the existence of the Golden Point somewhere in the Aethic block universe reproduces exactly the inferential form of the Accordance Principle that yields the stated-attributes of Nexic reasoning. Among the contingent natural histories one might find oneself at the end of, a fungal world, a dinosaurs-survive world, a wholly lemurine one, the one that must be conditioned on to reproduce the Nexic state of affairs we actually observe is the one in which the Golden Point exists.*

*Remark* 3.31 (A second robustness, and why the target's modal form was the right one). Principle 6 turns on a robustness: observerhood survives perturbations the record does not, so a conditioning indifferent to the difference cannot have produced the difference. The schedule channel of §5.7 exhibits a second target with the same defect, and it is nearer to hand than observerhood. *Survival to a finite horizon is robust in exactly that way.* A biosphere may survive any stretch one names without having climbed its schedule at all — by not being struck, at bare background, on the lucky branch that Remark 5.64 shows never leaves the support of a finite-horizon conditioning. So conditioning on having-lasted leaves the record unexplained for precisely the reason conditioning on observers did: it is satisfied by histories in which nothing was climbed.

What survives the objection is the asymptotic form, and the point worth extracting is that the target was already stated in it. Definition 3.5 gives Goldenness as ∃∀, some course admits *no* foreclosing continuation, which is a condition on the whole of a future and not on any initial segment of one. A reading of the Golden Point as a passage completed at some finite time would have inherited the robustness above and failed here. The modal form is therefore not a technical nicety of the definition; it is what makes the target capable of doing the work Principle 6 asks of it.

The cost is recorded rather than absorbed: this raises what the conditioning must be, and by §6.8 no one is entitled to know that it obtains.

### 3.8.1 Bounding the counterfactual weight

The reductio of 6 turns on a single inequality: that the counterfactual Aethae obtained by reopening Golden-Point prerequisites carry substantially more weight than the Aetherian Nexus. Stated bare, that is an assertion, and an objector may simply deny it — perhaps coal is common given a Carboniferous, perhaps the metallurgical sequence is near-forced given hands and fire. This subsection replaces the assertion with a construction, a bound, and a statement of exactly what empirical input remains outstanding.

**The construction.** Fix stated-attributes $a_1, \ldots, a_k$, each a prerequisite of the Golden Point and none a prerequisite of observerhood. Let $\mathcal{A}$ denote the Aetherian Nexus, in which every $a_j$ obtains, and let $\mathcal{A}_j$ denote the Aethus obtained by reopening $a_j$ alone to its logical complement, holding the remaining $k-1$ at their realized values.

**Lemma 3.32** (Disjointness; [**P**]). *The Aethae $\mathcal{A}_1, \ldots, \mathcal{A}_k$ are pairwise disjoint, and each is disjoint from $\mathcal{A}$. For $j \neq j'$, the pair $\mathcal{A}_j$ and $\mathcal{A}_{j'}$ disagree on* both *slot $j$ and slot $j'$; and $\mathcal{A}_j$ disagrees with $\mathcal{A}$ on slot $j$.*

**Proposition 3.33** (Weight ratio; [**P**]). *Write $\alpha_j$ for the probability that slot $j$ takes its realized value, conditional on the remaining slots taking theirs. Then*

$$\frac{W(\mathcal{A}_j)}{W(\mathcal{A})} = \frac{1-\alpha_j}{\alpha_j}, \tag{3.5}$$

*and by Lemma 3.32 the single-flip Aethae may be summed:*

$$R := \frac{W_\times}{W(\mathcal{A})} \;\geq\; R_1 \;=\; \sum_{j=1}^{k} \frac{1-\alpha_j}{\alpha_j}. \tag{3.6}$$

*Remark* 3.34 (What the bound does and does not require). Three features of (3.6) are worth isolating, because they are what make it usable.

*Independence is not required.* Only disjointness is, and that holds by construction. Dependence among the $a_j$ changes what the individual ratios *are*, the $\alpha_j$ become conditional rather than marginal, as stated, but not whether the terms may be added. This matters because the attributes at issue are plainly not independent, and an argument needing them to be would be unusable. *It is a lower bound, and deliberately a loose one.* The single-flip Aethae are a vanishing fraction of the counterfactual set: the full sum over all Aethae differing in at least one slot is $\prod_j \alpha_j^{-1} - 1$, larger by many orders. But the full sum requires the joint structure, and the reductio needs only that $R \gg 1$. An undercount that suffices is preferable to an exact figure that is contestable.

*Admissibility is governed by the Imperfection Magnitude Drop criterion.* A perturbation enters the sum only if it leaves the dependent structure intact — if reopening $a_j$ does not re-randomise what depends on it. A perturbation that triggers a drop does not yield a clean single-slot flip, and Lemma 3.32 fails for it. Coal in an inaccessible location is admissible on this criterion: it may be probabilistically catastrophic without re-randomizing the structure downstream. A relocation of the lineage's speciation to another continent is not, since the material that later mediates the outcome is not present there to be held fixed.

**How large must $R$ be, and how improbable must the attributes be?** The reductio requires only $R \gg 1$. Evaluating (3.6) for $k$ attributes each at probability $\alpha$:

| $\alpha$ | $k=1$ | $k=3$ | $k=5$ | $k=8$ |
|---|---|---|---|---|
| $10^{-1}$ | 9 | 27 | 45 | 72 |
| $10^{-2}$ | 99 | 297 | 495 | 792 |
| $10^{-3}$ | 999 | $3.0 \times 10^3$ | $5.0 \times 10^3$ | $8.0 \times 10^3$ |
| $10^{-4}$ | $1.0 \times 10^4$ | $3.0 \times 10^4$ | $5.0 \times 10^4$ | $8.0 \times 10^4$ |

The bar is far lower than the informal argument suggested. A *single* attribute at $\alpha = 10^{-3}$ delivers $R \sim 10^3$ unaided; no portfolio is needed, and the burden reduces to establishing one prerequisite as genuinely improbable. Correspondingly, the objection that the counterfactual sum might be comparable requires *every* candidate prerequisite to have $\alpha$ of order one, that coal placement, metallurgical sequence, and the rest are each more likely than not given the rest of the record, which is a considerably stronger claim than the objection is usually asked to bear.

**Estimating $\alpha$: the sister-case method.** The remaining empirical input is supplied by an instrument the companion already develops, and it should be used rather than reconstructed [5].

By the Theorem of Aimless Optimation, Optimation is the Cosmic Nexus conditioned on the attribute $H_N$ that an Aetherian exists in one's biosphere; and if an attribute $X$ is statistically independent of $H_N$, the two Nexae assign it the same weight. Independent attributes are therefore *controls*: their observed frequency estimates the Cosmic Nexus probability directly, unlifted by conditioning. The companion's worked case is the Catarrhini node — Old World monkeys furnish over a hundred extant species across twenty-five million years with no attainment of ape-level intelligence, and their fate being independent of $H_N$, that frequency transfers as the baseline for our own lineage at the same node.

This is exactly the quantity (3.6) requires. For an attribute $a_j$ admitting such a control, $\alpha_j$ is read off the control's frequency, and its contribution to $R_1$ follows.

**A corollary on once-only attributes, and the independence check that governs it.** A corollary of the same theorem bears on attributes observed exactly once, and it should be read as an inference rather than a consistency claim. Once a first occurrence has secured what the Golden Point requires, a *second* occurrence is statistically independent of $H_N$, it changes nothing about whether an Aetherian exists, and by the theorem it therefore retains its Cosmic Nexus weight and is not lifted. Optimation supplies what is required and does not overshoot.

The consequence is a discriminating observation. Write $\lambda$ for the unconditioned expected count across all opportunities. Under the hypothesis that $\lambda \ll 1$, Optimation must lift the attribute for $H_N$ to obtain, and lifts it to exactly one; *exactly one occurrence is what that hypothesis predicts*. Under the hypothesis that no lift was needed, occurrences are Poisson at rate $\lambda$ and exactly one is a coincidence of probability $\lambda e^{-\lambda}$. The likelihood ratio between them is $\left(\lambda e^{-\lambda}\right)^{-1}$:

| $\lambda$ | 1 | 3 | 5 | 10 | 20 |
|---|---|---|---|---|---|
| ratio favoring lift | 2.7 | 6.7 | 30 | $2.2 \times 10^3$ | $2.4 \times 10^7$ |

So a single occurrence tells strongly against the configuration having been *common* — $\lambda \geq 5$ is disfavored thirtyfold or worse — while discriminating only weakly between a genuinely minute $\lambda$ and one of order unity. Its evidential role is accordingly to exclude the objection that the prerequisites were unremarkable, not to supply the magnitude. The magnitude comes from the independent per-opportunity estimate, and where that gives $\lambda \ll 1$ the attribute contributes approximately $1/\lambda$ to $R_1$.

The corollary is only as good as its individuation, and this is where the difficulty sits. Rather than proceed directly to an estimate, we calibrate the instrument on two attributes for which the data are good. Both return nothing, and the pattern in *how* they return nothing is what fixes the strategy.

**Calibration I: coal.** Britain's coal is the obvious first candidate, and it fails at the endowment grain. Carboniferous and younger basins are known on virtually every continent, the Appalachians, the Ruhr, the Donets, the Kuznetsk, the Nord-Pas de Calais–Saar–Lorraine field, and others, giving eight or more on any reasonable individuation, and Britain's is not the most favorable, the Appalachian seams being unusually large and unusually exposed. With $\lambda \approx 8$ we obtain $\alpha = 1 - e^{-\lambda} \approx 0.9997$ and a contribution to $R_1$ of about $3 \times 10^{-4}$: nil.

The verdict is confirmed by the author whose thesis is most at stake. Allen observes that the Ruhr went unexploited not for want of coal but because Dutch peat held fuel prices down, removing the return on improving Rhine transport; and he draws the counterfactual explicitly, that had German coal been developed three centuries earlier the industrial revolution might have been a Dutch–German achievement rather than a British one [35]. He identifies Belgium, around Liège and Mons, as possessing a comparable ratio of wages to energy cost. So the claim that the path to the twentieth century ran through northern Britain was always about the *conjunction* of cheap energy with dear labor and the institutions to exploit both — never about the seams, which were widely available. A reader who recalls that claim should read it that way, and the estimator's verdict agrees with Allen rather than contradicting him, reached independently and from geology.

**Calibration II: domesticable megafauna.** A second candidate, drawn from a different literature, fails in the same manner. Diamond's count gives fourteen large mammal species domesticated of some hundred and forty-eight candidates, with thirteen of the fourteen wild ancestors confined to Eurasia [42] — a striking asymmetry, and the basis of a well-known thesis. But the relevant question for the estimator is not how the successes distributed; it is how many *continents* possessed a workable endowment. Diamond's own candidate counts are seventy-two for Eurasia against fifty-one for sub-Saharan Africa, which is not the profile of a rare attribute. Taking continents as the opportunities gives $\lambda \approx 2$ and a contribution of about 0.16: nil again.

Diamond's *conjunction* does better, megafauna together with an east–west principal axis and the crop package on one landmass gives $\lambda \approx 0.16$ and a contribution near 5.8, which is a real figure, and still four orders below what a single attribute would need to carry the reductio alone.

**Proposition 3.35** (The bound on *parallel* attributes; **[S]**)**.** *The two failures share a cause, and it is structural rather than accidental. Call an attribute* parallel *when its opportunities are a set of co-existing sites, landmasses, basins, continents, so that $\lambda = Np$ with $N$ the site count. For an attribute individuated at continental scale, $N \approx 5$. To obtain $(1-\alpha)/\alpha \sim 10^3$ then requires $\lambda \sim 10^{-3}$, hence a per-site probability of order $2 \times 10^{-4}$ — a claim that cannot* *responsibly be defended for any attribute of the form "this landmass possesses feature X". Deep time can deliver minute $\lambda$ for parallel attributes because its site count is astronomical; recent history cannot, because its site count is five.*

*Coal and megafauna are both parallel attributes. That, and not their recency, is why they failed.*

**Serial attributes escape the bound.** Proposition 3.35 is a claim about parallel attributes only, and the restriction is not a technicality — it identifies the wrong turn that coal and megafauna represent. An attribute may instead be *serial*: a conjunction of successive branch points within a single trajectory, each of which had to go one way for the next to be reached.

**Proposition 3.36** (Serial attributes; [**S**]). *Let an attribute consist of $m$ successive branch points, the $i$th resolved favorably with probability $p_i$ conditional on its predecessors. Then $\alpha = \prod_{i \le m} p_i$ and the contribution to $R_1$ is $\alpha^{-1} - 1$, which grows* exponentially *in $m$. At $p = 0.75$ throughout: $m = 8$ contributes about 9, $m = 16$ about 99, and $m = 25$ about $1.3 \times 10^3$.*

The opportunity structure is what differs. A parallel attribute is bounded by how many sites exist, and there are five continents. A serial attribute is bounded by how many branch points the trajectory contains, and a well-articulated historical episode contains many. Recent history is therefore not weak evidence; it was being individuated in the form that makes it weak. This is how the candidates should be stated. Not "Alexander existed", which is a parallel attribute of no force, but *that he survived and succeeded through the entire sequence*, the wound at the Granicus, Issus, Gaugamela, the near-fatal injury among the Malli, the absence of a disabling illness before consolidation, and the Hellenization surviving the immediate fragmentation of the empire, each a branch point with its own $p_i$. Likewise the Roman survival of the Second Punic War is a sequence, not a fact: the recovery after Cannae, Hannibal's failure to march, the holding of the Italian allies, Scipio's African campaign. Likewise the passage of Christianity from a provincial sect to an imperial religion, and its part in what followed. Each such episode is one attribute in (3.6), and each carries a contribution set by the length of its cascade rather than by the number of continents.

*Remark* 3.37 (What serial attributes cost). The gain in magnitude is paid for in the independence check, and the price should be stated plainly, since it is where such an estimate would fail. For each branch point one must hold that its failure had no adequate *substitute* — that had Alexander died at the Granicus, the Hellenization of the East does not simply proceed under another commander. History is rich in substitutes, and each $p_i$ that admits one contributes nothing, since the attribute is then not load-bearing for $H_N$ at that step. The estimator's discipline accordingly transfers from grain, where it sat for parallel attributes, to substitutability.

Two consequences follow. Cascade length must be counted in *substitution-free* branch points, not in dramatic incidents, which will reduce $m$ considerably below the narrative count. And the per-step probabilities must be conditional on the predecessors, since the steps are plainly not independent — a commander who has survived four battles is not a random draw at the fifth. Both corrections push $\alpha$ upward, and a careful estimate should expect the eventual contribution to fall well short of the naive product.

*Remark* 3.38 (Individuate the branch point by its role, not its occupant). The substitutability audit has a corollary that governs how each branch point must be *stated*, and getting it wrong is the most likely route to an inflated estimate. The counterfactual at each step must be taken over the *role*, not over the occupant. The question at the relevant branch point is not whether

Christianity in particular arose and spread, but whether *anything* arose that discharged the role it discharged — in the window, with the reach, and with the consequences for what followed. So individuated, the competing cults of the period are not rivals to the hypothesis but candidate fillers of the same role, and their availability raises $\alpha$ at that step rather than lowering it: where a role admits $\kappa$ independent candidates each of probability $q$, the role-probability is $1-(1-q)^\kappa$, not $q$.

Role individuation is therefore the *conservative* choice, and that is precisely its merit. It concedes the substitution objection in advance rather than meeting it afterward: a critic who replies that some other movement would have served is not refuting the estimate but restating the quantity already being estimated. What survives such an audit is a residue of steps at which no filler was available with comparable probability — and the claim, where a cascade is offered, is about that residue.

It also relocates where improbability may legitimately be found. It is not to be sought in the striking identity of the occupant, which is where narrative attention naturally goes, but in the narrowness of the window in which *any* occupant could have appeared. The transitional periods are the interesting ones for that reason, and the dramatic incidents largely are not.

*Remark* 3.39 (The limitation is present ignorance, not the structure). One further point about the status of these estimates, and it cuts in an unusual direction. The obstacle to evaluating $\prod_i p_i$ for any actual cascade is not that the quantity is ill-defined or that the method is unsound. It is that counterfactual history is not yet in a state to supply the $p_i$ with defensible error bars — the substitutability audits and the window widths that Remark 3.38 requires are matters on which the discipline has views but not measurements. The uncertainty is *ours*, and contemporary; it is not a property of the cascades.

Two things follow. The paper should not, and does not, offer a numerical cascade estimate: what it offers is the structure (3.6), the serial–parallel distinction, and the individuation discipline, with the arithmetic left for inputs that do not yet exist. And the expectation that such inputs will exist is a substantive bet rather than a certainty — one that quantitative counterfactual history will mature to the point of supplying audited branch-point probabilities. If it does, the enumeration becomes routine and the bound follows. If it does not, the weight claim remains structurally established and empirically unfilled, which is a weaker position than we would like and a stronger one than assertion **[O]**.

**An unnamed significance is a prediction, and it has a falsification test.** Remark 3.39 leaves the framework in a position that looks, at first, like an unfalsifiable auxiliary. We do not know what the establishment of a universalizing cult in the late-antique Mediterranean forced through downstream; the analysis nonetheless concludes that something Golden-relevant was forced through, on the strength of 3.25 alone. Stated carelessly this is the canonical bad move — every fluke acquires an unnamed significance, nothing counts against the framework, and the Accordance analysis becomes a device for dignifying whatever happened.

It is not that, and the reason is that the unnamed significance is a *prediction* with a definite content and a definite refutation. *Content.* For a cascade $C$ satisfying the two admissibility conditions, genuinely improbable, and load-bearing in the sense of Remark 3.38, the claim is that there exists a consequence of $C$ on which the probability of the Golden Point depends, and that it will be identified. The claim is made before the consequence is known, which is what makes it a prediction rather than a description.

*Refutation.* Exhibit a route to the Golden Point that does not require $C$. If the terminus is reachable without the cascade, then $C$ was not load-bearing for it, and the predicted consequence does not exist to be found. This is the substitutability audit of Remark 3.37 run at the *terminus* rather than at an intermediate — which is the only place it can be run when the intermediate is unnamed, and which is a harder audit rather than an easier one, since the chain to be traced is longer.

*And the criterion is diachronic.* A single unnamed significance is a debt; a growing ledger of them is a symptom. The framework is progressive if the ratio of identified to unidentified consequences improves as counterfactual history matures, and degenerating if the unnamed list only lengthens while the named one does not. We state this as the test we intend the program to be held to: *if the historiography improves and the significances remain unnamed, that counts against the framework and is not to be absorbed by it.* The passage of time is thereby made a test rather than an excuse, which is what the position in Remark 3.39 would otherwise lack.

One asymmetry should be noted, since it limits what confirmation is available. Failure to exhibit a $C$-free route is weak evidence at best, because such routes are counterfactual claims subject to the same immaturity that motivated the remark. The test is therefore one-sided in the ordinary way: it can refute a particular significance claim and cannot establish one, and the framework's standing rests on whether the identifications accumulate.

**The consequence: aggregate, but of serial attributes.** The reductio requires $R \gg 1$, not $R > 10^3$, and $R_1$ is a *sum*. Even under the corrections of Remark 3.37, a modest stockpile suffices: twenty attributes contributing about ten apiece give $R_1 \approx 200$, and a single well-attested cascade may supply as much on its own.

| attributes | each $\approx 5$ | each $\approx 10$ |
|---:|---:|---:|
| 10 | 50 | 100 |
| 20 | 100 | 200 |
| 40 | 200 | 400 |

So the program is an enumeration of serial contingencies, each articulated as a cascade and each audited for substitutability. The record supplies candidates readily — the military sequences above; the demographic rupture of the Black Death and its effect on the very wage structure Allen's account requires; and the theses of [42] beyond megafauna, several of which admit serial rather than parallel statement. What the two calibrations established is not that recent history is barren but that it must be interrogated along the trajectory rather than across the map.

**Why the negative calibrations are retained.** Two attributes were examined and both returned nothing. It would be tidier to omit them, and it would be a mistake. They demonstrate that the instrument has teeth. A method that never returns nothing cannot discriminate, and a reader entitled to suspect that the estimator was constructed to deliver a congenial answer is answered by two cases where it did not. They also fix the grain discipline *before* any positive estimate is offered, so that the eventual stockpile is credible rather than convenient. And they preempt the two objections a reader arrives with, that coal was surely the scarce thing, and that Diamond's asymmetry surely settles the matter, with the estimator's verdict rather than with assertion.

Most importantly, the pattern in the two failures is itself the finding. Had one failed and the other succeeded, the first would read as a red herring. Both failing, and failing for the same reason, is what establishes Proposition 3.35 and thereby the aggregation strategy. The negative results carry the argument. This is where the argument's remaining empirical debt sits, and it is now specific and bounded. What (3.6) establishes is the structure: given the $\alpha_j$, the bound follows. What Proposition 3.35 establishes is that no single recent attribute will supply it, so the debt is a stockpile of order twenty contingencies, each estimated by the sister-case method and each passing the independence check. The weight claim is accordingly no longer asserted but *conditionally computed* — reduced from a plausibility judgment about an unexamined sum to an enumeration of bounded terms, each obtained by an instrument calibrated in advance and each with its required magnitude stated **[O]**. What remains owed is the enumeration, and *that* is a research task rather than an argumentative gap: the terms are individually modest, the sources are identified, and the standard each must meet is fixed before any is computed.

*Remark* 3.40 (The conditioning is not teleological). A reader's first objection will be that conditioning on a future event to account for present observations is teleology: the Golden Point reaching backward to arrange its own prerequisites. The framework answers this at its foundations rather than here, and the answer should be cited rather than improvised.

The relevant result is the *indiscriminate time principle*, a corollary of the extrusion principle [4], whose content is a pair of mutual non-informations: knowing where the events an attribute's blanks refer to sit in space and time tells one nothing about when those blanks resolve for oneself, and knowing when they resolve for oneself tells one nothing about where they sit. Blankness is a property of the indexing Aethus, never of a clock reading — *blanks lack tense*, and an observer's unlearned past is exactly as open as their unformed future [43].

The objection therefore rests on a premise the framework denies. It assumes that the Golden Point's futurity makes it a different kind of object from an unresolved attribute of the past — that the not-yet-real is a category. On the Aethic account there is no such category: the Golden Point is a blank attribute relative to our Aethus, in precisely the sense that an unread outcome of the deep past is blank, and conditioning on it is the same operation in both cases. The asymmetry one feels is real but *entropic* rather than ontological: the future blocks discriminating information more stringently because it sits downstream of the entropic arrow, while the past merely happens to withhold it — and informational uncertainty is the same thing under either obstruction [4].

The mirror-image case matters, because it is a structure the reader can check without granting anything particular to this work. Newcomb's problem conditions on a *past* event, the predictor's sealed placement, which is blank relative to the agent's information state; and any treatment on which the agent's decision bears on that placement must decline the assumption at issue here, that a past occurrence is thereby a settled fact. Past-Alignment conditions on a *future* event that is blank relative to ours. Same operation, opposite direction, and by the indiscriminate time principle the direction is not a difference. A reader who accepts the one and rejects the other owes an account of which of the two mutual non-informations they mean to keep.[10]

What the Golden Point does here is therefore what any conditioning event does: it selects, and does not act. The Aetherian Nexus is not *caused* to contain rocketry by a later passage; the weighting conditioned on that passage assigns overwhelming weight to histories containing it, and we read the record we are in. Whether anything stronger than selection holds, whether the future cone is constrained rather than merely uninformative, is exactly what §3.9 declines

[10] The companion treatment is [26], which develops the Newcomb case on exactly this footing: determinacy indexed to an information state rather than to a position in time, so that an unread past outcome is a blank entry and blanks lack tense.

to settle, and the Void hypothesis is the position on which nothing stronger holds at all.

*Remark* 3.41 (Why this is not circular, and where it could fail)*.* The obvious objection is circularity: conditioning on the Golden Point and then finding its prerequisites is no achievement, since the prerequisites are what the conditioning selects for. The objection would land against a bare appeal to Past-Alignment, and it is worth being explicit that it does not land here, because the argument of 6 is *comparative* rather than confirmatory. Two candidate conditioning targets are placed side by side; they issue *different* predictions about the record, one indifferent to industrial prerequisites, one not, and the record discriminates between them. A comparison that could have gone the other way is not a circle.

Three ways it could fail, in decreasing order of seriousness. The weight claim remains the decisive one, though §3.8.1 has reduced it from an assertion to a single measurement: the bound (3.6) is proved, and one prerequisite at $\alpha \sim 10^{-3}$ suffices, so what would remove the contradiction is now the specific demonstration that *every* candidate prerequisite has $\alpha$ of order one. Second, the argument requires "observer" to be robust under exactly these perturbations; a sufficiently demanding notion of observerhood, one entailing industrial capacity, say, would collapse the two conditioning targets together and dissolve the reductio, though at the cost of making "observer" do work it is not usually asked to do. Third, the reductio establishes that the target is *not* observer-existence; that it is specifically the Golden Point is supported by Past-Alignment rather than forced by the contradiction, and some third target could in principle serve.

Both arguments are now in hand. The natural-history one carries the weakness Remark 3.41 flags, its robustness premise is contestable, and the human-history one of §3.7 does not; a reader who accepts only the second has accepted the principle, and the first is then the same inference extended into deeper time.

### 3.9 Extrapolating forward: three hypotheses

Past-Alignment is a claim about the past cone and is silent ahead. Three positions on the forward question are available, and this section states them without adjudicating.

**Definition 3.42** (Extrapolation hypotheses; **[O]**)**. Void.** The alignment of our past cone with the Golden Nexus's is a brute and uninfluential coincidence. Whether or not some connection obtains, no law or inferential rule constrains our future cone to align with the Golden Nexus's at the same spacetime position.

**Weak.** There is an explicit distribution over futures: the Nexus describing our future cone assigns each proper child Aethus a weight proportional to some function of its probability of the Golden Point — in the simplest case, that probability itself. Comparing two possible futures is then possible by an Accordance-like metric, their relative likelihood tracking their Golden-Point probabilities. Futures in which the Golden Point does not occur are either incomparable under the method or, in the strong reading, of probability zero relative to one's Aethus.

**Strong.** The Golden Aethus is already stated in one's own Aethus, and has been since the beginning of centric unfolding. On this reading the stated attributes of Nexic reasoning were conditioning on the Golden Point throughout; we occupy the Aethic block universe with a physical moving present some centuries before that event, while the conditioning that it occurs is already fully resolved and supplies exactly the natural-historical content readable off our past cone. Contingency remains in *which* route is taken, which stated attributes we come to gather, conditioned on their consisting with the Golden Point, but outright Aethic disjointness from the Golden Point is forbidden.

*Remark* 3.43 (Three questions the Strong hypothesis raises). If the Strong hypothesis holds, three metaphysical questions become pressing and none is answered here [**O**]: why future-cone Aethic information should be available at all; how and why that information is so intimately bound up with the initiation of centric unfolding, being perhaps its earliest and leading piece of Aethic content; and what the implications are for the entropic principle of Aethic reasoning and its account of how the past is distributed between past and future Aethic cones. Deriving either of the second or third from the other would be a natural line of attack.

*Remark* 3.44 (One consideration bearing on the forward question, which does not settle it). The three hypotheses are stated and not adjudicated, and this remark adjudicates nothing. It records a consideration that bears on the first of them and raises its price. By Remark 5.64, the climbing of a schedule is selected for by the asymptotic condition and by no finite-horizon one. If the record was produced under conditioning at all, which §3.6 argues and does not prove, then the conditioning event is an infinite-horizon one: a property of the entire future cone rather than of any initial segment of it.

The Void hypothesis holds that no law or inferential rule constrains our future cone. That sits awkwardly beside a conditioning event which *is* a property of that cone entire. The awkwardness is not a refutation, and the distinction that saves Void should be stated for it: a selection is not a law, and a reader may hold that the record is what asymptotic conditioning produces while denying that anything about the asymptote constrains what follows. What the consideration does is make that a conjunction rather than a single claim — Void must now carry both halves, where as originally stated it carried one. Remark 3.45 sketches a form in which it might carry them.

Nothing here supports Weak or Strong, which face their own difficulties (Remark 3.43); nothing here claims the conditioning obtains; and §6.8 forbids anyone from knowing that it does.

*Remark* 3.45 (How Void could look: a proposal, not an adjudication; [**O**]). Void is stated as a denial, that no law or inferential rule constrains our forward cone, and a denial has no ontology. Since Remark 3.44 raises its price, it is only fair to record that a positive form is available which pays that price rather than evading it. What follows is a proposal about how Void *could* be realized, offered without endorsement and without prejudice to the other two.

*The construction.* Let the conditioning event be region-indexed in its own statement: attributes whose referential events lie in the past cone are constrained as one conditioning would have them, attributes elsewhere as another. This is not two operations — a conjunction of region-indexed constraints is a single event in the product space, and conditioning on it is one act. Read as a gate: *if past-cone, condition on $A$; otherwise on $X$.* A smooth interpolation between the two regimes serves as well as a step and is the more natural object.

*The weakening that makes it work.* $A$ need not *be* the infinite-survival condition. It need only approximate it closely enough that Principle 5 is observed — and Past-Alignment is an empirical claim about what the record looks like, which more than one event can produce. So $A$ may be categorically distinct from infinite survival while agreeing with it to within observational reach, and Void's forward denial becomes the claim that $X$ departs from $A$ where nothing has yet checked.

*What that buys.* The natural objection is over-determination: every region is eventually some observer's past cone, so a later observer treats as past-conditioned what an earlier one treated as future-conditioned, and consistency would force $X \equiv A$ everywhere — collapsing the construction into a single conditioning. Under the approximation reading it does not bite. The two need only agree *approximately* on regions that are somebody's past, and approximate agreement is compatible with categorical distinctness. The objection bites only against the strict identity, which the construction does not assert.

*The constraint that remains.* Whatever indexes the gate cannot be a distinguished spacetime position: Proposition 3.20 lets the conditioning be anchored at any Aethus in the lineage and returns that anchor's continuation, and Remark 3.21 forbids privilege outright. The index must therefore be schematic — computed by every observer from their own anchor and yielding the same partition of their own cone. That is not an obstacle so much as a specification: an anchor-invariant index exists in this work already, since the goldening interpolation of §6.2.2 is age-relative and its windows are a fixed fraction of a biosphere's own history.

*And the coincidence need not be explained.* If the transition is quasi-abrupt and falls near our position, Void owes no account of why. That is not evasion within Void's own terms: its thesis is that such alignments are brute and uninfluential, and a hypothesis whose content is bruteness pays in brute facts by design. The fair scoring is that this is a cost by ordinary lights, three brute facts where the hypothesis as stated carried one, and not a cost by Void's, and a reader will weigh those differently depending on what they think explanation is for.

*Where it risks collapsing into Weak.* The smooth version's danger is that a graded conditioning strength over futures is a weighting of futures, which is Weak's position and not Void's. Whether daylight remains turns on whether the interpolation index is a function of Golden-Point probability. This work keeps those apart, Remark 5.25 insists $\mathbb{P}(\Lambda_\infty < \infty)$ is the probability of advancing rather than of not dying, so the gap is real but narrow, and a proposal of this kind should be checked against it before being relied on.

None of this adjudicates. It says only that Void has a form available in which it is a positive hypothesis about what the conditioning event is, rather than a refusal to say; and the form leaves open exactly what the rest of this subsection leaves open, which is what the conditioning actually is.

*Remark* 3.46 (Weak as the intermediate, and what each refusal commits it to; **[O]**). Symmetry requires the same treatment of the remaining hypothesis, and Weak turns out to be the most structurally constrained of the three — its position is fixed by two refusals, each carrying a commitment it cannot decline. *Refusing Void commits Weak to privileging the Golden Point.* If some rule constrains the forward cone, Weak's is indexed to Golden-Point probability and to nothing else — so the Golden Point is not one outcome among many but the quantity that sets the measure. This commitment is shared with Strong, and it is what separates both from Void: the forward question's first division is whether the Golden Point indexes anything ahead.

*Refusing Strong commits Weak to our contingent position doing work.* Under Strong the conditioning is already fully resolved and every anchor returns the same settled object, so where one stands is inferentially idle. Under Weak the distribution is over *one's own* proper children, so the anchor does real work. That second commitment needs care, because stated carelessly it collides with Remark 3.21. It does not. What non-privilege forbids is a metaphysically distinguished position at which the conditioning is anchored and nowhere else. Weak's anchor is *schematic*: every Aethus indexes its own distribution over its own children, by the same rule, and Proposition 3.20 says precisely that the conditioning may be anchored anywhere and returns that anchor's continuation. Weak is therefore not in tension with recursive anchoring but is the hypothesis that takes it forward.

Which admits a sharper statement of Weak's position than the definition gives. *Weak is the only one of the three under which recursive anchoring has forward content.* Under Void there is nothing ahead to anchor; under Strong anchoring makes no difference, since all anchors return the same resolved object; under Weak each anchor returns a different distribution over its own children, and the differences are the hypothesis's whole content.

*One consequence for the Aetherian Nexus, which requires a distinction the section has not yet drawn.* Principle 6 demotes observer-conditioning as an *explanation* of the record: observerhood is too robust under perturbation to have produced it. That leaves untouched a second role. Something must specify where "we" are, and Principle 5's own footnote already leans on the Aetherian Nexus for exactly this, noting that our Aethus is a proper child of it rather than the Nexus entire. *Target* and *index* are different offices, and failing at the first says nothing about the second. Under Weak the index does forward work, since the distribution is over the children of a position the index picks out — so the Aetherian Nexus recovers a standing under Weak that it has under neither of the others, without any part of the reductio being withdrawn [5].

*And the costs, since the other two have had theirs stated.* Weak owes an argument for its functional form: the definition says the weight is proportional to *some function* of Golden-Point probability, and any increasing function would serve, with nothing yet selecting one — so the hypothesis as stated is a family rather than a position. It owes an account of how the index is computed, which the schematic reading requires be available to every anchor alike. And its weighting is *relative*, a share of a total the hypothesis does not bound, so it inherits the caution of Remark 5.60 without amendment: that a future carries the largest share of the forward measure is not that it is likely, and comparison by an Accordance-like metric is comparative in exactly the sense that supplies no absolute reassurance. None of the three is supplied here.

### 3.10 Relation to the rest of this work

Two compatibility questions arise immediately, and one structural parallel is worth flagging as an open lead. *Compatibility with the vantage restriction.* §6 insists that Golden reasoning is a vantage and not an obligation — inferential, licensing conditional expectations rather than duties, partly on the ground that $Q$ is a measure nothing occupies. The Void and Weak hypotheses sit comfortably with that restriction. The Strong hypothesis does not: if the Golden Aethus is already stated and disjointness is forbidden, then the future-directed reasoning has a purchase the vantage restriction disclaims. A reader adopting Strong should understand that it buys something $\Phi 2$ deliberately gave away, and should re-price the argument accordingly.

*Compatibility with the epistemic ceiling.* §6.8 holds that no finite observation can certify a biosphere as Golden, since $Q$ and $\mathbb{P}$ are locally equivalent and separate only at infinite horizon. Past-Alignment appears to sit close to this, since it asserts an agreement between observable content and the Golden Nexus. The two are reconcilable, and the reconciliation is that they concern different cones: the ceiling forbids distinguishing $Q$ from $\mathbb{P}$ over *future* trajectories, while Past-Alignment compares *past* attribute content. Nothing in Past-Alignment certifies that we shall reach the Golden Point; it claims that our record is what conditioning on that event would have produced, which is a claim about behind us.

*An open structural parallel.* The trichotomy of Definition 3.42 is suggestively parallel to the regime trichotomy of §6.2.1 — trapped, critical, viable. In particular the Strong hypothesis resembles the viable regime, where $\mathbb{P}(T = \infty) > 0$ and $Q \ll \mathbb{P}$ makes the Golden Biosphere an ordinary sub-population one may belong to, while Void resembles the trapped case in which no passage occurs. Should the parallel be exact, the extrapolation question would not be independent of the anatomy question: which hypothesis obtains would be settled by $\rho$ and by the driver's scale function rather than by metaphysics, and Theorem 5.38 would decide it. We flag this as a lead and do not claim it **[O]**.

*Remark* 3.47 (The reversal of order). If the argument of this section is accepted, the framework's architecture inverts. Optimation is no longer the starting weighting but the first derived result — what Golden-conditioning looks like cropped to the observable cone. The Golden Biosphere is no longer characterized by survival asymptotics and then found interesting; it is the conditioning target, and the survival asymptotics of §5 are the account of what such a target consists in. The sections that follow are then not a sequence of separate results but the development of a single object from different sides: §5 gives its dynamics, §6 its ensemble structure and what may be inferred from occupying it, and §11 the institutional machinery that would determine whether a given biosphere reaches it.

## 3.11 Ledger

**Costs.** *Past-Alignment* (5) — [$\Phi$], the section's principal commitment; *falsifier:* an attribute of the record whose presence Golden-conditioning does not predict, or a Golden-relevant trait whose probability is undiminished under perturbation (Remark 3.24). *The weight claim* within 6 — **[S]**, asserted rather than computed, and the reductio's load-bearing step; *falsifier:* a demonstration that the counterfactual sum does not outweigh the Aetherian Nexus. *Robustness of observerhood* — [$\Phi$]; *falsifier:* a defensible notion of observer entailing the industrial prerequisites, which would collapse the two conditioning targets. *Rung-screening*, inherited with the substitution property [23], and carrying its license unchanged.

**Buys.** Optimation derived rather than posited, and derived from a comparison that could have failed. A conditioning target that explains what observer-conditioning cannot — why the record contains its industrial and technological prerequisites at all. An explicit statement of what is *not* claimed about the future, together with the three hypotheses that exhaust the forward question. And a reorganization of the surrounding work into the development of one object, stated in Remark 3.47.

**Inherited dependency.** Both arguments now depend on technoanatomy being a *process* rather than a parameter, for the reason given in Remark 10.8: under a fixed adverse $\rho$ the cascade pronounces the question closed before it is asked, and the section's claims are then vacuously true and useless. The dynamics of $\rho_t$ are open **[O]** (§5.6.3), so this section's arguments are conditional on a piece of the framework that is not yet built. That is a real exposure and is not reduced by the arguments' internal soundness.

**Not claimed.** That the future cone aligns with the Golden Nexus; that the Golden Point will occur; that any of the three extrapolation hypotheses is correct; and that the parallel with the regime trichotomy of §6.2.1 is more than suggestive.

# 4 Task-objects: individuation for a perturbed history

Every argument of §3 quantifies over perturbed natural histories. The dinosaurs never evolve; Africa develops otherwise; a Golden Trek contains no Earth. Each such argument asks whether some attribute survives the perturbation, and each therefore presupposes an answer to a question it never asked: what makes a perturbed Earth still an Earth, or a perturbed Aristotle still an Aristotle. Under an ontology of possible worlds the question has a standard answer. Under an ontology of Aethae it does not, and the perturbation arguments are ill-posed without a replacement.

This section supplies the replacement, which our earliest dated work already contains: the *task-object*, an object individuated not by perceived independence from other objects but by the contribution it makes to a designated root. The concept was arrived at independently and coincides with Kripke's distinction between rigid and contingent designators [44]; the departure from Kripke is not in the distinction but in what is done with it, and in a denial of the metaphysics his rigid designators presuppose (§4.3). Axiomatically the section develops Remark 5.7: the Golden-categorical premise supplies the root, and this section builds the objects the premise individuates, criterion there, construction here, one doctrine, and, under the Route World's Strong reading only, the section closes on the framework's ontological climax, the Golden Aethus (§4.6). Two liabilities are carried openly: the correspondence between objects and task-objects is many-to-many in both directions, so a selection principle is owed and only sketched (§4.4); and the reply to the standard identity objection runs through a commitment about consciousness whose cost is priced in §4.5.

## 4.1 The problem: perturbation requires individuation

Consider the argument of §3.5.2 in the form it was actually run. One perturbs the record, the dinosaurs persist, and observes that the capacity to build rocket vehicles does not survive. For this to be an argument rather than a stipulation, the perturbed history must remain comparable to ours: it must still contain a planet, a biosphere, a lineage, against which the missing capacity can be registered as missing.

That comparability is what ordinary object-talk cannot supply here. Perturb the record enough that no informed person would call the resulting planet Earth, and the question "does Earth still have rocketry?" does not become false; it becomes unasked. Yet the argument needs it asked, because the Imperfection Magnitude Drop it is trying to exhibit is precisely a drop *in the substituted planet's* contribution to the Golden Point.

The difficulty is sharpened by the framework's own commitments. Under an Aethic ontology there are no completed determinate histories across which an object could retain a categorical substance, so nothing plays the role that a rigid designator's referent plays in the possible-worlds picture (§4.3). The individuation must therefore come from somewhere else, and the proposal is that it comes from the root the analysis is already conditioned on — which is Remark 5.7 arriving as this section's solution: the definitional root of well-formed feature-terms and the conditioning target of the perturbation arguments are one object, and the coincidence is the design.

## 4.2 The definition

**Definition 4.1** (Task-object; [Φ]). Fix a root Aethus $\mathcal{R}$. A *task-object* relative to $\mathcal{R}$ is an object individuated not by its perceived independence from other objects but by the contribution it makes to $\mathcal{R}$. Where the root is the Golden Point, a *Task-Earth* is — for instance — the planet on which the makers of the Golden Point speciated, and the corresponding *task-humans* and *task-events* are individuated likewise.[11]

*Remark* 4.2 (The axiomatic root, and what contribution is). Two claims fix this definition's place in the framework, and both run in the same direction: from Remark 5.7 to here.

*Task-objects arise from the Golden-categorical premise.* The root parameter $\mathcal{R}$ is, at its origin and at its canonical value, the Golden Point, and the derivation order is premise first: Golden-categorical robustness is the criterion of well-defined reference, and the task-object is what the criterion *builds* — the object-form of Golden-categorical individuation. The parametric definition above is the abstraction from that case, not a neutral genus of which the Golden root is one species, and the distinction matters below.

*At the canonical root, contribution is a defined quantity.* "The contribution it makes to $\mathcal{R}$" is, for an arbitrary root, a phrase owed a semantics. At the Golden root the framework already possesses one, assembled from its own operations: complement the individuating attributes and read the displacement, the Imperfection Magnitude Drop of §3.1, graded by the $\varepsilon$-Departure apparatus (Definition 3.14) and priced by the investment bound (Proposition 7.6), so that an object's contribution to the Golden Point is the $G$-weight its complementation forfeits. Nothing analogous exists for an arbitrary root unless its inquiry supplies it. So the concept's well-posedness is native at the Golden root and *inherited* elsewhere, exactly where an analog referent for contribution is supplied — which is the precise sense in which the premise is the concept's conceptual root and not merely its first customer.

*The axiomatic order matches the genetic order.* The provenance footnote above records that Task-Earth was built in July 2022 in service of the Golden Point material it predates on paper; the premise converts the service into the source. That the concept's stated axiomatics now runs in the direction its construction actually ran is recorded deliberately: it is the difference between a root assigned in retrospect and a root that was generative, and the corpus's dated record is what makes the claim checkable rather than honorific.

None of this retracts the root-relativity stated next; it locates it. The *machine* is parametric in $\mathcal{R}$; the *semantics* is generative at the Golden Point.

**Root-relativity is by design, not by accident.** The root is a parameter. Task-object individuation is root-relative by construction: the machinery requires no particular root, and a different inquiry with a different root would individuate differently and legitimately — provided it supplies what Remark 4.2 shows the Golden root supplies natively, a formal referent for contribution. This matters for how the concept should be read outside this paper: nothing in the *machinery* is committed to prosperity, survival, or any other particular optimand. What is committed is a pair: that some root must be designated before individuation is well-defined, and that at the designated root contribution must mean something. The Golden Point is where this program obtains both at once, and where the concept was generated; other roots rent what it owns.

[11] *Provenance.* The concept and the Task-Earth example are ours, recorded 1 July 2022 and reproduced in the natural-historical development of August 2022 [25], which makes this the earliest dated item in the present corpus — predating the extrusion principle by some weeks and the Golden Point material it was written to serve by a month. The formalization below, the placement against Kripke, and the pricing of the identity objection are later.

**Why this is not a redescription of ordinary objects.** Ordinary objects and task-objects come apart in both directions, and the examples are ours. Earth 3.9 billion years ago is called Earth, despite differing from the present planet in continent configuration, atmospheric composition, and much else. A perfect duplicate in another star system would not be called Earth, despite differing in none of them. Neither verdict tracks contribution to a root, and both are stable features of ordinary usage — which is precisely why ordinary usage cannot serve where the analysis needs comparability across perturbation.

### 4.3 Relation to Kripke, and the departure

The distinction between a designator that picks out its object robustly under counterfactual variation and one that picks it out by a description which the object might not have satisfied is Kripke's [44]. The present concept was arrived at independently, by a different route, asking what continuity there is between Earth's persistence in the past and the prospects of a biosphere's near future, and the coincidence is worth recording as a convergence rather than claimed as a discovery.

The departure is twofold.

*First, in what is done with the distinction.* Kripke's conclusion is prohibitive: do not use contingent designators where rigid ones are required. The present move is instead to adopt the awkward category as both ontologically apt and operationally useful — the pattern by which physics adopted Newton's first law over the intuition of motion requiring a mover. That pattern is not general license, and should not be read as one: counterintuitiveness is usually a warning, conspicuously so in moral matters, and the claim is only that this is one of the uncommon cases in which the received intuition is out of step with the ontology, plausibly because what served human societies through evolution was never selected for tracking it.

*Second, and more radically, in a denial of the metaphysics.* The Aethic framework does not accept the objects to which rigid designators are supposed to point. With Aethae rather than possible worlds as the base unit, there is no categorical substance persisting across worlds for such a designator to track: by the Aethic reversal principle, the underlying determinate reality required to constitute such an object cannot be well-defined in the first place [4]. The scope of this denial should be stated carefully — *what is denied is Kripke's metaphysical presupposition, not his linguistics.* The modal argument survives untouched as a claim about how names function in language; the present claim lives a level below, about what that functioning tracks.

**Robustness recovered without substance.** What made rigid designation attractive was robustness under perturbation, and that is recoverable without the metaphysics. Placing several description classes of an object in Aethic superposition yields a designation that survives perturbation by landing in another class when one fails — robustness as a property of the superposition rather than of a substance beneath it. The strong form of the resulting position is that an Aethic ontology must make task-objects fundamental and admit rigid-designator talk only as non-arbitrary shorthand for them.

### 4.4 The two-way fan, and the selection problem

**Proposition 4.3** (Many-to-many in both directions; **[S]**)**.** *The correspondence between objects and task-objects is many-to-many. One object admits indefinitely many task-objects: Earth may be individuated as a spherical body composed largely of iron, or as a region within a star's* *habitable zone, and so on without limit. One task-object admits indefinitely many objects: fix Task-Earth as the speciation site of the Golden Point's makers, and every planet satisfying it qualifies — including one on which Hawaii never formed, which is not our Earth and is a Task-Earth.*

The fan is the concept's principal technical liability, since it means a task-object is not determined by the object it individuates, and some selection is required. Our own assessment is that the selection carries irreducible speculation, but speculation that might be linguistically systematized; the criterion offered is to prefer the task-object yielding the simplest or most informative interplay with the wider question the root encodes.

**The criterion is a gauge choice, and the paper already has one.** That criterion is recognizably a gauge fixing: one probes the freedom until the description aligns most economically, exactly as one selects coordinates in which a physical problem is cleanest. The comparison is ours and is substantive here, because §5.3 performs the same operation on the survival model — initial data there form a gauge orbit invisible to every asymptotic exponent, and the analysis proceeds by identifying what the freedom cannot touch. The two are the same methodological move at different levels: find the quantities the arbitrary choice leaves invariant, and state the theory in those.

This suggests where a systematization would come from, and the framework supplies a candidate it does not yet apply here. The construction for degree-of-existence over description classes [4] selects canonical classes as *fixed points*, those whose further refinements no longer improve description quality, which is exactly the shape of criterion the task-object selection problem needs. Whether that construction transfers is open **[O]**, and it is the first thing we would attempt.

At the canonical root a second instrument becomes available, and it is this work's own. Among the task-objects an object admits, the conditioning is not indifferent: it concentrates on the individuations that are load-bearing, those whose defining contribution the record shows *invested*, in the sense Proposition 7.6 prices, so "prefer the task-object whose contribution carries the weight" is a selection criterion the Golden root supplies for free, where an arbitrary root supplies none. It is offered at **[S]** beside the fixed-point candidate, not in place of it; the two would be expected to agree where both apply, and testing that agreement is part of the same open item.

## 4.5 The identity objection, and what answering it costs

The standard objection is Kripke's own. Aristotle might not have taught Alexander; he would have been Aristotle regardless; so the person is what the name tracks, and no task-human individuation is required. Under the present proposal the corresponding task-human, the man who taught Alexander in the sense of equipping him for some specified subsequent achievement, would simply not have been instantiated, and the objection says this gets the identity facts wrong.

The reply is not to deny the objection's premise but to run it out.

**The reply, as a reductio of the objection's own criterion.** Grant that identity is settled by continuity of consciousness through a common point of divergence, which is what the objection's force rests on. Under centric unfolding, any other person's self-describing Aethus is reachable by ascending to a sufficiently early ancestor Aethus, plausibly in infancy or before, wherever a substantial portion of the universe enters disagreeing superposition, and perturbing there. The objection's own criterion therefore identifies not merely the untutored Aristotle with Aristotle, but *every* person with every other: the set of human-pointing rigid designators collapses to a single equivalence class.

The structure of the reply should be read carefully, since its force depends on it. It is not a positive thesis that all persons are identical, offered for acceptance. It is a demonstration that the criterion the objection relies on, applied consistently under centric unfolding, cannot deliver the fine-grained identity facts the objection needs — so the objection cannot be used against task-object individuation without first supplying a criterion that survives its own generalization.

Two costs are recorded. The reply commits to reading continuity of consciousness through the framework's own treatment of centric unfolding [4], which is a substantive commitment and not a neutral one. And a reader may take the collapse as a reductio of the framework rather than of the objection; the reply is that the collapse follows from the objection's criterion together with centric unfolding as stated, so contesting it while retaining Aethic reasoning would require an extra-centric unfolding proposal, which nothing here supplies **[O]**.

### 4.6 The ontological completion: the Golden Aethus

Everything in this section so far has run at the level of reference: a criterion (Remark 5.7), its constructions (Definition 4.1), and a formal referent for contribution (Remark 4.2). This subsection states what the arc closes on when the Route World's Strong reading is added, and it should be said plainly at the outset both that this is the framework's categorical and ontological climax and that it is *conditional*: everything here consumes the Strong reading, and nothing earlier in the section does. A reader who declines the Route World keeps the semantics whole and loses only what follows.

**Definition 4.4** (Golden Aethus)**.** The *Golden Aethus* is the Aethus that states exactly the Golden attribute, a Golden biosphere in the future light cone, the Golden Point event as a future-referring stated attribute, with every other attribute genuinely blank. It is the *index* of the Golden Nexus: conditioning the Cosmic Nexus on the Golden Aethus's stated content is what the Golden Nexus *is* (§3.1). It is distinct from the Aetherian Nexus, which is the Aethus of our whole natural history: under the Strong reading the Aetherian Nexus is a weight-carrying descendant of the Golden Aethus — the record is stated content accreted within the Golden Aethus's blanks.

*Remark* 4.5 (The toggle: void and world)*.* Two measures stand one attribute apart, and the distance between them is the argument. The bare Cosmic Nexus absorbs no conditioning (§3.1), and its typical configurations are the high-entropy ones: unconditioned occurrence-weight is entropy-dominated, ordered macrostructure is exponentially atypical within it, and the companion's own reading of the bare measure's default tendencies is the same fact in Nexic vocabulary [5] **[S]**. Call that face what it is — the void. Now state the one attribute the Golden Aethus carries, and the weight redistributes onto exactly the histories that can *carry* it: a Golden biosphere is not statable over vacuum, so the conditioning concentrates on deep ordered records, biospheres, lineages, the whole furniture of a world. The existence-in-the-weighting of every ordered thing therefore toggles with a single stated attribute — void on one side of the statement, world on the other. And the toggle has a biosphere-scale mirror already on the books: it is $\mathbb{P}$ against $Q$

(Definition 6.1) run at cosmic register, the goldening of the universe rather than of a biosphere, which is why nothing in this subsection requires machinery the paper does not already own.

**Everything as a fluctuation on the Golden Aethus; [Φ], Strong-conditional.** Under the Strong reading, the toggle licenses an ontological statement, and it is the strongest sentence in this work: each thing, object, and circumstance in the universe is a *fluctuation on the Golden Aethus* — a weight-carrying excursion within its blanks, superposition content that the Golden Nexus renders load-bearing.

The word "fluctuation" is doing exact work, and the inversion inside it should be stated with care. Under the bare Cosmic Nexus, ordered structure is a fluctuation in the dismissive sense: improbable, incidental, owed nothing. Under the conditioning, the same structures are fluctuations still, they remain the variance-structure of a measure, excursions rather than substances, but they are no longer accidents: the conditioning concentrates on them because the stated attribute cannot be carried without them. What toggles is not fluctuation-status but *accident*-status. That is the precise sense of "where things arise from": things arise in the weighting, from the Golden Aethus, as its resolution-structure — and the framework has described this shape before. A biosphere-scale state is already, on the books, a weighted superposition of the poles (Remark 3.12, Remark 3.15); the present thesis is that description run on the universe's contents — the same shape at cosmic scale, a consilience rather than an addition.

Two consequences discharge obligations incurred earlier. First, the categorical claim: if things themselves arise from the Golden Aethus, then categories drawn non-arbitrarily should trace to the Golden Point — not because a semantics was chosen well but because the individuation being read off is the world's own. The Golden-categorical premise and the task-object machinery are, under Strong, not conventions but transcriptions, and the "intended all along" of Remark 4.2 acquires its full meaning: the criterion was tracking constitution. Second, the contribution referent becomes a constitution-measure: the $G$-weight a complementation forfeits (Remark 4.2) is, under Strong, how much of the thing there *is* — the analogy to read it by, flagged as structural and consuming no physics, being excitations over a ground state: objects stand to the Golden Aethus as excitations to a vacuum, the geometry being that of a conditioned measure's typical set and nothing more.

*Remark* 4.6 (The vantage itself: non-indeterminacy roots to the same point). The fluctuation thesis concerns what things *are.* A further extension, the author's, concerns the standpoint from which anything is determinately anything at all — and it is worth stating separately because it is what carries the ontological completion into registers as abstract as morality (§9.7). The companion's first postulate already makes determinate reasoning *indexical*: without a central Aethus there is no validity, no pruning, no context — the procedural statement that all logic runs relative to an index [4]. The question the postulate leaves open is which index a *universe's* worth of determinate content has. Under the Strong reading the answer is the one this subsection has already built: the contentful universe-scale index is the Golden Aethus, of which every determinate vantage in the record, the Aetherian Nexus, and our own Aethus within it, is a weight-carrying descendant. So the toggle of Remark 4.5 reads epistemically as well as ontologically: remove the conditioning and what remains is not merely the void of things but the void of *standpoint* — indeterminacy with no structure in any particular direction, in which anything goes because nothing is stated anywhere. What separates vantage and context from that abyss is the same object that separates world from void. (The view-from-nowhere tradition meets the framework exactly here, and is engaged at Remark 9.13: a literal nowhere is the abyss, so the honest question was never whether to view from an index but which index the viewing consumes [45].) Conceptual form, not only thing-ness, roots to the Golden Point — graded by the same ladder, priced by the same fences, and consumed nowhere earlier than this remark.

**A payoff: the fluctuation-observer pathology displaced.** One consequence is substantial enough to state separately, at **[S]**. Under a bare high-entropy measure conditioned only on the existence of an observer-moment, the typical observer is a *minimal* fluctuation — the Boltzmann-brain pathology of the cosmological literature [46, 47]: records deceptive, environments absent, induction self-undermining. Under conditioning on the Golden attribute the pathology is displaced by the toggle itself, with no added postulate: a minimal deluded fluctuation contributes nothing to carrying a Golden biosphere, so the conditioned weight concentrates on observers embedded in the deep, real, ordered structure the attribute requires — observers whose records are records. Two scope notes keep this honest. Observer-conditioning alone (Optimation in the bare sense) does not obviously achieve the displacement — it is the Golden attribute's structural demands that do the work, which is an argument *for* the Golden target from an unexpected quarter. And nothing here claims a solution to the cosmological measure problem: the claim is internal to the framework's weighting, and is priced as such. One further adjacency is recorded at **[O]** and not claimed: whether the record's thermodynamic asymmetry is itself induced by the future-referring conditioning, a future-boundary condition that tense-freedom renders legitimate, is open.

*The price, paid in full.* Four fences. *Conditionality, as a ladder*: under Strong, the thesis is the framework's ontology; under Weak, it holds of the record, the past cone is Golden-conditioned by Past-Alignment, and is silent forward; under Void it is unavailable, and the semantic premise of Remark 5.7 stands alone on its comparative defense, having consumed none of this. *Not idealism*: the Golden Aethus is nobody's mind; statedness is a property of an index and not of anyone's access (Remark 8.20), and every verdict here is realist in exactly that sense. *Not teleology*: arising-from is arising-in-the-weighting; the conditioning does not act, choose, or design, and the standing fences of §3 govern unweakened. *Scope*: the quantifier "essentially all" ranges over real-world feature-terms per Remark 5.7; abstracta are outside it by that remark's own scoping, and the categorical claim carries its own check — a non-arbitrary real-world category that resists every tracing to the root would break the quantifier, and exhibiting one is the invited falsification.

The arc of this work's categorical machinery, criterion, construction, referent, constitution, terminates here. This is, conditionally and at its stated price, where things arise from.

## 4.7 What this licenses

With individuation in hand, three things earlier in the paper become well-formed rather than merely persuasive. *The perturbation arguments.* §3.5.2 may compare our record with histories in which the dinosaurs persist, because both contain a Task-Earth and a task-lineage against which a missing capacity registers as missing. Without task-objects the comparison has no common term and the argument does not run.

*The substitution property.* The generalization across rungs [5] places a trait of ours in the conditioning seat and deduces that alternative histories fail its occurrence. That maneuver requires the trait to be identifiable in histories where it does not occur, which is a task-object identification and not an object one. *Imperfection Magnitude Drops.* A drop is a fall in contribution to the root under substitution [23]. Since a task-object just is an object individuated by that contribution, the drop is a change in a task-object's value rather than a comparison between incommensurable objects — which is what makes it a quantity at all.

**Order of development.** The dating is worth one line, because it inverts the order of presentation. Task-objects are dated 1 July 2022; the Golden Point material follows in August [25]; the extrusion principle and with it tense-free blankness follow weeks after that. The individuation scheme therefore came first and the ontology that vindicates it came second — the concept was constructed to serve a question about a biosphere's prospects, and only later found itself entailed by an ontology built for unrelated reasons. We record this as provenance rather than as evidence, but it does bear on the charge that task-objects are a device fitted to the argument they support.

### 4.8 Ledger

**Costs.** *Task-object individuation* (4.1) — [Φ]; *falsifier:* a perturbation argument of the kind §3 requires, shown to be well-formed under ordinary object individuation, which would make the apparatus idle. *The denial of rigid designators' referents* — [Φ], inherited from the Aethic ontology [4] and not argued here; *falsifier:* the ontology's. *The selection principle* — [**O**], owed and not supplied; the fan of Proposition 4.3 is real and a criterion beyond "simplest or most informative" is not given. *The identity reply* — [Φ], resting on continuity of consciousness read through centric unfolding, with the singleton collapse acknowledged as a cost a reader may decline to pay.

**Buys.** Well-formedness for the perturbation arguments, the substitution property, and Imperfection Magnitude Drops, none of which survives ordinary object individuation intact. A principled account of why ordinary usage fails here rather than an appeal to its being inconvenient. And a concept whose earliest dated statement precedes both the material it serves and the ontology that vindicates it. And, under the Strong reading only, the constitution reading of §4.6, with the fluctuation-observer displacement it purchases.

**Not claimed.** That Kripke's modal argument is mistaken; that all persons are identical; that the Golden Point is a privileged root *of the machinery* — its generative standing for the semantics (Remark 4.2) and, under Strong only, its constitutive standing for the ontology (§4.6) *are* claimed, and priced where stated; or that the selection problem is solved.

# Part II: The Survivorship Mathematics

Part II is the mathematics alone, and nothing normative enters it. The two-channel hazard is built on the Kolmogorov pair, the two-condition theorem separates the regimes, the schedule channel is developed as the third hazard, and the survivorship limit is taken to the Doob transform with its three modes (§5). Everything that the later parts consume is proved here at [**P**] or carried at a stated grade, so a reader who wants only the probability can stop when this part ends, and it is for that reader that the standalone extraction exists.

# 5 The model and its survivorship structure

This section is mathematics. It establishes the object the philosophical section reasons about, and it is organized by what that reasoning consumes rather than by order of discovery: the object (§5.2), its coordinates (§5.3), the verdict on survival (§5.4), and the structure of conditioning on survival (§5.8). Material that maps the terrain around the model rather than establishing it, horizon theory, the free-energy background, the waiting room and gain profile, the transfer computations, and the numerics, is placed in appendices, where it is available without standing in the way.

Claims are tagged **[P]** proved here, **[C]** classical or cited, **[A]** conditional on the soft-wall Ansatz (Ansatz 5.18), **[S]** scaling-level, **[O]** open. The tag **[C]** is load-bearing and frequent: the persistence exponent, the harmonic function of the conditioned process, the free-energy asymptotics of the background and the boundary classification of one-dimensional diffusions are all imported, and this section claims none of them. What it claims is the model, the coordinates, and the assembly.

## 5.1 Relation to the literature

Four bodies of work border this section, and the reader deserves a precise map of what is classical, what is adjacent, and what is claimed as new. *The classical layer.* The state process is Kolmogorov's 1934 diffusion [48], the primordial hypoelliptic example [49]. Its persistence theory is a mature subject with a clean lineage, McKean [50], Goldman [51], Lachal [52], Sinai [53], Isozaki–Watanabe [54], Denisov–Wachtel [55], surveyed in [56] and, as the *random acceleration process*, possessing an extensive physics literature reviewed in [57]. The conditioned (never-hitting) process, our limit object in §5.8, is constructed by Groeneboom, Jongbloed and Wellner [58]. None of this is ours; we import it.

*The adjacent frameworks.* Our survivorship interpolation belongs to two established genres. First, Theorem 5.75 is a *penalization* statement in the sense of Roynette–Vallois–Yor [59]: a limit of Wiener-type measures reweighted by a time-dependent functional, identified as a martingale change of measure — here the weight is survival itself, and the polynomial tail places it in the clockless class. Second, the mode classification of Theorem 5.81 sits against the theory of quasi-stationary distributions and $Q$-processes surveyed in [60]: classical QSD theory is at home in the exponential-tail class (our modes (I) and (III)), while the critical polynomial mode admits no proper QSD in fixed coordinates, its Yaglom object being self-similar. We use both frameworks as positioning, not as inputs; the proofs here are self-contained given the cited persistence theory.

*The applied layer: what this is a model of.* Stochastic hazard rates are not new, and the wrapper $S_t = \exp(-\int_0^t h)$ with $h$ a diffusion is entirely standard: it is the reduced-form, or intensity-based, framework of credit risk [61] and of stochastic mortality, where the force of mortality is routinely modeled by a diffusion, the mean-reverting Brownian Gompertz specification of Milevsky and Promislow [62] being a canonical instance. What distinguishes the present model is not the wrapper but *where the noise is placed*: standard practice diffuses $h$ or $\log h$, whereas we diffuse $-\frac{d}{dt}\log h$, one integration further out. The consequences are structural rather than cosmetic — $\log h$ becomes $C^1$, the state becomes degenerate and 3/2-self-similar, and the governing persistence exponent moves from $\frac{1}{2}$ to $\frac{1}{4}$ (Remark 5.9, §5.4.3). We stress the point because it locates the novelty precisely, and disclaims the rest.

*The conceptual ancestor.* The thesis of §5.8, that selection on survival makes the observed ensemble diverge systematically from the unbiased one, is not new in substance. It is the frailty phenomenon of Vaupel, Manton and Stallard [63] and Vaupel and Yashin [64]: when the frail are removed first the population hazard falls below the individual hazard, the gap widening with age, producing mortality deceleration at the population level with no corresponding deceleration for any individual. We claim no priority over that insight, and regard this section as exhibiting its *critical dynamical* analog. The differences are structural and are the point. Frailty is a static random variable fixed at birth and the analysis is a mixture computation over a population; here the randomness is a driver evolving in time, and the object is a limit of path measures singular with respect to the unbiased law (Theorem 5.80). Where the frailty literature computes how an observed hazard curve bends, the interpolation of §5.8 identifies the limiting law of the survivor's entire trajectory, together with the drift (5.50) that conditioning manufactures.

*What is claimed as new.* To our knowledge, the following are new at least in assembly, and we flag them as this section's contribution: the log-derivative placement of the noise, with its gauge analysis (§5.3.1) and the two-sided soft-wall analysis whose remaining gap is located exactly (Ansatz 5.18); the scale-function horizon trichotomy with mortal critical case (§A.2); the adverse-drift survival rate $3\nu^2/8\sigma^2$ with its front-load-and-coast optimal profile (Proposition A.10); and the survivorship interpolation read as a monotone flow with the tail index as its linear-response coefficient, classified across drivers into saturation, transformation and concentration (§5.8). Where a statement merely re-expresses known results in this frame, the tags and citations say so.

**Convention 5.1.** For positive functions, $f \sim g$ means $f/g \to 1$; $f \asymp g$ means $0 < \liminf f/g \leq \limsup f/g < \infty$; $f \lesssim g$ means $f \leq Cg$. Hatted quantities are nondimensionalized by the intrinsic timescale of Definition 5.20. $\Phi$ is the standard normal distribution function; $\mathbf{1}_A$ is an indicator; Leb is Lebesgue measure. All processes are defined on a filtered probability space satisfying the usual conditions, and $W$ denotes a standard one-dimensional Brownian motion.

## 5.2 The object

### 5.2.1 Setup: hazard, credit, and the Kolmogorov diffusion

**Definition 5.2** (Model)**.** Fix $\sigma > 0$ and initial data $(h_0, \mu_0, D_0) \in (0, \infty) \times \mathbb{R} \times \mathbb{R}$. Define the *improvement rate*, *credit*, *hazard*, and *cumulative hazard* by

$$\mu_t = \mu_0 + \sigma W_t, \qquad D_t = D_0 + \int_0^t \mu_s \, ds, \qquad h_t = h_0 e^{-(D_t - D_0)}, \qquad \Lambda_t = \int_0^t h_u \, du, \qquad (5.1)$$

and write $x_t = \log h_t$, so that $\mu_t = -\dot{x}_t$. We also set $I_t = \int_0^t W_s \, ds$, so that $D_t = D_0 + \mu_0 t + \sigma I_t$.

*Remark* 5.3 (Which quantity accumulates, and why it is not the logarithm)*.* One choice in the construction below is easy to make silently and worth making out loud, because this work returns to its consequence three times. Along any single path the natural additive ledger is the *cumulative* hazard $\Lambda$, and survival is $e^{-\Lambda}$ on that path. Across a superposition the two do not commute: $\Lambda$ is per-path and does not average, $\mathbb{E}[e^{-\Lambda}] \neq e^{-\mathbb{E}[\Lambda]}$ by Jensen, with the gap in the survivor's favour, while the *weight* does average, linearly, because averaging weights is what the framework's law of total probability is.

So the canonical accumulator is the weight and not its logarithm, and $-\log$ survival is a per-path reparametrisation rather than a quantity one may average over Aethae [4]. The Cox construction below respects this by keeping $\Lambda$ inside the expectation, which is why $\mathbb{P}(T > t) = \mathbb{E}[e^{-\Lambda_t}]$ and not $e^{-\mathbb{E}[\Lambda_t]}$. The same gap is what Remark 5.63 records for completion weight, what Remark 5.57 records for the criticality criterion, and what Remark 5.61 records for the companion's contrast. Four instances of one fact: a geometric mean is what a path experiences and an arithmetic mean is what an ensemble reports, and everything in this work that looks like optimism under averaging is that gap read from the wrong side.

**Definition 5.4** (Departure time; Cox construction)**.** Let $E \sim \mathrm{Exp}(1)$ be independent of $W$ and define $T := \inf\{t \geq 0 : \Lambda_t \geq E\} \in (0, \infty]$. Then $\mathbb{P}(T > t \mid \mathcal{F}_\infty^W) = e^{-\Lambda_t}$ and $\mathbb{P}(T = \infty \mid \mathcal{F}_\infty^W) = e^{-\Lambda_\infty}$, with the convention $e^{-\infty} = 0$. The *survival function* is $S_t = \mathbb{P}(T > t \mid \mathcal{F}_\infty^W) = e^{-\Lambda_t}$.

**Definition 5.5** (Viability)**.** The *viability event* is $\mathcal{I} := \{\Lambda_\infty < \infty\}$. On $\mathcal{I}$ the conditional survival probability $S_\infty = e^{-\Lambda_\infty}$ is strictly positive; off $\mathcal{I}$ it vanishes. Thus

$$\mathbb{P}(T = \infty) \;=\; \mathbb{E}\big[e^{-\Lambda_\infty}\mathbf{1}_{\mathcal{I}}\big] \;\leq\; \mathbb{P}(\mathcal{I}), \tag{5.2}$$

and we distinguish throughout between *viability* ($\mathbb{P}(\mathcal{I})$, the probability that immortality is achievable) and *immortality proper* ($\mathbb{P}(T = \infty)$).

*Remark* 5.6 (What the hazard is a hazard of). A reading question should be settled at the definition, because an accusation hangs on it. If the Cox event is glossed "biosphere death," then the framework has installed at its root exactly the primitives, organism, death, biosphere, it elsewhere claims to individuate from survivorship structure, and the accusation of smuggling would be fair. The gloss this work intends is different and should be stated plainly: the Cox event is a *departure-sufficient occurrence*, an event whose realization forecloses, and $h$ is the arrival rate of sufficiency, with nothing about cells, organisms, or death presupposed. Death-events are then the typical *occupants* of the departure-sufficiency *role*, which is where their operational anchoring (extinction events are what a record shows) survives intact — but the role, not the occupant, is what the model consumes, and the role is stated in the framework's own primitive, the closure of possibility.

The circularity this invites is apparent and dissolves by direction. The worry: the hazard is defined by reference to the Departure Point, and the Departure Point (§3.4.1) is built over the hazard's own events. The dissolution: the implication runs one way, and only one, by design. Base is defined by the analytic model; $\mu G$ is its closure under inevitability; the containment $\mu G \supsetneq$ Base (3.3) says the Cox events *suffice* for departure and are not necessary to it — so the hazard is an implies-departure object, never a departure-defining one, and the unfavorable modal cases are added by $G$'s recursion, never by re-writing the hazard. This is the same non-biconditional shape as the companion's third postulate, and deliberately so (Remark 3.6): where a biconditional would close a circle, a one-way implication that recurses to a wider net leaves the definitional order a straight line — events at the root, closure above them, and the arrow never reversing. The mathematics is untouched by any of this; what the remark fixes is the interpretation layer, and with it the answer the smuggling objection receives.

*Remark* 5.7 (Golden-categorical well-definedness; [Φ]). The preceding remark removes death and biosphere from the root; this one states what stands there instead, and it is entered as a premise at its own grade because it is a semantic thesis and not a theorem.

*The premise.* The well-definedness of real-world feature-terms takes the Golden Point, the asymptotic survivorship structure, as its first-principles root: a term is well-defined, in the sense this framework can certify, to the degree its reference is robust relative to that structure. The *biosphere* itself is the first application: it is individuated as the carrier of the survivorship structure, the unit whose continuations the fixed points of §3.4.1 classify, rather than presupposed as a biological given. And the *task-object* is not a second application but the premise's constructive form: §4 is where this premise builds its objects, the root Aethus of Definition 4.1

instantiated at the Golden Point being the case from which the concept arose and the case at which "contribution" has a formal referent (Remark 4.2). The premise and the task-object are one doctrine at two registers, criterion and construction, and the section develops the second. The premise generalizes a discipline the corpus already practices: the companion requires categorical definitions to survive transitions between neighboring Aethae, its natural-kind terms are rebuilt to that standard, "dinosaur" by anatomical superclade and mechanism-role rather than by roster or genome [5], and the present premise is that criterion's intended completion: robustness across Aethae matures into robustness under the Golden conditioning, Aethic-categorical into *Golden-categorical*, one criterion at two depths.

*The price, paid openly.* Three fences. The model's mathematics nowhere consumes the premise, the theorems run on whatever unit is plugged into them, so a reader who declines it loses only the answer to the smuggling objection at the level of reference, and nothing downstream. The premise is semantic, not idealist: well-definedness is not existence, and no feature waits on the framework to be. And the self-anchoring is acknowledged rather than hidden: rooting the framework's semantics in its own central object is a circle of exactly the kind every foundational scheme carries somewhere, and the claim made for this placement of it is comparative — the survivorship structure is a root the framework can actually individuate, where pretheoretic biology hands it the individuation problems of another science as axioms. The non-biconditional shape guards here too: features may admit other adequate definitions; the premise says where *this* framework's certificates come from, and only that, at [Φ] with the comparative claim its falsifier — exhibit a unit-individuation the framework needs that the survivorship root cannot supply and the premise fails where it stands. One forward pointer completes the placement: under the Route World's Strong reading the premise acquires an ontological completion, the categories trace to the root because the *things* do, developed and priced at §4.6, which this remark does not consume.

The pair $(\mu, D)$ is the *Kolmogorov diffusion* [48]:

$$d\mu_t = \sigma\, dW_t, \qquad dD_t = \mu_t\, dt, \qquad \mathcal{L} = \tfrac{\sigma^2}{2}\partial_\mu^2 + \mu\, \partial_D. \tag{5.3}$$

**Lemma 5.8** (Regularity)**.** *The free process $(\mu_t, D_t)$ is Gaussian with mean $(\mu_0,\, D_0 + \mu_0 t)$ and covariance $\left(\begin{smallmatrix} \sigma^2 t & \sigma^2 t^2/2 \\ \sigma^2 t^2/2 & \sigma^2 t^3/3 \end{smallmatrix}\right)$; in particular it admits the explicit smooth transition density written down by Kolmogorov [48]. More generally, $\mathcal{L}$ is hypoelliptic: with $X_1 = \sigma\partial_\mu$ and $X_0 = \mu\partial_D$ one has $[X_1, X_0] = \sigma\partial_D$, so Hörmander's bracket condition holds at first order [49]. Consequently the process killed at rate $e^x$ retains smooth densities, which is the regularity relevant to the killed problem of §5.2.6.*

*Proof.* Gaussianity and the moments follow from Lemma 5.21 below and linearity of (5.1) in $W$; the bracket computation is immediate. □

Three structural remarks frame everything that follows.

**No boundary in $h$.** The logarithmic parametrization makes $h > 0$ automatic: there is no absorbing or reflecting boundary in the hazard itself, and the entire problem concerns the growth of the integrated driver.

**Credit is spent only through negative rate.** $D$ is nondecreasing exactly while $\mu \geq 0$. Accumulated credit cannot be destroyed except by first driving the improvement rate negative;

this monotone coupling is the source of all horizon intuition, and its quantitative teeth are extracted in §A.1.

*Remark* 5.9 (Nonparametric-Bayes reading of the model). Definition 5.2 places an integrated Wiener process prior on $-\log h$ — equivalently, the classical cubic smoothing-spline prior of Wahba [65], since a prior whose second derivative is white noise is precisely one whose first derivative is Brownian. The model is therefore the canonical smoothness prior on the log-hazard rather than an exotic construction, and the results below may be read as the survival asymptotics that prior implies.

**Lemma 5.10** (Exact gauge role of the initial hazard)**.** *Write $\Lambda^{(h_0)}$ for the cumulative hazard with initial value $h_0$. Then $\Lambda_t^{(h_0)} = h_0\Lambda_t^{(1)}$ pathwise; hence $\mathcal{I}$ does not depend on $h_0$, while $\mathbb{P}(T=\infty) = \mathbb{E}[e^{-h_0\Lambda_\infty^{(1)}}\mathbf{1}_{\mathcal{I}}]$ is continuous and nonincreasing in $h_0$, strictly decreasing whenever $\mathbb{P}(\mathcal{I}) > 0$, with $\lim_{h_0\downarrow 0}\mathbb{P}(T=\infty) = \mathbb{P}(\mathcal{I})$ and $\lim_{h_0\uparrow\infty}\mathbb{P}(T=\infty) = 0$.*

*Proof.* The identity is (5.1); the limits follow by monotone and dominated convergence. □

Thus $h_0$ can never affect a dichotomy, only a value: it is the first and most complete gauge degree of freedom. The same will be shown for $(\mu_0, D_0)$ at the level of asymptotic exponents.

### 5.2.2 Primitives and the hazard

**Definition 5.11** (Two-channel model)**.** Fix $\sigma > 0$, $\epsilon > 0$, $n \in \mathbb{N}$, $\Gamma > 0$ and $\rho > 1$. Define

$$\text{driver:}\quad \mu_t = \mu_0 + \sigma W_t, \qquad [\mu] = \mathsf{T}^{-1},\ [\sigma] = \mathsf{T}^{-3/2}, \tag{5.4}$$

$$\text{muffling exponent:}\quad x_t = x_0 + \int_0^t \mu_s\, ds, \qquad [x] = 1, \tag{5.5}$$

$$\text{background:}\quad B_t = \sum_{i=1}^n e^{\epsilon W_t^{(i)}}, \qquad [B] = \mathsf{T}^{-1},\ [\epsilon] = \mathsf{T}^{-1/2}, \tag{5.6}$$

and let the total hazard be the sum of two channels,

$$h_t^{\text{tot}} = \underbrace{B_t\, \varsigma_b(x_t)}_{\text{crest}} + \underbrace{\Gamma\, e^{-\rho x_t}}_{\text{trough}}, \qquad \varsigma_b(x) := \frac{1+e^{-b}}{1+e^{x-b}} = \frac{1+e^{-b}}{2}\Big(1 - \tanh\tfrac{x-b}{2}\Big), \tag{5.7}$$

with *offset* $b > 0$. The normalizing factor $1 + e^{-b}$ is chosen so that $\varsigma_b(0) = 1$ exactly, whence $h^{\text{crest}}|_{x=0} = B_t$: the void limit is the background itself, for every $b$. Setting $b = 0$ recovers the unshifted form $B_t(1 - \tanh\frac{x}{2})$ Survival is $\bar{F}_t = e^{-\Lambda_t}$ with $\Lambda_t = \int_0^t h_u^{\text{tot}}\, du$, and the departure time $T$ is given by the Cox construction of the driftless case. There is *no* capacity parameter and *no* leak: $x$ is the undamped integral of $\mu$, exactly as in the driftless case — a stipulation about the *completed* regime, which Definition 5.49 exhibits as the $k = n$ boundary case of a staged leak. The same applies one derivative up: the Brownian driver of (5.4) is likewise a post-onset object, since an improvement rate is something an agent supplies and before the onset state there is none.

*Remark* 5.12 (Regimes). The three regimes of (5.7):

| regime | total hazard | reading |
|---|---|---|
| $x \gg b$ | $\approx B_t e^{\,b-x}$ | shielding; background passes through multiplicatively |
| $-k/\rho \ll x \ll b$ | $\approx B_t$ | *void*: the system contributes nothing |
| $x \ll 0$ | $\approx \Gamma e^{\rho\lvert x\rvert}$ | internal damage; $B$ has dropped out entirely |

The middle line is exact at $x = 0$: the normalization gives $h^{\mathrm{crest}}|_{x=0} = B_t$, so the void limit is the background itself. The offset $b$ places $x = 0$ *inside* the void rather than at its upper edge — see Remark C.4.

### 5.2.3 The four constraints

The model is answerable to four structural requirements. We state them as they were posed and record where each is met.

**Constraint 1** (Background reaches the total at positive $x$)**.** Met for $\rho > 1$, and *only* then: for $x \gg b$ the crest channel is $B_t e^{b-x}$, multiplicative in $B$, and by Proposition 5.13(ii) it dominates the trough at all large positive $x$ precisely when $\rho > 1$. This, not Constraint 2, is what the brittleness condition buys.

**Constraint 2** (Background drops out at negative $x$)**.** Met by Proposition 5.13(i), for *every* $\rho > 0$. Since $\Gamma$ is a primitive constant the trough contains no $B$ whatsoever, so at $x \ll 0$ the background disappears from the total entirely, not merely in its fluctuations.

**Constraint 3** (the driftless case undisturbed)**.** Met as a theorem, §D.2: both walls are $o(t^{3/2})$ and hence gauge.

**Constraint 4** (No smuggled primitives)**.** Met by Theorem C.7: with $B$ constant the total hazard is $h^{\mathrm{tot}}(0)e^{-\Psi(x)}$ for a single soft-kink gain $\Psi$, so the model is the driftless case's with the credit passed through a kink.

**Proposition 5.13** (Which channel dominates, and where; **[P]**)**.** (i) Negative $x$. *For every $\rho > 0$, $\Gamma e^{-\rho x}/\big(B\varsigma_b(x)\big) \to \infty$ as $x \to -\infty$, since $\varsigma_b$ is bounded while $e^{\rho|x|}$ is not. Hence $h^{\mathrm{tot}} \sim \Gamma e^{\rho|x|}$, free of $B$: Constraint 2 needs no condition on $\rho$ beyond positivity.*

(ii) Positive $x$. *As $x \to +\infty$ the crest is $\sim Be^{b-x}$ and the trough is $\Gamma e^{-\rho x}$, so the crest dominates for all large $x$ if and only if $\rho > 1$. For $\rho < 1$ the two cross at*

$$x_\times = \frac{k+b}{1-\rho}, \qquad k := \log\frac{B}{\Gamma}, \tag{5.8}$$

*beyond which the trough dominates and the background drops out* even for survivors.

*Proof.* (i) $\varsigma_b \le 1 + e^{-b}$ by Proposition 5.15, while $\Gamma e^{\rho|x|} \to \infty$ for any $\rho > 0$. (ii) Compare exponents $b - x$ and $-k - \rho x$; they are equal at (5.8), and for $\rho > 1$ the crest exponent is the larger for all $x$ exceeding a finite threshold. □

*Remark* 5.14 (What $\rho = 1$ is, and is not)*.* Three separate claims must be kept apart, and the first two are easily run together. *(a)* $\rho = 1$ is *not* the threshold for the background to drop out at negative $x$; that happens for every $\rho > 0$, by (i). *(b)* $\rho = 1$ *is* the threshold at which the background ceases to reach the survivor's hazard: for $\rho < 1$ internal damage overtakes the external channel beyond $x_\times$ and the system is effectively closed to the external world, violating Constraint 1. This is a sharp threshold in a structural parameter, and it concerns *openness*, not survival. *(c)* $\rho$ does *not* affect the survival exponent. For any $\rho > 0$ the dominant channel is an exponential in $x$, so the killing wall is a level of $x$ and the persistence problem is the driftless case's up to a rescaling of the credit by a constant. The exponent is $\frac{1}{4}$ throughout, which strengthens Constraint 3 rather than qualifying it.

### 5.2.4 Boundedness and the causal asymmetry

**Proposition 5.15** (The crest channel is bounded; **[P]**)**.** *$0 < \varsigma_b(x) < 1 + e^{-b}$ for all $x \in \mathbb{R}$, the supremum being approached only as $x \to -\infty$. Hence the external channel can* reduce *the background hazard without bound, but can* increase *it by at most the factor $1 + e^{-b}$, which tends to $1$ as $b$ grows.*

*Proof.* $(1 + e^{-b})/(1 + e^{x-b})$ is decreasing in $x$ with limits $1 + e^{-b}$ and 0. □

**Why this matters.** The slogan is ours: *strength can shield from the background but never amplify it*, and it is now enforced structurally rather than assumed. The earlier multiplicative form $B_t e^{-x}$ violated it — at $x < 0$ it multiplied the background without bound, which asserts that a failure of the system's own muffling magnifies the external world, a causal claim the ontology does not license. Amplification now arises only through the trough channel, which is internal damage: a distinct mechanism, correctly separated from the external one. Proposition 5.13 is the formal expression of that separation. Note how much the offset strengthens the slogan: at $b = 0$ the background could still be doubled, whereas for $b \gtrsim 6$ the permitted amplification is under a quarter of one percent. Shielding is unbounded; amplification is, for practical purposes, forbidden.

### 5.2.5 The construction as an iterated operator

The placement of the noise, on $\mu$, the negative logarithmic derivative of the hazard, rather than on the hazard or its logarithm, is the model's single most consequential choice, and this subsection states the sense in which it is not a choice at all.

**Definition 5.16** (The hazard operator)**.** For a positive function $f$ of time write

$$\mathcal{D}[f] \;:=\; -\frac{d}{dt}\log f. \tag{5.9}$$

Applied to the survival function it returns the hazard, and applied to the hazard it returns this model's driver:

$$\mathcal{D}[\bar{F}] = h, \qquad \mathcal{D}[h] = \mu. \tag{5.10}$$

The driver is therefore the *second iterate of the operation that defines the hazard in the first place*, and the model's noise sits at level two of a recursion whose level one is the hazard itself. This is a different statement from "we adopt a smoothness prior on the log-hazard and choose the integrated-Wiener member of the family," which is how the same object appears from the nonparametric-Bayes side, and it is a stronger one: no member of the family is being selected, because the ladder (5.10) has rungs and the model occupies one of them.[12]

[12] *Provenance, with its stages kept apart.* The operator (5.9) is ours, derived in May 2021 by a route worth recording because it is the physical one rather than the textbook one: examine $\frac{d}{dt}(1 - \bar{F})$ at $t = 0$, read the resulting quantity as an instantaneous *felt danger* propagating through the system, rewrite the survival function with that quantity as a parameter, replace the product $ht$ in the exponent by the time integral $\int_0^t h$, and solve for $h$ — which returns (5.9). The hazard thus arrives as a rate one can point at, not as a definition one is handed, which is the same difference that §5.2.5 turns on. The *second* application is later and its stages should not be run together: our record places the present stochastic use of $\mu$ on the hazard in autumn 2025, with an earlier and as yet unlocated use of a doubled operator on the survival function, in connection with the Golden Point rather than with hazard rates, somewhere in 2021–2023 and under a name of its own. That earlier use is author-attested and not verified here; if the journal entries are recovered, this footnote should be revised to cite them. The logistic deformation of §5.2.4 is later still, arising during the preparation of the present work from the causal constraint recorded there.

**Where the ladder stops.** The rungs are not equally hospitable, and the model sits on the last one that is analytically open. Noise at level one, $\log h$ Brownian, makes the credit Brownian and the persistence exponent $\frac{1}{2}$. Noise at level two, $\mu$ Brownian, the present model, makes the credit integrated Brownian and the exponent $\frac{1}{4}$ (Theorem 5.29). At level three the credit is a twice integrated Brownian motion, and no closed form for the exponent is known: existence and monotonicity are available for fractionally integrated processes [66, 56], and the persistence of iterated partial sums has been studied, but the value is open. Level one is close to a definition and level three is close to an open problem; level two is the rung on which the object is both nontrivial and solvable, and it is the rung the hazard construction reaches by being applied twice.

**The two channels, in operator terms, and why they differ.** Applying $\mathcal{D}$ to the two channels of (5.7) separately, at fixed background, gives

$$\mathcal{D}\big[h^{\text{trough}}\big] = \rho\,\mu, \qquad \mathcal{D}\big[h^{\text{crest}}\big] = \varsigma(x-b)\,\mu, \qquad \varsigma(u) = \frac{e^u}{1+e^u}. \tag{5.11}$$

The internal channel therefore preserves the operator relation *exactly*, with constant gain $\rho$; the external channel preserves it only up to a logistic weight running from 0 to 1.

The asymmetry is the causal constraint of §5.2.4 in its sharpest form, and it is stated as such. An exact relation $h = h_0 e^{-x}$ is unbounded above as $x \to -\infty$: it says that a failure of the system's own muffling multiplies the background hazard without limit, which is a causal claim the ontology declines — a biosphere may fail to shield itself from the external world, but it cannot thereby make that world more dangerous than it is. No such constraint applies inward. *One may harm oneself without bound; one may not amplify the world.* Hence the trade recorded in (5.11): the internal channel keeps the exact relation because nothing forbids it, and the external channel must be deformed because something does. The logistic is the minimal deformation purchasing boundedness, and the offset $b$ is how much of it is bought.

Three consequences are already in the text and are named here rather than derived again. The technoanatomy $\rho$ of Definition 5.32 *is* the internal channel's operator gain, $\mathcal{D}[h^{\text{int}}]/\mu$, which is a third and perhaps the cleanest answer to why that exponent deserves the name: it is the rate at which effort converts into change in self-inflicted hazard, protective when positive and wounding when negative. The void of §5.5 is the region where $\varsigma(x-b) \approx 0$, so that $\mathcal{D}[h^{\text{crest}}] \approx 0$ even for $\mu \neq 0$ — the operator is applied and nothing comes out, which is what Proposition C.8 measures from the other side. And the concealed gain $\Psi'$ of Corollary C.10 is exactly the hazard-weighted blend of the two relations in (5.11), which is why its range is $(0, \rho)$ and why it tends to 1 under shielding and to $\rho$ under damage.

### 5.2.6 Reduction to a persistence problem

**Proposition 5.17** (Feynman–Kac). *Let $u(t,x,\mu) := \mathbb{P}_{x,\mu}(T > t) = \mathbb{E}_{x,\mu}\big[e^{-\Lambda_t}\big]$. Then $u$ is the (classical, by Lemma 5.8) solution of*

$$\partial_t u = \frac{\sigma^2}{2}\partial_\mu^2 u - \mu\,\partial_x u - e^x u, \qquad u(0,\cdot,\cdot) \equiv 1. \tag{5.12}$$

*Proof.* Itô's formula applied to $s \mapsto u(t-s, x_s, \mu_s)\exp(-\int_0^s e^{x_r}\,dr)$ exhibits the right side as a martingale if and only if (5.12) holds; regularity is supplied by hypoellipticity of the killed generator. $\square$

The killing rate $e^x$ vanishes rapidly for $x \ll 0$ and diverges for $x \gg 0$: at exponent level it is an absorbing wall at $x = 0$, of thickness $O(1)$ in $x$, a factor $e$ in the hazard, which is negligible against excursions of size $t^{3/2}$. We elevate this to a labeled hypothesis, since it is the single non-rigorous reduction on which the tail asymptotics of §5.4.3 rest.

**Ansatz 5.18** (Soft-wall reduction)**.** Let $\tau := \inf\{t \geq 0 : D_t \leq 0\}$ denote the hard-wall passage time of the Kolmogorov diffusion. Then for every admissible initial datum there exist constants $0 < c \leq C < \infty$ and $t_0 < \infty$ with

$$c\,\mathbb{P}(\tau > t) \;\leq\; \mathbb{P}(T > t) \;\leq\; C\,\mathbb{P}(\tau > t), \qquad t \geq t_0,$$

after an $O(1)$ adjustment of the initial credit. Equivalently, $\log \mathbb{P}(T > t) = \log \mathbb{P}(\tau > t) + O(1)$.

**Status of the Ansatz.** One half is rigorous modulo a cited input. *Lower bound:* on the event $\{D_u \geq f(u)\ \forall u \leq t\}$ with $f(u) = 2\log\frac{2+u}{2}$ one has $\Lambda_t \leq h_0 e^{-D_0}\int_0^\infty \left(\frac{2+u}{2}\right)^{-2} du < \infty$, whence $\mathbb{P}(T > t) \geq e^{-C}\,\mathbb{P}(D \geq f \text{ on } [0,t])$; that a logarithmic moving boundary does not change the persistence exponent of an integrated process is the standard moving-boundary phenomenon (see the survey [56] and references therein; cf. also the robustness expressed by the universality results [66, 55]). *Upper bound:* from $\mathbb{P}(T > t) \leq e^{-K} + \mathbb{P}(\Lambda_t \leq K)$ and $\Lambda_t \geq h_0 e^{-D_0}\,\mathrm{Leb}\{u \leq t : D_u \leq 0\}$, the event $\{\Lambda_t \leq K\}$ forces $D$ to remain positive off a set of bounded Lebesgue measure. What is missing is the (plausible, unproved here) statement that *persistence with a bounded occupation tolerance has the same exponent as strict persistence.* Everything tagged **[A]** below depends on exactly this gap and nothing else; §F describes its direct numerical test.

The standing literature nearest the gap should be cited against it, because a referee will and should. Persistence with *partial survival*, each zero crossing killing with probability $p$, carries exponents that vary continuously with $p$ [67, 68], so "a softened wall leaves the exponent alone" is false in general, and the Ansatz cannot rest on softness as such. What it rests on is the structure of this softening: the killing here is rate-based and confined to a boundary layer of width $O(1)$ in the credit, against excursions of scale $t^{3/2}$ — a vanishing fraction of the process's range, where partial survival taxes every crossing at fixed price no matter the scale. The distinction is exactly what the numerics interrogate, and the measured constancy of the soft-to-hard ratio is evidence that the boundary-layer case sits in the exponent-preserving class the moving-boundary results [56, 55] delimit from the rigorous side.

### 5.2.7 The transfer, stated

**Theorem 5.19** (Transfer of the driftless case; **[A]**)**.** *In the two-channel model, for every admissible parameter set,*

$$\mathbb{P}(T > t) \;\asymp\; \underbrace{\mathfrak{h}\big(\mu_0 - \tfrac{\epsilon^2}{2},\ D_0 - \log n - b\big)}_{\textit{shifted initial data}} \cdot \underbrace{e^{-\Theta\left(n(b/\sigma)^{2/3}\right)}}_{\textit{room passage}} \cdot t^{-1/4}, \tag{5.13}$$

*with $\mathfrak{h}$ the persistence-harmonic function of the driftless case. In consequence: $\mathbb{E}[T] = \infty$ while $T < \infty$ a.s.; residual life is asymptotically Pareto($\frac{1}{4}$); quantiles grow as the fourth power of rarity; the effective population hazard is $1/4t$; and the survivorship interpolation, its pure Doob limit, its $s/4t$ linear response and its critical mode all hold verbatim, the limit being the same $Q$-process computed at the shifted data.*

*Proof.* Above the room the hazard is the driftless case's up to bounded constants (Lemma D.3); the wall is linear and absorbed exactly into the shifted data (Theorem B.4, with the offset $b$ contributing a further constant); the band collapses (Theorem D.4); the room contributes a bounded additive term to $\Lambda$, hence a multiplicative prefactor (Proposition D.6). The listed consequences are those of the driftless case, transported through (5.13). $\square$

**How close, in one sentence.** Exactly as close as it is possible to be: every difference between the two models is a shift of initial data or a bounded multiplicative constant, and we proved both to be gauge. The two-channel model buys its ontology, a positive fluctuating background, a bounded external channel obeying the causal asymmetry, an unbounded internal channel, an ordered pair of onsets and a waiting room, at a cost of precisely zero asymptotic exponents, and it repays part of the price by shrinking the one gap the earlier section had to assume.

## 5.3 Coordinates: gauge, strength, anatomy

### 5.3.1 Dimensional structure, self-similarity, and the gauge orbit

**Definition 5.20** (Units and intrinsic scales)**.** Assigning $[t] = \mathsf{T}$ gives $[\mu] = \mathsf{T}^{-1}$, $[\sigma] = \mathsf{T}^{-3/2}$, $[D] = 1$, $[h] = \mathsf{T}^{-1}$. The unique timescale constructible from $\sigma$ alone is $\sigma^{-2/3}$; the time-equivalents of the initial data are

$$t_\mu := \frac{\mu_0^2}{\sigma^2}, \qquad t_D := \Big(\frac{D_0}{\sigma}\Big)^{2/3} \quad (D_0 > 0); \tag{5.14}$$

and the unique dimensionless state coordinate on the quadrant $\{\mu > 0, D > 0\}$ is the *horizon coordinate*

$$\kappa := \frac{\sigma^2 D}{\mu^3}, \qquad \text{satisfying} \qquad \frac{t_D}{t_\mu} = \kappa_0^{2/3}. \tag{5.15}$$

Hatted variables are $\hat\mu = \mu\sigma^{-2/3}$, $\hat D = D$, $\hat h = h\sigma^{-2/3}$, $\hat t = t\sigma^{2/3}$.

**Lemma 5.21** (Second-order structure)**.** $\mathrm{Var}(I_t) = t^3/3$, $\mathrm{Cov}(W_t, I_t) = t^2/2$, *and for* $u \geq 1$, $\mathbb{E}[I_1 I_u] = \frac{u}{2} - \frac{1}{6}$.

*Proof.* By Fubini, $\mathbb{E}[I_a I_b] = \int_0^a \int_0^b (\alpha \wedge \beta)\, d\beta\, d\alpha$. For $a = b = t$ this is $2\int_0^t \int_0^\beta \alpha\, d\alpha\, d\beta = t^3/3$. Next, $\mathbb{E}[W_t I_t] = \int_0^t (\alpha \wedge t)\, d\alpha = t^2/2$. Finally, for $u \geq 1$, $\int_0^u (\alpha\wedge\beta)\, d\beta = \alpha u - \alpha^2/2$ for $\alpha \leq 1 \leq u$, and integrating over $\alpha \in [0,1]$ gives $u/2 - 1/6$. $\square$

**Proposition 5.22** (Exact self-similarity)**.** *Fix $\sigma$ and $c > 0$. If $(\mu, D)$ starts from $(\mu_0, D_0)$, then*

$$\big(\mu_{c^2 t},\, D_{c^2 t}\big)_{t\geq 0} \overset{d}{=} \big(c\,\mu'_t,\, c^3 D'_t\big)_{t\geq 0},$$

*where $(\mu', D')$ starts from $(\mu_0/c,\, D_0/c^3)$. Equivalently, $(I_{ct})_{t\geq0} \overset{d}{=} (c^{3/2} I_t)_{t\geq 0}$.*

*Proof.* Brownian scaling $(W_{c^2 t})_t \overset{d}{=} (cW_t)_t$, integrated once in time. $\square$

**Proposition 5.23** (The gauge orbit)**.** *The initial data $(h_0, \mu_0, D_0)$ affect asymptotics only through prefactors and timescales, never through exponents. Precisely:* (i) *$h_0$ is exactly gauge in the sense of Lemma 5.10;* (ii) *the deterministic contribution $x_0 - \mu_0 t$ to $x_t$ is affine in $t$, hence*

$(x_0 - \mu_0 t)/t^{3/2} \to 0$, *while* $\sigma I_t/t^{3/2} \overset{d}{=} \sigma I_1$ *is nondegenerate for every* $t$*: the initial data live in the subspace that the* $t^{3/2}$ *fluctuations cannot resolve;* (iii) *under the scaling of Proposition 5.22 the initial data transform as* $(\mu_0, D_0) \mapsto (c\mu_0, c^3 D_0)$*, so any scale-invariant asymptotic quantity is constant on the two-parameter orbit* $\{(c\mu_0, c^3 D_0) : c > 0\}$ *and, when finite and positive, is a function of* $\kappa_0$ *alone.*

*Proof.* (i) is Lemma 5.10; (ii) is immediate from Proposition 5.22 and Lemma 5.21; (iii) holds because the orbits of the scaling action on the open quadrant are exactly the level curves of $\kappa$. □

The sharp form of the slogan, that even the *prefactors* of survival asymptotics depend on the data only through a single equivalent timescale, is Proposition D.7 below.

### 5.3.2 The sign of the credit, and an exact half

One fact about the driftless model is used repeatedly below and is worth isolating, because it is the engine of a proof and not merely a description.

**Theorem 5.24** (Adverse configuration occupies exactly half of logarithmic time; **[P]**). *For the driftless driver, almost surely*

$$\lim_{S\to\infty} \frac{1}{S}\, \mathrm{Leb}\big\{s \in [0, S] : x_{e^s} < 0\big\} = \frac{1}{2}, \tag{5.16}$$

*so the set* $\{t : x_t < 0\}$*, the times at which the system's own muffling sits* above *rather than below its background hazard, has logarithmic density exactly* $\frac{1}{2}$*, for every parameter value and every initial condition.*

*Proof.* $Z_s := x_{e^s}/(\sigma e^{3s/2})$ is stationary, centered Gaussian and ergodic (the Lamperti argument of Theorem 5.26), so Birkhoff gives density $\mathbb{P}(Z_0 < 0) = \frac{1}{2}$ by symmetry. □

**What the exact half is for, and what it is not.** Two uses and one disclaimer.

*Use.* It is the engine of Theorem 5.28: where the driftless argument needed an unbounded integrand, the two-channel model needs only that the system spends a positive fraction of logarithmic time at hazard above background — and the fraction is exactly one half, for every parameter value. Boundedness of the hazard does not buy immortality; it changes which argument rules it out.

*Sharpening.* The no-viability proof of §5.4 used only the qualitative input $\mathbb{P}(Z_0 < -\varepsilon) > 0$; here the constant is exact. *Disclaimer, and it is the important one.* This is a statement about a *state coordinate*, not about technoanatomy. Every biosphere, however well constituted, spends half of logarithmic time with $x < 0$; the sign of the credit is a condition the system passes through and re-enters, not a property it has. Technoanatomy in the sense of Definition 5.32 is the exponent $\rho$, which governs what capacity *does* when acquired and does not fluctuate at all. Nothing about the frequency with which a biosphere is adversely positioned bears on whether its capacity is adversely directed, and the theorem should not be read as saying otherwise.

## 5.4 The verdict: certain departure, infinite expectation

*Remark* 5.25 (What $T$ is the time of). Throughout this section $T$ is the time of the *Departure Point* (Definition 3.5), the moment past which no continuation advances, not of biological death, and $h^{\mathrm{tot}}$ is that event's hazard. Death is one mechanism of foreclosure and irreversible loss of the capacity to advance is another; the analysis does not distinguish them, because for advancement they are the same event. It does, however, distinguish both from a mere lull: a biosphere that has stopped progressing but could still resume has not reached $T$. Where the classical results imported below speak of survival and first passage, the vocabulary is theirs and the referent here is departure. Read that way, $\mathbb{P}(\Lambda_\infty < \infty)$ is the probability of *advancing*, not of not dying — which is what the Golden Point was always a claim about.

### 5.4.1 Impossibility of immortality

**Theorem 5.26** (No viability; **[P]**)**.** *For every $(h_0, \mu_0, D_0, \sigma)$ in the driftless model* (5.1)*, $\Lambda_\infty = \infty$ almost surely. Consequently $S_\infty = 0$ a.s., $\mathbb{P}(\mathcal{I}) = 0$, $\mathbb{P}(T = \infty) = 0$, and $T < \infty$ a.s.*

*Proof.* Since $\Lambda_\infty = h_0 e^{-D_0} \int_0^\infty e^{-\mu_0 u - \sigma I_u} du$ and the prefactor is a fixed positive constant, it suffices to prove that the integral diverges a.s.

*Step 1 (Lamperti transform).* Define $Z_s := e^{-3s/2} I_{e^s}$, $s \geq 0$. As a linear functional of $W$, $Z$ is a centered Gaussian process. By Proposition 5.22, for each $r \geq 0$, $(Z_{s+r})_{s\geq 0} = (e^{-3(s+r)/2} I_{e^r e^s})_{s\geq 0} \overset{d}{=} (e^{-3s/2} I_{e^s})_{s\geq 0} = (Z_s)_{s\geq 0}$: $Z$ is stationary. By Lemma 5.21, for $s \geq 0$,

$$\mathbb{E}[Z_0 Z_s] = e^{-3s/2}\Big(\frac{e^s}{2} - \frac{1}{6}\Big) = \frac{1}{2} e^{-s/2} - \frac{1}{6} e^{-3s/2} \;\longrightarrow\; 0. \tag{5.17}$$

*Step 2 (ergodicity).* A stationary Gaussian process whose covariance vanishes at infinity is mixing [69], hence ergodic. *Step 3 (Birkhoff).* Fix $\varepsilon > 0$. Since $Z_0 \sim N(0, \frac{1}{3})$, $p_\varepsilon := \mathbb{P}(Z_0 \leq -\varepsilon) > 0$. By the continuous-time Birkhoff theorem, $\frac{1}{S}\,\mathrm{Leb}\{s \in [0, S] : Z_s \leq -\varepsilon\} \to p_\varepsilon$ a.s., so $\mathrm{Leb}\{s \geq 0 : Z_s \leq -\varepsilon\} = \infty$ a.s. *Step 4 (change of variables).* Let $A := \{u \geq 1 : I_u \leq -\varepsilon u^{3/2}\}$. Under $u = e^s$ the set in Step 3 maps into $A$ with $\mathrm{Leb}\{s : Z_s \leq -\varepsilon\} = \int_A \frac{du}{u} \leq \mathrm{Leb}(A)$, the inequality because $u \geq 1$. Hence $\mathrm{Leb}(A) = \infty$ a.s.

*Step 5 (divergence).* Choose $u_1 = u_1(\varepsilon, \mu_0, \sigma)$ so that $\sigma\varepsilon u^{3/2} - \mu_0 u \geq 0$ for $u \geq u_1$. Then

$$\int_0^\infty e^{-\mu_0 u - \sigma I_u}\, du \;\geq\; \int_{A\cap[u_1,\infty)} e^{\sigma\varepsilon u^{3/2} - \mu_0 u}\, du \;\geq\; \mathrm{Leb}\big(A \cap [u_1, \infty)\big) \;=\; \infty \qquad \text{a.s.} \qquad \square$$

**The asymmetry of exponential response.** The driver is symmetric; the response is not. A favorable excursion suppresses the integrand $e^{-D}$ multiplicatively toward zero, *bounded* benefit for $\Lambda$, while an unfavorable excursion of relative size $\varepsilon$ inflates it without bound. Exponential response to a symmetric driver is structurally hostile to survival; the mechanism is the sign-oscillation that makes low-dimensional random walks recurrent, aggravated by this asymmetry. Identifying exactly what a horizon must break, the symmetry of the driver, not the response, is the burden of §A.2.

### 5.4.2 No viability, by a different route

the driftless case's no-viability theorem was proved by exhibiting an integrand that *blows up*: on a set of infinite measure the credit satisfies $D_u \leq -\varepsilon u^{3/2}$, whence $e^{-D_u} \geq e^{\sigma\varepsilon u^{3/2}} \to \infty$. That argument is unavailable here whenever the crest channel acts alone, since $\varsigma_b$ is *bounded*. The conclusion nevertheless survives, by a weaker and more robust mechanism: a positive floor on an infinite set. The mechanism needs one lemma, and the lemma is worth stating in the generality it naturally has.

**Lemma 5.27** (Infinite sets defeat the driftless background; **[P]**)**.** *Let $W$ be a standard Brownian motion, $\epsilon > 0$, and let $S \subset [0, \infty)$ be a measurable set, possibly random but independent of $W$, with $\mathrm{Leb}(S) = \infty$ almost surely. Then*

$$\int_S e^{\epsilon W_u}\,du \;=\; \infty \qquad \textit{almost surely,}$$

*and consequently the same holds with the background $B_u = \sum_{i=1}^n e^{\epsilon W_u^{(i)}} \geq e^{\epsilon W_u^{(1)}}$ of (5.6) in place of $e^{\epsilon W_u}$.*

*Proof.* Conditioning on $S$, it suffices to treat deterministic $S$ of infinite measure. For each $T$, finiteness of $\int_{S\cap[T,\infty)} e^{\epsilon W_u} du$ is unchanged by the multiplicative factor $e^{\epsilon W_T}$, it is decided by $\int_{S\cap[T,\infty)} e^{\epsilon(W_u - W_T)} du$, and the integral over $S \cap [0, T]$ is a.s. finite, so the event $\{\int_S e^{\epsilon W_u} du = \infty\}$ lies in $\sigma(W_u - W_T : u \geq T)$ for every $T$, hence in the tail $\sigma$-field of the increments, and has probability 0 or 1 by Kolmogorov's zero–one law. Suppose the probability were 0. Then $\int_S e^{\epsilon W_u} du < \infty$ a.s., and applying the same conclusion to the Brownian motion $-W$ gives $\int_S e^{-\epsilon W_u} du < \infty$ a.s. as well; adding,

$$2\,\mathrm{Leb}(S) \;\leq\; \int_S \bigl(e^{\epsilon W_u} + e^{-\epsilon W_u}\bigr) du \;<\; \infty,$$

contradicting $\mathrm{Leb}(S) = \infty$. □

**Theorem 5.28** (No viability in the two-channel model; **[P]**)**.** *For every $\sigma, \epsilon > 0$, $n \geq 1$, $b \geq 0$, $\rho > 0$ and every $\Gamma \geq 0$, including $\Gamma = 0$, where the total hazard is bounded, one has $\Lambda_\infty = \infty$ almost surely. Hence $\bar{F}_\infty = 0$ a.s. and $T < \infty$ a.s.*

*Proof.* Discard the trough, which is nonnegative. On $\{x_u \leq 0\}$ we have $\varsigma_b(x_u) \geq \varsigma_b(0) = 1$, so

$$\Lambda_t \;\geq\; \int_0^t B_u \varsigma_b(x_u)\,du \;\geq\; \int_{S\cap[0,t]} B_u\,du, \qquad S := \{u : x_u \leq 0\}.$$

By Theorem 5.24 the set $S$ has logarithmic density $\frac{1}{2}$; writing $u = e^s$, $\mathrm{Leb}(S \cap [0, t]) = \int_0^{\log t} \mathbf{1}\{Z_s \leq 0\} e^s ds \to \infty$ a.s., since the integrand is positive on a set of density $\frac{1}{2}$ and $e^s \to \infty$. So $S$ is an $x$-measurable set of infinite measure, independent of the background's driving motions, and Lemma 5.27 gives $\int_S B_u\,du = \infty$ a.s., whence $\Lambda_\infty = \infty$. □

### 5.4.3 The survival tail and its consequences

**Theorem 5.29** (Persistence of integrated Brownian motion; **[C]**)**.** *Let $(\mu, D)$ start from $(m, 0)$ with $m > 0$ and let $\tau = \inf\{t : D_t \leq 0\}$. There is a constant $c_1 > 0$ such that*

$$\mathbb{P}_{(m,0)}(\tau > t) \;\sim\; c_1\Bigl(\frac{m^2}{\sigma^2 t}\Bigr)^{1/4}, \qquad t \to \infty. \tag{5.18}$$

*The exponent $\frac{1}{4}$, half the Brownian value, halved by one integration, is the persistence exponent of integrated Brownian motion, and its provenance is a chain worth recording precisely. McKean [50] computed the first-passage ("winding") law of the Kolmogorov diffusion; Goldman [51] derived the passage-rate asymptotic from McKean's law; Lachal [52] obtained the exact first-passage distributions; Sinai [53] initiated and settled, up to constants, the discrete counterpart $n^{-1/4}$ for the integrated random walk; Isozaki–Watanabe [54] supplied the sharp constants; and Denisov–Wachtel [55] proved the exact $n^{-1/4}$ asymptotics universally for centered integrated random walks with $2+\delta$ moments, via a discrete potential theory (see also the universality result of Aurzada–Dereich [66] for integrated processes). Starting states in the open quadrant obey the same power with a prefactor proportional to the harmonic function of Proposition D.8; this general-start form is established in [58], which recovers [51, 54] as special cases. For the surrounding theory see the survey [56] and, on the physics side, where the model is the* random acceleration process *and has an extensive literature of its own, the review [57].*

**The $m^{1/2}$ law is exact given the $m = 1$ asymptotic.** By Proposition 5.22 with $c = 1/m$ (and $\sigma = 1$), $\mathbb{P}_{(m,0)}(\tau > t) = \mathbb{P}_{(1,0)}(\tau > t/m^2)$, so once (5.18) is known for one $m$ it holds for all, prefactor scaling as $m^{1/2}$ exactly. The initial rate enters through its time-equivalent $t_\mu$ and nothing else.

**Claim 5.30** (Survival tail; [**A**])**.** *Under Ansatz 5.18, for every initial datum there are constants $0 < c_- \le c_+ < \infty$, depending on $(\hat{h}_0, \hat{\mu}_0, \hat{D}_0)$, with $c_- t^{-1/4} \le \mathbb{P}(T > t) \le c_+ t^{-1/4}$ for all large $t$; equivalently*

$$\mathbb{P}(T > t) \asymp t^{-1/4}, \qquad t \to \infty. \tag{5.19}$$

All of the following consequences are elementary integrations of (5.19); each inherits the tag [**A**]. Write $\alpha = \frac{1}{4}$.

**Proposition 5.31** (Consequences of the tail; [**A**])**.** *All of the following are elementary consequences of (5.19) by regular variation, with $\alpha = \frac{1}{4}$; we state them together rather than prove them severally, since each is a one-line integration and none is original.*

| *quantity* | *value* | *reading* |
|---|---|---|
| $\mathbb{E}[T^p]$ | *finite iff* $p < \frac{1}{4}$ | *even* $\mathbb{E}[\sqrt{T}] = \infty$*; heavier-tailed than Cauchy* |
| $\mathbb{E}[T]$ | $\infty$*, every initial condition* | *null recurrence: certain departure, infinite mean* |
| $\mathbb{E}[T \wedge \mathcal{T}]$ | $\asymp \frac{4C}{3}\mathcal{T}^{3/4}$ | *an observed mean is an artifact of the window* |
| $t_q$ *with* $\mathbb{P}(T > t_q) = q$ | $\asymp (C/q)^4$ | *quantiles grow as the fourth power of rarity* |
| $\mathbb{P}(T > \lambda t \mid T > t)$ | $\to \lambda^{-1/4}$ | *residual life Pareto(*$\frac{1}{4}$*), mean infinite at every age* |
| $r(t) = -\frac{d}{dt}\log \mathbb{P}(T > t)$ | $\approx 1/4t$ | *a Lindy law;* $\int^\infty r = \infty$*, so departure is certain* |

*Two remarks are worth more than the derivations. The truncated mean shows that any finite dataset reports a finite mean lifetime growing as the $3/4$ power of its own duration, so the quantity is not estimable. And the effective hazard $1/4t$ sits far from the mortality boundary, any positive exponent gives certain departure, but deep inside the infinite-mean regime, whose boundary is exponent $1$.*

## 5.5 Technoanatomy: the sign of the trough exponent

Section 5.2 introduced the trough channel $\Gamma e^{-\rho x}$ under the standing restriction $\rho > 1$, imposed so that the background would reach the survivor's hazard (Constraint 1). That restriction has never been examined, and lifting it turns out to supply the model's missing structural parameter. We lift it now: $\rho$ ranges over $\mathbb{R}$.

**Definition 5.32** (Technoanatomy)**.** The exponent $\rho$ is the *technoanatomy*[13] of the system: the degree to which increasing muffling capacity fails to be turned back against the system that acquired it. Reading the trough channel $\Gamma e^{-\rho x}$ as internal hazard:

[13] *Provenance.* The notion, its square–cube framing, and the exponential-against-linear scaling that §6.6.1 later derives as $\rho = -(\alpha - \beta)$ all appear in our journal three years before the present model, and the entry is reproduced here in full because it states the mechanism more compactly than the formal development does. Dated 23 July 2023, 1:42 pm, under the heading *Golden Biosphere Analog to the Square Cube Law*:

> "Just thought of the craziest idea! You know the quote about how advanced aliens might attack us like how boys throw stones at frogs? The implication I get from this is that such an alien biosphere is portrayed as just a scaled up version of our own so to speak in every aspect of society, such that we just get a biosphere which is far more advanced technologically, but interacts about itself in similar ways, such as they are precisely are to humans as humans are to ants.
>
> So, here's why I think this won't work:
>
> The idea is basically an analog of the square cube law to Astrobiology, working in a similar way to how you can't just scale an ant up to the size of an elephant. There has to be a transformation in the process instead. The idea, essentially, is that the capacity for us to destroy ourselves grows exponentially with technology, whereas the defense mechanisms put in place with our current society, given the tragedy of the commons and all, grows only linearly with technology. If we assigned each human comparatively limitless power, such that the advanced analog of 'boys' might attack humans with the same mindset as human boys attack frogs, then such an advanced civilization will implode as soon as it spawns. Why? Because the linear defense mechanisms of a human society would be immediately overtaken by the vast exponential issues propagated by such a technology.
>
> Anyways, so this demonstrates essentially two things: A) we will not remain constant in societal mindset with time, and B) we will be essentially forced through the keyhole of the Golden Point mindset long before we reach the capacity of supposedly treating humans like frogs to boys."

The entry must be read as its author wrote it, which is *conditionally on survivorship* — the conditioning having been the paradigmatic heart of the program since at least 2022, on our own record, and therefore predating this entry as well. Read that way, every one of its claims is recovered below; read unconditionally, two of them would appear to fail. The distinction bears directly on how much the entry anticipates, and read correctly the answer is all of it.

Six things are worth marking, since each became a separate part of the apparatus. The exponential-against-linear contrast is $\rho < 0$: destruction capacity growing as $e^{\alpha x}$ against enforcement growing more slowly is exactly (6.3), and the entry states the mechanism without the surface-versus-volume derivation that §6.6.1 supplies. The square–cube analogy is the source of the names *technostrength* and *technoanatomy*, so Remark 6.13's observation that the joke turned out to be the mechanism is a retrospective reading of this entry rather than a later discovery. "Implode as soon as it spawns" is Theorem 5.36. And "Golden Point" is the earlier name for what the present work calls the Golden Biosphere.

The entry's two numbered conclusions are the remaining pair, and both are theorems here once the conditioning is respected. Item (B) — that a biosphere is forced through the keyhole *before* attaining the capacity in question — is Proposition 5.42: the affordable duration of adverse anatomy is $\delta \lesssim \Gamma^{-1} e^{-|\rho| x}$, so conditional on being alive at large technostrength, adverse anatomy is essentially absent, and a biosphere cannot have arrived there without having already passed through. The keyhole is not a force acting on biospheres — unconditionally, Corollary 5.43 allows a biosphere simply to fail — but a property of the survivors, which is what the entry claims. Item (A) — that societal arrangement will not remain constant — is the anatomy selection measured in §6.3.2, where the conditioned distribution of $\rho$ shifts away from its prior. Neither is a prediction the model contradicts; both are among the results it was later built to state.

| | internal hazard as $x$ grows | reading |
|---|---|---|
| $\rho > 0$ | falls, $\propto e^{-\rho x}$ | capacity is patched as it is built |
| $\rho = 0$ | constant at $\Gamma$ | capacity neither patched nor weaponized |
| $\rho < 0$ | rises, $\propto e^{\lvert\rho\rvert x}$ | *the system eats itself* |

**Why this is the right home for the notion.** Two things recommend $\rho$ over the phase coordinate $\Theta$ of §5.3, with which it is easily conflated. First, $\Theta$ is a *state coordinate*: it fluctuates, is uniform under $\mathbb{P}$, and every system occupies every value of it half of logarithmic time (Theorem 5.24). Anatomy in the intended sense is not something a system passes through but something it *is*. Second, and decisively, the intended notion is a claim about what *strength does*, whether acquiring capacity raises or lowers one's own hazard, and that is a statement about a derivative, $\partial_x \log h^{\text{int}} = -\rho$, not about a position. The phase coordinate is retained under its own name; technoanatomy is $\rho$.

*Example* 5.33 (The asteroid, and why the channels must be two). The reason internal and external hazard cannot be folded into one term is best seen in a case where the same capability serves both. Consider a biosphere that becomes progressively better at deflecting asteroids. Every increment of that capability lowers the external hazard: the crest channel falls, and by (5.7) it falls multiplicatively in the background, exactly as it should.

But an apparatus that can redirect a body large enough to end a biosphere is, by construction, an apparatus that can *aim* such a body. The same increment therefore also creates a capability that did not previously exist and that is directed at the biosphere itself. What happens to the total hazard depends on nothing about asteroids and everything about whether the biosphere is so arranged that the new capability can be turned inward — which is precisely $\rho$.

With $\rho > 0$ the deflection apparatus is built into an arrangement that patches its own misuse faster than it creates the possibility of it, and the net effect of the capability is protective. With $\rho < 0$ it is not, and the biosphere has purchased a reduction in external hazard at the price of a larger increase in internal hazard — the trade being worse the more capable the apparatus. The example is not incidental to the model: it is why (5.7) carries two channels rather than one, and why the second is unbounded above while the first is not (Proposition 5.15). A single-channel model cannot represent a capability that cuts both ways, and every capability worth having does.

The dilemma is not this work's discovery, and the priority should be recorded where the example lives: Sagan and Ostro stated it for this very case — deflection capability is aiming capability, so the apparatus meant to remove the hazard becomes the larger hazard [70]. What the model adds is a functional form and a verdict: the dilemma is the two-channel structure itself, its resolution is a property of the biosphere's arrangement rather than of the apparatus, and the two-condition theorem is what their dilemma looks like once $\rho$ is a parameter rather than a worry.

*Remark* 5.34 (On not measuring $\rho$ yet: a deliberate reservation). The obvious next question is how one would measure $\rho$ for a real system. No procedure is offered here, and the omission is deliberate rather than a gap awaiting routine work. The reasoning is given here, partly because it is the framework's own reasoning applied to itself, and partly so that a later proposal can be judged against it.

*What is open is narrower than it may appear.* The exponent is already fully defined: $\rho = -\partial_x \log h^{\text{int}}$, the operator gain of the internal channel (5.11). What is absent is the *identification map*, which real configurations carry which values, and that is a different absence from a vague concept. It is the difference between having no definition of temperature and having no thermometer, and only the second is a thing one might reasonably decline to supply in haste.

*A premature proxy would be adverse anatomy applied to measurement.* A scored index for technoanatomy would be a capability acquired for the purpose of lowering hazard which, once acquired, becomes the target of optimization and thereby raises it: institutions would come to manage the index rather than the quantity, and the quantity would be free to worsen behind it. That is Goodhart's law, and in this framework's own vocabulary it is $\rho < 0$ with the measurement apparatus as the capability. Proposing an anatomy metric carelessly is therefore not merely a methodological risk; it is an instance of the failure the metric exists to detect.

*Two structural features make the delay affordable.* First, $\rho$ is a *derivative and not a level* — a rate of response of self-inflicted hazard to acquired capability, not a stock. It is correspondingly resistant to proxying by any static score, since to game it one would have to control how hazard responds to capability, which requires already possessing the property. Second, and more usefully, *the trichotomy of Proposition 5.35 turns on sign alone.* Trapped, critical and viable are separated by whether $\rho$ is negative, zero or positive, with no magnitude entering anywhere. An investigator may therefore work at sign level today, asking, of a particular capability in a particular system, whether its acquisition raises or lowers self-inflicted hazard, and obtain the regime classification, which is most of what the model is for, without committing to any index.

*And the right formalization is downstream, not absent.* If $\rho$ is eventually obtained as in §11.6, computed from institutional structure rather than scored onto it, then there is no index to manage, because the quantity is derived from what the system *is* rather than assigned to it by an assessor. That is the formalization this reservation is holding the place for. Reaching for a proxy in the interim would foreclose it, which is to say: it would trade a route that advances for one that does not **[O]**.

**Proposition 5.35** (The anatomy trichotomy; **[P]**)**.** *Write* $h(x) = B\varsigma_b(x) + \Gamma e^{-\rho x}$ *with* $B, \Gamma > 0$*.*

(i) *If* $\rho > 0$*,* $h$ *is strictly decreasing with* $h(x) \to 0$ *as* $x \to +\infty$*: capacity is unboundedly protective.*

(ii) *If* $\rho = 0$*,* $h$ *is strictly decreasing with* $h(x) \to \Gamma > 0$*: capacity is protective but against a floor.*

(iii) *If* $\rho < 0$*,* $h \to B(1 + e^{-b})$ *as* $x \to -\infty$ *and* $h \to \infty$ *as* $x \to +\infty$*, with a unique interior minimum. Writing* $A := Be^b$ *and* $|\rho| = -\rho$*, the minimum sits at*

$$x_* = \frac{1}{1+|\rho|} \log \frac{A}{|\rho|\Gamma}, \qquad h_* = A^{\frac{|\rho|}{1+|\rho|}}\, \Gamma^{\frac{1}{1+|\rho|}} \Big(|\rho|^{\frac{1}{1+|\rho|}} + |\rho|^{-\frac{|\rho|}{1+|\rho|}}\Big), \tag{5.20}$$

*in the regime* $x_* \gtrsim b$ *where the crest is exponential.*

*Proof.* $\varsigma_b$ is strictly decreasing with limits $1 + e^{-b}$ and 0; $e^{-\rho x}$ is decreasing for $\rho > 0$, constant for $\rho = 0$, and increasing for $\rho < 0$. (i) and (ii) are immediate. For (iii), approximating the crest by $Ae^{-x}$ above the offset, $h' = 0$ gives $Ae^{-x} = |\rho|\Gamma e^{|\rho|x}$, whence $x_*$; substituting back and collecting the common factor $A^{|\rho|/(1+|\rho|)}\Gamma^{1/(1+|\rho|)}$ gives $h_*$. Uniqueness follows since $h'' > 0$ wherever $h' = 0$. (Verified numerically to three significant figures at $b = 3$, $k = 6$, $\rho = -0.2$: predicted $x_* = 8.84$, $h_* = 1.744 \times 10^{-2}$; measured 8.88 and $1.757 \times 10^{-2}$.) ∎

**Theorem 5.36** (Non-positive anatomy is fatal, and fast; **[P]**)**.** *Let* $\rho \leq 0$*. Then:*

(i) Viability is zero, for the random background and pathwise. $\Lambda_\infty = \infty$ *almost surely, so* $\mathbb{P}(\Lambda_\infty < \infty) = 0$*.*

(ii) The tail is exponential at frozen background. *If $B_t \equiv B > 0$, then $h_* := \inf_x h(x) > 0$ and*

$$\mathbb{P}(T > t) \;\le\; e^{-h_* t} \qquad \textit{for all } t, \tag{5.21}$$

*whence $\mathbb{E}[T] < \infty$. For the random background no such floor exists, $\inf_t B_t = 0$ a.s. (Remark 5.37), and no exponential bound is claimed there; part (i) is what holds pathwise, its recurrent-driver case routed through Lemma 5.27.*

(iii) In either case no power-law structure survives*: neither the $t^{-1/4}$ tail, nor infinite life expectancy, nor Pareto residual life, nor the Lindy reading of §5.4.*

*Proof. (i)* If $x_t \to +\infty$ the trough $\Gamma e^{|\rho| x_t} \to \infty$; if $x_t \to -\infty$ the crest tends to $B_t(1 + e^{-b})$ and $\int_0^\infty B_t\, dt = \infty$ a.s. by Lemma 5.27 with $S = [0, \infty)$ (the earlier appeal to recurrence of the driving Brownian motions was informal). On any other path at least one channel is bounded below along a set of infinite measure, and the two halves are worth naming: either $\mathrm{Leb}\{t : x_t \ge 0\} = \infty$, where for $\rho \le 0$ the trough alone floors the hazard at $\Gamma e^{-\rho x_t} \ge \Gamma$ there, a deterministic floor on an infinite set, or $\mathrm{Leb}\{t : x_t \le 0\} = \infty$, where the crest with $S = \{t : x_t \le 0\}$ routes through Lemma 5.27 exactly as in Theorem 5.28. In every case $\Lambda_\infty = \infty$.

*(ii)* For $\rho = 0$, $h \ge \Gamma > 0$ independently of the background. For $\rho < 0$ at frozen $B$, $h_*$ is the interior minimum of Proposition 5.35(iii), strictly positive since both channels are; then $\Lambda_t \ge h_* t$ and $\mathbb{P}(T > t) = \mathbb{E}[e^{-\Lambda_t}] \le e^{-h_* t}$. □

*Remark* 5.37 (Why the exponential rate is not pathwise for the random background)*.* The scope of (5.21) deserves stating, since an earlier form of this theorem claimed the bound held pathwise and used no property of the driver. For $\rho = 0$ that is exact: the trough alone floors the hazard at $\Gamma$, whatever the background does. For $\rho < 0$ it is not, and the reason is that $h_*$ is a function of $B$. The two-channel background $B_t = \sum_i e^{\epsilon W_t^{(i)}}$ of (5.7) is random with $\inf_t B_t = 0$ almost surely, so there is no $B_0 > 0$ below which the background never falls, and no single $h_*$ floors the hazard along every path. What survives unconditionally is part (i), the cumulative hazard still diverges, since a vanishing crest requires $x_t \to -\infty$ which is exactly where the crest is at its largest, so the *verdict* is unchanged and only the *rate* is scoped.

Read correctly, then, (5.21) is an annealed or frozen-background statement, and the correct reading of $\mathbb{E}[T] < \infty$ is the same. Nothing downstream consumes the rate: §5.8 consumes the trichotomy and the two-condition theorem, and §9.4's trapped row consumes the qualitative shape, which part (i) supplies. Whether an unconditional exponential rate holds for the random background, plausibly yes, at a smaller constant, by an ergodic argument on $B$, is open **[O]**.

**What the trapped regime looks like.** The reading of $\rho < 0$ should be said plainly, because it inverts the intuition the rest of the model builds. With negative technoanatomy the *safest available configuration is not the strongest but the weakest*: $h \to B(1 + e^{-b})$ as $x \to -\infty$, so a system with bad anatomy is safest at background hazard with no capacity at all, and every increment of capacity beyond $x_*$ is net harmful. The system is not merely at risk; it is trapped between an external hazard it cannot shed without capacity and an internal hazard that capacity creates. Equation (5.20) prices the trap: the best achievable hazard $h_*$ is a geometric interpolation between the external scale $A$ and the internal scale $\Gamma$, weighted by anatomy.

## 5.6 Transience, and the two-condition theorem

Anatomy governs whether capacity protects. It does not by itself deliver survival, because protection must also be *acquired*, and whether it is depends on the driver. This subsection assembles the two into the model's central structural result.

### 5.6.1 The feedback condition

By Proposition 5.35, $\rho > 0$ makes $h \to 0$ as $x \to +\infty$; but $\Lambda_\infty < \infty$ additionally requires $x_t \to +\infty$ fast enough for $\int^\infty e^{-\min(1,\rho)x_t} dt$ to converge, and $x_t = \int_0^t \mu_s \, ds$ grows without bound only if the driver does not return. That is a transience condition on $\mu$, and the boundary classification of one-dimensional diffusions settles it exactly.

**What the theorem computes.** By Remark 5.25, $\mathbb{P}(\Lambda_\infty < \infty)$ is the probability that the biosphere *advances* — that some continuation available to it reaches the Golden Point. The two-condition theorem below therefore does not say what it takes to avoid dying; it says what it takes for a route to exist at all, and the two factors are the two things a route requires. That reading is what makes the trapped regime intelligible: with $\rho < 0$ the probability is identically zero, so a biosphere there has departed in the strict sense — not because it has stopped, but because nothing it can do advances.

**Transience is a feedback loop.** The canonical transient driver is

$$d\mu_t = \nu \, \mu_t \, dt + \sigma \, dW_t, \qquad \nu > 0, \tag{5.22}$$

whose drift is proportional to the state: *the rate of improvement grows with the improvement already achieved.* This is the self-reinforcing dynamic that the technological-singularity literature posits, written as a diffusion, and it is not an addition to the model but the member of the driver family whose scale function is bounded. The connection made in §5.6.2 is therefore not an analogy: the condition under which the model admits immortality *is* the condition under which the driver exhibits feedback.

### 5.6.2 The theorem

**Theorem 5.38** (Two conditions for viability; **[P]**)**.** *Let the driver be a regular diffusion on* $\mathbb{R}$ *with scale function* $s$*, and let* $\rho \in \mathbb{R}$ *be the technoanatomy. Then*

$$\mathbb{P}\big(\Lambda_\infty < \infty\big) \;=\; \underbrace{\mathbf{1}\{\rho > 0\}}_{\textit{anatomy}} \cdot \underbrace{\frac{s(\mu_0) - s(-\infty)}{s(+\infty) - s(-\infty)}}_{\textit{escape}}, \tag{5.23}$$

*the second factor, the driver's* escape probability*, read as follows. When at least one scale limit is finite, it is the displayed formula, with the conventions that* $s(+\infty) = \infty$ *(and* $s(-\infty)$ *finite) gives* 0*, and* $s$ *bounded above with* $s(-\infty) = -\infty$ *gives* 1*. When* both *scale limits are infinite the driver is recurrent and the factor is not determined by* $s$ *at all (Theorem A.5(iv)): it equals* $\mathbf{1}\{\bar{m} > 0\}$ *for positive recurrent drivers with integrable stationary mean* $\bar{m}$ *(Proposition A.6), it equals* 0 *for the Brownian driver (Theorem 5.26), and for null-recurrent drivers beyond Brownian it is open* ***[O]****. Both factors are necessary: neither good anatomy without feedback, nor feedback without good anatomy, admits viability.*

*Remark* 5.39 (The forward forcing: survival as asymptotically decisive evidence; sketch at **[P]**, reading at the ladder's grades). A configuration-level consequence of the theorem deserves its own statement, because it is the forward-time mirror of the conditioning argument's past-direction climb. Let the configuration, the sign $\rho$ together with the driver class, be drawn from any prior that gives positive mass to the viable set, the configurations for which the theorem's product is positive. Non-viable configurations have survival probabilities decaying to zero, exponentially in the trapped regime and polynomially in the critical one, while viable configurations hold a positive limit, so the conditional probability of viability given survival to time $t$ tends to one: there is a soft threshold in $t$, set by the slowest non-viable clock, past which a surviving system is almost certainly in the configuration the two-condition theorem describes. On the event of survival forever the statement is no longer asymptotic but outright, since $T = \infty$ entails $\Lambda_\infty < \infty$ almost surely and hence the viable configuration itself. In the author's words: each attribute intersection of the biosphere surviving through proper time $t$ tilts the disagreeing superposition of poles, the tilt becomes asymptotically decisive, and under full conditioning on indefinite survival the competence the Golden Point marks is not made likely but implied, there being a threshold beyond which surviving with non-Golden dangers of any kind still standing becomes arbitrarily unlikely. This is the to-the-future extension of the past-direction result that the same conditioning forces the evolutionary climb by which the Golden Point is reachable at all, the end-Cretaceous impact standing among the record's clearest instances as one of the bigger deadlocks which we fought through with Golden conditioning. Two fences hold unchanged. The mixture sketch is stated for the model's configuration space, with the identification of the viable configuration as the Golden Point's marker carried at the framework's usual grades and not at **[P]**; and the epistemic ceiling governs exactly as before, because this remark concerns the conditional law and not any observer's epistemic state, so survival so far, at any finite time, certifies nothing for the one surviving.

A third factor joins these once the schedule channel of §5.7 is carried: (5.41) prefixes the product with the probability that the schedule completes at all, which attenuates where the anatomy indicator annihilates. The theorem also scopes the Golden Tendency of §2.3: the answer survivorship selects at §2.4.2 purchases nothing toward the capacity this theorem separates from it.

*Proof.* On $\{\mu_t \to +\infty\}$: eventually $\mu_t \geq 1$, so $x_t \geq x_{t_0} + (t - t_0)$ and, for $\rho > 0$, $h(x_t) \lesssim e^{-\min(1,\rho)(t-t_0)}$, whence $\Lambda_\infty < \infty$. For $\rho \leq 0$, Theorem 5.36(i) gives $\Lambda_\infty = \infty$ almost surely, for the random background and regardless of the driver, so the first factor is necessary. On $\{\mu_t \to -\infty\}$, $x_t \to -\infty$ and $h \to B(1 + e^{-b}) > 0$ for every $\rho$, so $\Lambda_\infty = \infty$; on the recurrent event the outcome is governed by Proposition A.6, $\bar{m} > 0$ gives $D_t/t \to \bar{m}$ and $\Lambda_\infty < \infty$, while $\bar{m} \leq 0$ (nondegenerate) gives $\Lambda_\infty = \infty$ by its parts (b)–(c), with the Brownian driver settled by Theorem 5.26 and general null recurrence left open, which is the statement's scope clause. The hitting probabilities of $\pm\infty$ are the classical scale-function formula [71, 72]. □

**Corollary 5.40** (The solvable case; **[P]**). *For the feedback driver* (5.22)*,* $s'(v) = \exp(-\nu v^2/\sigma^2)$ *is integrable at both ends, and*

$$\mathbb{P}\big(\Lambda_\infty < \infty\big) = \mathbf{1}\{\rho > 0\} \cdot \Phi\Big(\frac{\mu_0\sqrt{2\nu}}{\sigma}\Big). \tag{5.24}$$

*The viability probability is thus a* product of two thresholds of different character*: a sharp one in the technoanatomy, at* $\rho = 0$*, which is a discontinuity; and a smooth Gaussian sigmoid in* *the initial improvement rate, of width $\sigma/\sqrt{2\nu}$, which sharpens to a step only in the small-noise limit.*

*Remark* 5.41 (Against the singularity literature). Equation (5.23) states, in the model's own terms, an objection to the singularity and techno-optimist literatures which is sharper than the usual one. Those literatures supply the second factor and are silent on the first: feedback is argued for at length, and whether capacity is turned back on its holder is not a parameter they carry. The model's verdict on that omission is not that it leaves the conclusion unsupported but that it inverts it. With $\rho < 0$, feedback does not merely fail to secure survival; it *accelerates the failure*, since $\mu_t \to +\infty$ drives $x_t \to +\infty$ and hence, by Proposition 5.35(iii), $h \to \infty$ superexponentially. Supplying feedback alone is strictly worse than supplying neither: a system with bad anatomy and no feedback sits near $h_*$, while one with bad anatomy and feedback runs to unbounded hazard as fast as its improvement compounds.

### 5.6.3 If anatomy itself fluctuates

Technoanatomy has been treated as a constant, which is an idealization and the model's most obvious place to give. Suppose instead $\rho = \rho_t$ is a process. Two consequences follow at scaling grade, and together they sharpen rather than blunt the event-horizon reading.

**Proposition 5.42** (The tolerance for adverse anatomy collapses with strength; **[S]**)**.** *Let anatomy dip to $\rho_t = -|\rho| < 0$ for a duration $\delta$ at a time when technostrength is $x$. The contribution to the cumulative hazard is $\asymp \delta\,\Gamma \mathrm{e}^{|\rho| x}$, so that such an excursion is survivable only if*

$$\delta \lesssim \Gamma^{-1} e^{-|\rho| x}. \tag{5.25}$$

*The affordable duration of adverse anatomy therefore decays* exponentially in technostrength.

**Corollary 5.43** (Recurrent anatomy is fatal; **[S]**)**.** *If $\rho_t$ is recurrent on $\mathbb{R}$, if anatomy returns to negative values infinitely often, then $\Lambda_\infty = \infty$ almost surely,* whatever the driver. *Each negative excursion occurring at technostrength $x_t$ contributes $\mathrm{e}^{|\rho| x_t}$, and along any path on which strength accumulates this diverges superexponentially. Viability therefore requires $\rho_t$ to be eventually positive almost surely, transient upward, or absorbed above some $\rho_c > 0$, and the two-condition theorem becomes a* two-transience *condition, with $\mathbb{P}(viable)$ a product of two scale functions rather than an indicator times one.*

**This is the event horizon the fixed model could not state.** Three readings, and the third is the one to keep.

*The threshold becomes an escape rather than a level.* With $\rho$ fixed, the anatomy condition is a sign test one either passes or fails from the outset. With $\rho$ fluctuating it is an event, anatomy escaping upward and not returning, which is what an event horizon is: not a place in the state space but a passage after which the past is unreachable.

*"Hard to regress past" becomes quantitative.* Regression is never impossible; what (5.25) says is that the affordable *duration* of it collapses exponentially in strength. A biosphere may be badly configured for a long while when weak and for almost no time at all when strong. *An ordering constraint falls out.* Since the cost of anatomical failure grows with strength while the benefit of capacity does not grow with anatomy, anatomy must be secured *before* strength accumulates. A biosphere that acquires capacity first and attends to its own arrangement afterwards is running

the two in the fatal order, and (5.25) prices exactly how fatal: by the time it turns to the problem, its margin for error has shrunk by $e^{|\rho|x}$.
The dynamics of $\rho_t$ are not supplied here, and supplying them is the model's principal open direction: an anatomy that degrades when capacity outruns the patching of it, $d\rho = (\text{patching}) - \lambda\, \dot{x}\, dt$ or similar, would make technoanatomy emergent rather than parametric. Whether the asymptotics of the resulting two-process system remain tractable is **[O]**.

**Where the earlier results now sit.** The trichotomy locates the rest of this section. The driftless model of §§5.4–5.8, certain departure, infinite life expectancy, the $t^{-1/4}$ tail, the soft horizon and the whole survivorship interpolation, is the case $\rho > 0$ with a *recurrent* driver: good anatomy, no feedback. It is the boundary between the trapped regime, where viability is zero and life expectancy finite at frozen background (Theorem 5.36), and the viable regime, where immortality has positive probability (Corollary 5.40). That is the same structural position the driftless case occupies within the scale-function classification of Appendix A, and it is not a coincidence: both trichotomies are the same statement resolved along different axes — one along anatomy, one along the driver.

### 5.6.4 The internal channel as a perturbation

The two channels admit a perturbative treatment in the internal coupling $\Gamma$, and it is worth recording, both because it supplies the model's one series expansion and because the series behaves in an instructive way at the anatomy threshold.

**Proposition 5.44** (First-order internal correction; **[P]** to first order, **[S]** for the convergence claim)**.** *Write $\Lambda_t = \Lambda_t^{\text{crest}} + \Gamma \int_0^t e^{-\rho x_s} ds$. Expanding the survival functional in $\Gamma$,*

$$\mathbb{P}(T > t) \;=\; \mathbb{P}_{\text{crest}}(T > t) \left[ 1 - \Gamma\, \mathbb{E}_{Q_{\text{crest}}}\Big[ \int_0^t e^{-\rho x_s}\, ds \Big] + O(\Gamma^2) \right], \tag{5.26}$$

*where $Q_{\text{crest}}$ is the crest-only survivorship measure of §5.8. The leading correction is therefore $\Gamma$ times the* survivor ensemble's *expected exposure to the internal channel, not the unconditioned ensemble's — the conditioning enters the correction rather than sitting outside it.*

*Proof of the first-order term.* $e^{-\Lambda} = e^{-\Lambda^{\text{crest}}}(1 - \Gamma \int e^{-\rho x} + O(\Gamma^2))$ pointwise; take expectations and divide by $\mathbb{E}[e^{-\Lambda^{\text{crest}}}]$, which converts the expectation of the bracket into a $Q_{\text{crest}}$-expectation by the definition of that measure. □

**The expansion fails exactly at the anatomy threshold.** Under $Q_{\text{crest}}$ the credit grows as $x_s \asymp \sigma s^{3/2}$, so the correction term behaves as $\int^\infty e^{-\rho\sigma s^{3/2}} ds$, which converges if and only if $\rho > 0$. Numerically at $\sigma = 1$: the integral over $[0, 60]$ is 0.90 at $\rho = 1$ and 6.7 at $\rho = 0.05$, but $2.2 \times 10^{10}$ at $\rho = -0.05$ and $1.0 \times 10^{60}$ at $\rho = -0.3$, with essentially all the mass in the far tail once $\rho < 0$.

The reading is given because it is a check on the model rather than a decoration. A perturbation series whose radius of convergence collapses at a parameter value is the standard signature of a genuine phase transition there, as opposed to a modeling artifact or a definitional boundary: the expansion fails because the object it is expanding around ceases to exist, not because the algebra becomes awkward. That $\Gamma$-perturbation theory breaks down precisely at $\rho = 0$ is therefore independent evidence that the trichotomy of §5.5 is a real division of the model and not a convenient way of describing it. It also explains why no uniform treatment of the two channels is available: for $\rho \leq 0$ the internal channel is not a perturbation of anything, being the dominant term at every large $x$.

*Remark* 5.45 (What is available in closed form, and what is not). A brief inventory, since the model is now large enough that the question arises. *Exact:* the viability product (5.23); the Gaussian sigmoid (5.24); the trapped-regime optimum $x_*$ and floor $h_*$ of (5.20); the gain minimum $\kappa(\rho)e^{-(\rho b+k)/(1+\rho)}$; the first-order term of (5.26). *Exact as an exponent, unknown as a constant:* the survival tail $Ct^{-1/4}$, whose prefactor involves the Groeneboom–Jongbloed–Wellner harmonic function and is not evaluated here. *Series:* the bias field expansion $(1-s/t)^{-1/4} = 1 + s/4t + O(s^2/t^2)$ of (6.1), in the interpolation parameter $s/t$; and (5.26), in $\Gamma$. *Variational rather than closed:* the adverse-drift rate $3\nu^2/8\sigma^2$ and its optimal profile, obtained from the Cameron–Martin action by Cauchy–Schwarz — the model's analog of an Euler–Lagrange computation, and the natural processing step whenever a conditioned path law is wanted at large deviation scale. *Not available:* the trapped regime's survivor tail beyond the exponential upper bound, and the asymptotics of any model in which $\rho$ carries its own dynamics.

## 5.7 The third channel: schedule

(5.7) carries two channels, and Remark 3.7 records that they do not exhaust the hazard of departure: $h^{\text{tot}}$ prices Base and nothing else, while the hazard of departure is the hazard of $\mu G$, which by (3.3) is at least as large and in general strictly larger. That remark calls the larger rate open **[O]**. This subsection is a first attempt at it. The claim is modest: the rate has a computable shape, the shape is a third channel, and the distribution it needs is one the companion already supplies.

### 5.7.1 A collision of vocabulary, settled first

**Convention 5.46** (Endogenous, internal, schedule)**.** The companion divides departure into *external*, inflicted from outside the biosphere, and *internal*, defined there as the complement: the remaining cases, all occurring within the biosphere's scope [5]. That category is a residue and is correspondingly wide; the companion's own instance of it is a biosphere too underdeveloped to reach a milestone given its resources and timing, which is a scheduling failure and not a self-inflicted one.

The internal channel of (5.7) is narrower: hazard the biosphere inflicts on itself with capacity it has acquired, whose response to that capacity is the technoanatomy $\rho$ of Definition 5.32.

The two senses are a category and one of its members. This work uses *endogenous* for the companion's residue and reserves *internal* for the channel:

$$\text{endogenous} = \underbrace{\text{internal}}_{\text{capacity turned inward}} + \underbrace{\text{schedule}}_{\text{derailment before capacity can be held}} + \cdots \tag{5.27}$$

The ellipsis is substantive. A category defined by exclusion is not shown exhausted by exhibiting two members, and nothing here claims these are the last.[14]

[14] The rename is a convention of this work and is offered rather than imposed; *non-external* and *endemic* would serve. One point of order regardless of the choice: the companion is deposited, so a reader arriving from the deposit will meet *internal doomsday* in the residue sense, and the convention is owed to bridge the two states of that text whether or not the term is changed there.

### 5.7.2 The schedule

**Definition 5.47** (Schedule; **[S]**)**.** A *schedule* is a sequence of $n$ gates to be passed in order. At gate $k$ there are $m$ competing exponential clocks with rates $\lambda_{k,1}, \ldots, \lambda_{k,m}$, the first corresponding to the realized option. Write

$$\lambda_k = \sum_{j=1}^{m} \lambda_{k,j}, \qquad \pi_k = \frac{\lambda_{k,1}}{\lambda_k}, \tag{5.28}$$

so that gate $k$ fires at rate $\lambda_k$ and resolves to the realized option with probability $\pi_k$. Write $k_t$ for the number of gates passed by time $t$. The *onset state* is $k = n$: the first state at which the muffling exponent $x$ of (5.5) can be held at all.

The rates are the primitives and $\lambda_k, \pi_k$ are read off them, which matters because it is the per-option rates a natural-historical account would estimate — not a tempo and a fallibility chosen separately. Stage dependence is retained deliberately. The reason the object is a *sequence* rather than a set is that each event's effect depends on the state its predecessor left: the climatic and ecological baton-passing that carried a lineage down from the trees is not a set of independent draws, and a constant $\pi$ would model it as one. Stage-dependent $\pi_k$ is the minimum that preserves the structure; state-dependent $\pi(\cdot)$ is what the picture actually asks for, and is not supplied here **[O]**.

One point of scope, since it is a natural worry and the construction already absorbs it. An event later in the sequence arriving before its predecessors needs no transition of its own: it is one of the $m$ options competing at the gate it arrives at, and resolving to it is a wrong resolution like any other. The $m-1$ non-realized options are not a homogeneous class, each carries its own rate by Definition 5.48 and its own consequence, so heterogeneity of failure is present without being labeled by type. What the chain does not carry is partial progress within a gate: it is Markov between them, which is a simplification and is chosen rather than derived.

The discreteness is a limitation of the same kind as calling Earth a sphere. Exotic structure in the ordering is not recoverable from a chain over $m^n$ articulable realizations; the shape of the dependence is, and that is what is wanted here.

### 5.7.3 Where the rates come from

Definition 5.47 takes the per-option rates as given. One specification of how they are drawn is worth carrying, because it determines more than it looks like it should.

**Definition 5.48** (Log-spread rates; **[S]**)**.** At each gate the $m$ rates are drawn afresh and independently with $\log \lambda_{k,j}$ *logistic*, linear across the bulk with both tails capped, since a shortest feasible time bounds the fast end and a longest tolerable one bounds the slow, and *which option is the realized one is independent of its rate*. The correct option may be the fastest of the $m$, or the slowest, and nothing about the draw favors either. The lognormal reading is the default and carries its own justification: a log-rate that is a sum of many independent multiplicative contributions is normal by the central limit theorem, so the law is inherited rather than chosen, and the capped variants are refinements for the extreme tails.

Three consequences, in ascending order of how much they change. *The waiting time is set by the correct option's own rate.* A gate fires at $\lambda_k = \sum_j \lambda_{k,j}$ and resolves correctly with probability $\pi_k = \lambda_{k,1}/\lambda_k$, so the expected number of attempts before a correct resolution is $1/\pi_k$

and each costs $1/\lambda_k$. The product is $1/\lambda_{k,1}$: however many wrong options crowd the gate, what a biosphere waits on is the realized option's own clock. Exposure per gate is therefore log-spread along with the rate, and the difference between a lucky gate and an unlucky one is orders of magnitude of waiting rather than a factor.

*The contrast between options is a random order of magnitude, with the sign withheld.* $\log(\lambda_{k,1}/\lambda_{k,j})$ is a difference of two log-uniform draws: symmetric about zero, spread across the range, and committing to no direction. That is the shape of the High-Contrast Earth Hypothesis, and the correspondence is exact rather than suggestive — the Hypothesis asserts a random order of magnitude between independently constructed alternatives and *deliberately withholds the sign*, with the Strong Accordance Principle supplying direction separately. (5.48) supplies the same shape from a mechanical premise about competing clocks, and the direction arrives the same way it does there: from conditioning, below. So the Hypothesis acquires a microfoundation at the gate level, and does not have to be imported as a posit at this scale — though what replaces it is another **[S]**, and the trade is one posit for a more checkable one rather than a derivation from nothing.

The correspondence should be stated at the right strength, since it is not an identity of distributions. By (5.38) the contrast a perturbation carries is $r\log_{10}(1 + (B + \Gamma)/\lambda)$, which in the regime supplying many orders (Remark 5.59) is $r$ times a *log-rate gap* — and under Definition 5.48 a log-rate gap is a difference of normals, hence normal, hence symmetric and spread across orders with no committed sign. The companion states the Hypothesis as uniform rather than normal. What the collapse consumes, however, is not the shape but two functionals of it, the typical contrast $\mathbb{E}[c]$ and the averaged one $-\log_{10}\mathbb{E}[10^{-c}]$, and both are defined for either law. So the construction is compatible with the Hypothesis in the sense the argument uses it, without reproducing its stated distribution.

The scale of the spread is a further matter, and the accurate thing to say is not that it awaits measurement. Nothing available to an observer inside a biosphere fixes what *sets* the range over which competing rates are distributed: one meets the distribution already running, with no vantage from which its parameters could be explained rather than merely estimated. That is §6.8's structure applied one level down, and the scale is scoped as uncertifiable for the reason $\rho$ is, rather than listed as a parameter someone has yet to supply.

*And conditioning selects the fast-correct branch, which is why the realized path looks optimal.* Exposure at gate $k$ is $1/\lambda_{k,1}$, paid at hazard $B + \Gamma$ by Proposition 5.52, so a branch's survival weight carries $\exp\{-(B+\Gamma)/\lambda_{k,1}\}$ at every gate. Branches in which the realized option happened to be slow pay exponentially for it. Conditioning on survival therefore selects, gate by gate, for histories in which the correct option was among the fastest available — not because anything chose them, but because the alternatives were charged for the wait.

That is the mechanism, and should be stated as one. Under the Cosmic Nexus the realized option is a random draw and is usually slow; under conditioning one is overwhelmingly likely to find oneself on a branch where it was fast at gate after gate. The appearance of an optimal path threaded through a combinatorial mess is what survivorship does to a log-spread rate landscape, and requires no optimization to produce it. This is the Optimal-Earth phenomenology arriving from the schedule's own machinery rather than being assumed alongside it [5, 23].

### 5.7.4 Incompleteness decays capacity

The first commitment is that until the schedule completes, the muffling exponent cannot be held. This departs from Definition 5.11, which states that there is no capacity parameter and no leak; the departure is deliberate and is confined.

**Definition 5.49** (Staged leak; [**S**])**.** Replace (5.5) and (5.4) by

$$dx_t = \mu_t\,dt - \gamma_{k_t} x_t\,dt, \qquad d\mu_t = -\nu_{k_t}\mu_t\,dt + \sigma\,dW_t, \tag{5.29}$$

with $\gamma_k, \nu_k$ nonincreasing in $k$ and $\gamma_n = \nu_n = 0$. The schedule of coefficients is *geometric*: each closed gate multiplies the leak by a constant factor,

$$\gamma_k = \gamma_0\,\varphi^k, \qquad \varphi = (\gamma_{n-1}/\gamma_0)^{1/(n-1)} < 1, \tag{5.30}$$

with $\gamma_0$ enormous, so that before any gate is closed the credit is pinned at the origin to a degree with no practical upper bound, and $\gamma_n = 0$ at completion.

The solution of the first equation is $x_t = \int_0^t e^{-\gamma_k(t-s)}\mu_s\,ds$: the credit is still the integral of the driver, accumulated additively, with each increment buffered by a multiplicative decay from the moment it lands. The leak is not an alternative to (5.5) but a discounting of it. The second equation is the same commitment one derivative up, and it is what the picture requires rather than an extra assumption. The driver $\mu = -\frac{d}{dt}\log h$ is an *improvement rate*, and a rate of improvement is something an agent supplies. Before the onset state there is none, so $\mu$ is not a free random walk either: neither the stock nor the rate is sustained, and the two dampings are one absence expressed at two levels. A biosphere's ancestors do not hold a small amount of technostrength — they hold none, and what the pair of equations says is that excursions away from zero are transient rather than accumulated.

At stage $k$ the deterministic part obeys $\frac{d}{dt}\log x = -\gamma_k$: capacity acquired before completion bleeds away at a rate each completed gate reduces. The leak is by hand, and making it emergent, $\gamma$ falling out of the schedule's own structure rather than being assigned to it, is the model's natural next target [**O**].

**What a closed gate buys, and why almost none of it is bought early.** (5.30) makes gates equal in $\log\gamma$ rather than in $\gamma$, and the consequence for the payoff is severe enough to state on its own. The credit a biosphere can sustain at stage $k$ scales as $1/\gamma_k = \varphi^{-k}/\gamma_0$, so it grows *geometrically* in gates closed. But the hazard depends on the credit *exponentially*, both channels of (5.7) do, by construction, so the relief a closed gate delivers goes as $\exp\{-\rho C\varphi^{-k}/\gamma_0\}$, doubly exponential in $k$. Closing the last gate is worth more than closing all the others together, by a margin that is not close.

With $\gamma_0$ enormous the reading sharpens. The early gates take a biosphere from a credit that is unimaginably pinned to one merely very pinned, which is no relief at all in the only currency the hazard reads. Practical relief arrives in the final stretch, and a schedule half-climbed is nearer in hazard to one not begun than to one completed. Set this beside Remark 10.5, which has stakes rising monotonically for combinatorial reasons — a late step carries the improbability of every step behind it. The present observation is a second and independent reason for the same monotonicity, arriving from the leak's geometry rather than the chain's combinatorics. Two arguments of different provenance agreeing on the profile is a better position than either alone.

*Remark* 5.50 (The existing model is the completed case)*.* $\gamma_n = 0$ is not a convenience. It makes Definition 5.11 the $k = n$ boundary case of (5.29) exactly, so nothing proved in §§5.4–5.8 is disturbed: the persistence exponent $\frac{1}{4}$, the trichotomy, the two-condition theorem and the survivorship interpolation are all statements about the completed regime, and this channel prefixes them rather than amending them.

The price is that the incomplete regime has its own asymptotics — and they can be given, which is the next proposition's business. The persistence problem is not the one §5.4.3 solves, and its exponent does not carry.

**Proposition 5.51** (The incomplete regime has no polynomial tail; **[S]**)**.** *Under* (5.29) *with* $\gamma_k, \nu_k > 0$ *held fixed, the pair* $(\mu, x)$ *is a two-dimensional linear system with both eigenvalues negative, driven by white noise; it is therefore ergodic with a stationary Gaussian law centered at the origin. Persistence of* $x$ *is then persistence of a stationary Gaussian process, which decays* exponentially*:*

$$\mathbb{P}\big(x \text{ stays positive to } t\big) \asymp e^{-\theta(\gamma_k,\nu_k)\,t}, \qquad \theta > 0. \tag{5.31}$$

*No power law survives. The rate* $\theta$ *is not given in closed form here; persistence rates for stationary Gaussian processes are generally not, and* (5.31) *identifies the class rather than the constant.*[15]

**What incompleteness costs, and what completion buys.** The contrast with the completed regime is not a change of exponent but a change of kind, and it is the sharpest thing the channel says about why the schedule is worth climbing.

§5.4 establishes that the completed model departs with probability one and yet has infinite life expectancy: the tail is $t^{-1/4}$, residual life is Pareto, quantiles grow as the fourth power of rarity, and the effective hazard is $1/4t$ — falling at every age. By (5.31) the incomplete regime has none of it. An exponential tail has finite expectation, memoryless residual life, quantiles growing logarithmically, and a hazard that does not fall at all. *Every consequence of Proposition 5.31 is a consequence of having completed*, and none of it is available before.

Two readings follow, and the second is structural. The Lindy structure is a *reward* rather than a property. A biosphere before its onset state is not on a fat tail at low altitude; it is on a thin one, and no amount of waiting improves its position, since an exponential is memoryless. What completing the schedule buys is not more of the same but a different tail entirely.

And the floor is the trapped regime's floor with one difference that matters. Theorem 5.36(ii) gives $\mathbb{P}(T > t) \le e^{-h_* t}$ for adverse anatomy at frozen background, and (5.31) gives an exponential floor for incompleteness. The two are structurally alike and disposed of oppositely: the anatomy floor cannot be left, since $\rho$ is a parameter, while the schedule floor is *closable* — $\gamma_k$ and $\nu_k$ vanish at completion, and the debt that stands between is payable whenever the criterion of §5.7.10 holds. An incomplete biosphere and a trapped one sit on the same kind of tail; only one of them can get off it.

A biosphere may of course be both, and the two floors then compose exactly rather than competing. If $h^{\text{tot}} \ge h_*$ then $\mathbb{P}(T > t) = e^{-h_* t}\,\mathbb{E}\big[e^{-\int (h^{\text{tot}} - h_*)}\big]$ identically, a constant floor factors out of the exponential functional, leaving the damped-persistence problem for the re-centered hazard with its own rate $\tilde{\theta}$. So $\theta_{\text{joint}} = h_* + \tilde{\theta}$: neither floor governs, they *add*, and the interventions are correspondingly separable. Completing the schedule removes $\tilde{\theta}$ and not $h_*$; moving $\rho$ removes $h_*$ and not $\tilde{\theta}$. A biosphere that is both trapped and incomplete must do both things, and doing either buys exactly its own term.

[15] Simulation at $\sigma = 1$, $8 \times 10^4$ paths. With $\gamma = 1$ and the driver *undamped* ($\nu = 0$) the decay is a power law, fitted slope $-0.476$, log–log correlation 0.9995, which is the damped randomly accelerated particle and is not the regime this definition describes. With the driver damped, the fit inverts decisively: at $(\gamma, \nu) = (1, 0.5)$ and $(1, 1)$ the correlations against $t$ are 1.0000 and 0.9999 with rates 0.222 and 0.312, against log–log correlations of 0.975 and 0.985.

### 5.7.5 The schedule gates the external channel

Completing the chain confers no capacity. It confers only the ability to *hold* capacity — which sounds like a small thing and is not, because of what (5.7) does at $x = 0$.

**Proposition 5.52** (An unheld $x$ leaves the crest at bare background; **[P]**)**.** *By the normalization of* (5.7), $\varsigma_b(0) = 1$ *exactly, so* $h^{\mathrm{crest}}\big|_{x=0} = B_t$*; and the trough does not vanish there either, since* $\Gamma e^{-\rho\cdot 0} = \Gamma$*. A biosphere that cannot hold x therefore faces the external background unshielded* and *pays the internal channel's full coefficient, at total* $h^{\mathrm{tot}}\big|_{x=0} = B_t + \Gamma$*. Its cumulative hazard is* $\Lambda_t = \int_0^t (B_u + \Gamma)\, du \to \infty$*, and* $\mathbb{P}(\Lambda_\infty < \infty) = 0$ *identically — for every parameter value, every anatomy* $\rho$*, and every driver. The configuration and its consequences are those of Proposition 10.2.*

*Remark* 5.53 (The external term is small only because the schedule succeeded)*.* Read at $x > 0$ the crest channel is the least of the model's worries: it is bounded above by Proposition 5.15, it is shielded without limit as capacity accumulates, and beside an unbounded internal channel it looks like one measly term. That reading presupposes what the schedule supplies. Run the dependency out. Shielding is the factor $\varsigma_b(x)$; $\varsigma_b$ shields only for $x > 0$; $x$ can be held only at $k = n$; and $k = n$ is reached through $n$ gates each of which can go $m - 1$ ways wrong. *So the survivability of the external channel is downstream of the schedule entire.* The cosmic background does not become more dangerous when the schedule fails; it becomes the only thing that happens, which is the sentence Remark 10.3 writes about a biosphere whose muffling agent has been removed.

That is not a coincidence of phrasing. *An incomplete schedule and the extinctionist's withdrawn world are the same configuration.* Both are $x$-cannot-be-held; both sit at bare $B_t$; both are Proposition 10.2's case; and both lie in $\mu G \setminus \mathrm{Base}$ by §3.4.1 — departed by configuration rather than by event, with no hazard yet fired. The channel therefore does not merely add a term. It exhibits the model's apparently smallest channel as conditional on its largest improbability.

### 5.7.6 Failure incurs debt, not weight

What a wrong gate costs can now be said, and the natural first answer is the wrong one. Treating a wrong gate as multiplying an onset weight makes that weight a primitive, a number the model is handed rather than one it produces, and leaves the cascade of §3.4.1 inexpressible, since a single multiplicative factor has nowhere to put the recursion by which an Imperfection Magnitude Drop's residual carries continuations of its own.

**Definition 5.54** (Debt; **[S]**)**.** A wrong gate does not suppress a weight. It *incurs debt*: $r$ further gate-successes are owed, from the state now occupied, before completion is reached. Write $Z_t$ for the outstanding debt, with $Z_0 = n$. Each firing consumes one gate and, with probability $1 - \pi$, adds $r$:

$$Z \longmapsto \begin{cases} Z - 1 & \text{with probability } \pi, \\ Z - 1 + r & \text{with probability } 1 - \pi. \end{cases} \tag{5.32}$$

*Completion* is the event $Z = 0$.

The natural-historical reading is the one that motivates it. Human extinction does not leave the biosphere holding a fractional prospect; it leaves it owing a further sequence, some other lineage must run its own improbable gymnastics from wherever the world now stands, and the onset state recedes by exactly that sequence's length. Debt postpones a deadline. It does not discount a prize.

$r$ need not be a primitive, and the model is better if it is not. Read the schedule as a tree whose paths each carry a definite number of events, differing between paths: a wrong turn then places the biosphere on a longer path, and *the excess length is the debt.* On that reading (5.32) is the induced dynamics on *events remaining* rather than a further posit, $r$ is a structural property of the tree rather than a parameter, and whether it correlates with the rates, assumed away above, becomes a question about how the tree is built. Nothing downstream changes, since the induced dynamics is (5.32).

Two things follow immediately and neither was available before. *The win condition is binary.* A branch either clears its debt or does not, and any branch that clears it completes — however deep the debt ran, and with no residual weight to carry. The grading that a weight formulation would posit emerges instead from the *distribution* over branches. *And the recursion is structural.* Debt spawns debt: the extra gates are gates, they can go wrong, and going wrong incurs more. That is the cascade's shape, and the object it makes is standard.

**Proposition 5.55** (The debt process is Galton–Watson; **[P]**)**.** *Under Definition 5.54, $Z$ is the population of a Galton–Watson process with offspring probability generating function*

$$f(u) \;=\; \pi + (1-\pi)\,u^{r}, \qquad \textit{mean offspring} \quad \varkappa \;=\; r(1-\pi). \tag{5.33}$$

*Completion is extinction of $Z$. Hence completion occurs almost surely if and only if $\varkappa \le 1$, and for $\varkappa > 1$ the probability of never completing is $1 - q^{n}$, where $q$ is the least root of $q = f(q)$ in $[0,1]$.*

*Remark* 5.56 (A phase transition in the schedule's own structure)*.* $\varkappa = r(1-\pi) = 1$ is a sharp threshold, and it is not a restatement of anything already in the paper. Below it, debt clears with probability one: however unlucky a biosphere is, it owes a finite sequence and eventually pays it. Above it, a positive fraction of branches accumulate debt without bound, cycling into ever-deeper obligation, each new gate as fallible as the last, and those branches never reach the state at which $x$ can be held. The transition sits at $r = 1/(1-\pi)$: a biosphere may tolerate large debt per failure if failures are rare, or frequent failures if each is cheap, and the product is what decides.

The threshold is structurally the anatomy threshold's sibling. $\rho = 0$ divides configurations by whether acquired capacity protects; $\varkappa = 1$ divides them by whether incurred debt is payable. Both are sign tests on a structural parameter, both are sharp, and neither is visible in any magnitude the system displays.

*Remark* 5.57 (Under log-spread rates the criterion is the geometric mean; **[P]**)*.* Definition 5.48 redraws the rates at every gate, so $\pi_k$, and with it $\varkappa_k = r(1-\pi_k)$, is not a deterministic sequence but an i.i.d. random one. The debt process is then a branching process in a *random* environment rather than a varying one, and the criterion changes accordingly. By the Smith–Wilkinson criterion, completion is almost sure exactly when

$$\mathbb{E}\big[\log \varkappa\big] \;<\; 0, \tag{5.34}$$

the *geometric* mean of the offspring means, and not when $\mathbb{E}[\varkappa] < 1$.

The two disagree, and the direction of the disagreement matters. By Jensen $\mathbb{E}[\log \varkappa] \le \log \mathbb{E}[\varkappa]$, so a schedule may complete with probability one while its *arithmetic* mean offspring exceeds unity. Using $\mathbb{E}[\varkappa]$ is therefore not a conservative simplification but a wrong answer in the pessimistic direction: it condemns schedules that in fact clear.[16]

One caution about what the criterion certifies. (5.34) settles whether the debt clears *eventually*; it does not say when. Under a random environment the probability of not having completed by stage $n$ decays at the tilted rate $\gamma^* = -\min_{0\le s\le 1}\log\mathbb{E}[\varkappa^s]$, which is bounded above by $|\mathbb{E}\log\varkappa|$ and can fall far below it: for the $\varkappa \in \{0.1, 5\}$ environment above, $\gamma^* = 0.0158$ at $s^* = 0.092$ against $|\mathbb{E}\log\varkappa| = 0.347$, a factor of twenty-two.[17] Certain completion and timely completion are therefore different questions, and only the first is settled here **[O]**.

The reading is the one this work keeps arriving at. A geometric mean is what a typical realization experiences; an arithmetic mean is what an ensemble average reports. The quenched quantity governs, exactly as Remark 5.63 says it does for the completion weight, and here it governs the dichotomy itself rather than only the rate.

**Unrecoverable failures, and a doom weight from the schedule alone.** Definition 5.54 makes every failure recoverable at a price. Some plausibly are not, an event arriving before its prerequisites may foreclose rather than cost, and the natural encoding is $r = \infty$, which is to say a *defective* offspring law. Let a gate resolve correctly with probability $\pi$, recoverably with probability $q_1$ at debt $r$, and unrecoverably with probability $q_2 = 1 - \pi - q_1$. Then

$$f(u) = \pi + q_1 u^r, \qquad f(1) = \pi + q_1 < 1, \tag{5.35}$$

so $u = 1$ is not a fixed point and the least root of $u = f(u)$ lies strictly below one *whatever* the recoverable branch does. Completion probability is bounded away from one by $q_2$ alone, and subcriticality of the recoverable branch does not rescue it.

Two consequences. The schedule now produces a positive doom weight unaided, a nonzero probability of never reaching the onset state, computed from $(\pi, q_1, q_2, r)$ and nothing else, which is a second concrete instance of the graded object Remark 5.62 discusses. And the question whether foreclosing options dominate acquires two answers rather than one: at fixed mean offspring the second factorial moment enters the survival bounds, so they may dominate the *probability* of non-completion while leaving the mean untouched.[18]

### 5.7.7 The offset weighting, emergent

The constraint on this formulation was that it reproduce the companion's offset weighting rather than replace it. It does, and the derivation also says what the offset is *made of.* By Proposition 5.52 a biosphere pays $B + \Gamma$ while incomplete — the background unshielded, and the internal channel at its bare coefficient. Debt is therefore paid in hazard: clearing $Z$ gates at rate $\lambda$ costs $D/\lambda$ in exposure, and a branch's survival weight is $e^{-(B+\Gamma)\cdot(\text{total gates fired})/\lambda}$.

[16] Simulation, Poisson offspring, 4000 trials each. With $\varkappa$ equal to 0.1 or 5 with equal probability, $\mathbb{E}[\varkappa] = 2.55$ and $\mathbb{E}[\log\varkappa] = -0.348$: the arithmetic criterion calls it supercritical, the geometric criterion subcritical, and completion is observed with frequency 1.000. Where the two criteria agree, $\varkappa \in \{0.5, 3\}$ and $\varkappa \in \{0.9, 1.6\}$, both geometrically supercritical, completion frequencies are 0.738 and 0.690.

[17] The rate identity is the standard large-deviation form for subcritical branching in a random environment; the polynomial correction depends on the subcriticality class.

[18] Checked against direct simulation of (5.32) with a killing branch, $4 \times 10^4$ trials each: at $(\pi, q_1, r) = (0.85, 0.10, 3)$ the least root is 0.9306 against 0.9310 simulated; at $(0.70, 0.25, 2)$, 0.9046 against 0.9040; at $(0.60, 0.35, 2)$, 0.8571 against 0.8557.

**Proposition 5.58** (Completion weight as a least fixed point; **[P]**). *Let* $s := e^{-(B+\Gamma)/\lambda}$ *be the survival factor per gate-time. Where* $\lambda$ *is itself drawn per gate, the correct single-*$\lambda$ *surrogate for the mean log-weight is the* harmonic *mean* $\big(\mathbb{E}[\lambda^{-1}]\big)^{-1}$*, since* $-\log u$ *is to first order linear in* $1/\lambda$ *— which for a lognormal law is* $\lambda_{\mathrm{geo}} e^{-s_\lambda^2/2}$*, smaller than the geometric mean, so passage costs and contrasts both exceed what geometric-mean substitution would give. The expected completion weight from a single owed gate is the least root* $u \in [0,1]$ *of*

$$u \;=\; s\,f(u) \;=\; s\big(\pi + (1-\pi)u^r\big), \tag{5.36}$$

*and from* $n$ *initial gates it is* $u^n$*. For small* $B/\lambda$*,*

$$-\log u \;=\; \frac{B+\Gamma}{\lambda}\cdot\frac{1}{1-\varkappa} + O\Big(\big(\tfrac{B}{\lambda}\big)^2\Big), \qquad \varkappa = r(1-\pi) < 1. \tag{5.37}$$

*Proof.* (5.36) is the total-progeny functional equation for a Galton–Watson process weighted by $s$ per individual. For (5.37), write $u = 1-\varepsilon$ and expand: $1-\varepsilon = (1-(B+\Gamma)/\lambda)(1-\varkappa\varepsilon)+O(\varepsilon^2)$ gives $\varepsilon(1-\varkappa) = (B+\Gamma)/\lambda$. The factor $1/(1-\varkappa)$ is the expected total progeny per individual, as it must be. □

**What the offset is made of, and where the recursion went.** Two effects separate cleanly, and the separation is this subsection's return.

*Contrast per unit of depth.* A single wrong gate adds $r$ gates, each exponential at rate $\lambda$, so the added waiting is $\mathrm{Gamma}(r,\lambda)$ and the weight multiplies by $\mathbb{E}[e^{-hT}] = (1+h/\lambda)^{-r}$ with $h := B+\Gamma$. Matching to the companion's $10^{-c_{\mathrm{eff}}}$,

$$c_{\mathrm{eff}} \;=\; r\,\log_{10}\!\Big(1+\frac{B+\Gamma}{\lambda}\Big), \tag{5.38}$$

exactly and with no approximation. The deterministic-exposure reading, $r/\lambda$ of waiting at hazard $B+\Gamma$, giving $(B+\Gamma)r/\lambda\ln 10$, is the $\lambda \gg B+\Gamma$ limit of it and *overstates* the contrast elsewhere, since $\log(1+x) < x$. The direction of that error is worth naming: it makes the companion's collapse look easier than it is, so (5.38) is what should be used. The High-Contrast Earth Hypothesis's "extremely many orders" is thereby not a posit but a reading: one gets many orders when the background is high, the debt large, or the tempo slow.

*Depth accumulated.* The recursion does not change (5.38); the branching factor $1/(1-\varkappa)$ cancels out of the per-depth contrast exactly. What it changes is how much depth there is: expected total gates fired per owed gate is $1/(1-\varkappa)$, so a biosphere near criticality accumulates depth without the contrast per unit of it worsening at all. The recursion inflates the exposure, not the exchange rate — and inflates it without bound as $\varkappa \to 1$.

The separation identifies two independent levers. A biosphere may be badly placed because each failure is expensive ($c_{\mathrm{eff}}$ large) or because failures compound ($\varkappa$ near one), and the two are not the same defect.

*Remark* 5.59 (Contrast wants slow gates; survival wants fast ones)*.* (5.38) has two limits and they pull opposite ways, which is a structural fact about the channel rather than a defect of it. Where gates are fast against the standing hazard, $\lambda \gg B+\Gamma$, the contrast is $c_{\mathrm{eff}} \approx r(B+\Gamma)/\lambda\ln 10$ — small. Wrong turns cost little, passage is cheap, and alternatives are barely suppressed: there is nothing to collapse a sum over.

Where gates are slow, $\lambda \ll B+\Gamma$, the contrast is $c_{\mathrm{eff}} \approx r\log_{10}\big((B+\Gamma)/\lambda\big)$ — a *log-rate gap*, hence many orders of magnitude whenever the gap is large. But the same slowness prices survival at $(\lambda/(B+\Gamma))^r$ per wrong turn, which is astronomically small. So the regime supplying the companion's contrast is the regime in which the realized history had to be enormously lucky, and the two are one fact seen from either end. That is not a tension to be resolved but the Miracle phenomenology in rate coordinates: high contrast between natural histories and low probability of the realized one are not independent observations.

*Remark* 5.60 (The collapse is a claim about shares, not about likelihood). A companion point, and the one most easily misread. The sum-collapse establishes that the realized ordering carries a share of order unity *of the total*. It says nothing whatever about the size of that total, and under a log-scale randomization the total is minute. Two facts about extremes make this precise. The best of $m$ draws from a spread log-rate law is not far above the field: the maximum of $m$ normals sits $\sqrt{2\ln m}$ deviations above the mean asymptotically, and only 1.54 deviations at $m = 10$ — so the optimum is not qualitatively better than a typical option, it is the same law read at its edge. And the optimum still faces the standing hazard: with $\lambda \ll B+\Gamma$ the weight of even the best option at a gate is $\approx \lambda/(B+\Gamma)$, orders below one, and $n$ gates multiply those orders.

A worked instance makes the two magnitudes visible together. Take $\log_{10}\lambda$ normal with mean 15 decades below $B+\Gamma$ and deviation 3, with $m = 10$, $n = 20$, $r = 3$. The best option at a gate sits 4.6 decades above typical; the weight of the all-best path is about $10^{-208}$; a single wrong turn costs $c_{\text{eff}} \approx 31$ decades; and the collapse condition $c_{\text{eff}} > \log_{10}(n(m-1)) = 2.26$ is met roughly fourteen times over.[19]

So relative dominance is overwhelming and absolute weight is $10^{-208}$, and the two are not in tension — both follow from the same spread, which is what makes one alternative dominate its rivals and what makes all of them improbable. The reading to avoid is therefore explicit: *the highest-weight ordering is not the likely ordering.* Nothing in the companion's apparatus, and nothing here, converts a share of a small number into a large one. This is the same discipline §6.7 applies to the survivor's own position, and it applies with more force here because the numbers are worse.

*Remark* 5.61 (Which contrast the collapse consumes). A random contrast does not enter the companion's sum-collapse the way a deterministic one does, and the difference has to be recorded. An alternative at depth $\Delta$ is suppressed by $10^{-\sum_{i\le\Delta} c_i}$, so with independent draws the per-tier factor is $(\mathbb{E}[10^{-c}])^{\Delta}$ and the binomial closes at $\big(1+(b-1)\mathbb{E}[10^{-c}]\big)^n$. The operative quantity is therefore $-\log_{10}\mathbb{E}[10^{-c}]$, not $\mathbb{E}[c]$, and by Jensen the first is strictly smaller. The gap is severe rather than technical: for $c$ uniform on $(0, 2M)$ the averaged contrast is $\log_{10}(2M\ln 10)$, growing only *logarithmically* in the spread; for $c$ normal with mean $\mu$ and deviation $s$ it is $\mu - s^2\ln 10/2$, negative once $s \gtrsim \sqrt{2\mu/\ln 10}$.[20]

The resolution is the one this work keeps reaching. The averaged contrast is dominated by alternatives no one encounters; the claim the collapse is used for concerns the share carried by the *realized* configuration, which is quenched. One refinement to "runs on $\mathbb{E}[c]$": the quenched tier weight is a sum of order $\big(n(b-1)\big)^{\Delta}$ lognormal terms and is extreme-value dominated at

[19] Direct simulation, $2\times10^5$ draws; the extreme-value theory agrees to two decimals, $\mu + 1.5388s = -10.38$ against $-10.39$ observed.

[20] At $\mathbb{E}[c] = 10$: uniform on $(0, 20)$ gives 1.66; normal with $s = 2$ gives 5.45; normal with $s = 3$ gives $-0.36$. Direct sampling understates the collapse, four million draws return 1.25 for the last, because $\mathbb{E}[10^{-c}]$ is carried by a tail sampling does not reach, which is itself the point.

small $\Delta$, so the binding constraint sits at the shallowest tier and carries a fluctuation allowance,

$$\bar{c} \ > \ \log_{10}\bigl(n(b-1)\bigr) \ + \ \frac{s\sqrt{2\ln\bigl(n(b-1)\bigr)}}{\ln 10}, \tag{5.39}$$

relaxing toward the $\mathbb{E}[c]$ form as $\Delta$ grows, since the fluctuation scales as $\sqrt{\Delta}$ against a mean linear in $\Delta$. At $s=3$, $n=20$, $b=10$ the threshold moves from 2.26 to 6.45, which $\mathbb{E}[c]=10$ still clears. The companion's own hedge, that its deterministic use is the union-bounded envelope of an almost-sure claim, points the same way. So the collapse survives randomization in the quenched reading and fails in the annealed one, and a reader wanting the annealed statement needs a contrast far larger than $\mathbb{E}[c]$ suggests **[O]**.

*Remark* 5.62 (This is the graded cascade, in one concrete case). Remark 3.16 keeps the cascade's Boolean recursion and the $\varepsilon$-grading of Definition 3.14 in separate layers, and records what would join them: a fixed point of an $M$-valued doom operator, called open **[O]** and named there as the natural next question.

(5.36) is an instance of exactly that object. The map $u \mapsto s\,f(u)$ is monotone on $[0,1]$, its least fixed point is the graded analog of $\mu G$'s least fixed point on the Boolean lattice, and the grading it carries is a weight rather than a label. The completion probability is therefore *computed* as a fixed point rather than assigned, which is what the two-layer remark asks for.

The claim should be taken at its size. This is a single-type process carrying one number, not the $M$-valued operator over the full lattice of states the general construction would need; what is exhibited is the first concrete case and the shape it confirms. What the case does supply is the general construction's *definition*, and the choice is not the obvious one. The Boolean cascade characterizes $\mu G$ modally: every course admits a foreclosing continuation, which is a fact about the state space. A graded version cannot be built on that, because a modal fact carries no measure. It can be built on the dynamical reading, *all paths lead to* the Departure Point, rather than all courses make it exist, since that is a hitting statement, and the measure of paths hitting a set is canonically the *least* fixed point of the associated operator. (5.36) is exactly such a hitting measure, which is why it arrived as a least fixed point without anyone imposing it.

Two things follow for the general construction. The operator to build is the one whose fixed point is the hitting measure of the departure set, and Kleene iteration from $\perp$ reaches it wherever the weight algebra's multiplication is Scott-continuous. And the projection back to the Boolean layer is a *threshold* rather than a linear map, Boolean $\mu G$ is where the surviving weight is zero, not where the graded value takes a particular magnitude, so the consistency requirement is $\mathrm{eval}(u) = \mathbf{1}\{u=0\}$, and it should be imposed at definition time rather than proved afterwards. Whether the construction so specified generalizes to the lattice remains open **[O]**.

*Remark* 5.63 (Which rate an observer should use). The completion weight $u^n$ is an expectation over branches, and near criticality it is dominated by the rare branch that incurred little debt: by (5.37) the *mean* total progeny is $1/(1-\varkappa)$, while the median sits far below it. An observer reasoning about their own history should therefore not use the mean. Expected completion weight is carried by branches that got lucky early, and a biosphere is not entitled to assume it is on one — the same quenched–annealed discipline the reference sheet records for the survivorship transform.[21]

[21] The same gap is recorded for the survivorship transform in Appendix E; the two are instances of one phenomenon rather than one result applied twice.

### 5.7.8 The composite

The schedule does not win Golden-Point probability. It wins the probability of reaching the one state at which $x$ stops decaying, and everything the model already proves takes over from there. That gives the total a two-stage shape:

$$\mathbb{P}(\text{advance}) = \underbrace{\mathbb{P}\big(Z \text{ clears}\big)}_{\text{schedule: (5.36)}} \times \underbrace{\mathbb{E}\Big[\mathbb{P}\big(\Lambda_\infty < \infty \mid X_\tau\big)\Big]}_{\text{the model, from the completion state}}, \tag{5.40}$$

with $\tau$ the completion time and $X_\tau = (\mu_\tau, D_\tau)$ the *model's* state at that moment — the Kolmogorov pair of Definition 5.11, and not the schedule's debt $Z$, which is zero at $\tau$ by construction. Factoring the second stage by Theorem 5.38,

$$\mathbb{P}\big(\Lambda_\infty < \infty\big) = \underbrace{u^n}_{\text{schedule}} \cdot \underbrace{\mathbf{1}\{\rho > 0\}}_{\text{anatomy}} \cdot \underbrace{\frac{s(\mu_0) - s(-\infty)}{s(+\infty) - s(-\infty)}}_{\text{escape}}. \tag{5.41}$$

**The three factors are not of one kind.** Anatomy *annihilates*: $\rho \le 0$ gives zero, not a small number, by Theorem 5.36. The schedule *attenuates*: a wrong gate costs debt, and debt cleared is completion, so the factor is drastically minute and never quite zero — which is what it is to be an Imperfection Magnitude Drop in the companion's exact sense [5]. So the schedule factor is the residual $\varepsilon$ of Definition 3.14 computed rather than named, and the channel's connection to the cascade is an identification rather than an analogy.

**The decomposition is hierarchical, not a product.** (5.40) should be read exactly. It is not an independence factorization, $X_\tau$ depends on how long the debt took to clear, so the two factors are not independent, but a hierarchical decomposition, and what makes it usable is that the influence runs one way. The schedule's law is autonomous: $\lambda_k$ and $\pi_k$ consult the gate count and nothing else, so the schedule may be solved first, without reference to the driver. The leak then carries the schedule's verdict into the driver through $\gamma_{k_t}$, and the model runs from the state that leaves. Schedule, then driver, then model, with no arrow back.

That one-directionality is what a state-dependent schedule threatens, and it threatens it in only one of the two ways such dependence can be read. Let $\pi$ depend on the schedule's own *configuration*, which gates are closed, in what order, with what debt outstanding, and the schedule remains autonomous, since configuration is generated by the schedule itself. It becomes a multitype branching process, with types the configurations and criticality governed by the top Lyapunov exponent of products of mean matrices rather than by the scalar criterion of §5.7.10, and the hierarchy survives intact. Let $\pi$ instead depend on the *driver*, on $x_t$ or $\mu_t$, and the schedule consults what it was supposed to precede. The hierarchy collapses, (5.42) becomes a genuine control problem, and the optimal history must trade action spent early against exposure shortened.

The baton-passing picture that motivates state dependence is of the first kind: what an event does depends on the state its predecessor left, which is configuration. So the version the natural-historical reading requires costs a multitype upgrade and not the control problem. Whether the driver *also* influences gate outcomes is a modeling question about the world rather than a mathematical one, and the answer adopted here is that it does not. Gate outcomes are stochastic in the direct sense: what makes an option the realized one is a fact about the onward schedule it opens, not about the credit held when it fires. The hierarchy therefore holds, (5.42) does not become a control problem, and the multitype upgrade is the whole cost of state dependence.

One refinement that adoption invites is recorded rather than pursued. If an option is realized in virtue of the ease of the gates it opens, then correctness is not a label attached to one of the $m$ branches but a *derived* property of the subtree below it — which makes the schedule a tree carrying a value function rather than a sequence with a marked option, and makes the realized path something survivorship picks out rather than something given in advance. That is consistent with Definition 5.48, whose independence claim concerns an option's *own* rate and not its descendants'. Whether the tree formulation reproduces the sequence one at the level of (5.36) is open [**O**].

*Remark* 5.64 (Only the asymptote selects the schedule). The channel prices what climbing costs. Whether anything *selects* for climbing is a separate question, and the answer separates the two conditioning events this work uses.

*Finite horizons do not select it.* Condition on $\{T > t\}$ for any finite $t$. A biosphere that never completes its schedule sits at bare background by Proposition 5.52 and survives to $t$ with probability $e^{-Bt} > 0$. The un-climbed branches therefore carry positive weight under every finite-horizon conditioning, and the lucky route, survive by not being struck, and never pay the schedule at all, remains in the support throughout.

*The asymptote does.* Condition instead on the viability event $\mathcal{I} = \{\Lambda_\infty < \infty\}$ of Definition 5.5. By Proposition 5.52, $\mathbb{P}\big(\mathcal{I} \mid \text{schedule incomplete}\big) = 0$ *exactly* — not small. The exclusion is at the level of support rather than of weight, and it is the only conditioning in this work that achieves it. So the schedule is selected for by the asymptotic condition and by nothing weaker. The crossover is worth seeing quantitatively, because it is where the balance actually sits. Survival at bare background is $e^{-Bt}$; survival once completed and critical is $\asymp Ct^{-1/4}$ by Claim 5.30. A polynomial beats an exponential eventually and always, so completion wins in the end for every $C > 0$; but the crossover $T_*$ solving $e^{-BT} = CT^{-1/4}$ recedes as $C$ falls, and $C$ is set by the state completion leaves (Remark 5.65). A biosphere completing into a poor state finds the schedule worth having climbed only at horizons that run away toward the asymptote.

The reading is the one that matters for what this channel is for. *Nothing about surviving a finite stretch gives a reason to climb.* Only conditioning on there being no last moment makes the climb pay — which is why the schedule appears in the record of a biosphere whose history was conditioned, and would not appear in the record of one that had merely lasted.

*Remark* 5.65 (The schedule supplies the initial data the model assumes). (5.40) makes an identification that should be stated plainly: *the completion state is the initial datum* that Definition 5.2 takes as given. The model begins where the schedule ends, and the schedule delivers not a point but a distribution over points — shaped by how long the debt took to clear and how far $x$ decayed while it did.

By Proposition 5.23 the initial data are gauge: they act through prefactors and timescales, never through exponents. So the schedule cannot move the persistence exponent $\frac{1}{4}$, cannot move the trichotomy, and cannot move the two-condition theorem. What it sets is the prefactor — and by Theorem 5.29 the prefactor is proportional to the persistence-harmonic function $\mathfrak{h}$ of the starting state. That is the same $\mathfrak{h}$ the Golden Score computes (Proposition 9.2), and §9.3 records as its principal cost that no map exists from a decision to a point in $(\mu, D)$. The schedule produces exactly such points. Whether decisions bearing on a biosphere's schedule, on $\lambda$, on $\pi$, on debt incurred, thereby acquire computable harmonic values is the first place a candidate identification map might be found. We flag it as a lead and claim nothing **[O]**.

*Remark* 5.66 (The variational form, stated and not solved). The second factor of (5.40) is a conditioned path problem, and Remark 5.45 already names the tool: the adverse-drift rate and its front-load-then-coast profile come from the Cameron–Martin action by Cauchy–Schwarz, the model's analog of an Euler–Lagrange computation and the natural processing step wherever a conditioned path law is wanted at large-deviation scale.

The composite asks for that computation across both stages. One seeks the path minimizing

$$S[\mu] \;=\; \int_0^\infty \left[\frac{\dot{\mu}_t^2}{2\sigma^2} \;+\; h\big(x_t, k_t\big)\right] dt \tag{5.42}$$

subject to (5.29), to the debt dynamics (5.32), and to $\Lambda_\infty < \infty$: a least-action path through a landscape whose running cost is the hazard and whose constraint is the schedule. The pathfinding reading is exact rather than figurative — the action is a length and the optimal history a geodesic in it.

Two features of the answer are predictable without solving it, and both are checks on a solution's behavior. The running cost during incompleteness is $B$ and is *independent of the driver*, since $x$ is not held and $\varsigma_b(0) = 1$; so the schedule stage contributes a term the driver cannot optimize against, and the variational freedom lives entirely after $\tau$. And because (5.40) factorizes, the optimal post-completion profile is the existing one, started from the distribution Remark 5.65 describes rather than from a chosen point.

Whether the stages remain separable once $\lambda$ or $\pi$ depends on the state, which the baton-passing picture requires, is the substantive question and is open **[O]**. If they do not, (5.42) becomes a genuine control problem rather than a factorized one, and the optimal history would trade early action against schedule exposure in a way the present separation forbids.

### 5.7.9 The internal channel is probably a schedule too

One structural claim precedes the open list, because it is the channel's clearest suggestion about the rest of the model and because it is an inference the Continuity Principle licenses rather than a speculation laid beside it. Principle 4 holds that hazard-lowering across mediations is one activity and not a family of analogs. The schedule represents pre-onset hazard as gates. If continuity is taken at the width Remark 3.19 states it, then post-onset internal hazard should be representable the same way, and the trough's smooth exponential should be a coarse-graining of gates not yet identified.

*Remark* 5.67 (The trough's exponential is geometric compounding; **[S]**). Take $h^{\text{int}}$ proportional to the count of *unpatched failure modes* carried at capacity $x$. Suppose each increment of capacity both creates modes and passes gates that patch them, and write $q$ for the fraction left unpatched per unit of capacity. Then $\text{unpatched}(x) = \text{unpatched}(0)\, q^{\,x}$, so

$$h^{\text{int}}(x) \;=\; \Gamma q^{\,x} \;=\; \Gamma e^{-\rho x}, \qquad \rho = \ln(1/q). \tag{5.43}$$

So geometric compounding of per-gate patch rates *would* produce the exponential of (5.7), with $\Gamma$ the count at $x = 0$. The construction is offered at that strength and no more: it exhibits a mechanism yielding the observed form, not evidence that this is the mechanism. (5.7) remains the paper's statement of the internal channel, nothing downstream is rewritten in gate coordinates, and the construction's use is heuristic — it says what one would look for, and becomes a derivation only if the gates are named.

Three things follow, and the second is the reason to record this. *The anatomy trichotomy becomes a compounding test.* Proposition 5.35 divides on the sign of $\rho$; under (5.43) that is $q$ against 1 — patching outpacing creation, matching it, or falling behind. *And it is then the same test the schedule already runs.* Remark 5.56 calls $\varkappa = 1$ the anatomy threshold's sibling. Under this construction they are not siblings but one test applied twice: both ask whether a compounding factor exceeds unity, and they differ only in what indexes the compounding — gate count for the schedule's debt, capacity for the trough's unpatched modes. That is what continuity predicts, and finding it is a check on the principle rather than a use of it.

*The construction agrees with the one the paper already has.* §6.6.1 derives $\rho = -(\alpha - \beta)$ from boundary scaling — an extractive domain growing as $e^{\alpha x}$ against enforcement growing as $e^{\beta x}$. Under (5.43) that is $q = e^{\alpha-\beta}$, so $q > 1$ exactly when $\alpha > \beta$, and the two microfoundations return the same condition by different routes. Neither is derived from the other.

What is missing is the thing the reformulation is for: *which events*. The schedule channel names its gates as natural-historical transitions, and the internal channel's gates would be whatever a biosphere must get right for acquired capacity not to be turned inward — institutional, presumably, and §11.6 is where a candidate list would come from. Until they are named, the smooth rate is the correct representation, and the reason needs precise statement: a smooth hazard is the *annealed* description of an unidentified jump process, so by the discipline of Remark 5.63 it flatters the typical history. An observer using (5.7)'s trough to reason about their own biosphere is using a mean carried by branches that patched luckily **[O]**.

### 5.7.10 What is open

*Depth-dependent debt.* $r$ and $\pi$ are constant, so all failures cost alike. Two quantities must be kept apart before that is repaired, since they are easily run together and only one of them is ordered by position. *Stakes*, what failing forfeits, rise monotonically along the chain, by Remark 10.5: a late step carries the improbability of every step behind it. *Debt*, what recovering costs, does not. An early gate may carry enormous $r$: a biosphere without its moon, or with one a third the size, owes a compensating sequence for axial stabilization and tides at the *first* gate rather than the last, and it is not obvious that any finite sequence discharges it. Severity is not ordered by position, and a model that assumed it was would misplace the worst gates.

Repairing this makes $Z$ a branching process in a varying environment, $\varkappa_k = r_k(1 - \pi_k)$ a sequence rather than a number, and the criterion a condition on *products* rather than on any single term: writing $S_k = \prod_{j<k} \varkappa_j$, completion is almost sure when $\sum_k S_k^{-1} = \infty$ and has positive failure probability when the sum converges, under a uniform second-moment condition on the offspring laws. For constant $\varkappa$ this recovers $\varkappa \leq 1$. The reading is worth having: one catastrophic gate does not doom a schedule whose remaining gates are benign enough, since it is the geometric mean of the $\varkappa_k$ that decides, a single spike is survivable in a way a raised floor is not, with the caveat that under Definition 5.48 the environment is random rather than merely varying, so the operative criterion is (5.34) and the sum below describes the deterministic case. The question of whether a deeper drop costs more *per unit of debt* is answered there rather than here, and answered negatively: the rates are redrawn at each gate and correctness is independent of them, so severity is neither constant nor monotone in depth but resampled, and it is the distribution rather than any trend that carries the structure. The same harmonic sum gives the rate and not merely the dichotomy: two-sided bounds of Agresti type put the probability of not having completed by stage $n$ comparable to $\left(\sum_{k \leq n} S_k^{-1}\right)^{-1}$, up to the second-moment factor, so

the quantitative form comes free with the criterion and is what a bridge to Proposition 5.51's timing statements would consume [**O**].

*State-dependent gates.* $\lambda(\cdot)$ and $\pi(\cdot)$ are what the baton-passing picture requires and what Remark 5.66 identifies as deciding whether the composite factorizes. *The leak is assigned rather than emergent*, and the incomplete regime's asymptotics remain unknown (Remark 5.50). *And the parameters are unmeasured.* $B$, $\lambda$, $\pi$, $r$ and $n$ are what the companion's serial-attribute program (Remark 3.37) would supply — and (5.38) makes that program's task easier rather than harder, since $c_{\text{eff}}$ need no longer be estimated directly but follows from three quantities a historian might plausibly bound. Remark 5.34's reservation applies with full force throughout: a scored schedule index would be a measurement apparatus optimized against, which is the failure the channel exists to price.

### 5.7.11 The modern-era term, formalized

The three passages that follow this subsubsection carry the schedule's modern-era argument in prose — the live-term hypothesis, the identification of accordance-legible golden causality with the term's operation, and the assignment of the term's last leg to the technoanatomy — and a reader is owed the formal content underneath them before the prose is read. The argument of the subsection as a whole then runs in one line: the schedule is a sequence of gated clocks (Definition 5.47) with log-spread rates (Definition 5.48) whose incompleteness leaks the credit (Definition 5.49) and gates the external channel (Proposition 5.52); the term's contribution to the hazard of Departure is the derailment hazard defined next; the hypothesis that the schedule is not yet exhausted is entered on evidence rather than principle; under it the remaining gates act on the anatomy rather than the credit, with a settlement law the conditioning front-loads; and none of this disturbs the ceiling, whose residual the last passage classifies.

**Definition 5.68** (Derailment hazard, and anatomy-directed gates; [**S**])**.** With the schedule in the state $k_t$ of Definition 5.47, the *derailment hazard* is the instantaneous rate at which any wrong option at the open gate fires first,

$$\sigma_t \;:=\; \lambda_{k_t}\bigl(1-\pi_{k_t}\bigr), \tag{5.44}$$

and it is the schedule's contribution to the hazard *of the Departure Point* in the implication sense of Remark 5.6: a derailment is a Departure-sufficient occurrence carrying the Aethic weight of the derailed branch, which under the extra-successes reading adds gates to the schedule rather than closing it outright. The term is high exactly when a gate with a large total rate and a small favorable share stands open, and that is the formal content of the phrase "high stakes toward landing the next Miracle." Gates $k \le n$ are *credit-directed*: their closure lowers the leak of (5.30). Gates $k > n$, if any, are *anatomy-directed*: their resolution does not act on the credit but on the anatomy, the option $j$ that fires at gate $k$ contributing an increment $\Delta_k^{(j)}$ with $\Delta_k^{(1)} > 0$ for the realized-favorable option and the wrong options adverse, so that

$$\rho_t \;=\; \rho_0 + \sum_{n<k\le k_t} \Delta_k^{(j_k)}, \qquad \rho_\infty \;=\; \lim_{t\to\infty} \rho_t, \tag{5.45}$$

$j_k$ being the option that fired at gate $k$. The two-condition theorem reads on $\rho_\infty$.

**Proposition 5.69** (Front-loaded settlement; sketch at [**P**])**.** *Under Definition 5.68 and the live-term hypothesis, conditioning on survival tilts every anatomy-directed gate toward its favorable* *option and tilts the times of favorable resolutions earlier, so that the conditional law of $\rho_\infty$ stochastically dominates the unconditioned law and the conditional law of the crossing time to $\rho > 0$ is stochastically earlier, both in the likelihood-ratio order.*

*Sketch.* Two paths identical except at one anatomy-directed gate differ in accumulated hazard by the integral of $\Gamma(e^{-\rho^{\mathrm{adv}} x_t} - e^{-\rho^{\mathrm{fav}} x_t}) > 0$ over the time the two anatomies differ, so the favorable path's survival weight exceeds the adverse path's by the exponential of that integral; conditioning on survival therefore reweights each gate's resolution by a factor increasing in the favorability of its increment, and each resolution time by a factor decreasing in the time, which is the likelihood-ratio tilt stated. The one-gate argument iterates over the gates since the tilts compound multiplicatively along a path. The exact conditions, including the finiteness of hazard accumulated before crossing and the leverage factor $x_t$ that makes the tilt vanish while the credit is small, are the window theorem's debt (§5.7.11). $\square$

*Remark* 5.70 (The third term is not exhausted; **[S]** with imports at [$\Phi$]). The working assumption until this remark was that the schedule channel belonged mostly to the record: the evolutionary ladder climbed, the term's modern residue mergeable into the internal and external channels. The correction the author supplies is more careful than a reversal of that assumption, and it should be stated at its exact strength. The continuity principle does not decide whether the present lies before or after the last prerequisite; it is silent on whether one should or should not have been born after the last major Miracle. What it decides is conditional: *if* prerequisites remain, their operation runs continuously through the present, since nothing in the conditioning marks the modern day as a privileged moment at which the schedule closed. The live third term is therefore a hypothesis, not a default that continuity licenses, and the prior practice's error was symmetrical: it treated exhaustion as the default, which continuity licenses no more than it licenses liveness. What the hypothesis has in its favor is not a principle but evidence, and the next paragraph says which.

The author's central move is then an identification, and it is what supplies the evidence just promised. What the accordance principle under Golden conditioning renders legible as golden causality in the modern record, the specific accordance-yielding attributes physically observable as they occur, is not a separate phenomenon that happens to be visible; it is the third term operating. The standing conviction that golden causality is more visible than the internal and external channels would suggest is on this reading not a resolution artifact of the attribute level (Remark 5.82); it is channel identity, accordance-legible concentrations being schedule steps landing. This sharpens that remark's open question into an identification carried at **[S]**, and it carries a scoreable texture, entered here with the conditions of its failure: if the identification holds, the rarity audit should find modern concentrations clustering on prerequisite-shaped structures rather than spreading uniformly over the record, and a uniform spread would count against it.

The class of events in question is the sequential-achievement class [73], and its citable historical exemplar keeps the discussion honest about magnitudes: Newton's occurrence when he occurred, the quantile of intellectual horsepower struck at that point of the process, was an event of low probability relative to the alternatives, individually improbable and collectively required, which is the signature of a schedule step. The general claim leans on the class, and on the expectation that most of its modern members have not yet been identified; no single member is load-bearing.

One candidate modern member should be named exactly once, because naming it is the only way to fence it. The production of the present framework by its date is itself an event whose probability was low relative to the alternatives, given how many things had to go right to lead to it, and it is therefore a formally admissible candidate instance of the class. It is entered here as hypothetical only and expressly not load-bearing: the fluke hypothesis may simply be true of it, and the occupancy fences of Remark 9.12 bar the reflexive claim in any case. The class carries the argument, and no member of it does.

None of the riskiness a live third term adds changes the conditional picture, and this is the author's retention clause. The forward forcing of Remark 5.39 reads on this channel too: under conditioning at the asymptotic future, in character much like a stationary-action principle, the conditional law concentrates on schedules that complete, because reaching the Golden Point outweighs the third term's hazard once the conditioning is asymptotic. The prerequisites still standing are, under that conditioning, nailed with conditional probability approaching one, which retains in the forward direction exactly what the Optimation climb established in the backward one.

**The last leg falls on the anatomy (design at [S], one sketch at [P], debt at [O]).** The reason the third term was tangled with the internal channel in the first place deserves to be named, because it was a formal problem and not a confusion. An active schedule registers the technostrength $x$ as incomplete, and an $x$ still being established undercuts the internal term's own analysis, which reads its muffling against a mature credit; a live third term therefore appeared to break the very channel it feeds, and merging the term's modern presence into the other two was the workaround. The preceding passage removes the workaround without removing the problem, since on the live-term hypothesis the term operates now, and given that hypothesis the resolution is forced rather than merely permitted: the $x$-directed component of the schedule must already sit in its exponential-tail regime by the modern day, with $x$ robust, so the term's remaining incompleteness before the Golden Point has to act somewhere other than the credit.

The author's proposal is that it acts on the technoanatomy directly, and the proposal is best stated together with a fact about $\rho$ that the record disguises. The technoanatomy is movable at every time with equal ease; what varies across history is not its mobility but its *leverage*, which scales with the credit, since the internal term $\Gamma e^{-\rho x}$ is insensitive to $\rho$ when $x$ is small and grows in sensitivity as $x$ grows, $\partial h/\partial\rho$ carrying a factor of $x$. In the deep past the anatomy therefore appeared frozen because the technopower that would have exposed it was low, and the honest description of the premodern record is that the species rode on a technoanatomy inherited from the Stone Age without its value being tested, each culture carrying a version of its own, the versions merging into a single biosphere-level sign only when the industrial era coupled them and raised $x$ to where the sign began to matter. The last leg of the schedule is then this: with leverage now high, the remaining pushes act on $\rho$ as a trajectory, and the interval between the present and the Golden Point is exactly the crapshoot of pulling this off, embodied in technoanatomy contributions, the doing of things right or the failing to. The modern-day constraint that motivated the old merge is satisfied rather than violated: the internal hazard remains high on the whole historical timeline's present because the leaning of $\rho$, now that it counts, remains to be settled, which is consonant with the scaling observation of §6.4.1, a present-configuration scale-up collapsing, worse than ideal now yet with significant potential for improvement.

What "settled" means under this design is fixed by one sketch, stated at **[P]** because it is an easy likelihood-ratio statement once the model is in hand. Let $\tau$ be the time at which $\rho$ crosses to its favorable side along a path, and compare two paths identical except that one crosses at $\tau_1$ and the other at $\tau_2 > \tau_1$. The earlier crosser accumulates less hazard by $\Gamma\int_{\tau_1}^{\tau_2}\left(e^{-\rho_{\mathrm{adv}}x_t} - e^{-\rho_{\mathrm{fav}}x_t}\right)dt > 0$, so its survival weight exceeds the later crosser's by the exponential of that quantity, and conditioning on survival reweights the unconditioned law of $\tau$ by a factor decreasing in $\tau$: the conditional law of the crossing time is stochastically earlier than the unconditioned one, in the likelihood-ratio order. This is the *survivorship prefers early change* statement, and it has a corollary that should be read carefully because the record tempts a misreading of it. From inside any surviving history that has not yet crossed, the conditional hazard of crossing is elevated, so every present looks like the moment at which the shot must be taken. That appearance is real as a statement about the conditioned law and misleading as a statement about capacity: the species can move $\rho$ just as much later as now, and what changes with delay is only that fewer surviving histories have it happening late, more survivors sprouting from the histories in which hazard was lowered earlier. The feeling that our shot is now is therefore a corollary of the preference for early change, present at every vantage the conditioning produces, and not evidence that the present vantage is special; this is the mechanism behind the refusal recorded at Remark 9.12, which forbade the occupant claim before its cause was known. Settlement, accordingly, is not a freezing of $\rho$ but the point past which the conditional mass has largely moved, and $\rho_\infty$ is the limit along surviving paths of a quantity that never stops being movable. The formal statement of both the process and its front-loaded settlement is Definition 5.68 and Proposition 5.69.

The design wires into the standing structure at four points. The two-condition theorem reads on the settled value: viability becomes the indicator of $\rho_\infty > 0$ times the escape factor with $\rho_\infty$ random, so the modern era is the era in which the indicator's argument is being determined, by pushes that could in principle have come at any time and are, under conditioning, front loaded. The fluctuating-anatomy corollary (Corollary 5.43) is the bridge case, fluctuation before settlement. The preceding passage's identification gains its mechanism, the accordance-legible steps being the pushes. And the forward forcing extends to the settlement law: under asymptotic conditioning the law of $\rho_\infty$ concentrates on the favorable side, for the reason Remark 5.39 states, and the preference for early change is that concentration's temporal profile.

The formal debt this passage enters, at **[O]**, is the $\rho$-dynamics window theorem it presupposes, now specified by the sketch above: the leverage statement made exact, conditions on the crossing-time law and on the pre-crossing fluctuation under which the frozen-$\rho$ analysis survives as the limit along surviving paths, and the hazard accumulated before crossing remaining finite under stated tail conditions so that the theorem's conditional structure reads on $\rho_\infty$ unchanged. This is the best well-idealised model the author can presently state, and it is entered as design rather than result.

*Remark* 5.71 (The third term and the ceiling; **[S]**, with the theorem untouched)*.* The natural next question, and the one a referee should ask, is whether a live and observable third term breaches the epistemic ceiling, since watching schedule steps land successfully amplifies credence that the asymptotic conditioning is the explanation, and under the conditioning one does well on all three terms by deduction. The answer is that the theorem is untouched while the texture changes, because the ceiling bounds certification and never bounded credence. Locally the two measures are equivalent, so every finite audit outcome has positive probability under both and no bounded observation decides the matter; but the likelihood ratio between them is free to climb without limit, and the amplification the question describes is exactly that climb. Indeed the deduction is the singularity read from the evidence side: the conditioned law predicts systematic schedule success where the unconditioned law predicts it rarely, each audited success multiplies the ratio, the divergence of the accumulated evidence is the mutual singularity at infinity, and its unreachability at finite times is the ceiling. The live third term does not breach the structure; it populates it, supplying the concrete observable carrier along which the permitted climb occurs.

What the term genuinely adds is evidence that survives the mere-survival discount. Survival to the present is blunted by observer conditioning, as the fence on Remark 5.39 records, but the manner of the successes is finer-grained than survival: their prerequisite shape, their accordance structure, and the technological-condition specificity to which observer-conditioning is provably indifferent, which is where the comparative force of Proposition 3.25 lives after its restatement. The schedule channel is therefore the channel in which admissible evidence accumulates, the place where the deflation "we would observe this regardless" fails. The discipline is retained in full: amplification proceeds only at the audit's rate, route-deletion established case by case against the retrospective-target null (Remark 3.27), the clustering texture of Remark 5.70 standing as the scoreable check, and "the only explanation" read throughout as "favored over the stated rivals," at **[S]**.

One consequence for the extended model is entered with the window theorem's debt rather than asserted. If settlement occurs at a finite time and the settled value were observable, an adverse settlement would be a finite-time event of positive probability under the unconditioned law and probability zero under the conditioned one, so the extended model may permit certification in the uncontrolled direction while the consoling direction remains uncertifiable forever, preserving and sharpening the one-sidedness that Remark 6.19 already records. Two provisos attach: local equivalence before settlement is to be verified for the extended model, and $\rho_\infty$ is a structural parameter whose direct observability is itself an idealization. Nothing here is consumed by the ceiling's standing dependents; the dependency runs toward the window theorem.

*Remark* 5.72 (The extreme-induction regime; **[S]**, regime-conditional). The preceding remark describes a climb; this one classifies its asymptote. Suppose the skew of Aethic weight toward the Route World continues through enough Aethic time that the Route World comes to hold arbitrarily close to the whole of the weight. The ceiling never closes, by the theorem, and the question worth a remark is what the residual gap has become by then. The author's answer is that it converges categorically on the classic problem of induction: the live doubt remaining is of the same class as the doubt that the sun does not rise tomorrow, given the establishing of one's lifetime as the induction base [74]. This is offered as class identity rather than analogy, since the remaining ways of being wrong are the Humean ways, and those attach equally to every piece of our most reliable ontological understanding. The phenomenology follows: human inference auto-interprets questions of that class as closed, because everything we treat as known already carries exactly that residual, so the ceiling's line, though never crossed, comes to feel crossed, and the feeling is not an error about the theorem but a correct classification of what the residual has become.

What the regime licenses is a parity claim, and what it does not license is certification, and the two must be kept apart at the usual price. Once the residual is Humean, refusing closure here while granting it to gravity is an isolated demand for rigor, and the framework may say so; the theorem meanwhile stands untouched, the determinacy and the access split exactly as Remark 6.19 left them. And the regime itself is conditional in a way the discipline will not waive: the skew must actually have accumulated, at the audit's rate, under the preceding remark's rules, so the classification is earned by the climb rather than assumed at its foot. The clause about human auto-interpretation is a psychology-of-inference observation and is entered as such, at

[**S**], alongside the rest.

## 5.8 Conditioning on survival

### 5.8.1 The interpolation is a monotone flow of upward tilts

**Definition 5.73** (Survivorship interpolation; bias field)**.** For $t > 0$ set $\mathbb{P}^{(t)} := \mathbb{P}(\,\cdot \mid T > t)$, and analogously $\mathbb{P}^{(t)}_\tau := \mathbb{P}(\,\cdot \mid \tau > t)$ for the hard-wall functional; $\mathbb{P}^{(0)} = \mathbb{P}$ is the unbiased law. For $0 \le s < t$ the *bias field* on the window $[0, s]$ is the Radon–Nikodym derivative

$$\beta_t(s) := \frac{d\mathbb{P}^{(t)}}{d\mathbb{P}}\Big|_{\mathcal{F}_s} = \mathbf{1}_{\{T>s\}}\,\frac{\mathbb{P}_{X_s}(T > t-s)}{\mathbb{P}_{X_0}(T > t)}, \qquad X_s = (\mu_s, D_s), \tag{5.46}$$

the second equality by the Markov property; likewise for $\mathbb{P}^{(t)}_\tau$ with $\tau$ in place of $T$. The interpolation at finite $t$ is thus globally absolutely continuous with respect to $\mathbb{P}$, with density $\mathbf{1}_{\{T>t\}}/\mathbb{P}(T > t)$ on $\mathcal{F}_\infty$.

**Lemma 5.74** (Survivorship tilts upward; [**P**])**.** *In the driftless model, couple two initial data $(\mu_0, D_0, h_0) \le (\mu_0', D_0', h_0)$ (componentwise in the first two arguments) synchronously, with the same $W$. Then pathwise $\mu_t' - \mu_t = \mu_0' - \mu_0 \ge 0$, $D_t' - D_t = (D_0' - D_0) + (\mu_0' - \mu_0)t \ge 0$, hence $h' \le h$, $\Lambda' \le \Lambda$, and both $\mathbb{P}_x(T > t)$ and $\mathbb{P}_x(\tau > t)$ are nondecreasing in $(\mu_0, D_0)$ (and nonincreasing in $h_0$). Consequently every bias field* (5.46) *is, on $\{T > s\}$, a nondecreasing function of the state $X_s$: at every conditioning horizon, survivorship bias is an upward tilt in rate and credit, and the interpolation is a monotone flow of such tilts.*

*Proof.* Immediate from the displayed pathwise inequalities and Definition 5.4. □

### 5.8.2 The critical limit is the pure Doob transform

**Theorem 5.75** (Existence and identification of the limit; [**C**], soft version [**A**])**.** *Assume the pointwise refined tail asymptotics $\mathbb{P}_{(\mu,d)}(\tau > t) \sim c\,\mathfrak{h}(\mu, d)\,t^{-1/4}$ of Theorem 5.29 and Proposition D.8, together with the martingale property $\mathbb{E}_x[\mathfrak{h}(X_s)\mathbf{1}_{\{\tau>s\}}] = \mathfrak{h}(x)$ furnished by [58]. Then for every $s \ge 0$ and every bounded $\mathcal{F}_s$-measurable $\Psi$,*

$$\mathbb{E}\big[\Psi \mid \tau > t\big] \;\longrightarrow\; \mathbb{E}_Q[\Psi], \qquad t \to \infty,$$

*where $Q$ is the time-homogeneous Markov law with*

$$\frac{dQ}{d\mathbb{P}}\Big|_{\mathcal{F}_s} = \mathbf{1}_{\{\tau>s\}}\,\frac{\mathfrak{h}(X_s)}{\mathfrak{h}(X_0)}\;: \tag{5.47}$$

*the* pure *Doob $\mathfrak{h}$-transform, at eigenvalue one. Under Ansatz 5.18 the same limit holds for $\mathbb{P}^{(t)} = \mathbb{P}(\,\cdot \mid T > t)$.*

*Proof.* Write $f_t := \mathbf{1}_{\{\tau>s\}}\mathbb{P}_{X_s}(\tau > t-s)/\mathbb{P}_{X_0}(\tau > t)$, so that $\mathbb{E}[\Psi \mid \tau > t] = \mathbb{E}[\Psi f_t]$. By the Markov property $\mathbb{E}[f_t] = 1$ for every $t$. Factorizing,

$$f_t = \mathbf{1}_{\{\tau>s\}}\,\frac{(t-s)^{1/4}\,\mathbb{P}_{X_s}(\tau > t-s)}{t^{1/4}\,\mathbb{P}_{X_0}(\tau > t)}\Big(1 - \frac{s}{t}\Big)^{-1/4} \;\longrightarrow\; f := \mathbf{1}_{\{\tau>s\}}\frac{\mathfrak{h}(X_s)}{\mathfrak{h}(X_0)} \quad \text{a.s.,}$$

using the assumed asymptotics at the (fixed, random) starting point $X_s$ and at $X_0$, and $(1 - s/t)^{-1/4} \to 1$. The martingale property gives $\mathbb{E}[f] = 1 = \lim \mathbb{E}[f_t]$, so by Scheffé's lemma $f_t \to f$ in $L^1(\mathbb{P})$, and $\mathbb{E}[\Psi f_t] \to \mathbb{E}[\Psi f] = \mathbb{E}_Q[\Psi]$ for bounded $\Psi$. That (5.47) defines a consistent, time-homogeneous Markov family is the standard theory of Doob transforms by invariant harmonic functions; time-homogeneity with no eigenvalue factor is possible precisely because $\mathfrak{h}$ is invariant (eigenvalue one) for the killed semigroup. □

*Remark* 5.76 (Why survivorship has no clock here; **[P]**). The mechanism deserves isolation. For *regularly varying* tails the kinematic factor $\mathbb{P}_x(\tau > t - s)/\mathbb{P}_x(\tau > t) \to 1$ for each fixed $s$: the age of the conditioning drops out at leading order, and the limit is an un-tilted $h$-transform. For *exponential* tails $\mathbb{P}_x(T > t) \approx C(x)e^{-\theta t}$ the same factor converges to $e^{\theta s}$, and survivorship acquires a clock — the eigen-tilting of mode (III) in Theorem 5.81. The polynomial class is exactly the class in which survivorship bias is, asymptotically, a change of measure with no time-dependence. In the taxonomy of penalizations of Wiener measure [59], the survival weight $e^{-\Lambda_t}$ thus lands in the clockless (martingale-normalizable) class precisely because its expectation is regularly varying.

**Proposition 5.77** (Interpolation rate; the tail index as linear-response coefficient; **P/C**)**.** *Under the hypotheses of Theorem 5.75, for fixed $s$ as $t \to \infty$,*

$$\beta_t(s) \;=\; \mathbf{1}_{\{\tau>s\}}\, \frac{\mathfrak{h}(X_s)}{\mathfrak{h}(X_0)} \Big(1 - \frac{s}{t}\Big)^{-1/4} \big(1 + o(1)\big), \qquad \Big(1 - \frac{s}{t}\Big)^{-1/4} = 1 + \frac{s}{4t} + O\Big(\frac{s^2}{t^2}\Big). \tag{5.48}$$

*The natural interpolation parameter is therefore $s/t$, and the persistence exponent $\frac{1}{4}$ is the linear-response coefficient of survivorship bias: the leading finite-$t$ correction to the survivor's universe is $s/4t$ per window of length $s$. Approach to the limit is polynomial — the critical signature, contrasting with the exponential approach of mode (I) and the exponentially growing tilt of mode (III) below.*

*Proof.* Insert the assumed asymptotics into (5.46) and expand. □

### 5.8.3 The limit object: survivorship drift and promotion to transience

**Theorem 5.78** (Survivorship drift; **[P]** given **[C]** inputs)**.** *Under $Q$ the state solves*

$$d\mu_t = \sigma^2\, \partial_\mu \log \mathfrak{h}(\mu_t, D_t)\, dt + \sigma\, dW_t^Q, \qquad dD_t = \mu_t\, dt, \tag{5.49}$$

*for a $Q$-Brownian motion $W^Q$: conditioning adds drift only in the noisy coordinate. With the representation* (D.6)*,*

$$b_Q(\mu, D) = \frac{\sigma^2}{\mu}\, G(\kappa), \qquad G(\kappa) := \frac{1}{2} - 3\kappa\, \frac{F'(\kappa)}{F(\kappa)}, \tag{5.50}$$

*and:* (i) $b_Q \geq 0$ *everywhere (survivorship never pushes the rate down), by Lemma 5.74 applied to $\mathfrak{h}$;* (ii) $G(0^+) = \frac{1}{2}$*, since $\mathfrak{h}(\mu, 0^+) = c_1' \mu^{1/2}$ by Theorem 5.29;* (iii) $G(\kappa) \to 0$ *as $\kappa \to \infty$, since $\mathfrak{h}(0^+, d) = c_2 d^{1/6}$ is independent of $\mu$ at leading order. Statements (ii)–(iii) use the regularity of the profile $F$ furnished by the explicit formulas of [58].*

*Proof.* For invariant harmonic $\mathfrak{h}$, the transformed generator is

$$\mathcal{L}^{\mathfrak{h}} f = \mathfrak{h}^{-1} \mathcal{L}(\mathfrak{h} f) = \mathcal{L} f + \sigma^2 \big(\partial_\mu \log \mathfrak{h}\big)\, \partial_\mu f,$$

since only $\partial_\mu^2$ is second order in (5.3); this is (5.49). Formula (5.50) is the logarithmic derivative of (D.6) using $\partial_\mu \kappa = -3\kappa/\mu$. Positivity: synchronous coupling gives $\mathbb{P}_{(\mu,d)}(\tau > t)$ nondecreasing in $\mu$, hence $\partial_\mu \mathfrak{h} \geq 0$ wherever the derivative exists. The boundary values follow from the quoted prefactor laws: $F(0) = c_1' \in (0, \infty)$ and $\kappa F'(\kappa)/F(\kappa) \to 0$ as $\kappa \to 0$ by smoothness of $F$ at the origin; as $\kappa \to \infty$, $\mathfrak{h} \sim c_2 d^{1/6}$ forces $\mu\,\partial_\mu \log \mathfrak{h} \to 0$. $\square$

**Corollary 5.79** (Wall drift; promotion to transience; **P/C**). *Near zero credit ($\kappa \to 0^+$) the survivorship drift reduces to $\sigma^2/2\mu$, so that in this regime the conditioned rate solves*

$$d\mu_t = \frac{\sigma^2}{2\mu_t}\,dt + \sigma\,dW_t^Q.$$

*Two cautions attach to this formula.* (i) *This is the Bessel drift of index $\delta = 2$, not $\delta = 3$: the exponent $\frac{1}{2}$ in $\mathfrak{h}(\mu, 0^+) \propto \mu^{1/2}$ is half the exponent* 1 *of the classical harmonic function $x$ for Brownian motion killed at the origin, the halving being exactly that of the persistence exponent from $\frac{1}{2}$ to $\frac{1}{4}$ under one integration.* (ii) *Under $Q$ the coordinate $\mu$ is not autonomous, by* (5.50) *its drift depends on $\kappa$, hence on both coordinates, so the comparison is a statement about the* form *of the drift in a limiting regime, not an identification of processes.*

*The correct Pólya statement is therefore not a Bessel index but a recurrence class, and at that level it is unconditional: under $\mathbb{P}$ the credit is null recurrent, returning to its initial level a.s. (Theorem 5.26) with infinite expected return time (Corollary 5.31), whereas under $Q$ it is transient, $D_s \to \infty$ a.s. (Theorem 5.80). Conditioning on survival promotes the credit from the recurrent to the transient class, the Pólya* 2D $\to$ 3D *transition, with no literal change of dimension anywhere. Dimension remains, as in §A.2, the wrong order parameter for the state dynamics; what survivorship alters is the recurrence class of the* ensemble.

**Theorem 5.80** (Locally ours, globally elsewhere; **P/C**). (i) *$Q(\tau = \infty) = 1$.* (ii) *$\mathfrak{h}(X_s) \to \infty$ $Q$-a.s. as $s \to \infty$; moreover $D_s \to \infty$ $Q$-a.s. [58].* (iii) $\liminf_s D_s = -\infty$ *$\mathbb{P}$-a.s.* (iv) *Hence $Q|_{\mathcal{F}_s} \ll \mathbb{P}|_{\mathcal{F}_s}$ for every finite $s$, with the explicit density* (5.47), *while $Q \perp \mathbb{P}$ on $\mathcal{F}_\infty$: the survivor's universe is locally absolutely continuous with respect to ours on survivors, and globally disjoint from it. The exit from the measure class is completed only at $t = \infty$; at every finite conditioning the bias is the bounded tilt of Definition 5.73.*

*Proof.* (i) $Q(\tau > s) = \mathbb{E}_{\mathbb{P}}[\mathbf{1}_{\{\tau > s\}}\mathfrak{h}(X_s)]/\mathfrak{h}(X_0) = 1$ for every $s$ by the martingale property. (ii) Under $Q$, $1/\mathfrak{h}(X_s)$ is a nonnegative supermartingale: $\mathbb{E}_Q[1/\mathfrak{h}(X_{s+r}) \mid \mathcal{F}_s] = \mathbb{P}_{X_s}(\tau > r)/\mathfrak{h}(X_s) \leq 1/\mathfrak{h}(X_s)$; it therefore converges $Q$-a.s., and since $\mathbb{E}_Q[1/\mathfrak{h}(X_s)] = \mathbb{P}_{X_0}(\tau > s)/\mathfrak{h}(X_0) \to 0$, Fatou forces the a.s. limit to vanish, i.e. $\mathfrak{h}(X_s) \to \infty$; the sharper statement $D_s \to \infty$ is part of the description of the conditioned process in [58]. (iii) is contained in the proof of Theorem 5.26 (the set $A$ there is unbounded, and on it $D_u \leq D_0 + \mu_0 u - \sigma\varepsilon u^{3/2} \to -\infty$). (iv) Local absolute continuity is (5.47); mutual singularity on $\mathcal{F}_\infty$ follows since $\{\lim_s D_s = +\infty\}$ has $Q$-probability one and $\mathbb{P}$-probability zero. $\square$

### 5.8.4 The three modes of survivorship

**Theorem 5.81** (Modes of the interpolation; **P/C/S**). *As $t \to \infty$ the survivorship interpolation exhibits exactly three asymptotic modes across the driver classes of §A:*

(I) Saturation ***[P]*** *(favorable drivers, $\mathbb{P}(T = \infty) > 0$): since $\{T > t\} \downarrow \{T = \infty\}$,*

$$\left\|\mathbb{P}^{(t)} - \mathbb{P}(\,\cdot \mid T = \infty)\right\|_{\mathrm{TV}} \;\leq\; \frac{2\,\mathbb{P}(t < T < \infty)}{\mathbb{P}(T > t)} \;\longrightarrow\; 0 \quad \textit{on } \mathcal{F}_\infty,$$

*and every bias is bounded by $1/\mathbb{P}(T = \infty)$: survivorship saturates at a bounded, globally absolutely continuous reweighting. No new universe.*

(II) Transformation ***[C/A]*** *(critical drivers; the driftless model): the limit is the pure Doob transform of Theorem 5.75, locally absolutely continuous and globally singular (Theorem 5.80), approached at the polynomial rate of Proposition 5.77. The new universe is entered, but only at infinity.*

(III) Concentration ***[S]*** *(adverse drivers, exponential tails; e.g. Proposition A.10): on fixed windows the limit is the eigen-tilted transform of Remark 5.76 with clock $\theta = 3\nu^2/8\sigma^2$, while at macroscopic scale the conditioned path law collapses onto the deterministic optimal profile of Proposition A.10(iii): $\sup_{v\in[0,1]}\left|\mu_{tv}/t - \frac{|\nu|}{4}v(2-3v)\right| \to 0$ in $\mathbb{P}^{(t)}$-probability. Survivorship becomes a law of large numbers for the rare: the survivor ensemble is asymptotically deterministic at leading order.*

*The three modes are in bijection with the horizon trichotomy (Theorem A.5, Proposition A.6), and they constitute a strictly finer invariant: modes (II) and (III) share $\mathbb{P}(\mathcal{I}) = 0$ but are separated by the interpolation — by whether the survivor's universe is approached polynomially, without a clock, or exponentially, with one.*

*Proof.* (I) is continuity of measure from above plus the displayed total-variation estimate, whose numerator is $2\mathbb{P}(\{t < T < \infty\})$ and whose denominator is bounded below by $\mathbb{P}(T = \infty) > 0$. (II) collects Theorems 5.75 and 5.80 and Proposition 5.77. (III) is the large-deviation sketch of Proposition A.10(iii) read under the conditional measure, together with Remark 5.76; elevating it to a full proof requires the sharp prefactor asymptotics $\mathbb{P}_x(T > t) \sim C(x)e^{-\theta t}$, which we do not establish here. □

**The verdict.** The section's question can now be answered in one paragraph. The survivorship interpolation $t \mapsto \mathbb{P}^{(t)}$ is a monotone flow of upward tilts (Lemma 5.74), globally absolutely continuous at every finite $t$, whose asymptotic character is the sharpest invariant of the system and takes exactly three forms (Theorem 5.81). The driftless model realizes the critical form: the limit exists, is the pure Doob transform by the persistence-harmonic function, carries no clock (Remark 5.76), is approached at rate $s/4t$ with the tail exponent as linear-response coefficient (Proposition 5.77), is locally equivalent on survivors to the unbiased law yet globally singular (Theorem 5.80), and differs in generator by the universal survivorship drift (5.50), equal to $\sigma^2/2\mu$ at the wall and carrying the credit from the recurrent to the transient class (Corollary 5.79). The gauge structure of §5.2 persists to the end: the initial data cancel from the limit density (5.47) up to the normalization $\mathfrak{h}(X_0)$, and the survivorship drift is universal. The event horizon absent from state space is therefore real, but it is a feature of the ensemble, not of the state: it is crossed by the conditioning parameter $t$, which is to say, by survival itself, and never by the system.

**The same refinement the companion performs, in the path measure.** One structural agreement with the companion is worth recording, since it was reached independently on both sides and neither development notes it. The companion derives classical survivorship bias as the Accordance Principle under *two coarsenings*: the observer tilt flattened from a graded weighting to a two-valued indicator, and that indicator read off whole histories rather than off single attributes [5, 24]. Restore the grading and relocate the bearer, and the classical bias refines into the principle.

The interpolation of this subsection performs the same two refinements in the path measure. The crude object is $\mathbf{1}_{\{T>t\}}/\mathbb{P}(T > t)$, an indicator, borne by the whole trajectory, and the limit object is $\mathfrak{h}(X_s)/\mathfrak{h}(X_0)$, a graded weight borne by the state at each time. So the tilt is regraded from indicator to continuum, exactly as the companion regrades it, and its bearer moves from the completed path to the local state, exactly as the companion moves it from the completed history to the attribute. What Theorem 5.75 proves is that the refinement is not a modelling choice here but the limit the crude object converges to, which is a stronger statement than the companion is in a position to make for its own case.

Two things follow, and the second is the reason to record it. Survivorship conditioning and observer conditioning are the same operation refined in the same two directions, in the static and dynamic registers respectively — so the frailty ancestry of §5's literature note and the companion's coarsening result are two views of one fact. And an agreement of this shape is evidence of the kind neither development can manufacture for itself: the refinements were derived on one side from a measure-theoretic limit and on the other from a typicality principle, with no common step, and they arrived at the same pair **[S]**.

### 5.9 Summary of the model section

The system is a Kolmogorov diffusion in $(\mu, D)$ with exponential killing; at exponent level it is the persistence problem for integrated Brownian motion. Initial data are gauge: they act only through an equivalent timescale $t_*$, and the one intrinsic coordinate is $\kappa = \sigma^2 D/\mu^3$ (§5.2). For the driftless model, viability is impossible for every initial condition (Theorem 5.26), yet life expectancy is infinite: $\mathbb{P}(T > t) \asymp (t_*/t)^{1/4}$, with Pareto($\frac{1}{4}$) residual life, quantiles growing as the fourth power of rarity, and an effective Lindy hazard $1/4t$ — null recurrence, the two-dimensional case of the Pólya analogy (§5.4). No horizon exists there: the crossover is a maximally soft quarter-power law smeared over orders of magnitude, and recovery cost is sublinear in credit. Where genuine horizons exist, the order parameter is the boundedness of the driver's scale function, the horizon profile *is* the normalized scale function (Theorem A.5), the recurrent class is decided by the stationary mean with mortal critical case (Proposition A.6), and the solvable unstable-OU driver realizes the profile exactly as a Gaussian sigmoid $\Phi(\mu_0\sqrt{2\nu}/\sigma)$ of width $\sigma/\sqrt{2\nu}$, a step function only in the small-noise limit (§A). Under adverse drift the survival tail is exponential with rate $3\nu^2/8\sigma^2$, carried by a front-load-then-coast optimal fluctuation. The destination is the survivorship interpolation (§5.8): a monotone flow of upward tilts $\mathbb{P}^{(t)}$ whose critical limit is the pure Doob transform by the persistence-harmonic function, approached without a clock at rate $s/4t$, the tail exponent as linear-response coefficient, locally absolutely continuous on survivors yet globally singular, and generated by the universal survivorship drift, equal to $\sigma^2/2\mu$ at the wall: conditioning carries the credit from the recurrent to the transient class, the Pólya promotion effected by survival rather than by dimension. Across drivers the interpolation has exactly three modes, saturation, transformation, concentration, in bijection with the horizon trichotomy and strictly finer than the survival dichotomy. A horizon is not a place. Where it exists in state space it is a ratio of scales; and where it does not, it is real nonetheless — located in the ensemble, crossed not by the state but by the conditioning parameter, which is to say by survival itself.

One channel stands outside that account and prefixes it. The schedule of §5.7 prices what must go right before the muffling exponent can be held at all: failure incurs debt rather than discounting a weight, the debt process is Galton–Watson with a criticality at mean offspring one, and the completion weight is the least fixed point of (5.36) — from which the companion's per-perturbation contrast emerges as $Br/\lambda \ln 10$ rather than being assumed. Its consequence for the rest is structural: an unheld $x$ leaves the crest at bare background, so the external channel, the smallest term in (5.7), is survivable only because the schedule completed, and the state at which it completes is the initial datum everything above takes as given.

*Remark* 5.82 (What the hazard rate is a projection of; [Φ] on the companions). The hazard rate this section models is not the ontology; it is a projection of it. The richer object is Aethus-valued attribute structure, and the relation is exactly the one the companion program exhibits when exotic structure over Aethus values crunches to hazard rates at the interface with survivorship mathematics [5]: the rates are an intermediate quantity standing between the attribute-level structure and the survival analysis, faithful to what the analysis needs and silent about nearly everything else. On the framework's own expectation it is the attribute level that a mature successor engages directly, with rates retained as the projection one computes when the question is survivorship and only survivorship.

A worked example fixes the picture. The event of nuclear war enters the internal channel as one hazard term, yet the event itself is heterogeneously composed of many Aethic attributes, the conditions that jointly establish it, and an episode such as the Cuban missile crisis reads in this vocabulary as the triggering of nearly all of the needed preconditions with a terminal miss: followed in Aethic time along the moving present, the total event's rate spikes as each precondition lands and collapses at the end because the last stages did not occur. The over-idealised model the author proposes treats the preconditions as intervals, each carrying an entry rate and an exit rate, with the rates of each interval heavily dependent on which others are open, where the dependence is carried in the Aethic jointness-weight sense rather than as causal linkage; that dependence is what allows successor preconditions to fire on timescales of hours once enough others stand open, though not in general. Nothing in this is reductionist, and by the companion's fundamental theorem of empirical Aethae it could not be: the Attribute Set design leaves the decomposition chosen, so the nuclear-war event is itself the total, one decomposition is useful to this model, others are or are not, and no cut is more real than the event it cuts [4].

One program-level question is entered here as open **[O]**, because this remark is its natural site: how much of what the rate projection renders indiscernible is legible as golden causality when the explicit attributes are examined. The accordance principle under Golden conditioning already permits identifying, by the rarity of observed events, that certain structures carry it intensely (§7), and the attribute-level version of that audit is future work with a stated instrument, and Remark 5.70 sharpens the question into an identification. Nothing in Part II consumes this remark; it prices the model's ontology and points outward, at the companions' grades.

### 5.10 Where these results are used

This section has been developed as a self-contained account of a hazard process, and is readable as one. It is not, however, freestanding within this work: each of its principal results has a consumer elsewhere, and naming them is the quickest way to see why the section is here rather than in a paper of its own. *The two-condition theorem* (5.38) is the paper's most-used result. It supplies the regime trichotomy on which §6.2.1 scopes the General/Golden distinction — and which, as recorded there, restricts the measure-theoretic claim to the critical case. It supplies the dominance argument of Remark 10.4, where the extinctionist option and the settled-at-$x_*$ option are shown to carry viability probability zero alike. And its anatomy factor is what §11 exists to compute.

*Technoanatomy* (Definition 5.32) is consumed twice over. §6.6 identifies extractive organization with $\rho < 0$, which is what converts the Extraction Hypothesis from a vocabulary match into a claim about the sign of an exponent; and §11.6 takes the derivation of $\rho$ from institutional dynamics as its object, together with an account of why the machinery there does not yet reach it. *The trapped-regime theorem* (5.36) does the work in §10.1, where adverse anatomy is shown fatal independently of the driver — the step that makes the argument against extinctionism turn on forfeiture rather than on comparative hazard.

*The tolerance collapse* (5.42) yields the ordering constraint invoked in §10.1.4: since the affordable duration of adverse anatomy decays as $e^{-|\rho|x}$, anatomy must be secured before capability accumulates, and a system that attends to its arrangement late does so with the least margin it will ever have. *The gauge results* of §5.3 recur in a different register in §4.4, where the selection of a task-object among the many that individuate a given object is a gauge fixing of the same shape: find the quantities the arbitrary choice leaves invariant, and state the theory in those.

*And the survivorship interpolation* of §5.8 is what §6 is about in its entirety — the Doob limit there being the Golden Biosphere, and the local absolute continuity of Theorem 5.80 being the epistemic ceiling that §6.8 turns against this work's own conclusions as firmly as against its targets. *The schedule channel* (§5.7) is consumed in three places and supplies one thing to each. It discharges the open rate of Remark 3.7, at the cost recorded there. It supplies Remark 3.16's $M$-valued fixed point in one concrete case. And by Remark 5.65 it produces the distribution over initial data that §9.3 needs and does not have — which is a lead toward that section's principal open cost rather than a discharge of it.

# Part III: Golden Reasoning and the Route World

Part III states what conditioning on the Golden Point licenses and what it forbids. General and Golden are distinguished and the refutations run (§6); the epistemic ceiling fixes what no finite observer certifies; retrocausal accordance is derived as the instrument, with the finale paradox resolved as the companion's designed relay (§7); and the Route World carries the strong reading with its fences and its acting argument (§8). What Part IV consumes from here is an inference discipline together with its limit, namely that reachable is never certifiable.

## 6 Golden reasoning

The preceding sections were mathematics. This one is philosophy, and says so. Its method is to let the mathematics *limit the possibility space* and then to reason philosophically inside what remains. Two streams of load-bearing claims therefore run through it, and are tagged accordingly: mathematical results carry the tags of §5, while philosophical premises carry [Φ] and are numbered $\Phi1, \ldots, \Phi5$, each graded in the corpus's own vocabulary, stipulation, declaration, conjecture-grade, premise, imported result, and each priced with its falsifier in the ledger of §6.9. Nothing is smuggled: where the argument turns on a premise rather than a theorem, the premise is named, argued for, and the thing whose exhibition would break it stated.

The thesis is that the fundamental categorical division among world-scale living systems is not Pre-Civilization versus Civilization but *General* versus *Golden*, $\mathbb{P}$ versus $Q$, and that this division, unlike the other, is not a threshold anyone chose. The ideal category is empty, but its practical form is not: a biosphere of age $t$ is $\varepsilon$-Golden over windows of length $4\varepsilon t$, so that Goldenness is bought by having lasted and by nothing else. From this follow: the rejection of the Kardashev scale, not because its thresholds are arbitrary (though they are) but because the model admits a terminal stage gated by a variable the ladder cannot index; the rejection of the violent-aliens thesis, conditional on one named premise; and a strict epistemic ceiling which limits this section's own conclusions as much as its targets'.

Two tag systems meet in this section and should not be confused. §5 grade mathematical claims by how well established they are: **[P]** proved, **[C]** classical or cited, **[A]** conditional on the soft-wall Ansatz, **[S]** scaling-level, **[O]** open. The companion program grades commitments instead by *what sort of thing they are*, exposed postulate, conjecture-grade, assumption, standing license, schema premise, and prices each with the thing whose exhibition would break it [5, 22, 23]. The two are orthogonal rather than rival, and this section uses both: the mathematical tags for what it inherits from §5, the companion's vocabulary for what it adds. Each premise below accordingly carries its kind in its statement, and §6.9 assembles the whole into a ledger in the companion's two-column format, on the companion's own principle that a priced cost which could never be wrong is not a cost but a decoration.

## 6.1 The referent problem

*Premise* $\Phi 1$ (The biospheric referent; *methodological stipulation*). The appropriate unit of analysis for questions about the long-run trajectory of world-scale living systems is the *biosphere*: the totality of the living and technical activity associated with a world, considered as one system.

The argument is one of referential stability across the relevant timescale. "Species" fails backward: the system that became us was not a species for most of its history, and speciation is not the joint along which the relevant quantity, total hazard-bearing organization, varies. "Civilization" fails backward for the same reason and worse, since it names a phase rather than a system, so that using it as the unit builds a phase transition into the vocabulary before any analysis has occurred. "Planet" fails forward: it is a substrate, not a system, and there is no reason the system should remain confined to one. "Biosphere" is the only available term that refers to the same thing at every point of the trajectory, which is exactly what an asymptotic analysis requires.

This is not a terminological preference. A unit that changes referent partway through cannot appear in a limit theorem. §5 concern the $t \to \infty$ behavior of a single system; the vocabulary must survive the limit or the theorems cannot be applied at all.

**What this already costs Kardashev.** Kardashev's scale [75] classifies *civilizations* by energy capture. Whatever its other merits, the choice of unit forecloses the move this section makes: one cannot ask what the universe looks like from the vantage of a long-survived biosphere if the vocabulary presumes a civilization, since "civilization" carries semantic content, agency, society, purpose, that the analysis is supposed to derive rather than assume. The unit is doing work that ought to be done by argument.

## 6.2 General and Golden

**Definition 6.1** (General and Golden biospheres)**.** A *General Biosphere* is a biosphere considered under $\mathbb{P}$, the unconditioned law of §5. A *Golden Biosphere* is a biosphere considered under $Q$, the survivorship limit — the Doob $\mathfrak{h}$-transform of §5, the $t \to \infty$ limit of conditioning on $\{T > t\}$. The family $\mathbb{P}^{(t)} = \mathbb{P}(\cdot \mid T > t)$ interpolating them is the *goldening* of the biosphere.

*Remark* 6.2 (The interpolation is a family; $Q$ is a measure)*.* The distinction in that last sentence is substantive, and a specific error follows from letting it slide. $\mathbb{P}^{(t)}$ is a *family* indexed by $t$: each member is a different measure, and the index moves. $Q$ is a single measure. So quantities defined by conditioning behave differently along the two. Under any fixed measure the conditional probability of a fixed event is a martingale in the filtration — so under $Q$ it is one. Watched *along* the interpolation it is not, because the conditioning tightens as $t$ advances and there is no single measure for the martingale property to hold under; an extinction weight so watched acquires a downward drift that is an artefact of the moving index and not a fact about the process.

The framework's own reading of the difference is the cleaner one. Passing to $Q$ is not a reweighting applied progressively as survival accumulates; it is the conditioning made part of the base state, which in the companion's terms is the statement of a future-referring attribute in present books [4] — available because statedness is a property of the index rather than of the block's clock, and Remark 3.22 is the consequence drawn out. The Doob transform is what that operation looks like analytically, and the reason $Q$ is a measure where $\mathbb{P}^{(t)}$ is only a family is that the former has already absorbed its condition and the latter is still acquiring one.

Three features of this definition do work that no threshold-based division can do. *First, the split is categorical and unchosen — in the critical regime.* When $\mathbb{P}(T = \infty) = 0$, as it is under a recurrent driver with $\rho > 0$, the laws $Q$ and $\mathbb{P}$ are *mutually singular* on the infinite-horizon $\sigma$-algebra: there is an event — the credit tending to infinity — of $Q$-probability one and $\mathbb{P}$-probability zero. General and Golden are then not two regions of one population separated by a line someone drew but different measures, the difference established by a theorem rather than a stipulation. Pre-Civilization versus Civilization has no such status, being a threshold in a continuous variable with both the variable and the value contested.

The scoping is not a technicality and §6.2.1 states it in full: outside the critical regime the singularity fails, and with it the unchosenness. This is the section's strongest claim and it is regime-specific. *Second, in the critical regime nothing occupies the Golden category — as an ideal.* By the mode classification of §5.8.4, that regime is the critical one: the survivor's law is approached but entered only at infinity. Every actual biosphere is somewhere on the interpolation $\mathbb{P}^{(t)}$, at finite $t$, and none is ideally Golden. This would make the category inapplicable were it not for the practical form of §6.2.2, which is occupiable, and which is what Golden reasoning actually assumes of anything.

*Third, the approach is quantified.* The bias field of §5 gives, on a window of length $s$ at conditioning horizon $t$,

$$\beta_t(s) = \mathbf{1}_{\{T>s\}} \frac{\mathfrak{h}(X_s)}{\mathfrak{h}(X_0)} \Big(1 + \frac{s}{4t} + O(s^2/t^2)\Big), \tag{6.1}$$

so that goldening proceeds at rate $s/4t$ with the persistence exponent $\frac{1}{4}$ as its linear-response coefficient. "The further out you look, the more Golden it gets" is not a slogan here; it is (6.1).

### 6.2.1 Golden reasoning in the three regimes

The two-condition theorem partitions the model, and Golden reasoning has a different character in each cell. Since the apparatus of this section was built inside one of them, the partition is stated before anything is built on it.

| regime | $\mathbb{P}(T = \infty)$ | what General/Golden becomes |
|---|---|---|
| *trapped*: $\rho \le 0$ | $0$, $\mathbb{E}[T] < \infty$ | concentration; $Q$ collapses onto one path |
| *critical*: $\rho > 0$, recurrent | $0$, $\mathbb{E}[T] = \infty$ | **mutually singular; the split is categorical** |
| *viable*: $\rho > 0$, transient | $> 0$ | $Q \ll \mathbb{P}$; an ordinary sub-population |

Three consequences, and the middle one costs this section something. *Trapped.* With adverse anatomy, viability is zero and life expectancy is finite at frozen background (Theorem 5.36), and survivors concentrate on the optimal-fluctuation path: mode (III) of §5.8.4, where the conditioned ensemble is asymptotically deterministic. There is a General/Golden distinction, but the Golden ensemble has no randomness left to study and no biosphere reaches it.

*Critical.* This is where the rest of this section lives, and it is the only cell in which the categorical claim holds. Death is certain but life expectancy infinite; $Q$ is entered only at infinity and is mutually singular with $\mathbb{P}$; the interpolation $\mathbb{P}^{(t)}$ is non-trivial, and everything from (6.1) through the epistemic ceiling is a statement about it. *Viable.* With good anatomy and feedback, $\mathbb{P}(T = \infty) > 0$ and conditioning on immortality is an *ordinary conditional probability*: $Q \ll \mathbb{P}$ globally, total-variation convergence holds, and the Golden Biosphere is a sub-population of the General one — a set of paths one could in principle be a member of, with no measure-theoretic novelty and no unchosen categorical split. Mode (I), saturation.

The consequence for this section's central claim should be stated without softening, because it is a real limitation. *The measure-theoretic recasting of the survivor category is available in exactly one of the three regimes.* Where survival is genuinely achievable, General and Golden differ by a reweighting rather than in kind, and the categorical division reduces to the ordinary distinction between a population and a favored part of it. The framework's most distinctive structural claim thus holds precisely where immortality does *not*, and a reader who believes we inhabit the viable regime should regard §§6.2–6.8 as describing a case they think we are not in.

Two things blunt the loss without removing it. Which regime obtains is decided by $\rho$ and by the driver's scale function, neither of which is observable (§6.8), so no one is presently entitled to the belief that we inhabit the viable regime. And the critical regime is not a knife-edge in the pejorative sense but the boundary the other two are defined against: it is where the apparatus is sharpest precisely because it is where the two failure modes meet.

Two wirings the partition now carries, added where later parts made them load-bearing. *Reachability sorts with the regimes.* In the viable regime the Golden Point is reachable at positive probability under $\mathbb{P}$ itself — conditioning is not what makes it reachable but what makes it *conditionable-on* where reachability degenerates: critical, where the event is null yet modally alive and the Doob limit is required; trapped, where there is nothing left to condition on. *And the partition prices the field's central metric.* Ranking worlds by event-probability misorders them in both directions across this table — the undisturbed world minimizes every event ledger and is already departed, while the critical regime is condemned at probability one and is where everything the survivorship theory computes resides (Remark 2.19, §10.1, §3.4.6). The regime an event metric writes off is the regime this section was built for.

### 6.2.2 Practical and Ideal Golden Biospheres

The ideal category is empty, but the questions one asks of a biosphere are not questions about its infinite horizon. They concern windows — the next millennium, the next galactic orbit. On a window the distinction collapses to a tolerance.

**Definition 6.3** (Practical and Ideal)**.** The *Ideal Golden Biosphere* is $Q$. A biosphere of age $t$ is a *Practical Golden Biosphere to tolerance* $\varepsilon$ *on windows of length* $s$ if $\mathbb{P}^{(t)}$ and $Q$ agree to within $\varepsilon$ on the $\sigma$-algebra $\mathcal{F}_s$.

**Proposition 6.4** (Practical Goldenness is a ratio, not a level; **[S]**)**.** *By the bias field* (6.1)*, the density of* $\mathbb{P}^{(t)}$ *with respect to* $Q$ *on* $\mathcal{F}_s$ *is* $(1-s/t)^{-1/4}(1+o(1))$*, so the two laws deviate at order* $s/4t$*. Hence*

$$\varepsilon\text{-Golden on windows of length } s \qquad \Longleftrightarrow \qquad t \gtrsim \frac{s}{4\varepsilon}, \tag{6.2}$$

*and a biosphere of age* $t$ *is* $\varepsilon$*-Golden for every question concerning a window shorter than the* Golden horizon $s_\varepsilon(t) = 4\varepsilon t$*. The tag is **[S]**: the* $s/4t$ *scale follows from the leading term of* (6.1)*, but the exact total-variation constant depends on the* $o(1)$ *structure and is not established here.*

*Remark* 6.5 (The invariant)*.* The Golden horizon is a fixed fraction of age: $s_\varepsilon(t)/t = 4\varepsilon$, independent of $t$. *Every surviving biosphere is Golden over a constant fraction of its own history, and the fraction is four times one's tolerance.* There is no threshold in strength, in energy, in technology, or in any state variable whatever; the only thing that buys Goldenness is having lasted, and what it buys is proportional. To be 1%-Golden over the coming millennium requires twenty-five thousand years behind one.

This sharpens §6.4.1: the objection to Kardashev is not merely that its thresholds are placed arbitrarily, but that the operative threshold does not live in the space it measures.

*Remark* 6.6 (The premise of Golden reasoning is weaker than it looks)*.* This answers the natural suspicion that Golden reasoning assumes something optimistic. It does not. By (6.2), practical Goldenness to any fixed tolerance follows from age alone; and age is exactly what the question already supposes. To ask what our descendants are like across any substantial future is already to condition on there *being* descendants across that future, and that conditioning by itself confers practical Goldenness over windows proportional to the horizon supposed. The assumption is not that the future goes well. It is that there is one — and the model converts that supposition, and nothing further, into the whole apparatus.

The corollary for the present is deflationary and should be stated plainly: we are not practically Golden to any interesting tolerance, and Golden reasoning does not claim we are. It describes the ensemble our descendants would be drawn from in those futures where they exist.

*Premise* Φ2 (Golden reasoning is a vantage, not an obligation; *scope declaration*)*.* Golden reasoning is the practice of re-analyzing questions about biospheres by taking $Q$ as the reference measure rather than $\mathbb{P}$. It is *inferential*: it licenses conditional expectations, not duties.

Φ2 governs what the apparatus outputs; it is silent on the separate question of how an agent grounds duties the apparatus never issued, for which see the selection rule of §2.2.

*Remark* 6.7 (Why this is not longtermism)*.* The objection that any future-weighted reasoning collapses into "act in the interest of the future, as you happen to conceive it" does not apply, and the reason is structural rather than rhetorical. A normative longtermism requires future beings whose interests ground obligations. Golden reasoning has no such beings available: $Q$ is a measure that nothing occupies in the critical regime (Definition 6.1), where moreover $\mathbb{P}(T = \infty) = 0$. In the viable regime the objection would have purchase and is met differently, by $\Phi 2$ alone rather than by the vacuity of the reference class. What $Q$ supplies is a *conditional distribution*, from which one may infer what long-survived systems are like without thereby acquiring any duty toward them. The distinction is the ordinary one between a likelihood and a value, and it is preserved here by the mathematics rather than by stipulation. Whether one ought to care about the far future is a question this section does not address and does not need. §9.6.4 closes the concession made above, by exhibiting an ordering difference that survives a normative reading.

Three consumers of this distinction now sit downstream and should be visible from here. The finitude tradition's dissolution runs entirely on the Practical face: unbounded lifespan as a convergent series of mortal moments, hazard strictly positive at every instant, forever, is what deprives the tedium and finitude objections of their target (Remark 2.17). The Ideal pole's epistemic status is fixed by the ceiling: never certified, only approached, which is why every operative claim in this work runs through the Practical register (§6.8). And the invariant above is the quantitative face of the Lindy step, having lasted as the only purchase, proportionality as the price, consumed where the Route World's evidence method is built (§8.3).

## 6.3 Technostrength and technoanatomy

**Definition 6.8** (The two coordinates)**.** *Technostrength* is the muffling exponent $x_t$: the capacity the biosphere has accumulated, positive when its own organization holds its hazard below background and negative when it holds it above. *Technoanatomy* is the exponent $\rho$ of Definition 5.32, the sign of $-\partial_x \log h^{\mathrm{int}}$: whether acquiring capacity lowers or raises the biosphere's hazard *to itself*.

The two are of different kinds, and keeping them apart is the whole of what follows. Technostrength is a *state coordinate*: it fluctuates, and by Theorem 5.24 every biosphere, however well constituted, spends exactly half of logarithmic time with $x < 0$. Technoanatomy is a *structural property*: it does not fluctuate at all, it says what capacity does when acquired, and $\rho < 0$ names the biosphere whose own strength is turned against it, which is what Definition 5.32 means by eating oneself. *Adverse strength* is $x < 0$; *adverse anatomy* is $\rho < 0$, and only the second is a defect.

Two results of §5 are worth restating in this vocabulary because they are counterintuitive. *Technostrength is not cumulative — in the critical regime.* Under a recurrent driver $x$ oscillates on scale $t^{3/2}$ and returns to every level infinitely often: it is not a stock that accumulates but a magnitude that fluctuates, so a framework treating it as monotonically increasing has the wrong kind of object. Under a *transient* driver it does accumulate, and the ladder picture is locally right — which is the concession §6.2.1 makes precise and prices.

*Every biosphere is adversely configured half the time.* By Theorem 5.24 the set $\{t : x_t < 0\}$ has logarithmic density exactly $\frac{1}{2}$, for every parameter value and every biosphere. Adverse *strength* is therefore not a defect distinguishing some biospheres from others but a condition all of them occupy half of logarithmic time — which is precisely why it cannot be what "bad anatomy" means, and why the two must be kept apart.

### 6.3.1 What conditioning does to each coordinate

**Numerical methods.** All figures reported in this subsection, in §6.3.2, in Remark 10.4, and in the numerical check under Proposition 5.35 were produced by direct Monte Carlo on the two-channel model, using the exact Gaussian update of Lemma F.1 for the driver-credit pair and forward Euler for the background exponentials, at step $\Delta = 10^{-3}$. Conditioning on $\{T > t\}$ was performed by rejection, particles were killed by the Cox rule and the survivors reweighted uniformly, so the conditioned estimates are $\mathbb{P}^{(T)}$ estimates and not $Q$ estimates, exactly as the column headings below say. Standard errors on the tabled probabilities are binomial and never exceed 0.004; the median-$x$ and quantile-$\rho$ entries carry bootstrap errors of comparable size. *Every* $\mathbb{P}(\rho > 0 \mid \textit{survived})$ *figure is prior-dependent*: $\rho \sim \mathrm{Unif}(-1, 3)$ was chosen to place a quarter of the mass on adverse anatomy and has no other justification, and the $0.75 \to 0.84$ movement should be read as the direction and rough magnitude of selection under that prior rather than as a calibrated number. Code and seeds are available on request.

The following are simulation results for the two-channel model of §5 at $\sigma = 1$, $\epsilon = 0.05$, $n = 3$, $b = 3$, $\rho = 2.5$, $k = 6$, conditioning horizon $T = 4$, $N = 1.2 \times 10^5$ particles.

| | General ($\mathbb{P}$) | Conditioned ($\mathbb{P}^{(4)}$) |
|---|---:|---:|
| $\mathbb{P}(x < 0)$ (adverse strength) | 0.501 | 0.0037 |
| median $x/\sigma T^{3/2}$ | $-0.001$ | $+1.085$ |
| $\mathbb{P}(\rho > 0)$, $\rho \sim \mathrm{Unif}(-1, 3)$ | 0.750 | 0.839 |
| 5% quantile of $\rho$ | $-0.80$ | $-0.44$ |

The two $\rho$ rows depend on the prior $\mathrm{Unif}(-1, 3)$ placed on anatomy, which is a choice and not a result; the $x$ rows do not.

*Remark* 6.9 (Truncation versus rescaling). Two numbers, and they differ in kind rather than in force. Adverse strength is suppressed by a factor of 137 — abruptly, and within a few multiples of the intrinsic timescale. Technoanatomy is selected too, but *gradually*: $\mathbb{P}(\rho > 0)$ moves from 0.75 to 0.839 at the same horizon, reaching 0.909 only by $T = 6$, with the negative tail truncating steadily (5% quantile $-0.80 \to -0.44 \to -0.14$).

The asymmetry is structural rather than incidental, and it follows from what the two objects are. Strength is a state coordinate, so conditioning pushes it out of a half-space it can and does re-enter — fast, and reversibly. Anatomy is a fixed structural parameter, so it can be selected only through the accumulated hazard cost of never changing — slow, and irreversibly. Anatomy is the slower and the more decisive: by Theorem 5.36, adverse anatomy is eventually fatal with certainty, while adverse strength is a condition every biosphere occupies half of logarithmic time and survives.

### 6.3.2 Does conditioning select technoanatomy?

The measurement above concerns technostrength, a state coordinate; the claim the argument needs is about technoanatomy, a structural parameter, which therefore cannot be selected *within* a trajectory. It can be selected across an ensemble, and that is directly computable: place a prior on $\rho$, evolve, and reweight by survival. Taking $\rho \sim \mathrm{Uniform}(-1, 3)$, so that $\mathbb{P}(\rho > 0) = 0.75$ a priori:

| $T$ | $\mathbb{P}$(survive) | $\mathbb{P}(\rho > 0 \mid \text{survived})$ | 5% quantile of $\rho$ |
|---|---|---|---|
| 2 | $2.8 \times 10^{-3}$ | 0.745 | $-0.80$ |
| 4 | $2.2 \times 10^{-4}$ | 0.839 | $-0.44$ |
| 6 | $1.9 \times 10^{-4}$ | 0.909 | $-0.14$ |

Survivorship does select for positive technoanatomy, and the adverse tail truncates steadily. But the selection is *gradual* where the strength truncation was abrupt: adverse strength falls by a factor of 137 at $T = 4$, while $\mathbb{P}(\rho > 0)$ has moved only from 0.75 to 0.84 at the same horizon. Anatomy is the slower and the more decisive: by Theorem 5.36 an adverse $\rho$ is eventually fatal with certainty, while adverse strength is a condition every biosphere occupies half of logarithmic time and survives.

**The coordinates stay separable.** Strength and anatomy cannot fuse under conditioning, and the reason is structural rather than measured: $\rho$ is a parameter of the model and $x$ a functional of the path, so no amount of conditioning makes one a function of the other. They therefore remain distinct diagnostic questions rather than two aspects of a single "advancement" — which is exactly what a one-dimensional scale cannot represent.

## 6.4 What Golden reasoning refutes

### 6.4.1 The Kardashev scale

The usual complaint against Kardashev's scale [75] is that its thresholds, planetary, stellar, galactic energy capture, are arbitrary. That complaint is correct but weak, since a scale may be useful with arbitrary units. Golden reasoning supports three stronger objections, of which the first is a theorem.

*(i) The terminal stage exists, and is gated by a variable the scale cannot see.* In the driftless model $\Lambda_\infty = \infty$ almost surely and no absorbing state of safety exists; in general one does, and locating it is what makes the objection bite. By Theorem 5.38,

$$\mathbb{P}(\text{viable}) = \mathbf{1}\{\rho > 0\} \cdot \frac{s(\mu_0) - s(-\infty)}{s(+\infty) - s(-\infty)},$$

so there *is* an absorbing state of safety — and reaching it requires two conditions, of which Kardashev's scale tracks at most one. The escape factor is the accumulation of capacity, which energy capture does index, however crudely. The anatomy factor is not on the scale at all: $\rho$ does not appear in energy capture, is not monotone in it, and cannot be inferred from it.

The objection is therefore not that the ladder has no top but that *climbing it is neither necessary nor sufficient for reaching the top.* Worse, by Remark 5.41, a biosphere with $\rho < 0$ that climbs fast does not merely fail to arrive: its hazard diverges superexponentially in the very capacity the scale is measuring. A scale indexed on capture alone cannot distinguish the ascent that arrives from the ascent that is a fall. That last point is the vulnerable-world hypothesis [76] restated with a continuous parameter in place of a discrete draw, and §6.7.3 sets out what is and is not shared.

*(ii) The indexed quantity is not monotone.* Technostrength is stationary and recurrent in logarithmic time (Definition 6.8). A ladder presumes a quantity that ratchets. Energy capture may of course ratchet while technostrength does not — but then energy capture is not tracking the quantity that governs survival, which returns us to (iii). *(iii) A magnitude index cannot*

*represent categorical selection.* By Remark 6.9, survivorship truncates technoanatomy while merely rescaling technostrength. A one-dimensional index of magnitude has no coordinate in which to express a deleted half-space. It is not that Kardashev measures the less important variable, that claim is false, and was corrected above, but that it measures the one whose selection is quantitative, and is structurally unable to express the one whose selection is categorical.

**On Dyson spheres.** Dyson's proposal [77] is often read as the natural Type II signature. Under Golden reasoning it is better read as a *hypothesis about technoanatomy* disguised as one about technostrength: it asserts not merely that a biosphere could capture stellar output but that a long-lived one would be organized so as to want to. That assertion is exactly what §6.6 disputes, and it should be recognized as an assertion rather than an extrapolation.

*Remark* 6.10 (The expansionist mirror: grabby selection)*.* One modern argument deserves separate notice because it reaches the Kardashev picture from selection reasoning — this framework's own instrument. The grabby-aliens model observes that loud, expansionist civilizations dominate what is *visible*, and runs anthropic timing arguments under that weighting [78]. Read beside this section, that is a rival conditioning: weight by volume occupied and the typical survivor is an expander; weight by persistence and the two-condition theorem makes expansion-rate orthogonal to the factor that decides. The two conditionings are not in contradiction, one asks what is seen, the other what survives, but they license opposite extrapolations from the same record, and conflating them reproduces the Kardashev error at the level of selection effects: visibility-weighting smuggles in the assumption that the seen and the persisting are the same ensemble, which is exactly what the trough channel denies. The framework's quarrel is therefore not with the grabby model's arithmetic but with any normative or predictive reading of it that takes loudness for viability; §6.8 bounds what either conditioning can claim to certify.

### 6.4.2 The violent-aliens thesis

The thesis in view holds that intelligence evolves preferentially in predators, hence that advanced biospheres are likely to be predatory or violent. We engage the position rather than any proponent.

> "It was the saying of Bion, that though the boys throw stones at frogs in sport, yet the frogs do not die in sport but in earnest." — Plutarch, *Moralia* [79]

The line is Bion of Borysthenes', preserved by Plutarch, who deploys it against hunting for sport; it has since been borrowed by astrobiological writing as the image of what a technologically superior biosphere might do to an inferior one. It is the right epigraph for this subsection because it names the assumption at issue more exactly than the conquest analogies usually offered. The serious form of the violent-aliens thesis is not that an advanced biosphere would hate us. It is that we would be *beneath its consideration* — that its power could be spent across the boundary between itself and us casually, without the expenditure ever rising to the level of a decision. The stones are thrown in sport. That is the claim, and any reply pitched at malice misses it.

The image also fixes what the reply must not do. It must not argue that such a biosphere would be kind, since kindness is not what the thesis denies; nor that it would have outgrown violence, since sport is not violence in the relevant sense. What it must do is address the *scaling*: whether a biosphere can be arranged so that power is spendable in that manner and still arrive at the scale where the spending would matter. Our own route to the present framework began from exactly this observation, three years before the model existed, and §5.5 records the entry.

Before the model is brought to bear, a further placement is needed, because the *conclusion* here is the one most likely to be mistaken for a new one and it is not. The conclusion, that anything old enough to encounter must be peaceable, is a standing argument in SETI, sometimes called the *technological adolescence* argument, and it long predates this paper: a civilization reaching the capacity to destroy itself must either become socially responsible or fail to persist, so whatever survives has already made that transition. Cooper documents the argument and its currency, and also supplies the standing rebuttal: that it conflates altruism with amiability, that it is typically asserted rather than derived, and that it presumes a trajectory of moral improvement for which there is no evidence [6]. Nothing below claims the conclusion as new. What is at issue is whether a mechanism can be supplied that survives Cooper's objection.

The inference under dispute has the form: predatory origin $\Rightarrow$ predatory maturity. Golden reasoning denies the implication, on the grounds that it neglects the conditioning that must be applied before any encounter can be contemplated. Any biosphere one could meet is old; any biosphere that is old is drawn from $\mathbb{P}^{(t)}$ for large $t$, not from $\mathbb{P}$; and $\mathbb{P}^{(t)} \to Q$, under which technoanatomy is truncated. Whatever selection operated at origin, a second and stronger selection operates over the trajectory, and it is the second that determines what survives to be encountered.

*Premise* $\Phi 4$ (Origin does not transfer; *premise*). Facts about the selective regime that produced intelligence in a biosphere do not, without further argument, transfer to the selective regime that governs that biosphere's persistence at world scale. The two regimes act on different units, at different scales, over different timescales.

$\Phi 4$ is not exotic; it is the ordinary observation that a trait adaptive for an organism in a population need not be adaptive for a biosphere in a universe. What the model adds is that the second regime is not merely different but *stronger* in the relevant sense: it deletes a half-space of configurations, and it does so with a force measured in §6.3.1.

What the model contributes to the frogs case in particular is a reply that never mentions disposition. A biosphere that can spend power casually across a self/other boundary is one that maintains such a boundary and enforces it — and by §6.6.1 the enforcement of a boundary is interface-like while the domain it encloses is volume-like, so the arrangement carries $\rho < 0$ by (6.3). By Corollary 6.11 its viability probability is then exactly zero. The stone-throwers are not ruled out because they would have grown kind; they are ruled out because the organization that permits stone-throwing at that scale is the organization the scaling argument says cannot reach that scale. The frogs survive not by the boys' mercy but by the boys' having imploded before attaining the requisite size — which is, in a form Bion would have recognized, the square–cube law again.

Three further features distinguish this from the folk argument, and they are exactly the three points Cooper's rebuttal presses on. *It is not a moral claim.* What is selected is $\rho$, the sign of $-\partial_x \log h^{\mathrm{int}}$ — a structural property of how a biosphere is arranged, carrying no implication that its members are kind, and applying identically to a biosphere with no members in the relevant sense at all. The conflation of altruism with amiability has nothing to attach to. *It is not asserted but measured.* §6.3.2 exhibits the selection on $\rho$ and finds it *gradual*, 0.75 to 0.91 over the horizons computed, where the folk argument implicitly treats the filtering as complete. *And it is not a trajectory.* Nothing here says biospheres improve; $\rho$ does not change at all in the fixed model, and what conditioning does is reweight an ensemble, not move a system along a path. The selection is Darwinian in form rather than developmental, which is what the folk version gets wrong and Cooper is right to object to.

This much, however, only establishes that predatory maturity is not *inherited*. To conclude that it is positively *selected against*, one needs the identification of predation with negative technoanatomy. That identification is a philosophical premise, and it is the subject of the next subsection. Everything in §6.7 is downstream of it.

### 6.4.3 Trajectory arguments for internal neglect

A third target belongs here, and it lies inside the biosphere rather than out among the aliens. Longtermism in its strong form holds that what most matters about our actions is their effect on the very long run [33]; and a recurring corollary within that literature is that resources directed at the welfare of some populations do less for the long-run future than the same resources directed elsewhere, so that broad investment in the worse-off is not what a future-oriented agent should fund. We engage the corollary as a position and name no one, as with the violent-aliens thesis.

The position is already under sustained criticism, and it is worth being exact about how the present objection differs, because the difference is what makes it worth adding. The standing critiques are *external*: they fault longtermism on ethical and political grounds — as an ideology with damaging real-world effects and a flawed underlying moral theory [80], and as a doctrine whose cosmic accounting licenses the discounting of present suffering [81]. Those objections deny the framework's premises. The objection developed here does the opposite: it *grants* the long-run criterion entirely, and shows the position failing by that criterion, on the framework's own terms and in its own currency. If it succeeds it is therefore not an alternative to the ethical critiques but a complement of a different kind — an internal result rather than an external one.

Stated at its strongest it is not a callous argument but an optimization: interventions differ in long-run leverage, a future-oriented agent should allocate to high-leverage interventions, and if the leverage of some interventions is low then funding them is a loss measured against the future. Nothing in that reasoning is invalid. The question is whether the leverage calculation is complete. Golden reasoning says it is not, and the missing term is technoanatomy. No new premise is required, and this is the point worth making: a position of this kind is $\Phi 3$ with the boundary drawn *inside*. Writing off part of a system is the maintenance of a self/resource distinction across an internal interface, a region of the biosphere to which protection is not extended, which is structurally the same object as the extractive boundary of §6.6.1, and it inherits the same scaling. Enforcement of an internal boundary is interface-like; the interior it excludes is volume-like; so by (6.3) the arrangement carries $\rho < 0$, and Corollary 6.11 applies verbatim.

The consequence is that the position is self-undermining in its own currency, and sharply rather than rhetorically. By Corollary 6.11(ii) a biosphere with $\rho < 0$ has viability probability *exactly zero* however much capacity it acquires; by (i) its viability is zero where a $\rho > 0$ biosphere's is positive; and by Remark 5.41 acquiring capacity faster makes its position worse, since $x \to \infty$ with $\rho < 0$ drives hazard to infinity superexponentially. An argument that trades internal cohesion for external capacity, offered in the name of the long-run future, is therefore an argument for a shortened one — not because neglect is distasteful, which the model cannot see, but because the boundary it requires is the structure the model identifies as fatal.

The thought experiment that makes this concrete is worth stating, because it isolates the variable. Hold a society's technoanatomy fixed at whatever value its present arrangements realize, and scale its technostrength to galactic magnitude. If $\rho \geq 0$, nothing goes wrong. If $\rho < 0$, the hazard does not rise gently but spikes as $e^{|\rho| x}$: at every scale the same arrangement supplies more micro-opportunities for the system to act against itself, and each opens further ones. The elephant-sized ant of Remark 6.13 is exactly this, and the square–cube law is exactly why.

Four fences, because this is the section's most contestable application. It refutes a *class* of positions, any that trades internal cohesion for external capacity, and not any author's wider view, nor longtermism as such: a longtermism that valued internal cohesion would be untouched, and the argument is in that sense a constraint on longtermist reasoning rather than a refutation of it. It is downstream of $\Phi 3$ and inherits its conjecture grade; reject $\Phi 3$ and this goes with it. And it establishes nothing about what anyone ought to do, by $\Phi 2$: what it establishes is that a particular argument for neglect, evaluated by its own long-run criterion, fails on that criterion.

## 6.5 A prediction outside cosmology

Most of what this section has established is a re-derivation. The conclusions of §6.4 were available before it — the Kardashev scale has long been thought crude, the peaceability of long-lived civilizations is a standing SETI argument [6], and the mechanism behind $\Phi 3$ has precedents in the collapse and existential-risk literatures [82, 76]. What the section offers is that they follow from one premise with a named exponent, which is a unification rather than a discovery.

Unification is a respectable contribution, but it carries an obligation that a discovery does not. A framework which only reproduces what it was built to reproduce is a reformulation, and the way to tell the two apart is whether anything falls out that was not among the inputs. This subsection states the clearest such consequence, in a domain the framework was not designed for and where it can be checked.

**The claim.** The asymmetry measured in §6.3.1, adverse strength truncated abruptly, technoanatomy selected gradually, is not a fact about biospheres. Nothing in its derivation mentions one. It is a consequence of conditioning a two-coordinate system on survival, where one coordinate is a functional of the path and the other a parameter of the model, and it should therefore appear wherever that structure does.

> *Cross-domain prediction.* In any population subject to survivorship selection, carrying a fluctuating state coordinate alongside a fixed structural parameter that governs whether the state's growth raises or lowers hazard, conditioning on survival should **truncate** the state coordinate, deleting a region it can otherwise re-enter, and **shift** the structural parameter, progressively and without deleting anything, the two effects being comparable in magnitude and opposite in kind.

Candidate domains are not exotic: firms with a fluctuating balance-sheet position and a fixed governance structure; institutions with a fluctuating resource base and a fixed constitutional arrangement; lineages with a fluctuating population size and a fixed life-history strategy. In each the two coordinates are separately measurable, old members are selected from many, and the comparison the prediction asks for is between surviving and full cohorts rather than between a model and the sky.

**What would count against it.** Three outcomes would tell against the framework, and they should be stated plainly, since a prediction whose failure conditions are vague is not doing the work this subsection is meant to do. If the selection acted *categorically on the structural parameter and quantitatively on the state coordinate*, the asymmetry inverted, the account of why the two behave differently would be wrong, since that account turns on which of them can be re-entered.

If *neither* coordinate showed selection in a population where survivorship is known to operate, the framework would be describing a mechanism that does not act at accessible scales, whatever its formal standing. And if the structural parameter showed selection *as abrupt* as the state coordinate, the distinction drawn in §6.3 between a state and a structure would not be doing the work claimed for it, and the argument of §6.4.2, which rests on that distinction to answer the charge of positing a moral trajectory, would lose its reply.

**Two qualifications.** The prediction is weaker than it may appear, in two respects that should be recorded rather than left for a reader to find. It is a claim about *shape*, not magnitude. The framework does not predict how large either effect should be in any particular domain, since that depends on the hazard scale, the horizon, and the prior spread of the structural parameter — none of which the framework supplies. What it predicts is that one effect is a truncation and the other a shift, which is a qualitative signature and correspondingly easier to satisfy by accident.

And it is not *unique* to this framework. Any account on which survival depends on a structural property will predict some selection on that property; what is distinctive here is the predicted *contrast* in kind between the two coordinates, and only the contrast is diagnostic. A domain exhibiting selection on both, with no difference in character, would be consistent with much and would confirm little.

With those qualifications the prediction stands as the section's answer to the question of whether its unification is more than a restatement. We do not know the answer; the point of stating the prediction in this form is that someone could find out.

This prediction no longer stands alone, and the family it belongs to is the paper's falsifiable perimeter, listed once so a reader can audit it whole. Physical and statistical register: the present cross-domain claim; the executed numerics, whose stated falsifiers did not fire (§F); the mortality-deceleration signature, population hazard decaying as a universal $c/t$ under criticality, against the plateau fingerprint of static frailty [83], carried in the standalone extraction's discussion; and the Tainter route to measuring $\alpha - \beta$, the one posited inequality, from collapse-era returns data (Remark 6.12, [82]). Physical-provisioning register: prepositioned prerequisites measured against a habitability-conditioned base rate, with the census of systems as the eventual instrument (Remark 8.16). Transfer register: the hard-steps machinery re-run under the Golden target, open and specified (§6.7.1, [84]). Reception register, scoreable rather than lab-testable and flagged as such: the phase-transition prediction and the species-criterion test it doubles as (Remark 9.11, Remark 10.1). Registers differ; the discipline does not — each member is entered with the conditions of its own failure.

## 6.6 The Extraction Hypothesis

*Premise* $\Phi 3$ (Extraction Hypothesis; *conjecture-grade*). Organization built on unbounded extraction, consuming the resources of a locale and moving on, has *negative technoanatomy*: $\rho < 0$

in the sense of Definition 5.32, so that internal hazard rises with acquired capacity rather than falling with it.

The premise is now a claim about the sign of an exponent rather than an identification between a social form and a vocabulary, and this is worth pausing on, because it is what the anatomy parameter buys. Before, one had to be told what “bad technoanatomy” meant and then persuaded that extraction was an instance. Now Φ3 says something with a truth value fixed by the model: that for extractive organization, $-\partial_x \log h^{\mathrm{int}} < 0$. The consequences follow mechanically rather than interpretively, and they are severe.

**Corollary 6.11** (What Φ3 costs an extractive biosphere; **[P]** given Φ3)**.** *If extractive organization has $\rho < 0$ then, by Theorem 5.36 and Theorem 5.38:* (i) *its viability is zero and, at frozen background, its life expectancy is finite with $\mathbb{P}(T > t) \le e^{-h_* t}$ for the floor $h_*$ of* (5.20) *(Theorem 5.36(ii));* (ii) *its viability probability is* exactly zero*, whatever feedback it achieves — the anatomy factor in* (5.23) *annihilates the escape factor;* (iii) *it inhabits the trapped regime of §6.2.1, where none of the Lindy structure holds and the survivor ensemble concentrates rather than diversifying; and* (iv) *its safest available configuration is not its strongest but its* weakest*, since $h \to B(1 + e^{-b})$ as $x \to -\infty$ while $h \to \infty$ as $x \to +\infty$.*

Item (ii) is the one to hold onto. An extractive biosphere does not merely have poor prospects; it has *no* prospect of viability, and acquiring capacity faster makes its position worse rather than better. Item (iv) states the same thing from the other end and is the sharpest form of the thesis: for a system that turns its strength on itself, every increment of strength beyond $x_*$ is a net loss, and the counsel of the model is to stop.

This is the section’s single load-bearing philosophical premise. It is not a theorem and is not presented as one. We argue for it.

### 6.6.1 The argument from boundary cost

Extraction requires a maintained distinction between the extracting system and the extracted resource: a boundary across which value flows one way. The boundary is not free. It must be enforced — against diversion, against defection, against the tendency of a system’s own subunits to treat the system as their resource, which is precisely what a tumor does. Now consider how the two sides of this ledger scale with technostrength, and note that the anatomy parameter lets the scaling be written down rather than gestured at. Let capacity be indexed by $x$. The domain over which extraction occurs grows with reach and is volume-like, $V \sim e^{\alpha x}$; the boundary across which enforcement is exercised is interface-like, $S \sim e^{\beta x}$. The claim is $\beta < \alpha$: volume outgrows surface. The unenforced fraction of the domain is then $1 - e^{-(\alpha-\beta)x} \to 1$; measured against the enforcement available to contain it, the unprotected extractive interior grows as $V/S \sim e^{(\alpha-\beta)x}$, and internal hazard with it:

$$h^{\mathrm{int}} \sim e^{(\alpha-\beta)x} \qquad \Longrightarrow \qquad \rho = -(\alpha - \beta) < 0. \tag{6.3}$$

So the argument does not merely conclude that extraction is bad anatomy; it *names the exponent.* Technoanatomy is minus the excess of the extractive domain’s growth rate over the boundary’s, and the more reach an extractive organization acquires per unit of enforcement, the worse its anatomy is. Hence:

> *As a biosphere’s technostrength grows, the cost of maintaining the self/resource distinction grows faster than the capacity to maintain it, and the shortfall is the exponent $\alpha - \beta > 0$.*

This mechanism is not original to the present work, and §6.7.3 records its ancestry: Tainter's account of collapse from the declining marginal returns of complexity [82] reaches the same interior optimum from archaeological evidence, and Bostrom's vulnerable-world hypothesis [76] reaches a compatible conclusion from technological risk. What the present treatment adds is the exponent (6.3) and the hazard-theoretic consequences that follow from its sign.

Past some scale, the cheapest available target of an extractive subsystem is the biosphere itself — not through malice but through gradient, exactly as a tumor exploits the body's own supply mechanisms rather than building its own. Extraction at scale therefore turns inward, and in the model's vocabulary this is precisely the *trough* channel of §5: internal damage, unmediated by the environment, unbounded above, and with adverse anatomy $\rho < 0$ ensuring that its costs outrun the crest channel's gains per unit.

*Remark* 6.12 (The one posited inequality, and what would settle it)*.* Everything in (6.3) follows from the scalings once $\beta < \alpha$ is granted, and that inequality is asserted in a clause. It is the whole load, and it should carry its own accounting rather than travel inside a derivation that looks computed.

*What it says.* That enforcement is interface-like while the extractive domain is volume-like — so that reach purchased at rate $\alpha$ can be policed only at rate $\beta < \alpha$, and the shortfall is the exponent. The geometric intuition is the square–cube law and it is genuinely suggestive; it is not an argument, because the objects here are institutions rather than solids and nothing forces an institutional cost to scale like a surface.

*What would defeat it.* Enforcement mechanisms that scale with the domain rather than with its boundary, of which there are real candidates. Internalized norms are carried by the units themselves and are volumetric by construction; distributed verification scales with participants; and an immune system, the biological analogue the tumor image invites, is precisely a policing mechanism distributed through the volume it protects, which is why organisms defeat the naive surface bound the image would impose on them. If any such mechanism achieves $\beta \geq \alpha$ at scale, $\rho$ need not be negative for an extractive organization and $\Phi3$ fails as a general claim. The premise's force is therefore that no known arrangement has achieved it *at the scales in question*, which is weaker than the derivation's presentation suggests and is the honest form.

*What would establish it.* Not the geometry but the record. Tainter's series are the right kind of evidence, since declining marginal returns on complexity is $\beta < \alpha$ measured rather than posited: an administrative cost growing faster than the domain it administers is exactly the shortfall, and the archaeological cases supply the growth rates [82]. Fitting $\alpha - \beta$ to those series, rather than citing Tainter as a convergent conclusion, which is all §6.7.3 presently does, would convert $\Phi3$ from a premise into a measurement, and would supply the first empirical estimate of a technoanatomy anywhere in this work. That is a tractable piece of work, it is the most direct route from this framework to data, and it is open **[O]**.

*Scope of the exposure.* The two-condition theorem, the trichotomy, and everything in §5 are untouched by this: they take $\rho$'s sign as given and say what follows. What rests on $\beta < \alpha$ is the identification of *extractive organization* as a case of $\rho < 0$ — that is, $\Phi3$ and the consequences §6.6 draws from it, and nothing else.

*Remark* 6.13 (The names were the mechanism)*.* The argument just given is the square–cube law. That is the law from which "technostrength" and "technoanatomy" were named — initially as a joke about giants. The joke turns out to be the mechanism: a biosphere whose extractive reach has outgrown its capacity to remain one system is an elephant-sized ant, and collapses for the same reason. The model's brittleness exponent $\rho$ is the quantitative form of "how badly does the mismatch hurt".

### 6.6.2 Objections

*Objection* 6.14 (Bounded extraction). A biosphere might extract only from what is genuinely external and never grow past the scale at which boundary cost bites.

This is granted, and it is not a counterexample but a concession of the point. A biosphere that halts its extractive growth at a sustainable scale is not the unboundedly expanding entity whose possibility is at issue; it is not a Kardashev trajectory. $\Phi 3$ concerns *unbounded* extraction, and the objection concedes that unbounded extraction is unavailable.

*Objection* 6.15 (Cheap boundaries). Sufficiently advanced technique might make boundary enforcement scale with volume rather than surface.

This is the serious objection and we do not refute it. In the notation of (6.3) it is the claim $\beta \geq \alpha$: enforcement scaling at least volumetrically, so that $\rho \geq 0$ and the whole corollary lapses. That is an empirical claim about the scaling of coordination costs, it is where $\Phi 3$ is falsifiable, and stating it as an inequality between two exponents is the chief gain of the anatomy parameter — the objection and the premise now disagree about a measurable number rather than about a description. We note only that every known instance of large-scale organization, organisms, ecosystems, institutions, exhibits $\beta < \alpha$, and that the burden of the exception lies with whoever claims it.

*Objection* 6.16 (Assuming the conclusion). Defining bad technoanatomy as "that which raises hazard" and then asserting extraction raises hazard makes the argument circular.

The charge is now easy to answer precisely, and the anatomy parameter is what makes it so. The model defines $\rho$ as $-\partial_x \log h^{\text{int}}$ and proves what follows from its sign; that is a theorem about an exponent and mentions extraction nowhere. $\Phi 3$ claims that a particular form of organization realizes a negative value of that exponent, and argues for it from boundary scaling. Two distinct objects, one proved and one argued, and neither borrowed from the other. The identification is supplied from outside, by §6.6.1, and could be false — it would be false if $\beta \geq \alpha$. The division of labor is exactly the one this section's method requires: mathematics limits the possibility space, philosophy says which region of it we occupy.

*Remark* 6.17 (What would falsify $\Phi 3$, and what would not). Three falsifiers, and they are *operational* in the companion's sense, checkable in domains that exist, rather than requiring a capability the argument itself says no one has [23]. $\Phi 3$ fails if $\beta \geq \alpha$ in (6.3), if enforcement is shown to scale at least volumetrically in reach; if an extractive organization at scale is exhibited stable over timescales long compared with its own doubling time; or if internal hazard is observed to *decrease* with technostrength in an extractive system, which is $\rho > 0$ measured directly and refutes the premise outright. The first is the live question, and it is live in institutional, ecological and organismal data rather than only in cosmology.

What would *not* falsify it: the bare possibility that extraction might turn out fine, or a hypothetical technology stipulated to make boundaries cheap. The claim is about scaling, scaling is measurable wherever large-scale organization exists, and a falsifier must therefore exhibit the scaling rather than assert its negation.

## 6.7 Consequences

The following are downstream of $\Phi3$ and inherit its status. Each is falsifiable in principle; none is currently testable. *(a) Encountered biospheres are not extractive.* Not because predation is impossible but because extractive organization is a transient: it may be instantiated, but the conditioning under which any encounter occurs selects against it. A predatory biosphere "copy-pasted" into the galaxy would be a configuration of the $\mathbb{P}$-ensemble, not the $Q$-ensemble, and would not persist to be met.

*(b) Absence of megastructures is not evidence of absence.* If $\Phi3$ holds, stellar-scale extraction is a signature of biospheres on their way out, not of biospheres that lasted. Searches keyed to such signatures [77] are therefore keyed to the wrong tail of the distribution, and their null results carry less evidential weight against the existence of old biospheres than is usually assumed.

*(c) Old biospheres are Lindy — and here the novelty is narrow, so it is stated narrowly.* From the survival tail $\mathbb{P}(T > t) \asymp Ct^{-1/4}$ (§5): residual life is asymptotically Pareto($\frac{1}{4}$) in units of attained age, and the effective population hazard is $1/4t$. The longer a biosphere has persisted, the longer it may be expected to persist, in exact proportion — while still departing with probability one.

That conclusion is *not* new, and it would be a misrepresentation to present it as such. Lindy longevity for technological civilizations is an active subfield: Balbi and Ćirković model the termination rate as a Weibull hazard and note that a decreasing rate yields a Pareto-like fat tail of exactly Lindy character [7]; Balbi and Grimaldi investigate Lindy's law for technosignature longevity directly, and show it can overturn the expectation that a first detection be long-lived [8]; Kipping and collaborators reach related conclusions about the age of a first contact from lifetime distributions [9]; and Ord surveys the mechanisms from which a Lindy effect can arise at all, including one that conjures it from an uninformative prior [10, 85].

What this section adds to that literature is one thing only. In each of those treatments the power law is *parameterized*: the exponent is a free shape parameter, chosen for tractability or fitted, and the question studied is what follows from a given value of it. Here the exponent is an *output* — $\frac{1}{4}$, and it would have been $\frac{1}{2}$ had the noise entered the log-hazard rather than its derivative (§5). The claim is therefore not that biospheres are Lindy, which is others', but that the Lindy exponent is fixed by the smoothness of whatever drives the hazard, and is in that sense a measurable structural fact rather than a modeling choice. Whether that addition is worth much depends entirely on whether the driver's smoothness is ever ascertainable, which this section does not claim.

*(d) The age distribution of extant biospheres is heavy-tailed.* Quantiles scale as the fourth power of rarity. Any sample of biospheres found at a given epoch is dominated by the old, and moments of order $\geq \frac{1}{4}$ diverge: the sample mean age of such a population is an artifact of the observation window, growing as $\mathcal{T}^{3/4}$.

### 6.7.1 Relation to anthropic reasoning

Golden reasoning stands in an unusual relation to the observation selection literature [86, 87, 34], and the relation deserves precision because it is the section's clearest methodological contribution. Gott's Copernican argument [87] derives a predictive distribution for a system's remaining duration from the premise that the observation time is uniformly distributed over the system's lifetime. The Doomsday argument [86, 34] proceeds similarly from a reference class and an indifference principle. In both, the age distribution is an *input*, justified by a principle of indifference.

In Golden reasoning the age distribution is an *output*. The tail $t^{-1/4}$ is not postulated; it is the persistence exponent of integrated Brownian motion, which follows from where the noise is placed in the hazard (§5) and nothing else. The reference class is not chosen; it is the conditioned ensemble $\mathbb{P}^{(t)}$, which the model constructs. The exponent could have been $\frac{1}{2}$, it is, for a mean-reverting driver, and which it is depends on the smoothness of the driver rather than on any epistemic principle.

The same contrast holds, and holds more sharply, against the technosignature-longevity literature, which is the nearer neighbor and is already model-based rather than indifference-based [7, 8, 9]. Those treatments carry a hazard function with a free shape parameter, Weibull $k$, or a Pareto exponent $\alpha$, and study the consequences of its value. The disagreement is therefore not about method but about where a parameter comes from: there it is chosen, here it is forced by the driver. Ord's survey is the useful corrective in both directions [10], since it shows how many distinct mechanisms deliver a Lindy effect — which is a warning that observing Lindy behavior would badly underdetermine the mechanism producing it, and hence that the exponent, not the effect, is where any inferential weight can sit.

One further neighbor in this family should be placed because its instrument is reusable. The hard-steps analysis, Carter's originally, given its modern Bayesian form by Snyder-Beattie, Sandberg, Drexler and Bonsall, infers from the timing of evolutionary transitions, under observer-conditioning, that the completed steps were individually improbable [86, 84]. That is timing run as an instrument on a conditioned record, and nothing in it is proprietary to the observer target: run under the Golden target the same instrument reads transition timings as evidence about route-narrowness, the quantity the invested reading prices (Proposition 7.6), and the hard-steps literature's machinery transfers whole, with only the conditioning attribute exchanged. Executing that transfer is a concrete open item **[O]**, and it is the nearest thing this section has to a ready-made empirical program built by others.

This does not vindicate the Copernican arguments; it relocates the disagreement. Where they require a defense of indifference, Golden reasoning requires a defense of the model. That is a better place for the burden to sit, because a model can be wrong in identifiable ways.

*Premise* $\Phi5$ (Rarity of origins; *imported result*). Biospheres are rare: the difficulty lies in origination, not in persistence past origination.

$\Phi5$ is the rare-earth position [88], and it is not imported from the general literature but from our companion program, where it is derived rather than assumed: the Nexic framework [5] argues that the astrobiological Copernican principle is inconsistent with the mediocrity commitments that were supposed to support it [22], and re-reads the natural-historical record accordingly [23]. Its interaction with the rest of this section is worth making explicit, because the combination is more coherent than either part alone. If origins are rare and survival is Lindy, then the observed silence requires no filter ahead of us: it is explained by the scarcity of starts, not by the destruction of the started. Golden reasoning and rare-earth are complementary — the first says that what persists is not extractive and therefore not loud, the second says there is little of it. Neither requires the Great Filter to lie in our future [89, 90].

### 6.7.2 Relation to the Nexic framework

The relation to that companion program is closer than a shared conclusion, and stating it precisely is worth a subsection, because the two halves turn out to partition the same problem.

*The same move, in opposite temporal directions.* The Accordance Principle constrains a realized natural-historical attribute $A$ against its complement $A'$, with $H$ the production of observers, by $P(H \mid A')/P(H \mid A) \lesssim P(A)/P(A')$: realized improbability must be compensated by observer-relevance [22]. Golden reasoning performs the same reweighting on a trajectory rather than an attribute, the Doob density $\mathfrak{h}(X_s)/\mathfrak{h}(X_0)$ tilting the path measure by the probability of the conditioning event downstream. Both are Bayes applied to a selection event; they differ in what is conditioned on and in which direction time is read. Accordance conditions on observers *having arisen* and reads backward over the record; Golden reasoning conditions on survival *to a future horizon* and reads forward over the trajectory. The two are complementary rather than redundant: retrodictive and predictive conditioning on the same kind of event.

*The companion names the degenerate case.* This is not a resemblance noticed from outside. The foundation paper itself identifies survivorship reasoning as the coarsest limiting case of its own constraint: restrict the state space to a single history's population and degrade the graded observer-condition $H$ to a binary survival event $S$, and the identity behind the principle reads $P(A \mid S)/P(A' \mid S) = \big(P(S \mid A)/P(S \mid A')\big) \cdot \big(P(A)/P(A')\big)$ — the textbook survivorship correction [22]. What the present section supplies is that limit developed *dynamically*: not a correction applied at one attribute but a measure on paths, with a limit object $Q$, an exponent $\frac{1}{4}$, and an interpolation between the two.

*Golden reasoning supplies what the companion declares itself silent about.* The Optimal-Earth paper states as a principled limitation that its framework is structurally silent on filters ahead: its constraint conditions on observer-production, which our own case has already secured, so no bound it issues bears on what becomes of observers afterward [23]. That is precisely the gap this section fills, because conditioning on $\{T > t\}$ is conditioning on a *future* event. The absence of a terminal stage, the Lindy residual life, the $t^{-1/4}$ tail and the goldening interpolation are all statements about what becomes of observers afterward. The two wings therefore partition the timeline at the observer: Accordance behind, Golden reasoning ahead.

*One identification, flagged rather than assumed.* The connection is a shared logical form plus one substantive identification, and the identification is not free. The companion's cosmic weighting is a primitive measure over partial information states; the $\mathbb{P}$ of §5 is a path measure induced by a specific model. Reading the two as one object requires taking the model's trajectory space as a family of partial states and its $\mathbb{P}$ as the cosmic weighting restricted to them — a move neither paper performs. Absent it, what is shared is the logical form, which is worth stating and is not an entailment. Nothing in this section's results depends on the identification being made.

*An apparent clash, dissolved.* The Optimal-Earth Conjecture [23] holds our record to be observer-optimal on effectively any attribute one can interrogate, while §6.2.2 holds that we are not practically Golden to any interesting tolerance. These do not conflict, and the reason is the temporal split just drawn: the first conditions on observer-production and grades the realized past, the second conditions on survival to a future horizon and measures distance from $Q$. A record may be optimal in every respect Accordance interrogates while its bearer sits at small $t$ on the goldening interpolation. Optimality behind is not Goldenness ahead.

### 6.7.3 Relation to the existential-risk and collapse literatures

Two established positions occupy ground adjacent to Φ3 and the two-condition theorem, and both were found after those results were written. They are recorded here because they narrow what is claimed as new — and because, on inspection, they support the premise rather than threatening the paper. *Collapse from declining returns.* Tainter's account of societal collapse [82] holds that complexity is adopted as a problem-solving strategy, that its marginal returns decline, and that past a threshold further complexity is pure cost, at which point collapse becomes likely. Structurally this is the U-shape of Proposition 5.35(iii) arrived at from archaeology thirty-six years earlier: a benefit-minus-cost curve in capacity with an interior optimum, beyond which acquisition is net harmful. The correspondence is close enough that the boundary-cost argument of §6.6.1 should be read as a re-derivation of Tainter's mechanism in hazard-theoretic terms rather than as an independent discovery of it.

Three differences remain, and they are what this section adds. Tainter's declining quantity is *economic and energetic return*; ours is *hazard*, so the two curves are about different things even where their shapes agree. Tainter's threshold is qualitative; ours is $x_*$ of (5.20), in closed form and with a computed floor $h_*$. And Tainter's account contains no analog of the escape factor, so it cannot state the two-sided result of Theorem 5.38 — that capacity growth is simultaneously *necessary* for viability and, absent anatomy, fatal.

*The vulnerable world.* Bostrom's hypothesis [76] holds that continued technological development will at some point attain capabilities making civilizational devastation extremely likely unless civilization exits what he calls the semi-anarchic default condition, with stabilization requiring greatly amplified capacities for preventive policing and global governance. The mapping into the present vocabulary is direct and should be stated without hedging: *the semi-anarchic default condition is low technoanatomy, and Bostrom's stabilization conditions are the raising of $\rho$.* Quantitative development of the hypothesis has begun, using survival-function methods on the urn model [91], which is the same class of object as §5.4.

What survives the comparison is narrower than the anti-acceleration result of Remark 5.41 might suggest on first reading, and is stated narrowly. The vulnerable-world hypothesis is an *existence claim about a discrete draw*, that the urn contains a black ball, whereas $\rho$ is a *continuous scaling relation* with an exponent derived from boundary geometry. And the vulnerable-world hypothesis is one-sided: development is dangerous. The two-condition theorem is two-sided, requiring the same quantity that it makes lethal, and the product structure of (5.23), in which one factor annihilates the other, has no counterpart there.

*Why this strengthens rather than weakens the section.* $\Phi 3$ is conjecture-grade and argued from a single line of reasoning. It now has two independent precedents reaching a compatible conclusion from unrelated evidence: one archaeological and economic, one technological and policy-facing. A premise supported by convergence across literatures that share no premises is in better standing than one argued from scratch, and the ledger's grading of $\Phi 3$ should be read in that light — still conjecture-grade, since none of the three derivations is a proof, but no longer solitary.

## 6.8 The epistemic ceiling

We close with a limitation which constrains this section's own conclusions at least as much as its targets'.

**Theorem 6.18** (No finite observation certifies Goldenness)**.** *$Q$ is locally absolutely continuous with respect to $\mathbb{P}$: on every finite window the two measures are mutually absolutely continuous on the survival event, with the explicit density of §5, finite on every finite window. They become singular only on the infinite-horizon $\sigma$-algebra. Consequently no observation of bounded duration can establish that a biosphere is Golden rather than General, and no bounded evidence can place a system on the interpolation $\mathbb{P}^{(t)}$ at one $t$ rather than another beyond the resolution the bias field permits.*

*Remark* 6.19 (What "in principle" may and may not mean here). The theorem's quantifier is *bounded duration*, and prose elsewhere in this work occasionally compresses that to *in principle*. The compression should be read narrowly, for two reasons.

*It is a limit on inference, not on determinacy.* Nothing above says there is no fact about whether a biosphere is Golden. It says no finite record distinguishes the measures, which is compatible with the fact being fully settled — and in the Route World of §8 it is settled, since statedness is a property of an Aethus and not of its bearer's access to it (Remark 8.20). Determinate and inaccessible is this framework's normal condition, not a special pleading.

*And the unavailability is one-sided.* The measures are singular on the infinite-horizon $\sigma$-algebra, so the infinite record does distinguish them — but that record is available only to a biosphere that persists to obtain it. A General biosphere departs at finite $T$ and never holds it; a Golden one could, in a limit it never finishes reaching. So the ceiling is not symmetric ignorance. *No biosphere ever learns that it is General*, and a Golden one learns it only asymptotically — the same shape as the fixation pole's unprovability in §5.7 and the affirmative answer's stability in §2.4. This framework's uncertifiable propositions are uncertifiable in the direction that would have been consoling. The live third term's interaction with this structure, amplification without breach and a possible sharpening of the one-sidedness, is priced at Remark 5.71, and the climb's asymptotic phenomenology, the extreme-induction regime, at Remark 5.72.

**What now stands on the ceiling.** This subsection is short because the theorem is; its weight is elsewhere, and the load map belongs at the site. Five results in this work stand directly on the ceiling. The Precipice engagement's third delta — existential security decomposes, and known-to-be-low is the half the ceiling forbids (§3.4.6). The anti-dogma delta of the reduction-refusal — dogma requires certifiable favor, and a structure that proves no one certifies the favorable side cannot function as a salvation guarantee (§9.10). The Kierkegaard consilience — the third route is held rather than possessed, commitment under proven uncertainty (§9.9). The node criterion's discipline — verdicts on commitment primitives are disciplined, never certified (§9.7). And the arrival-timing refusal — no one certifies the occupant (Remark 9.12). Two companions complete the map: silent forfeiture is the ceiling's factual twin, what cannot be certified also does not announce itself (Remark 10.6), and one statement is owed here because it is consumed everywhere: *reachable is never certifiable.* In the viable regime the Golden Point is reachable under the plain unconditioned law, positive probability, no conditioning required, the survivorship flow merely saturating onto the honest conditional, and the ceiling stands regardless: a biosphere can be on the favorable side of the dichotomy while no finite observation ever places it there. Reachability lives in the model; certification is what the model withholds; and the pairing is this framework's epistemology stated as measure theory, local equivalence on every window, singularity only at the horizon, with no translation step between the mathematics and the moral.

**What this costs each party.** Against the violent-aliens thesis: no finite observation of a biosphere, however extensive, could establish that it is predatory in the asymptotically relevant sense, so the thesis cannot be confirmed by encounter. Against *this section*: no finite observation could establish that an encountered biosphere is *ideally* Golden either. The claims of §6.7 are claims about a conditional distribution, not predictions about any individual system, and they must not be read as the latter.

The ceiling is, however, specific to the ideal. Practical Goldenness (Definition 6.3) is a finite-window property, and Theorem 6.18 does not forbid establishing it: mutual singularity is an infinite-horizon phenomenon, and on $\mathcal{F}_s$ the two laws are equivalent with bounded density. What obstructs the practical version is not structure but sample size — one would need an ensemble of biospheres, and we have one. That is a contingent barrier rather than a theorem, and the two kinds of limitation are worth keeping apart: the ideal category is unknowable to any finite observer, the practical one merely out of reach.

This symmetry is not a weakness of the position but a consequence of the structure that the position is built on, and we would regard its absence as a warning sign. An argument from survivorship that permitted confident identification of survivors would have proved too much.

### 6.9 What the section costs and what it buys

The apparatus is priced here in the companion's format [23]: costs enumerated with grade and falsifier, buys enumerated after, and the whole kept short enough to audit.

**Costs.** Six load-bearing commitments, no more, and their grades are not uniform.

1. *The biospheric referent* (Φ1) — **methodological stipulation**. *Falsifier:* exhibit a unit of analysis referring to the same object at every point of the trajectory while carrying less semantic content than "biosphere". The stipulation is that none exists; one counterexample retires it, and the theorems would then be applied in that unit instead.
2. *Vantage, not obligation* (Φ2) — **scope declaration**, and a self-imposed restriction rather than an assumption doing work. *Falsifier:* exhibit a normative conclusion derivable from the apparatus without adding a value premise. Its failure would be an expansion of the section's reach, not a defeat — but it would defeat the claim that Golden reasoning is not longtermism, which is why it is stated here rather than assumed.
3. *The Extraction Hypothesis* (Φ3), **conjecture-grade**, argued from boundary cost but not derived, and the section's principal exposure, though no longer a solitary one: §6.7.3 records two independent precedents [82, 76] reaching compatible conclusions from unrelated evidence, which is convergence rather than proof but improves the premise's standing. In its present form it is the claim $\rho < 0$ for extractive organization (Definition 5.32), which is why Corollary 6.11 follows from it mechanically and why §6.4.3 needs no further premise. Its position in the argument is exact and worth stating in the ledger: it is not a premise of any theorem, but the *identification* that connects the theorems to their application, so its failure does not falsify anything in §5 — it *empties* §6.7, leaving the selection results intact and silent about predation. *Falsifiers:* the three of Remark 6.17, all operational.
4. *Origin does not transfer* (Φ4) — **premise**. *Falsifier:* a mechanism by which the selective regime that produced intelligence in a lineage transfers to the regime governing a biosphere's persistence at world scale. Note this is the weaker of the two anti-violence commitments and can stand alone: Φ4 alone establishes that predatory maturity is not *inherited*, and only Φ3 adds that it is selected *against*.
5. *Rarity of origins* (Φ5) — **imported result**, carried at the companion's grade and not re-argued [5, 22, 23]. *Falsifier:* the companion's, and nothing this section adds.
6. *The inherited mathematical apparatus* — **[A]** and below. Every quantitative claim here, the $t^{-1/4}$ tail, the Lindy residual life, the $s/4\varepsilon$ criterion, the measured truncation, inherits

the tags of §5, in particular the soft-wall Ansatz. *Falsifier:* the numerical tests specified there, chiefly a fitted exponent robustly away from $-\frac{1}{4}$.

One further entry belongs in this column, though it is not a premise: *the section's novelty is narrower than its apparatus.* The mathematics of §5 is classical or routine, the $\frac{1}{4}$ exponent is McKean's, the background construction is an application of known results on sums of random exponentials, the Lindy conclusion of §6.7 is established prior work [7, 8, 9, 10]; and the mechanism behind Φ3 has precedents in the collapse and existential-risk literatures [82, 76, 91]; and the conclusion of §6.4.2 is the standing technological-adolescence argument of SETI, already current and already rebutted [6]. What is claimed there is a mechanism that survives the rebuttal, not the conclusion. What is claimed as new is the assembly and three specific items: the measure-theoretic recasting of the survivor category, the practical/ideal ratio criterion, and the epistemic ceiling. A reader who finds any of those three anticipated should discount the section accordingly, and the surrounding apparatus provides no independent support for them.

One accounting note closes the column, and it governs what all six costs are costs *of.* The unconditional list is short, and the ledger states it in full: that conditioning selects, that adverse anatomy is fatal (Theorem 5.36), and that viability requires two conditions rather than one (Theorem 5.38). The mutual singularity of $Q$ and $\mathbb{P}$ is *not* on that list; it is critical-regime only. What is priced is every sentence connecting that structure to biospheres, aliens, or the future. No rhetorical weight anywhere in this section is permitted to exceed the grade of Φ3, and where a consequence is stated in §6.7 it is stated as downstream of it.

**Buys.** Against those six, the section returns the following.

The categorical division is *unchosen*, in the critical regime: General and Golden are mutually singular measures there, so the split is established by a theorem rather than by a threshold someone placed, which is what Pre-Civilization/Civilization and every Kardashev boundary cannot say. §6.2.1 records the price, the claim is unavailable in the viable regime, and the entry is worth exactly what it is worth after that subtraction. Practical Goldenness is *computable*: $t \gtrsim s/4\varepsilon$, with the invariant $s_\varepsilon(t)/t = 4\varepsilon$, so the question "how Golden is this" has an answer with quantities in it. The premise is *weaker than it looks*: age alone confers practical Goldenness, and age is what the question already supposes, so the apparatus assumes there is a future rather than that the future goes well. The age Lindy *exponent* is derived rather than parameterized — a narrower buy than it first appears, and stated as such in §6.7, since Lindy longevity for civilizations is established prior work [7, 8, 9, 10] and what is added is that the exponent is fixed by the driver's smoothness rather than chosen. Against the indifference-based anthropic tradition [86, 87, 34] the contrast is larger, the age distribution there being postulated outright; but the nearer neighbor is already model-based, and the net gain is one parameter rather than a paradigm. The epistemic ceiling is *symmetric*, constraining the section exactly as tightly as its targets. And the section *fills a declared gap* in the companion: conditioning on $\{T > t\}$ is conditioning forward, which is what [23] states its own constraint cannot do.

One entry is a positive expectation rather than a dissolution, and it is entered deliberately, since an apparatus whose entire return is relief from other people's difficulties is answerable to nothing. The truncation/rescaling asymmetry of §6.3.1 is not a fact about biospheres; it is a fact about survivorship conditioning on a two-coordinate state. It should therefore appear in *any* domain carrying survivorship selection and an orientation-like coordinate alongside a magnitude-like one — firms, institutions, lineages, any population where the old are selected from the many. The prediction is specific: the orientation coordinate should show a deleted region among survivors, the magnitude coordinate a shifted distribution of roughly preserved shape, and the two divergences should be comparable in magnitude while opposite in kind. That is checkable outside cosmology, on data that exist, and it would embarrass the apparatus if the selection turned out to act on magnitude categorically and orientation quantitatively instead. It is entered here as an expectation and not a result.

### 6.10 What this section does not claim

The fence is stated hard, on the companion's discipline that an apparatus which prices what it claims must also refuse what it does not [23]. It is not claimed that any observed system is Golden, ideally or practically; by Theorem 6.18 the ideal version is unknowable to any finite observer (Remark 6.19) and the practical version is out of reach for want of an ensemble. It is not claimed that the companion's cosmic weighting and the path measure $\mathbb{P}$ of §5 are the same object; that identification is flagged as unperformed in §6.7.2 and nothing here depends on it. It is not claimed that any individual biosphere, ours included, will behave as §6.7 describes; those are statements about a conditional distribution. No normative conclusion is claimed ($\Phi$2). And the companion's natural-historical program, its conjectures, its bounds, its paleontological predictions, is not claimed, imported, or re-argued; $\Phi$5 enters as a stated result and nowhere else.

One thing remains on the far side of the fence, because a fence is legible only if what it excludes can be seen. A program built on this section would inherit three commitments, stated here as refusable rather than as claims. It would be committed to the boundary-cost scaling of $\Phi$3 showing up in institutional and ecological data, not merely in argument. It would be committed to the truncation/rescaling asymmetry appearing in survivorship-selected populations generally. And it would be committed to the exponent's dependence on noise placement, $\frac{1}{4}$ for integrated drivers, $\frac{1}{2}$ for Brownian ones, being visible in any domain where hazard dynamics can be measured directly. None of the three is established here, all three could be found false without touching a line of §5, and that is the point of naming them.

### 6.11 Section summary

The unit is the biosphere ($\Phi$1), because no other term survives the limit the theorems are taken in. The fundamental division is General versus Golden, $\mathbb{P}$ versus $Q$, and it is categorical without being chosen, since in the critical regime the two are mutually singular; but nothing occupies the Golden category *as an ideal*, which is a measure to reason from rather than a state to be in, and this is what distinguishes Golden reasoning from longtermism ($\Phi$2). The practical form is occupiable and is what the section actually assumes: a biosphere of age $t$ is $\varepsilon$-Golden over windows of length $4\varepsilon t$, a constant fraction of its own history, so that the premise of Golden reasoning is not that the future goes well but merely that there is one. The two coordinates are technostrength, a state coordinate, and technoanatomy, a structural parameter, and conditioning treats them differently in kind: adverse strength is truncated abruptly, 137-fold, while anatomy is selected gradually, $\mathbb{P}(\rho > 0)$ moving only from 0.75 to 0.84 over the same horizon. The second is the slower and the more decisive, since adverse anatomy is eventually fatal with certainty while adverse strength is a condition every biosphere occupies half of logarithmic time and survives. The Kardashev scale fails not because its thresholds are arbitrary but because the terminal stage it points at is gated by a technoanatomy the scale cannot index, so that climbing is neither necessary nor sufficient for arriving — and, with adverse anatomy, climbing fast is itself the failure. The violent-aliens thesis fails at the inference from origin to maturity (Φ4), and fails further if extraction is bad technoanatomy (Φ3) — which we argue from boundary cost, recovering the square–cube law from which the names came, and which we mark as the section's one load-bearing premise, with its falsification conditions. Downstream: encountered biospheres are not extractive, megastructure nulls are weak evidence, and old biospheres are Lindy with a derived exponent rather than an assumed one — which is where Golden reasoning improves on Copernican anthropics, by making the age distribution an output rather than an input. With rarity of origins (Φ5), the silence needs no filter ahead. And by the epistemic ceiling, none of this can be confirmed by encounter — a limitation that binds this section exactly as tightly as it binds what the section argues against.

# 7 Retrocausal accordance: the principle, the argument, and the target

The Accordance Principle constrains one's past: a present that would be a fluke relative to one's Nexus is evidence against that Nexus. Run forward, it constrains one's future, and this section develops that forward form as one cluster: the definition of *retrocausal* that keeps it clear of teleology, the principle in both its forms, the worked example at human scale on which a reader can check it, and the argument it powers — that the future attribute accounting for our occupying the present rather than deep time must be correlated with the whole of modern development to a degree with no ordinary analogue, and that indefinite biosphere survivorship is the candidate that fits.

The cluster is graded with more care than the rest of this work, not less, because it sits nearer to teleology than anything else here. The principle is [Φ] at the companion's grade; the argument's quantitative core is [Φ]; the identification of its conclusion with the Golden Point is [**S**], an inference to the best explanation, and says so; and the epistemic ceiling applies to all of it.

## 7.1 The word, before the principle

**Definition 7.1** (Retrocausal conditioning)**.** *Retrocausal*, in this work, means *conditioned on an Aethus lying in one's future light cone*, and nothing further. Golden Optimation is retrocausal in exactly this sense and in no other.

**What the stipulation does and does not buy.** The definition is deliberately dry, and the dryness is what keeps it inside Remark 3.40's discipline. Nothing propagates backward; no event acts on its own past; the verbs remain *is conditioned*, *is reweighted*, *is selected*, and the forbidden ones, *aims*, *seeks*, *progresses toward*, are as forbidden here as anywhere else in this work.

What should nonetheless be said plainly is that the stipulation lands closer to the classical notion than a purely evidential reading usually does, and for a structural reason rather than a rhetorical one. In a setting where causation is fundamental, conditioning on a future event is uncontroversially inference and uncontroversially not causation, because there is a further fact about what caused what. This framework has no such further fact: causation is emergent from the Aethic structure rather than primitive, and determinacy is indexed to an information state rather than to a date, so that an unresolved future outcome is a blank entry and blanks lack tense (Remark 3.22, [26]). With no primitive causal relation underneath, the distinction between *influence* and *conditioning* has nothing to consist in beyond the conditioning itself. The reading offered here is therefore the deflationary one, and the proximity to the inflationary one is recorded as a feature of the framework rather than as a claim about the world.

## 7.2 The principle, in two forms

The companion's Accordance Principle constrains one's past: a present that would be a fluke relative to one's Nexus is evidence against that Nexus. Run forward, it constrains one's future.

**Principle 7** (Retrocausal Accordance; [Φ])**.** Let $A$ be a realized attribute of one's Aethus with canonical alternative $A'$ its complement, and let $B$ be an attribute of one's future cone. Then

$$\frac{\mathbb{P}(B \mid A')}{\mathbb{P}(B \mid A)} \lesssim \frac{\mathbb{P}(A)}{\mathbb{P}(A')}, \tag{7.1}$$

weights taken under the Cosmic Nexus, with $\lesssim$ carrying the companion's exceedance modality and no free tolerance: for any $\kappa \geq 1$,

$$\Pr\left[\frac{\mathbb{P}(B \mid A')}{\mathbb{P}(B \mid A)} > \kappa \frac{\mathbb{P}(A)}{\mathbb{P}(A')}\right] \leq \frac{1}{1+\kappa}, \tag{7.2}$$

the probability taken over the realization [24]. The motto: *one's present cannot be arbitrary with respect to one's future.*

The principle is the Accordance Principle with one substitution, and the substitution is the whole of what is new. There, the conditioning attribute is the production of an observer and lies behind one; here it is an attribute of the future cone. Everything else transfers unchanged — the canonical alternative is the complement, so nothing is analyst-supplied; the draw is the same Copernican one; and by Bayes (7.2) is exactly the statement that the realized member carries $B$-conditioned weight below $1/(1+\kappa)$, which for $\kappa \geq 1$ at most one member of a pair can do. What licenses the substitution is that the framework indexes determinacy to an information state rather than to a date, so a future-cone attribute is available to condition on in the same sense a past-cone one is, (Remark 3.22).

Two readings to keep apart. *Not a promise:* the principle does not say one's flukes will pay off. It says futures in which they do not are disfavored in the stated ratio, which is an inference from the fluke and not a claim about what the future will do — Remark 3.40 applied here. *And not a tolerance:* the $\lesssim$ prices violations rather than forbidding them, and (7.2) fixes the entire tail profile, so a $\kappa$-fold exceedance is not a counterexample but an event of probability at most $1/(1+\kappa)$.

**The exceedance form is the multiplicity guard.** This bears directly on the look-elsewhere worry raised above, and supplies more of an answer than was there credited. Because (7.2) bounds each interrogated pair's violation probability, linearity bounds the expected number of $\kappa$-fold violations among $n$ interrogated pairs by $n/(1+\kappa)$ — an exact noise floor, supplied by the principle's own tail profile rather than imposed from outside [24]. Scanning a hundred attributes, one should expect up to ten spurious nine-fold flags and one spurious ninety-nine-fold flag; a trace standing clear of that floor is forced, and one that does not is not.

So fluke-hunting in a Route World is not the undisciplined activity §8 first suggested. It has the same multiple-comparison discipline every inference of the kind owes, with the correction computed rather than chosen. The concession the companion records applies here with full force and is the operative caution: *n must count the attributes implicitly scanned and not merely those explicitly interrogated*, since a striking attribute is interrogated because a scan flagged it. In a Route World the scan is one's whole attention, and a correct $n$ is correspondingly enormous — which is a reason the identifiable traces must be conspicuous by a wide margin before they are worth anything, and not a reason there are none.

**Principle 8** (Retrocausal Accordance, bound form; [Φ])**.** For an Aetherian whose Aethus is $H_1$, and a stated attribute $A \supset H_1$ that is unlikely under the Cosmic Nexus, the probability of centric unfolding to a future Aethus containing stated attribute $B$ is bounded above by the probability of $A$ given $B$ under the Cosmic Nexus:

$$\mathbb{P}(B \mid H_1, N_0) \;\lesssim\; \mathbb{P}(A \mid B, N_0) \qquad (H_1 \subset A,\ \mathbb{P}(A \mid N_0) \ll 1), \tag{7.3}$$

with $\lesssim$ carrying the exceedance modality of (7.2).

The two forms are one principle read from either side. (7.1) is the ratio form, and connects to the companion; (7.3) is the bound form, and is the one on which the motto reads literally — futures that would leave one's present a fluke are capped in weight by how far they would leave it one. The derivation of the second is the companion's own argument run forward. If $A$ is exceptionally unlikely with respect to a future Aethus $H_1 \cap B$, then one's currently standing in $H_1$, for which $A$ has occurred, is already a fluke with respect to that future; and since one's future Aethae will eventually be one's own, and so warrant the first postulate on one's behalf, a present that is a fluke relative to any future one might reach violates the mediocrity constraint. The weight of unfolding into such a future is accordingly constrained in proportion to its dubiousness [5].

### 7.3 A worked case, at the scale a reader can check

The examples elsewhere in this work are cosmic. A principle that works only at cosmic scale is suspect, and the following runs it on a single attribute of a single life, which is the grain at which a reader can hold it against their own experience.

In grade school I was told, on a few occasions, that I had an exceptionally strong presentational speaking ability. Holding that in mind while deriving Nexic reasoning in 2021[22], I had a thought that struck me as odd and then as forced. *If I go my entire life without ever needing*

[22] A provenance note, so that the example is not misdated. The presentational-ability trait is what I used to *formalize* the retrocausal accordance principle in 2025, when a long-standing intuition resurfaced unprompted after a presentation I gave at a summer program. The 2021 version was not presentation-based, or not solely: it ran on a blend of particularly immense flukes I had witnessed through childhood, each of them plausibly conditioned on my later being able to produce these frameworks at all, and to varying degrees of confidence. The reasoning was the same; the attribute I hung it on was not. Two further notes belong here rather than in the body. First, the principle has failed in my hands: on a day in April 2026 I ran a set of guesses through it that came out wrong, and for an instructive reason — every inference had been conditioned on a particular result being important that turned out to be trivial. Human error in the Route World's leap of faith is as available as it ever was, and the failure is recorded because a worked failure is worth more to a reader than any success [92, 93]. Second, that failure raised a question the paper's ontology makes askable and the classical one cannot: is the counterfactual in which the result was never posted one in which it was *trivial all along*, or one in which the triviality is *joint* with the posting in this Aethic reality? Here "trivial" is a trait of the initial conditions of human mathematical knowledge rather than of the mathematics itself. Aethically both are live, and the question is not idle, since it bears on whether Golden causality's demotion of an event is a fact about the event or about the branch. The journal record of the episode, with the surrounding personal circumstances, is held in the corpus's archive material and is not reproduced here.

*that skill, does my having it now not represent an immense fluke with respect to that life?* Take the set of lives I might live in which strong speaking ability is never called on. Living the one I am in, where I have it, would be a decided fluke in that set. So, as an effective proof by contradiction to keep this waking moment from being a fluke, I would have to infer that somewhere in my future at least one task requiring strong speaking ability was coming. I had constrained an attribute of a future Aethus of mine from nothing but my own standing fluke content. And I appear to have been right, since I now know I needed exactly that ability to write and present the papers you are reading.

Two things about the example, and the second is the important one. *What it displays.* It is the retrocausal accordance principle applied to one attribute: the fluke is *having the ability*, the future attribute is *a task requiring it*, and (7.3) says the weight of futures in which no such task comes is capped by how much of a fluke the ability would be under them. Every Aetherian may constrain their own future this way from their own present, and the principle's content is exactly that this is legitimate — not that the constraint is tight.

*What it does not test.* The confirmation is retrospective, and a single retrospective confirmation is one draw. The principle predicts that standing flukes will *tend* to pay off, at the exceedance grade, $\kappa$-fold failures at probability at most $1/(1+\kappa)$, and one instance neither confirms nor refutes a tendency. The example is included because it is where the principle is checkable in kind, not because it is evidence of the principle's truth; a reader who runs it on their own standing flukes and finds it wanting has learned something the paper's cosmic cases could not have taught them.

**Two directions, one of them deductive.** The retrocausal accordance principle and Golden Optimation (§8.1) bear on the same fluke traits from opposite sides, and conflating them is the error most worth guarding against here. Golden Optimation runs *from* the conditioning target *to* the trait: if the Golden Point required it, one has it. That is deduction under the conditioning, and it explains *why* one has the trait. The retrocausal accordance principle runs *from* the trait *to* the future: having it, one is constrained toward futures in which it pays. That is inference, at exceedance grade, and it explains nothing about why one has the trait — only what one's having it bears on.

The converse of the deductive direction does not hold, and this work is careful never to run it. A fluke one possesses need not have been required by anything: to argue from "I have this trait" to "the Golden Point required it" would be to run the deduction backward, and nothing here does. What *does* hold is the forward form only, required, therefore had, which is why in a Route World the question one would face before a notoriously difficult feat is not *will I achieve this?* but *is this genuinely required, and am I the one to do it?* That is a different question, and it remains a leap because the requirement may simply not hold, in which case even in a Route World the feat is not probabilistically privileged as an Aethic event. Remark 8.22 says the rest: no one can determine whether their own feat is among the requirements, and the reformulated question is therefore not one anyone can answer in their own favour.

## 7.4 The finale paradox, and the argument that solves it

**The paradox.** Why does one find oneself in the modern day rather than at any earlier point in Earth's natural history? A uniform draw over the record lands in the last century with probability of order $10^{-8}$; and the compounding of the companion's doomsday channels worsens this, since at net hazard $\lambda$ the survival weight is halved by $\ln 2/\lambda$ and drops toward zero thereafter, so that a position late in a surviving history is rarer still. Being *here* is a fluke of enormous magnitude on the received reading, and the received reading has no resource to discharge it. Call this the *finale paradox*. The retrocausal accordance principle discharges it, and the discharge derives the shape of what one must be conditioned on.

*Remark* 7.2 (The paradox's two faces, and the relay it marks)*.* The paradox stated above is the Golden-side face of a question the companion program raises on its own account, and the pair should be seen together, because the relation between them is this work's point of contact with the companion rather than an echo of it.

*The companion's face.* Nexic reasoning reads a record conditioned on observerhood and finds Optimation in it. That reading raises a question the companion names and explicitly declines to answer from the Accordance Principle alone, on *hypotheses non fingo* grounds: *when does Optimation cease to apply?* [4]. The question is not idle housekeeping. Observer-conditioning is conditioning on an event in one's past cone, so the tilt it supplies is exhausted once the event has occurred, and a conditioning that is exhausted must have a terminus — yet nothing in the companion's apparatus locates one, and anything one proposed would be stipulated. The companion therefore carries the question open, and marks it as the frontier.

*The dissolution.* Golden conditioning does not answer the question; it removes the presupposition that generates it. If the operative conditioning event lies in the *future* cone, then Optimation is not a process running over an interval but the past-cone restriction of a conditioning that is not exhausted at all, and there is no terminus to find because there was never a duration to terminate. Remark 8.4 is the mechanism stated exactly: a required fluke is Golden Optimation before the observer's present reaches it and Optimation in the companion's sense afterwards, same event and same conditioning, with only the observer's position changed. So "when does Optimation stop" asks for the date at which a shadow ceases to be cast by an object still standing. What replaces the question is an account of what Optimation *is* from first principles, the past-cone shadow of a future-cone conditioning, which is not a smaller answer than the one sought but a different and prior one.

*What this makes the paradox.* It is the designed relay point between the two developments: the place where the companion's open frontier is handed across and dissolved rather than solved, and the reason the two programs are one apparatus read in two temporal directions (Remark 8.5). A reader tracking the corpus should take this subsection as the join, and not as a further application of the companion's machinery within the companion's scope.

*And it fixes what the magnitude below is for.* Two things are being done in this subsection and they carry different exposures. The *dissolution* is structural: it consumes the direction of the conditioning and nothing quantitative, so it survives any revision of the draw discussed at Ansatz 7.3. The *filter* of Proposition 7.4 is quantitative and does not: its force is the smallness of a ratio, and a different weighting over positions gives a different bound. A reader who rejects the draw therefore keeps the relay and loses the filter, which is the same one-way dependency this work observes elsewhere, applied here inside a single subsection.

*One consequence runs back to the companion.* The companion's ancestral-agent discussion rests part of its methodological caution on the finale question's being open — the frontier's openness is what makes premature verdicts about which eras Optimation covers illegitimate there [4]. Under the dissolution the caution is not withdrawn but re-grounded, and strengthened: there is no era in which the conditioning fails to reach, so a verdict resting on Optimation's having powered off before some ancestral stage is ill-posed rather than merely unsupported, its presupposition having failed. Whether the companion should be revised to state it that way is for the companion **[O]**.

**Ansatz 7.3** (The positional draw over the record; [Φ])**.** The weighting against which the present is assessed as improbable is *duration-uniform* over the record, hazard-weighted by the companion's doomsday channels: it is the elapsed-time measure that delivers $\mathbb{P}(A \mid N_0)/\mathbb{P}(A' \mid N_0) \sim 10^{-8}$, driven lower by the compounding.

**What the draw costs, and the bracket in both directions.** Ansatz 7.3 is posited, and the reason for naming it is that the Cosmic Nexus does not obviously deliver it. The companion's weighting runs over Aethae, observer-situations, and not over elapsed time [5, 24], and the two need not agree even in order of magnitude. The bracket runs both ways, which is why the ansatz is a real posit rather than a formality.

*Downward.* Restrict the answer space to Aethae capable of posing the question, an interrogation-indexed reading of the observer tilt, and such situations are concentrated exactly where the questioner is. The ratio deflates toward order one, the bound (7.5) goes slack, and the filter selects nothing. *Upward.* Weight instead by observer-situations across the whole record, with the pre-human ones counted as the companion's own treatment of ancestral agency declines to demote them, and the ratio is very much *worse* than $10^{-8}$: the filter tightens, and the argument's conclusion strengthens rather than fails.

So the ansatz is not a convenience chosen to make the argument work, one natural alternative sharpens it and the other dissolves it, and the honest statement is that the filter's force is exactly as secure as the weighting, which is presently not derived. *Falsifier and repair.* The repair is a derivation of the positional weighting from the Copernican draw: which Aethae populate this attribute's answer space, and at what occurrence weights. Until it exists the filter is [Φ] on the ansatz, and a reader who declines the ansatz holds Remark 7.2's dissolution intact **[O]**.

**The argument, in four steps.** *Step 1: the modern-day attribute set.* Let $A$ be the conjunction of stated attributes about our biosphere that apply exclusively to one's living in the present as opposed to any earlier stretch of natural history — that the hottest and coldest temperatures in the observable universe both exist on Earth [94, 95]; that the first vertebrate to pass the Kármán line was born within the last century [96]; that unlocking dormant genes by engineering is suddenly possible [97]. Any parent $A_i$ of $A$ is a subset of these developments.

*Step 2: retrocausal accordance forces a future requisite.* By (7.3), for each $A_i$ there is more likely than not some future stated attribute $B_i$ for which $A_i$ is a near-necessary prerequisite as against one's having been born far back in deep time. This is an optimisation query. For an $A_i$ containing only a fraction of the developments, having *no* future $B_i$ that takes the rest as necessary is unlikely by the principle; for the whole of $A$, a handful of developments necessitated by no future is permitted by the exceedance margin. So there is some $A_j$ for which the significant majority of modern developments root to a single future stated attribute $B_j$.

*Step 3: the shape of $B_j$.* $B_j$ must render those developments categorically necessary as prerequisites, against their not having occurred across deep time. It is therefore not merely correlated with human capacity to manipulate the environment; it is essentially the stated attribute with the strongest possible correlation to it — a maximally connected graph in which one future attribute takes every modern development as a total necessity.

*Step 4: the suppression bound.* This is the quantitative core, and it is a proposition.

**Proposition 7.4** (The accounting future must be almost perfectly correlated; [Φ]). *Let $A$ be the conjunction of attributes distinguishing the present from any earlier stretch of the record, $A'$ its complement, occupying some earlier stretch, and let $B$ be a future attribute for which Principle 7 is satisfied. Partition $B$ into a component $C$ almost perfectly correlated with $A$ and a component $C'$ approximately independent of it, with weight $\alpha$ on the second. Then, since $A' \cap C$ is near-invalid and $A'$ is independent of $C'$,*

$$\mathbb{P}(A \mid N_0 \cap B) \;=\; \frac{(1-\alpha)\mathbb{P}(A \mid N_0)}{(1-\alpha)\mathbb{P}(A \mid N_0) + \alpha\,\mathbb{P}(A' \mid N_0)\mathbb{P}(C' \mid N_0)}, \tag{7.4}$$

*and requiring the present to be no less weighted than the alternative gives*

$$\alpha \;\lesssim\; \frac{\mathbb{P}(A \mid N_0)}{\mathbb{P}(A' \mid N_0)}. \tag{7.5}$$

The bound's force is in its magnitude. A uniform draw over the record places the last century at some $10^{-8}$, and the exponential compounding of the companion's doomsday channels drives the ratio further down — at net hazard $\lambda$ the weight is already halved by $\ln 2/\lambda$, and ten such half-lives put the ratio near $10^{-11}$. So $B$ must track $A$ to within one part in $10^8$ or far tighter: *any* independent component of $B$ larger than that opens a channel through which the vastly more numerous earlier positions dominate, and $\mathbb{P}(A \mid N_0 \cap B)$ collapses with the full rapidity the ratio implies.

**What the bound is evidence for, and what it is not.** Read as a constraint the proposition is strong and the reading is narrow: whatever future attribute accounts for one's occupying the present rather than an earlier stretch must be correlated with the totality of modern development to a degree with no ordinary analogue. That is a filter on candidates, and most candidates fail it — a peripheral future development correlates with some of the record's recent attributes and is independent of most, which is exactly the $\alpha$ the bound forbids.

The step from there to the Golden Point is weaker and should be marked as such. It is that the Golden Point satisfies the filter more plainly than the alternatives on offer, since indefinite persistence of the biosphere is the one candidate that takes essentially every capacity the record has recently acquired as prerequisite rather than as incident. That is an argument from the absence of a better candidate, and it inherits the exhaustiveness problem Remark 8.21 records: the filter is computed, the survey of what passes it is not.

Two further cautions. The partition into $C$ and $C'$ is a modelling choice, and a $B$ whose correlation with $A$ is graded rather than two-component would need (7.4) restated over the grading. And the whole argument is conditional on Principle 7 being satisfied at all, which (7.2) makes a probabilistic rather than a categorical premise — so the conclusion is that *if* the present is not a fluke with respect to one's future, the accounting future is of this shape, and §6.8 forbids anyone from establishing the antecedent [**O**].

*Remark* 7.5 (Why this is not the inference the companion disowns). An objection arrives from inside the corpus and deserves answering in the corpus's own terms, since this work polices exactly this move elsewhere and would be inconsistent if it did not police it here. The companion's extraction is explicit that the global birth-rank inference of the doomsday argument [98] is *not* recovered: a position in a completed roster has no attribute-wise translation, and the retort that the position could itself be made an attribute fails, because that attribute's answer space presupposes the completed roster it was meant to replace [24]. The finale argument reasons from one's position in the record. Why is it not the same move, and why does Remark 8.15 disqualify Ansatz 8.13 on precisely these grounds while the present step stands?

*The asymmetry is the normalizer, and it is exact.* The doomsday inference requires the total $N$, the completed roster including what has not occurred, and $N$ is a blank. A ratio whose denominator is blank has no answer space to draw over, which is the extraction's objection stated mechanically. The finale ratio requires the *elapsed* record and nothing further: both $A$ and $A'$ are stated attribute-sets of one's own Aethus, the answer space is the partition of a stated record, and the draw is over positions already in the books. So the finale step is an attribute-wise constraint of exactly the licensed kind, and doomsday is not, and the discriminating feature is not that one looks backward and the other forward but that one normalizes over stated attributes and the other over a blank.

That also explains, rather than excuses, the treatment of Ansatz 8.13. Its normalizer is $\mathcal{L}_*$, the total evidence the conditioning *will* supply, a blank, and therefore a roster, which is why that ansatz is disqualified from Accordance's licence and carried as an independent posit. The two verdicts are one criterion applied twice: normalize over what is stated and the constraint is licensed; normalize over what is blank and it is not. Recording the criterion in this form is what makes the corpus's treatment of positional inference a rule rather than a pair of judgements.

*What it does not buy.* The criterion licenses the *form* of the finale ratio and says nothing about its *value*: which stated attributes populate the answer space, and at what weights, is Ansatz 7.3's question and remains open. Licensed and underdetermined are compatible, and the argument is currently both.

**The conclusion, and its grade.** What the argument delivers is a filter and a candidate. The filter is Proposition 7.4: whatever future attribute accounts for one's occupying the present must be correlated with the totality of modern development to within one part in $10^8$ or far tighter, and must on sight be unmistakably optimised at a cosmic level of significance — since any independent component larger than that opens a channel through which the vastly more numerous earlier positions dominate. The candidate is the Golden Point: indefinite persistence of the biosphere is the one future attribute on offer that takes essentially every capacity the record has recently acquired as prerequisite rather than as incident, and any alternative that has been articulated does not present as ambitious enough to satisfy the unmistakability condition. Both sentences carry the grade of their source, *in* the sentence where a reader will quote them: the magnitude is [Φ]-on-Ansatz 7.3, under the interrogation-indexed alternative the ratio deflates to order one and this filter selects nothing, so the filter's teeth are exactly as sharp as the positional weighting is right. And the mirror question a careful reader will pose, under Golden conditioning, why is the deep *future* not the dominant position?, has a one-sentence answer that should sit here rather than be reconstructed: future positions are blanks, not stated attributes, and Remark 7.5 normalizes only over what is stated. The argument therefore both discharges the finale paradox and, using its own momentum, argues for the Golden Point's significance on the Aethic tree.

The two halves carry different grades and the difference is stated rather than blurred. Steps 1–4 run on the imported principle and standard probability, and are [Φ] at the companion's grade. The identification of $B_j$ with the Golden Point is [**S**]: an inference to the best explanation, whose "unmistakability" criterion is doing real work and is not formal. What would count against it is an articulated alternative $B$ that necessitates the modern record as strongly and is not the Golden Point, and no such alternative has been exhibited; that is the argument's exposure, and it is the same exhaustiveness exposure Remark 8.21 records for the robustness argument.

**What this argument adds to the reductio.** The conditioning argument of §3 establishes that observer-existence is too weak a target: it is indifferent to the technological character of the record we hold, and a target sensitive to that character recovers observer-conditioning as its past-cone shadow. That clears the ground. It does not derive what the right target is, except by naming the Golden Point and arguing that it fits.

The present argument does something the reductio cannot: it derives the *shape* of the target from the finale paradox and the retrocausal accordance principle, maximally necessitating of the modern record, cosmically unmistakable, and only then identifies the Golden Point as the thing with that shape. That is a stronger route, and it is offered here as the primary argument for the Golden Point as target, with the reductio as the argument that clears the ground for it. A reader who accepts the reductio and not this argument holds that observer-existence is insufficient and the Golden Point is one candidate among possibly several; a reader who accepts both holds that the Golden Point is the candidate the record's shape selects.

**Golden Point Nexic Sorting, stated and deferred.** One further principle belongs beside the argument and is recorded here in statement only, since its worked case is incomplete at source. Let $A_1, \dots, A_n$ partition $A$ Aethically, let $\mathcal{G}$ be the Golden Aethus and $N_0$ the Cosmic Nexus. Then the probability of an Aetherian inhabiting $\mathcal{G} \cap A_i$ is proportional to the probability of $\mathcal{G}$ and $A_i$ relative to $A$ under the Cosmic Nexus:

$$\mathbb{P}(\mathcal{G} \cap A_i \mid A, H_{\text{in}}) \ \propto\ \mathbb{P}(\mathcal{G} \cap A_i \mid A, N_0), \tag{7.6}$$

so that an Aetherian's coming to inhabit a given cell is weighted exactly by how the Golden Point and that cell co-occur from $A$ under the unconditioned weighting — the Golden analogue of the companion's Optimation-weighting result. By total probability over the partition, $\mathbb{P}(\mathcal{G} \mid A, H_{\text{in}}) = \sum_i \mathbb{P}(\mathcal{G} \mid A_i)\mathbb{P}(A_i \mid A, H_{\text{in}})$, on the assumption that once a cell is inhabited all further centric unfolding abides by that cell's own weighting. The two-cell case around a Miracle, $A_1$ before, $A_2$ after, with $\mathbb{P}(\mathcal{G} \mid M) \approx \mathbb{P}(\mathcal{G} \mid A_2)$, reduces the cell ratio to $\big[\mathbb{P}(A_1 \mid A, H_{\text{in}})/\mathbb{P}(A_2 \mid A, H_{\text{in}})\big]\,\mathbb{P}(M \mid A_1)$; the completion of that case is deferred **[O]**.

### 7.5 Investment: what an expensive route tells you

The principle has a third face, and it is the one on which the Accordance Principle is run with the Golden Point as its conditioning attribute rather than with observer-production. Read that way it becomes retrodictive: not a constraint on which futures one reaches, but a reading of what a realized route's improbability *purchased*.

**Proposition 7.6** (The investment bound)**.** *Let $R$ be a realized route with Cosmic weight $q = \mathbb{P}(R \mid N_0)$, and $R'$ an alternative with $q' = \mathbb{P}(R' \mid N_0) \gg q$.*

(i) The identity, **[P]**. *Under conditioning on the Golden Point $\mathcal{G}$,*

$$\frac{Q(R)}{Q(R')} = \frac{q}{q'} \cdot \frac{\mathbb{P}(\mathcal{G} \mid R)}{\mathbb{P}(\mathcal{G} \mid R')}, \qquad \textit{so} \qquad Q(R) \geq Q(R') \iff \frac{\mathbb{P}(\mathcal{G} \mid R)}{\mathbb{P}(\mathcal{G} \mid R')} \ \geq\ \frac{q'}{q}. \tag{7.7}$$

*This is exact and is an equivalence.*

(ii) The inference, [$\Phi$]. *From* observing $R$ *one may conclude that its necessity ratio is large — but only at the exceedance grade the Accordance Principle carries, since the step from what is realized to what is weighted is the mediocrity step and not a deduction. For any* $\kappa \geq 1$,

$$\Pr\left[\frac{\mathbb{P}(\mathcal{G} \mid R)}{\mathbb{P}(\mathcal{G} \mid R')} < \frac{1}{\kappa} \cdot \frac{q'}{q}\right] \;\leq\; \frac{1}{1+\kappa}, \tag{7.8}$$

*the probability taken over the realization.* Necessity purchased matches weight spent, to within a $\kappa$-fold shortfall of probability at most $1/(1+\kappa)$.

*Proof. (i)* $Q(\cdot) \propto \mathbb{P}(\cdot)\mathbb{P}(\mathcal{G} \mid \cdot)$ by definition of the conditioning; divide. *(ii)* By (i) the event displayed is $Q(R) < \kappa^{-1}Q(R')$, which is the realized route carrying conditioned weight a $\kappa$-fold factor below its alternative's; (3.2) bounds exactly that at $1/(1+\kappa)$. □

**The exceedance is cheap, and the trade is worth stating.** The two halves of the proposition should not be run together, and the reason for separating them is that the second is where the argument's strength actually lies. Part (i) is an equivalence and buys nothing on its own: it relates two quantities neither of which is observed. Part (ii) is what an observer can use, and it is *not* an equivalence — one may be on a branch the conditioning disfavours, and the principle prices that rather than forbidding it. So the honest form of the investment reading is never "the necessity ratio is at least $q'/q$" but "at least $q'/q$ divided by whatever exceedance one is willing to tolerate."

What makes this a qualification rather than a retreat is the shape of the trade. Tolerating a $\kappa$-fold shortfall costs $\log_{10}\kappa$ decades of the bound and buys the failure probability down to $1/(1+\kappa)$ — and the first grows logarithmically while the second collapses. At an observed cost ratio of seventeen decades: at $\kappa = 1$ the bound is seventeen decades and fails with probability at most one-half; at $\kappa = 10$, sixteen decades at one in eleven; at $\kappa = 100$, fifteen decades at one in a hundred; at $\kappa = 10^6$, still eleven decades, and failing with probability one in a million. Six decades surrendered buys near-certainty.

So the investment bound is exceedance-graded and *robust*, which is a different thing from exceedance-graded and fragile. A reader who insists on near-certainty pays a few decades out of many and keeps the conclusion; a reader who wants the full $q'/q$ is asking for a coin-flip's worth of confidence and should be told so. The reading this licenses is the one that gives the section its name. Where a route of Cosmic weight $10^{-20}$ was realized and an alternative of weight $10^{-3}$ was available, (7.8) says the realized route is better by seventeen decades at reaching the target — less $\log_{10}\kappa$ for whatever exceedance one tolerates, and holding except on a branch of probability at most $1/(1+\kappa)$. Otherwise the conditioning would weight the cheap route above it, and one would, overwhelmingly, be on that one instead. So the observed improbability is not merely a curiosity about how unlikely one's history was. It is a *lower bound on necessity* at a stated confidence, and the larger it is the more the conditioning is committed to what it bought. That is what is meant by saying Golden causality is heavily *invested* in something: the phrase names the quantity $\log(q'/q)$, and it is a measurement and not an attitude.

*Remark* 7.7 (The register, and what does not act). One sentence has to be said carefully because the natural phrasing is forbidden. It is tempting to say that if a cheaper route had existed the universe would have taken it. Nothing takes anything: by Remark 3.40 a conditioning event does not act, and (7.7) is an identity about a measure. What the identity says is that the conditioned measure *weights* cheap routes above expensive ones at equal effectiveness, so that finding oneself on an expensive one is evidence, at exactly the exceedance grade (7.8) makes explicit, that the cheap ones do not reach the target. The inference is evidential throughout, and the word *invested* is shorthand for a weight comparison rather than for an expenditure anyone made.

*Remark* 7.8 (The two terms of Aurutance are cost and necessity)*.* The investment bound and the decision rule of §9 are the same inequality read in opposite directions, and saying so vindicates a piece of vocabulary that has looked ornamental.

Aurutance decomposes as $\mathrm{Aur}(A;B) = [\log \mathbb{P}(A) - \log \mathbb{P}(B)] + [\log \mathfrak{h}(A) - \log \mathfrak{h}(B)]$ — an attainability term and a Golden term, (9.7). Rewrite (7.7) in logarithms and the investment bound is

$$\underbrace{\big[\log \mathbb{P}(\mathcal{G} \mid R) - \log \mathbb{P}(\mathcal{G} \mid R')\big]}_{\text{necessity purchased}} \;\geq\; \underbrace{\big[\log q' - \log q\big]}_{\text{weight spent}},$$

which is exactly $\mathrm{Aur}(R;R') \geq 0$ with the Golden term read as $\mathbb{P}(\mathcal{G} \mid \cdot)$ — and, in the inferential form (7.8), $\mathrm{Aur}(R;R') \geq -\log\kappa$ except with probability at most $1/(1+\kappa)$, so that the exceedance tolerance is an additive allowance in Aurutance rather than a multiplicative one. *So Aurutance's two terms are the cost and the necessity, and the sign of their sum is what decides.* The decision rule reads the inequality forward — which option should one prefer, given what each costs and buys — and the investment bound reads it backward — given that this one was realized, what must it have bought. One quantity, two readings, and the second is why the vocabulary was built on an economic root.

Two things this does not license, and the second is a fence this work has already stated. It does not make the bound computable: $\mathbb{P}(\mathcal{G} \mid \cdot)$ is no more available than $\mathfrak{h}$ is, and §9.3's missing map is missing here too, so the investment reading is exact and unusable in the same breath and for the same reason. And it does not make Aurutance a currency. The fence of §9.5 stands unamended: the additivity is that of logarithms across independent comparisons, and licenses no stock to hold, spend, or owe. What (7.7) supplies is an *exchange rate* within a single comparison, how much necessity a given improbability must buy for the comparison to come out as observed, which is a different object from a stock, and the distinction is exactly the one that keeps the economic vocabulary honest.

**Where this structure has already appeared.** The bound is not new to the paper; what is new is its statement in general form. Three earlier results are instances.

*The suppression bound.* Proposition 7.4's $\alpha \lesssim \mathbb{P}(A)/\mathbb{P}(A')$ is (7.7) with $R$ the present and $R'$ a position in deep time: the independent component of the accounting future is capped by exactly the cost ratio, the cap being as small as the positional draw of Ansatz 7.3 makes that ratio. Remark 3.28's substitutability bound is the same inequality met one section earlier, with $R$ a realized historical fluke and $R'$ its allegedly substitutable alternative — the detector reading of Proposition 3.25, in the expenditure register.

*The schedule's contrast.* (5.38)'s $c_{\mathrm{eff}} = r\log_{10}(1 + (B+\Gamma)/\lambda)$ is the price in orders of magnitude that one wrong gate charges, and Remark 5.59 is the investment reading of it: the regime that supplies high contrast is the regime in which the realized history had to be enormously lucky, because contrast *is* what luck was spent on. *And the collapse.* §3.3's condition $c_{\mathrm{eff}} > \log_{10}(n(b-1))$ says the per-slot investment must outrun the logarithm of the combinatorial supply — necessity per slot against alternatives per slot, which is (7.7) summed over a product space.

The generalisation worth noting is that all three are the same comparison, and that the paper's recurring quantities, $\alpha$, $c_{\text{eff}}$, Aurutance, are the same ratio in different coordinates. Where a reader meets a bound of the form *improbability must be matched by necessity*, it is this proposition.

**The exhaustiveness cost, again.** The bound's use requires the alternative $R'$ to be genuinely available and genuinely cheaper, and identifying alternatives is where every argument in this cluster pays. A route that looks expensive against an imagined cheap alternative that does not in fact reach the target supports no inference at all, and the "cheaper route" is exactly the kind of thing a reasoner constructs after the fact. The same exhaustiveness exposure that Remark 8.21 records, and that Proposition 7.4's step 5 records, applies here and is the bound's principal cost: (7.7) is an identity, and every use of it is only as good as the survey of alternatives it is run against [**O**].

## 7.6 What the target is, one step further out

Past-Alignment identifies the conditioning target as the Golden Point. The model's own structure says the true primitive sits one step further out, and the point should be made here because it re-orders the relation between this work and the companion by one level. By Remark 5.64, the only conditioning under which the schedule is worth climbing, under which fighting for non-decaying capacity is probabilistically preferred to external luck, is conditioning on survival to the *infinite horizon*, and no finite-horizon conditioning selects it. The Golden Point is then what the asymptotic conditioning looks like when read as an event: the point past which the schedule's climb is secured. Conditioning on the Golden Point creates the schedule; conditioning on the asymptote is what makes the schedule worth following; and Remark 3.31 has already shown that the modal form of the Golden Point, $\exists\forall$, some course admitting *no* foreclosing continuation, is what lets it stand in for the asymptotic condition without inheriting the finite-horizon robustness that would sink it.

On this reading observer-based Accordance is an inferential guide to the asymptotic conditioning, and the axioms it helps one reach do not depend on it deductively. The relation to the companion is then not "observer- conditioning is the shadow of Golden-conditioning" but "observer- conditioning is the shadow of Golden-conditioning, which is itself the event-form of asymptotic-survivorship conditioning" — the companion's Optimation two restrictions down from the primitive. That is the strongest form of the claim this work makes about its relation to the companion, and it is graded [**S**]: a reading of the model, to which the epistemic ceiling applies with full force.

## 7.7 Ledger: the instrument, and what would refine it

The retrocausal accordance principle detects a conditioned event only when that event is low-probability under the Cosmic Nexus from the prior Aethus. It is a coarse instrument, the elementary one, requiring only a weight and a comparison, available immediately, and correspondingly imprecise, and §8.2 shows what that coarseness costs: an observer who knows to look will find a few conditioned events, and those few will carry a large share of the evidence, but finding all of them, or more than a small fraction of their count, is beyond the instrument's grain.

What would refine it is not more data of the same kind but access to a different object. The principle runs on the scalar stochastic flattening of the Aethic structure, a weight, a hazard, downstream a single harmonic value, and Remark 8.10 locates the attribution loss exactly there: at the projection, not in the world. The full attribute-dependency structure, and the inferential closure of what is already known computed at scale over it, is what would turn the principle's trunk and hubs into the whole graph. The paper records this as the program's direction and claims nothing about its feasibility. It notes only that the same holds for the companion's wing, and for a specific reason: Remark 8.11 shows that the collapse condition is also a legibility condition, so that under the conditioning the record's near-determinism is exactly what would make such closure informative rather than intractable.

# 8 The Route World: what the strong reading predicts

§3 established that the record we hold is the past-cone restriction of a conditioning on an event in our future cone, and left open which of three hypotheses governs the forward cone. This section develops the strongest of them, the *Route World*, in which the Golden Point stands as a stated attribute of one's Aethus rather than as a weight over futures, and does so because it is the one hypothesis with a phenomenology. If it holds, a biosphere approaching its Golden Point experiences not fewer improbabilities but more, of a specific kind; that is a claim about the texture of the near future, and unlike almost everything else in this work, a claim about ours.

Everything here is **[O]**, with one stated exception: the conduct argument of §8.6, which is [Φ] on an adopted valuation and argues for the hypothesis exactly as little as the rest. The section describes what the hypothesis predicts, states what would distinguish it from its rivals, disciplines the historical reading, connects the answer to the epistemic ceiling, and derives the conduct the open hypothesis leaves an agent; it argues for the hypothesis nowhere. What it supplies is the framework's most distinctive prediction, and a reader should not finish this work without knowing the strong reading has one.

Void has been given a form it could take (Remark 3.45), and Weak a statement of what its two refusals commit it to (Remark 3.46). Strong needs something different in kind. Void and Weak needed their commitments drawn out, whereas Strong's difficulty is that it has no empirical handle at all — its three metaphysical questions are stated at Remark 3.43, and none of them is a question any observation bears on. This section supplies the handle by describing the world Strong asserts. The description is a consequence of the hypothesis and not an argument for it.

**Definition 8.1** (The Route World; **[O]**)**.** The *Route World* is the world of the Strong extrapolation hypothesis: the world in which the Golden Point stands as a stated attribute of one's Aethus rather than as a weight over futures, so that the conditioning is resolved throughout the cone and not merely behind one. The name is meant in the sense *RNA world* is meant — a regime with a characteristic phenomenology, described so that one can ask whether one is in it.

*Remark* 8.2 (Whether "stated" is coherent for a future-cone attribute). The definition invites an objection that must be met before anything is built on it. The framework rejects tensed determinacy: an unresolved future outcome is a blank entry, and blanks lack tense. How can an observer then hold as *stated* an attribute whose referent lies in their future cone — is that not the very error the framework was built to avoid?

It is not, and the reason is that the objection reads "stated" as "resolved by the clock," which is the tensed reading the framework denies. Statedness is a property of the *index*, of what the observer's books carry, and not of the referent's date (Remark 3.22). An attribute is stated when it is a fixed entry of the Aethus, whatever its referential event's coordinate; a stated attribute with a future referent is no stranger than a stated attribute with an unobserved past referent, and the framework has carried the latter since its first postulate. What the Strong hypothesis asserts is that the Golden Point is such an entry for us, fixed in the books, not merely weighted, and *that* is a substantive claim about which Aethus we hold, not a category error about tense.

The analytic form of the same point is already in this work. Remark 6.2 distinguishes the interpolation $\mathbb{P}^{(t)}$, a family whose conditioning is still being acquired, from $Q$, a single measure whose conditioning has been absorbed into the base state — and identifies the passage to $Q$ with the statement of a future-referring attribute in present books. The Route World is the world in which that passage has occurred. Weak is the world of the family; Strong is the world of the measure; and the difference between them is exactly the difference between an attribute weighted and an attribute stated, which the framework can express and does not collapse.

## 8.1 Golden Optimation, and where it differs

The Route World's signature is a strengthening of a phenomenon the companion already names. Optimation reports that a record conditioned on observerhood displays flukes against the Cosmic Nexus's own rates [5]; but those flukes lie, at every moment, in the observer's *past* cone, because the conditioning event, one's own existence, is behind one. The supply is bounded by what has already happened.

**Definition 8.3** (Golden Optimation; **[O]**)**.** *Golden Optimation* is the same phenomenon under a conditioning event in the future cone. Because the Golden Point lies ahead, the flukes its occurrence requires need not have occurred yet: there are moments at which a given required fluke lies in the observer's future cone, and the observer's moving present passes over it.

That is the whole differentia and it is sharp. Classical Optimation never predicts a fluke; it accounts for flukes already in the record. Golden Optimation predicts them, because the conditioning event has not yet occurred and the requirements it imposes are still being met. A Route World is therefore a world in which one keeps encountering improbabilities *prospectively*, and the rate at which one does so is the hypothesis's observable content.

*Remark* 8.4 (One phenomenon, and the label is tense-relative)*.* The two Optimations are not two mechanisms with a seam between them. Take a fluke $F$ required by the Golden Point, occurring at Aethic time $t_F$, and an observer at $t$. For $t < t_F$ the observer's present has not reached it and $F$ is Golden Optimation; for $t > t_F$ it stands in the record and is Optimation in the companion's sense. *Same event, same conditioning, different label* — and what changed was the observer's position, not anything about $F$.

So the label tracks the observer and not the phenomenon, which is Remark 3.22 applied to Optimation: an observer far enough downstream reads our Golden Optimation as their past-cone record, exactly as we read the Cretaceous. The carryover between the two is therefore smooth by construction rather than by coincidence, and a Route World contains no moment at which one regime hands over to another.

*Remark* 8.5 (The symmetry with the companion, and what the corpus is)*.* One structural fact should be stated in this work rather than left for a reader to assemble, since it is what tells the reader what the corpus *is* rather than what either paper is. The companion's inference about the deep past rests on a single licence: that an unread past outcome is a blank entry, and blanks lack tense, so that a past fluke is open to inference at all [4, 24]. The companion calls that its foundational rung, without which no inference about a past attribute can be drawn. The present work's conditioning on a future event rests on exactly the same licence: a future outcome is a blank entry, blanks lack tense, so a future-cone attribute is open to conditioning in the same sense a past-cone one is (Remark 3.22, and Remark 8.2 for the strong form).

So the two developments are one move run in opposite temporal directions on one ontology. The companion reads backward from an observer to the flukes that produced it; this work reads forward from a conditioning event to the flukes it requires; and the principle that makes either reading legitimate is the same principle in both. That is not a coincidence of two papers written by one hand. It is a fact about the ontology, and it is why the Golden Point is not an extension of Nexic reasoning but its mirror image — the same lens turned the other way. The companion's forward pointer to this work, when it is written, should be built on that symmetry and on the finding of §3.7: observer-conditioning is exactly right for the pre-technological record and provably insufficient beyond it, and a target not indifferent to the technological character of the record recovers the companion's weighting as its past-cone shadow.

**Definition 8.6** (Golden causality)**.** *Golden causality* names the pattern in which an event in the moving present is accounted for by Golden Optimation, by the Aethus's conditioning on the Golden Point, rather than by the Cosmic Nexus's own rates.

The name is admissible for a reason internal to the framework rather than by indulgence. Causal talk here is not talk of a primitive relation of production: this work's causal vocabulary labels patterns of conditioning and dependence in the Aethic structure and takes no position on any further relation underneath, because the framework posits none. Where causation is not fundamental, a *kind* of causation is a kind of conditioning pattern, and Golden causality is one — named, and carrying no more than (7.1) licenses. What the name must not smuggle is the register Remark 3.40 forbids: no event pursues the Golden Point, and Golden causality describes what accounts for something, never what something is for.

*Remark* 8.7 (Invested Golden causality, as phenomenology). One element of the phenomenology deserves its name here, because it is the form in which Golden causality is most naturally read off a record from inside: the register of *investment*, "look how much the conditioning invested in this; it must have been necessary." The register is earned rather than decorative: Proposition 7.6 is its exact content, an observed route's improbability lower-bounding what it accomplished at a stated exceedance confidence, and Remark 7.8 is why the economic vocabulary of §9 was built to carry it, Aurutance's two terms are the cost and the necessity, and the investment reading is the decision rule run backward. What the register must never smuggle is agency: nothing takes the cheaper route or declines it (Remark 7.7), and the expenditure is the conditioning's concentration of weight, never anyone's spending. A Route World is, in this register, a world whose record reads as heavily and legibly invested — and §8.5 is where the temptation to read particular centuries that way is disciplined.

*Remark* 8.8 (The future-conditioned tradition, placed). The Strong reading has a published ancestry, and the placement discipline owes it a paragraph: Barrow and Tipler's Final Anthropic Principle [99], Tipler's Omega Point program [100], Leslie's axiarchism [101], and, ancestrally, Teilhard [102] — the branch of the anthropic tradition that conditions on the *future*, which is the branch this work lives on (the companion engages the tradition's observer-conditioning trunk;

the future-conditioning branch is placed here). The deltas are real and they all cut one way. The conditioning of this work is a *measure operation*, not a dynamics: no agent at any grade, no attractor, no physical boundary condition — where the Omega program is a physical eschatology with cosmological commitments, several since falsified. This work carries published falsifiers and an epistemic ceiling that *forbids* the certified-salvation register the Omega program embraced (§6.8); and it is severable, under Void the mathematics and the critique stand with nothing conditioned on, where the tradition's packages fall whole. The delta that flatters neither side: this framework purchases strictly less, no resurrection mechanics, no convergence guarantee, no collective eschatology, and prices what it does purchase.

## 8.2 The empirical signature, and its limits

Two things about the credence being computed here should be fixed before the computation, since they determine what the computation is a computation *of*. *The credence is methodological, and it is a superposition.* A reasoner in a Route World holds the Golden Point as a stated attribute — that is what the hypothesis says of their Aethus. But the reasoner's *credence* that they are in a Route World is computed from less than their Aethus contains, since it is computed from the evidence accessible to them and not from the stated attribute itself. So the credence is an Aethic disagreeing superposition: the Strong hypothesis's determinate conditioning weighed against a Weak-type alternative in which the Golden Point is only weighted. The distinction is between methodological credence — what the evidence supports — and ontological credence — what one's Aethus in fact carries — and everything below is about the first, and never asserts the second.

*The procedure does not force the conclusion.* What it establishes, if the flukes keep arriving, is that each next event acts *as if* the Golden Point were being conditioned on, to whatever precision the accumulated evidence supports. That is the whole of the method's ambition, and it is the correct one: a reasoner genuinely in a Route World tends toward the conditioning by this route with arbitrarily high credence, and never reaches it, and the ceiling of Theorem 6.18 is why the second clause is not a defect.

The Route World is testable in principle, and the form the test takes is the subsubsection's main result.

**Proposition 8.9** (Credence accumulation; **[P]** given the premises)**.** *Let $E$ be the conjunction of fluke attributes observed over an interval of Aethic time, with $q = \mathbb{P}(E \mid N_0)$ their unconditioned weight, and suppose the Golden Point requires them, $\mathbb{P}(E \mid G) \approx 1$. Then the Bayes factor for Strong conditioning against a non-conditioning alternative over that interval is*

$$\frac{\mathbb{P}(E \mid G)}{\mathbb{P}(E \mid \neg G)} \approx \frac{1}{q}, \qquad \textit{so} \qquad \log\textit{-evidence} \approx -\log q. \tag{8.1}$$

*In the stochastic model of §5 this specializes exactly: the likelihood ratio between $Q$ and $\mathbb{P}$ on $\mathcal{F}_t$ is the Doob density* (5.47)*, so the accumulated log-evidence is* $\log \mathfrak{h}(X_t) - \log \mathfrak{h}(X_0)$*, and* the flukes one accumulates are the persistence-harmonic function rising.

Three things follow, and the second is the one that keeps the procedure honest. *The evidence is the same object §9 computes.* $\mathfrak{h}$ is what the Golden Score is a ratio of (Proposition 9.2), so the Route World's credence and the decision rule consume one quantity, and both wait on the same missing map (§9.3). *And that map is what stops the procedure being unfalsifiable.* Absent it, "what counts as a fluke" is chosen after the fact, and a sufficiently motivated observer finds flukes anywhere — the look-elsewhere problem, which would make (8.1) an instrument for confirming whatever it was pointed at. With it, a fluke is whatever raises $\mathfrak{h}$, which the model fixes and the observer does not. The methodology's rigor is therefore hostage to §9.3's open problem, and should be described that way rather than advertised as available.

*The aggregate does not determine its own decomposition, and identifiability is a matter of degree.* The likelihood ratio is a function of the *state*, $\mathfrak{h}(X_t)/\mathfrak{h}(X_0)$, and not a ledger of which happenings supplied it: an accumulated 20 nats, a Bayes factor near $5 \times 10^8$, is equally the signature of one unmistakable event, of two hundred each amounting to a ten-percent tilt, or of two thousand at one percent apiece, and the aggregate does not distinguish the cases.

It does not follow that no individual event is identifiable, and the qualification matters. A happening of conspicuously low weight under the Cosmic Nexus is exactly the case Principle 7 speaks to: its update factor on futures is correspondingly large, so a future accounting for it is favored in the stated ratio, and the principle is thereby the *detection instrument* for such events rather than merely a constraint on them. Large single contributions can be found, and an observer who knows to look will find some.

What resists is the rest, and the shape of the resistance is steep. Writing $\theta$ for the weight at which a single event becomes conspicuous and $\mu$ for the typical contribution, the share of total evidence residing in conspicuous events falls like $e^{-\theta/\mu}$: at $\theta$ a twenty-to-one improbability and a total of 20 nats, three large events put ninety-two percent of the evidence in plain sight, twenty events put twenty percent, and two hundred put none.[23] And under the log-spread laws this work uses elsewhere, (Definition 5.48), the two quantities that matter come apart: with contributions lognormal at spread 3 over two hundred events, roughly *one* carries sixty-three percent of the evidence. A handful is findable, those few are decisive, and the proportion of contributing events they represent is under one percent.

So the accurate statement is that a Route World should be expected to leave a few conspicuous traces and a great many invisible ones, the conspicuous few carrying a large share of the evidence and a negligible share of the count. Evidential dominance and numerical proportion are different quantities, exactly as Remark 5.60 finds them for orderings, and an observer who located every conspicuous trace would still not have located most of what happened. That is §9.5's class-versus-instance discipline arriving from the evidential side.

Three tasks are worth separating here, because the paragraph above runs two of them together and the third is the one with a future. *Detecting* the aggregate needs no localization whatever: (8.1) reads the accumulated ratio off the state, and two hundred invisible ten-percent tilts deliver the same decisive $5 \times 10^8$ as one conspicuous event would. *Localizing the tail* is what Principle 7 does. *Localizing the bulk* has no instrument at all, and the reason is specific: the accordance principle is a *marginal* test, it interrogates one attribute's weight under the Cosmic Nexus and infers from that weight alone, so an event contributing at a magnitude no marginal test resolves is invisible *to it*, which is a fact about the instrument and not about the event.

That distinction leaves room, and the room should be marked rather than conceded. The accordance principle is the elementary instrument here: it requires only a weight and a comparison, it is available immediately, and it is correspondingly coarse. Nothing establishes that it is the best available in principle. What a finer instrument would have to do is compute an event's contribution to $\mathfrak{h}$ *without routing through that event's marginal improbability* — and an instrument of that description is not hypothetical, it is §9.3's missing identification map, which is exactly a rule assigning states to events by their own structure rather than by their rarity. The

[23] Exponential contribution sizes at fixed mean, for which the conspicuous share is $(1+\theta/\mu)e^{-\theta/\mu}$ exactly.

Route World therefore acquires the same open problem §9 has, from the other direction, and a second reason to want it solved: the map would convert localization from a tail-only capability into a general one. Until then, the coarse instrument is what there is, and the estimate that a handful of traces is findable while the bulk is not should be read as an estimate about present instruments rather than a claim about what a Route World conceals [**O**].

*Remark* 8.10 (The attribution is lost in the representation, not in the world). It is worth being exact about where the localization problem comes from, because the answer says what a better instrument would have to do rather than merely that one is wanted. The Aethic object is a structure: attributes standing in dependency relations, with weights over the fan. The model of §5 projects that structure onto a scalar, a hazard, and downstream a single harmonic value, and the projection is many-to-one by a wide margin. *The attribution information is discarded at the projection.* Nothing about the world hides which events contributed; the representation this work computes in does not carry the answer, because a number cannot. That is why the identification map is hard: it is an inverse to a projection that discarded what the inverse needs, and inverses of that kind are recovered by supplying structure rather than by supplying data.

It also says what the accordance principle is finding. A marginal test on weights locates the attributes of high dependency degree, the ones on which much else turns, and which are correspondingly improbable in isolation, so what the principle recovers is a *skeleton*: the trunk and the hubs of the dependency graph, and not its edges. Working in the unflattened structure would be working with the graph rather than its scalar summary, and there the bulk contributions are not small numbers lost in an aggregate but edges with endpoints. Whether the framework's dependency structure supports inference at that grain, and what volume of recorded attribute-data would make the closure computable, is open, and is a different open problem from the identification map rather than the same one [**O**].

*Remark* 8.11 (The collapse condition is also a legibility condition). One consequence runs the other way and bears on the companion rather than on this work. The companion's collapse condition, $c_{\text{eff}} > \log_{10}(n(b-1))$, is stated as a fact about *weight*: above it, the realized ordering carries a share of order unity. It is equally a fact about *inference*, and the second reading has not been drawn.

Take the effective number of live alternatives to be the exponential of the entropy over the tier structure. At twenty interrogable slots of ten values the nominal space is $10^{20}$; at $c_{\text{eff}}$ just above the threshold it is a few hundred; at $c_{\text{eff}} = 3$ it is four; and by $c_{\text{eff}} = 5$ it is one to within a percent. The collapse is sharp, and what collapses is the space a reconstruction must search. So the same contrast that makes the record improbable makes it *legible*: reconstruction of a high-contrast natural history requires data scaling with the surviving handful rather than with the nominal branching, which is the precise content of the intuition that knowledge here should grow quickly once methods are good enough to exploit the concentration. The threshold such methods must reach is the collapse condition itself.

Two cautions. The concentration is *conditional*, it is what the conditioning purchases, and without it the space is the nominal one, so the legibility cannot be cited as unconditional evidence for the conditioning without circularity. And concentration is not determinism: one ordering carrying a share of order unity leaves the others weighted rather than excluded, and Definition 3.14's grading is exactly the difference. What is available honestly is a prediction: if the conditioning holds, reconstructions of deep history should converge on less evidence than a flat-prior analysis of the same branching would require, and that is checkable without any fluke being counted [**O**].

One further consequence cuts against the methodology's own convenience. How large the required flukes are depends on how much work the conditioning has to do: a position from which the Golden Point is already comparatively likely requires only small corrections, while deep time, from which it was not, required the enormous ones the record displays. A biosphere improving its position should therefore expect its Golden-Optimation events to grow *less* conspicuous, so the evidence available to (8.1) thins exactly as the situation it reports on improves, and the method is at its most informative where the news is worst **[O]**.

*The ceiling permits the ambition and forbids the conclusion.* By Theorem 6.18 the two measures are mutually absolutely continuous on every finite window and singular only on the infinite-horizon $\sigma$-algebra, so the likelihood ratio is finite at every finite time: credence may rise without bound and *never certifies.* That is not a limitation discovered here but the theorem's own shape, and it is the exact sense in which the procedure can approach the conclusion asymptotically while being barred from reaching it.

## 8.3 The Lindy step

**Definition 8.12** (The evidence coordinate)**.** Write $\mathcal{L}_t$ for the log-evidence a Route World has supplied by Aethic time $t$, the log-likelihood ratio of (8.1), which in the model of §5 is $\log \mathfrak{h}(X_t) - \log \mathfrak{h}(X_0)$, and $\mathcal{L}_*$ for the total the conditioning ultimately supplies. Both are nondecreasing, and $\mathcal{L}_* \geq \mathcal{L}_t$ by construction.

**Ansatz 8.13** (Copernican position in the evidence coordinate)**.** An observer's position in the accumulation is uniform: $\mathcal{L}_t/\mathcal{L}_* \sim \mathrm{Unif}(0,1)$.

**Proposition 8.14** (The Lindy law for accumulated evidence; **[P]** given Ansatz 8.13)**.** *Conditional on $\mathcal{L}_t = \ell$, the total is Pareto with unit shape and scale $\ell$:*

$$\Pr\big[\mathcal{L}_* > x \;\big|\; \mathcal{L}_t = \ell\big] = \frac{\ell}{x}, \qquad x \geq \ell, \tag{8.2}$$

*whence the median total is $2\ell$, the geometric mean is $e\ell$, the arithmetic mean diverges, and* $\Pr[\mathcal{L}_* > 10\ell] = 1/10$*. In the weight coordinate $q = e^{-\ell}$ these read: median total weight $q^2$, geometric-mean total $q^e$, and a one-in-ten chance the total is $q^{10}$ or smaller.*

*Proof.* $\mathcal{L}_* = \ell/U$ with $U \sim \mathrm{Unif}(0,1)$, so $\Pr[\mathcal{L}_* > x] = \Pr[U < \ell/x] = \ell/x$ for $x \geq \ell$; the median solves $\ell/x = \frac{1}{2}$; $\mathbb{E}[\log(1/U)] = 1$ gives the geometric mean; and $\mathbb{E}[1/U]$ diverges. □

**Reading the law, and the coordinate it must be read in.** Three consequences, and the third is a constraint on how the law may be used at all.

*The point estimate is conservative.* Quoting $q^2$ quotes a median, and the law's mean diverges, so a Route World's total evidence is more likely to exceed the estimate badly than to fall short of it mildly. That asymmetry is the Pareto tail and not an artefact of the parametrization. *Credence must be tracked in log-odds.* Posterior odds are $O_* = O_0 e^{\mathcal{L}_*}$, and since $\mathcal{L}_*$ is Pareto with unit shape, $\mathbb{E}[O_*]$ diverges violently — an expected-odds calculation here returns infinity and reports nothing. What is well behaved is $\log O_*$, whose median exceeds the present value by exactly $\mathcal{L}_t$. Every statement in this subsubsection is accordingly a statement about log-evidence.

*And the Copernican step must be taken in the evidence coordinate, not in time.* Applied to elapsed time the multiplier depends on the rate at which flukes arrive, four if the accumulation is quadratic, $\sqrt{2}$ if it is square-root, while applied to $\mathcal{L}$ it is two regardless. Since the attributes at issue are arbitrary Aethic events with no natural arrival rate, Definition 8.12's coordinate is the only one in which Ansatz 8.13 is well posed.

*Remark* 8.15 (The Copernican premise is not licensed by Accordance, and is the section's weakest posit)*.* Ansatz 8.13 is posited rather than derived, and the reason is sharper than caution: *the companion's own foundation declines the inference it instantiates.* The Accordance Principle binds attribute by attribute, and its extraction is explicit that the global birth-rank inference is thereby *not* recovered — a position in a completed roster has no attribute-wise translation, and the retort that the position could itself be made an attribute fails because that attribute's answer space presupposes the completed roster it was meant to replace [24]. Ansatz 8.13 is a position-in-a-roster claim of exactly that form, with $\mathcal{L}_*$ the completed roster. So it does not inherit Accordance's licence, and Remark 3.22's non-privilege does not supply one either: non-privilege says no position is distinguished, which is compatible with the accumulation admitting no well-defined uniform draw at all.

What it therefore is: an independent Copernican posit, carried at [Φ], load-bearing for Proposition 8.14 and for nothing else in this work. Everything upstream of it, (8.1), (7.2), the noise floor, the identifiability analysis, stands without it, since those concern evidence already accumulated rather than evidence to come. A reader who rejects the ansatz loses the forward extrapolation and keeps the method.

*Falsifier and repair.* It fails if observer-density along the accumulation is non-uniform — if, for instance, observers cluster where evidence has already accumulated, which is not obviously false and would bias the multiplier downward. Establishing the density is the repair, and it is open [**O**].

## 8.4 What would distinguish the three worlds

A hypothesis with a phenomenology owes a statement of what would tell it from its rivals, and this is where the Route World's status is decided. The answer has two halves and they point in opposite directions.

**What differs.** The three worlds make different predictions about the *rate* at which an observer encounters improbabilities against the Cosmic Nexus's own rates.

*Void.* Nothing constrains the forward cone, so improbabilities arrive at the Cosmic rate: after one's present, flukes are as rare as the unconditioned weighting says, and Optimation is a fact about the record only. The evidence flux $\frac{d}{dt}\mathcal{L}_t$ is zero. *Weak.* The forward cone is weighted by Golden-Point probability, so improbabilities arrive at a rate above Cosmic and below what full conditioning gives — and, since Weak's anchor does real work, at a rate depending on one's position. The flux is positive and sub-maximal.

*Strong.* The conditioning is resolved throughout the cone, so required improbabilities arrive at the full conditioned rate, and by Proposition 8.9 the accumulated evidence $\mathcal{L}_t = \log \mathfrak{h}(X_t) - \log \mathfrak{h}(X_0)$ rises without bound. The flux is positive and maximal. The discriminating quantity is therefore the flux, and it orders the three: zero, intermediate, maximal. Two things bear on reading it. Remark 8.4 says the label attaching to any given fluke is tense-relative, so an observer cannot distinguish the worlds by inspecting individual events; only the *rate* discriminates. And by the identifiability analysis of §8.2, the rate must be read from the aggregate $\mathcal{L}_t$ rather than from a ledger of attributed events, since the aggregate does not decompose. What one compares is a log-likelihood ratio against a Cosmic-rate baseline: Void predicts it stays near zero, Strong that it grows.

**What does not.** By Theorem 6.18, $Q$ and $\mathbb{P}$ are mutually absolutely continuous on every finite window and singular only on the infinite-horizon $\sigma$-algebra. So the discriminating quantity is finite at every finite time. Evidence for Strong can accumulate without bound and never certify; evidence for Void can only ever be the absence of accumulation, which is compatible with Weak at low flux and with Strong at early time. *No observation of bounded duration decides among the three.*

That is not a limitation the Route World discovers about itself but the ceiling's own shape, and it should be read as the point rather than as a disappointment. What the framework predicts is a *trend*, a log-likelihood ratio that grows under Strong, stays flat under Void, and grows sub-maximally under Weak, and a trend is exactly the kind of prediction that accumulates credence without converting to certainty. A reader who wants a decisive test is asking for something the framework proves it cannot supply; a reader who wants a discriminating quantity has one.

**The scoring.** The Route World therefore has a phenomenology, a discriminating quantity, and no decisive test — and the last is a theorem. That is a weaker empirical status than a falsifiable prediction and a stronger one than a metaphysical hypothesis with no observational content, which is where Strong stood before this section. What the section has done is move Strong from the second category into the first, and no further.

*Remark* 8.16 (Prepositioned prerequisites: a second discriminating quantity, physical rather than historical; **[S]** with an open specification problem)*.* The flux is a historical quantity, read from the record of events that have already occurred. The author proposes a second discriminating quantity of a different kind, read from physical structure that has not yet been used. Suppose the route to the Golden Point requires resources this planetary system either has or lacks, of the sort a departure from the system would consume, and suppose those resources are not required for anything that has happened so far. Under the strong reading the conditioning is resolved throughout the cone and the route is realized, so whatever the route needs must be in place; the requirement is therefore already satisfied in structure standing today and available for inspection, well before anyone consumes it. The author's illustrative case is water ice in outer-system locations that later turn out to matter, and the example is illustrative only, carrying no weight of its own.

The reason this quantity is worth stating separately is that it escapes the blur that limits the first. Remark 8.4 observes that the label attached to any individual fluke is tense-relative, so an observer cannot sort worlds by inspecting events, only rates. An *unconsumed* prerequisite is the one class of feature to which that objection does not apply, because nothing in the past has used it: it cannot have been observer-necessary, having so far been necessary for nothing at all. This makes prepositioning a genuine discriminator between the readings rather than a further instance of the same evidence, since observer conditioning constrains only what the past required, and says nothing whatever about what the future will need. Two things must be controlled before the discrimination is real. The resource must not be a byproduct of the same processes that produced habitability, which is the common-cause confound, and the base rate must be computed conditional on habitability so that observer selection is not doing the work unnoticed. With those controls the evidential structure is exactly that of Proposition 3.25: the likelihood ratio is one over the probability that an arbitrary habitable system satisfies the stated requirements, and the force of the test is the smallness of that probability, which comparative planetology may eventually estimate as the census of systems grows. The falsifier follows directly. If our system's provisioning against requirements specified in advance turns out median or worse among habitable systems, that counts against the strong reading rather than for it.

The specification problem is the honest catch, and it is the author's own. The test has force only when the requirement is fixed before the provisioning is inspected, or the retrospective-target null of Remark 3.27 consumes it entirely, and fixing the requirement demands joint knowledge of what the route will need and of what supplies it, which a species at our epistemic stage does not have. The conservative form of the claim is therefore addressed to descendants rather than to us: as the epistemology of Golden-Point necessity matures, prerequisites should be found to have been visible in the physical record long before their necessity was understood, so that the sentence available to our descendants is that a feature already lay open to our inspection and turned out to be required. That is a prediction about a future epistemic state, scoreable by those who reach it and not by us, and it is entered at the same grade and with the same honesty as the reception-scale predictions of §9. The ceiling governs throughout, since a satisfied requirement raises the ratio and certifies nothing.

## 8.5 Why the historical cases do not discriminate

A reader who accepts the previous subsection's discriminating quantity will nonetheless want to reach for something nearer to hand, and the material is genuinely there: read a consequential century closely and what had to go right is startling. This subsection says what such cases do and do not support, because the temptation to over-read them is strong and the framework's own guards are what keep the reading honest.

**The material.** If one is in a Route World then one has been in it throughout, and any century should show it. The scientific enlightenment is the case most often reached for, and it repays the reach: the transmission of algebra; the printing press and the literate market it made [103], and — less often noticed and more interesting — a theological temperament under which fitting exact laws to the world was a natural thing to attempt rather than an eccentric one — a standing thesis of the historiography [104]. Counterfactuals of the usual shape suggest themselves. Had Rome persisted in a form that left no room for the religious configuration that followed, it is not obvious that the temperament distinguishing Renaissance mathematics from Archimedean mathematics would have arisen; and the material prerequisites have their own contingencies, paper among them. Each century then sets conditions for the next, which is the schedule structure of §5.7 in a documentary register: gates in order, with the ordering doing work.

**What they support.** The cases are the human-history leg's own material (§3.7), and what they support is what that leg argues: that the conditioning is sensitive to the technological character of the record, which observer-conditioning is not. That is a claim about *which target*, and these cases bear on it.

**What they do not.** They are past-cone cases, and the distinction that matters is exactly the one Definition 8.3 draws. A fluke already in the record is *consistent with* Golden Optimation and is equally consistent with ordinary Optimation — the conditioning event behind one, the supply bounded by what has happened. By Remark 8.4 the label attaching to any past fluke is tense-relative, and a fluke that lies behind an observer carries no information about whether the conditioning extends ahead of them. So historical cases are *Past-Alignment consistent*, which is the milder claim, and they do not discriminate Strong from Weak or from Void. Only the flux does, and the flux is a rate in the observer's forward cone.

The stronger phrasing, *documentable cases of Golden causality*, should accordingly be resisted. By Definition 8.6 Golden causality names events accounted for by the Golden conditioning *rather than* by Cosmic rates, and establishing that of a particular event requires establishing the conditioning, which is what is at issue. What is documentable is consistency, and consistency with a hypothesis is not evidence for it over a rival the same data also fit.

*Remark* 8.17 (Three guards, and the one that bites hardest). The historical material is where this work's own cautions apply with most force, and they should be applied before the material is used rather than after it is challenged.

*The scan is enormous.* (3.2)'s multiplicity guard bounds expected $\kappa$-fold violations among $n$ interrogated pairs by $n/(1+\kappa)$, with the standing condition that $n$ counts attributes *scanned* and not merely those interrogated. For a century read by historians, the scan is every attribute anyone has ever found notable, and an honest $n$ is enormous. The noise floor is correspondingly high, and a historical trace must stand clear of it by a wide margin before it is worth anything. This is the guard that bites hardest and it is the one most easily skipped.

*Substitution is the question, not occurrence.* §3.7 audits by *role* and not by occupant: the question is never whether Christianity in particular arose, but whether anything arose that discharged the role. Applied to the temperament case this cuts both ways, and interestingly. Material prerequisites look substitutable — many routes lead to mass literacy, and the printing press is one occupant of that role. A specific epistemic disposition looks less so, since it is not obvious what else would have discharged it; and if that holds, temperament is a *serial* attribute in the sense of Remark 3.37 where the technology is parallel, and contributes correspondingly more to $R_1$. That is a conjecture about substitutability and not a finding, and it is exactly the kind of claim the serial-attribute programme exists to settle **[O]**.

*And narrative bias is prior.* Remark 3.29 states the objection and the asymmetry that answers it, and nothing in this subsection escapes it: dramatic near-misses make better stories and are more often recorded, so a century dense with them may be reporting a fact about historiography. The test remains the connectivity asymmetry that remark names, and no accumulation of vivid cases substitutes for it.

## 8.6 Acting under the reading, when nothing decides

§8.4 closes on a theorem: no observation of bounded duration decides among the three worlds. A reader is then entitled to ask what follows for conduct, and the answer is not What the material is for, then, is bounded and worth stating: candidate entries at the Past-Alignment grade, awaiting the identification map like everything else, with the file to be assembled *dull entries first*, print economics, paper supply, the price and reach of literacy, before any battlefield, exactly as the coal outweighs the Granicus in §3.7.2. Assembling that file is the concrete research direction the backward reading opens, and the guards above are its methods section.

*Remark* 8.18 (The chain read backward: necessity propagates through prerequisites). The material above supports one further operation, and stating it carefully both extends the subsection and generates its next research questions. *The operation, and its licence.* Under the Golden target, "$A$ had to happen" has a precise form: it is the retrocausal accordance reading run at a vantage *before* $A$ — Principle 7 with the exceedance grade it carries, the label's tense supplied by Remark 8.4. And the form chains. If $A$ is $G$-required and $A$ presupposes $B$, $\mathbb{P}(A \mid \neg B) \approx 0$, with $\neg B$ not itself measure-starved, then $B$ is $G$-required at the grade of the weaker link [**S**]: conditional necessity propagates backward through presupposition, attribute by attribute, never period by period.

*The worked chain.* The record supplies the instance, and its middle link is already in this work's books. The wage structure on which Allen's account of mechanization rests is downstream of the demographic rupture of the Black Death, the serial-contingency program of §3.7 carries exactly this candidate, and the Renaissance's dependence on the same rupture is a standing historiographical thesis [105, 106]. So two independent routes into modernity pass through one fourteenth-century shock, and the chain then runs backward on its own licence: the shock presupposes the configuration it struck, the demographic density, the trade routes that carried transmission, the institutional structure whose rupture did the work, so the medieval period *up to those attributes* inherits the requirement; and that configuration presupposes, in part, the fall-shape of Rome already entered above as route-deletion. Under the Golden target, then, the medieval period had to happen — in the only sense the framework licenses: at the grain of the attributes the chain actually consumes, and at the grade of its weakest link.

*The question each link generates.* "Up to which attributes" is not a rhetorical flourish but the audit of Remark 8.17 applied link by link: the requirement attaches to the *role*, a demographic rupture of sufficient depth to reset labor scarcity and institutional equilibrium, and whether *Yersinia pestis* in that century was its only available occupant is the substitution question, open [**O**] and now precisely posed. The invested reading of Proposition 7.6 prices the answer's stakes: the conditioning concentrates on the cheapest sufficient shocks, so if cheaper occupants of the rupture-role existed and did not run, the realized route's cost is evidence they were not in fact sufficient.

*The Europe question, posed at its correct grain.* The chain as realized runs through Europe, and the association invites two errors that should be foreclosed in the same sentence that admits the fact. The association is *occupancy* until audited: whether comparable rupture-to-reconfiguration routes stood open elsewhere, the same pandemic struck the Islamic world; China carried its own demographic shocks, is the role-level question, and the framework supplies its form and no verdict. And necessity is not merit: by Remark 3.40 and Remark 7.7 nothing took the European route or declined another, no people is ranked by a conditioning's concentration, and a route's being required speaks to the narrowness of what reaches the target, never to the worth of what occupied it.

*The dark-investment fence, stated at full strength.* This remark's least comfortable consequence should be owned rather than softened: under the Golden target a mass-death event is conditionally required, and the framework must say exactly what that does and does not mean. It is a claim about route-narrowness, that among $G$-reaching continuations from the thirteenth century, those through a rupture of that depth carry nearly all the weight, and it is the invested reading at its darkest, enormous realized cost as evidence of how few cheaper routes there were. It licenses no providence, no "for the best," no valorization of catastrophe, and no comfort: the conditioning does not act, and an accounting is not an absolution. A reader who feels the sentence "the Black Death had to happen" as monstrous is reading its register correctly; what the framework adds is only that the monstrousness is located in the narrowness of the routes, not in any agency that chose them. The whole remark is conditional on the Golden target and on historiographical inputs carried at [**C**]/[**S**]; under Void it is history, unconditioned and unrequired.

"nothing." It is a dominance argument, and it does not require believing the Route World obtains.

**The argument.** Stated in steps, so each can be refused separately.

1. *The Route World's phenomenology is categorical, not enumerative.* What distinguishes it from the standard world is a rate and a structure, prospective improbability, tense-relative labelling, evidence accumulating as $\mathcal{L}_t$, and not a list of instances. So it can be adopted as a reading and applied to circumstances not on the list.
2. *If one is in the Route World, the world's phenomenology is that one.* True by definition of the hypothesis.
3. *One wants one's interpreted phenomenology to match the world's.* This is the epistemic norm the whole work runs on and is assumed rather than argued.
4. *So if one is in the Route World, one should read and act under Route-World phenomenology.* From 2 and 3.
5. *Where the Departure Point obtains, the adopted ordering ranks nothing.* By §9.6.5: past $\mu G$ the viability event has probability zero, $Q$ is undefined as a conditional measure, and every quantity indexed to it is undefined rather than small. This is a claim about an index and not about the world, and it is $\Phi$6-conditional throughout.
6. *Pruning the Aethae in which departure obtains leaves exactly the Route World.* Conditioning on non-departure is conditioning on the viability event $\mathcal{I}$, and by Remark 6.2 that conditioning *is* the passage from the family $\mathbb{P}^{(t)}$ to the measure $Q$ — the conditioning absorbed into the base state, which is the statement of a future-referring attribute in present books. The Route World is the world of the measure. So the residue is not merely Route-World-like; it is the Route World, and the identification is exact.
7. *Therefore the states in which one's conduct is ranked at all are exactly the states in which step 4 applies.*
8. *Therefore one should act under Route-World phenomenology now*, since every alternative state is one the adopted ordering reads as a single point.

**What the conclusion is.** *Methodological adoption, not ontological credence.* The argument does not say one is in the Route World, and supplies no evidence that one is; it says that the states in which the question of how to act has an answer are the states in which the Route-World reading is the correct one. That distinction is the one §8.2 draws between what the evidence supports and what one's Aethus in fact carries, and this argument moves only the first.

The shape is the forced-choice argument of §9.6.5 generalised from a single choice to a reading. There the safe course was scored at zero because the survival recommending it lay inside an indifference class; here every departed world is scored at zero for the same reason, and what remains is not a small good against a large one but a comparison the ordering can make against comparisons it cannot. Nothing unbounded enters, and no small probability is multiplied by a large payoff.

*Remark* 8.19 (The complacency objection, and the two answers it gets)*.* The objection is immediate: if one acts as though one is on a conditioned route, does one not thereby relax — believing oneself saved, and so ceasing to do the work the conditioning is supposed to require?

There are two answers, and they are of very different strengths. *The framework's own answer, which is a theorem.* The inference from "I am on a conditioned route" to "I may relax" requires that one's own effort be identifiable as inessential to the conditioning — and §8.7 establishes that nobody can determine whether their own contribution is among the requirements, since by Theorem 6.18 no finite observation places a system on the interpolation at all. The premise the complacency inference needs is therefore unavailable to any finite observer — which is a stronger statement than *unverified* and a weaker one than *indeterminate*, and Remark 8.20 is why the difference matters here. And Remark 2.14 closes the other route to it: an individual is not a lever on their biosphere's hazard, so "my effort is dispensable" and "my effort is decisive" are equally unavailable, and the Route World supplies no licence to either. *Acting under the Route-World reading is compatible with no relaxation of effort whatever*, and the reading itself is what forbids the inference.

*The author's empirical bet, which is not.* A second answer has been offered and should be recorded at its own grade rather than folded into the first: that populations do not in fact behave this way — that individual instability is averaged out at scale, and that a competent society interposes barriers between any one actor and catastrophic capacity, so that the complacent reading could not be acted on even where it were held. The second half is at least legible in the framework's own terms rather than an appeal to hope: an arrangement in which no acquired capability can be turned inward by any single hand is what $\rho > 0$ *is* at the grain of institutions (Definition 5.32), so the wager is precisely the wager that one's biosphere carries favourable anatomy at that grain — which makes it statable, and not thereby true. It is a claim about how human institutions actually work, offered as a wager rather than a result, with a concrete falsifier, evidence that awareness of the hypothesis degrades collective self-preservation rather than being absorbed by the aggregation and the barriers, and it sits uncomfortably beside §11, which models institutional balance as something that fails under exactly the adversarial loading the wager assumes away. The paper records the bet with its falsifier and does not endorse it [Φ]; the first answer is the one the argument leans on, and it leans on nothing empirical.

**Which step to attack.** The argument's steps are not equally exposed, and a reader deciding where to push should be told which is which. Steps 2 and 4 are definitional or immediate. Step 6 is a theorem — the identification of the pruned residue with the Route World is Remark 6.2, and if it fails, the family and the measure were never distinct and much else fails with it. Step 3 is an epistemic norm this work assumes and does not defend; a reader who wants their interpretation *not* to track the world has other problems.

The exposed step is 5, and it is exposed in one specific way. The indifference past $\mu G$ is an indifference *of the adopted ordering*, and an agent who holds values the ordering does not encode, who cares about suffering in a departed world, say, is not indifferent there, and for them the pruning removes states that still matter. The argument is therefore Φ6-conditional in the same sense as everything downstream of §2.2, and it is weaker than it looks for readers whose evaluative commitments outrun the one premise this work asks them to adopt. That is not a defect to be repaired but the scope the argument has, and step 1 is the other place to push: if the Route World's phenomenology is only the sum of its instances, it cannot be adopted as a reading and step 4 has nothing to apply.

## 8.7 Fences

*Remark* 8.20 (Determinate is not accessible, and why the distinction cannot be relaxed here). An objection is available against the fences below, and it is good enough that stating it is better than hoping no reader finds it. It runs as a reductio. *The objection.* The fences say that whether one's own contribution is among the conditioning's requirements is a thing no observer can establish. But the Route World is precisely the hypothesis that the conditioning is *resolved* — the Golden Point stated rather than weighted, throughout the cone. If it is resolved, the requirements are settled; if they are settled, there is a fact about whether any given contribution is among them. So either that fact is available, in which case the fences fail and the complacency inference goes through — or it is not available even in the resolved case, and one wonders what "resolved" was supposed to mean. Push the first horn and the Route World undermines the conduct it recommends; push the second and it looks like a hypothesis with no content. Either way, so the objection concludes, the Route World is incoherent and its incoherence is a proof that we are not in one.

*The answer, and it is not a concession.* The objection equivocates on "available." Statedness is a property of an *Aethus*, of what the world's books carry, and the framework is explicit that an Aethus is deductively closed and indexed without regard to whether its bearer can compute over it [4]. So in a Route World the requirements are determinate *in the books* and remain inaccessible *to the reasoner*, and no contradiction arises, because those are different predicates of different objects. What Theorem 6.18 forbids is not there being a fact but *certification of it from observation*: $Q$ and $\mathbb{P}$ are mutually absolutely continuous on every finite window, so no finite record distinguishes them. That is a limit on inference and not on determinacy.

*Two corrections the objection earns.* First, the fences should not have said *in principle*, and above they no longer do: the measures are singular on the infinite-horizon $\sigma$-algebra, so an observer of unbounded duration would distinguish them, and what the ceiling supplies is unavailability to any *finite* observer. Second, the resolved case is not contentless. What "resolved" means is Remark 6.2's distinction, one is under the measure $Q$ rather than the family $\mathbb{P}^{(t)}$, and that has consequences the section spends its length on, including the evidence flux that orders the three worlds. A hypothesis can be substantive and its truth still uncertifiable; this work's own epistemic ceiling is the standing example, and it applies to this work's conclusions as firmly as to its targets.

*And the structure is one the section already carries.* Remark 8.10 makes the same distinction on the evidential side: the attribution information is discarded *at the projection*, not in the world, so which events supplied the accumulated evidence is determinate and not recoverable. Determinate-and-inaccessible is the framework's normal condition rather than a special pleading introduced to save the fences. *And a route that closes the leverage without the diagnosis.* Even were the objector granted the availability premise arguendo, no reductio against the Route World follows, for a structural reason independent of the equivocation above: the ceiling is a property of the model under every hypothesis of §8.4, Void, Weak, and Strong alike, so a premise that falsifies all three worlds at once selects among none of them. What an exhibited certification would refute is the mutual-absolute-continuity structure itself, at the site where its empirical exposure lives (§F), and the Route World would stand or fall with its rivals rather than before them.

*Remark* 8.21 (The robustness argument, and why it is not relied on). A stronger conclusion suggests itself and should be marked as the weakest thing here. If each conservative alternative, conditioning on some nearer, more peripheral future event rather than on the Golden Point, is one whose own passage will dissolve whatever objectivity it seemed to have, then only a target at the asymptote remains stable under the motion the argument itself describes; and one may be tempted to conclude that nothing *but* the Golden Point can be coherently conditioned on.

Two reasons not to lean on it. It requires that the peripheral alternatives be exhaustively surveyable, which is the same exhaustiveness the companion's own arguments purchase carefully and this one assumes. And it is the argument's only step that is not a likelihood computation, so a reader who accepts (8.1) and (7.1) and rejects this paragraph loses nothing else. The accumulation is what gives the Route World teeth; the robustness claim is a conjecture stated beside it [**O**].

*Remark* 8.22 (What the Route World does not license). One reading has to be closed explicitly, because it is the natural one and it is dangerous. If required flukes occur, then, the reading goes, an individual contemplating some improbable feat should ask not whether they will succeed but whether the Golden Point requires their succeeding, and proceed accordingly. Nothing in this work supports that, and two of its results forbid it. By Remark 2.14 an individual is not a lever on their biosphere's hazard: what they hold is a position, not a purchase, and Golden Optimation is a statement about which histories one finds oneself in rather than about what any agent can bring about. And by Theorem 6.18 nobody can determine whether their own feat is among the requirements, since no finite observation places a system on the interpolation at all. The premise of the reading is therefore unavailable to any finite observer, which is not the same as there being no fact of the matter (Remark 8.20).

What survives is weaker and is the defensible form: a Route World is one in which improbabilities are more common than the Cosmic Nexus's rates predict. It is not one in which any particular improbability is owed to any particular person, and an agent who acts on the stronger reading is not applying this framework but discarding §6.8. Remark 9.7 fences the neighboring case in the same terms and for the same reason.

**The phenomenology extended: from events to constitution.** The Route World as developed in this section is a claim about *events*: which occurrences carry weight, how the record reads under the conditioning, what the forward flux does. Its Strong reading supports one further extension, developed where the categorical machinery lives rather than here: from the structure of events to the constitution of things — the universe's contents as weight-carrying excursions on the Aethus that states the conditioning attribute (§4.6). Nothing about the Route World's own claims strengthens or weakens by the extension, and its fences govern there unweakened; the pointer is recorded here because a reader of this section should know the hypothesis's full reach before weighing it.

*Remark* 8.23 (A consequence for periodization). One application is worth recording because it is where the Route World meets Principle 4 and returns something concrete. Golden Optimation running over the ancestral record and over the present is one process under one conditioning, so there is no categorical transition between the environments of natural history and those of the present — which is continuity's own claim, arrived at from the conditioning side. The present is nonetheless distinguished in *stakes*, by Remark 10.5, and the two are compatible for the reason §2.5 gives: a monotone sequence distinguishes every later point over every earlier one without distinguishing any point categorically. The right image is a field potential that is high here because the Golden Point is near ahead — and the misreading to guard against is that a high potential marks a step. It does not: a potential rises continuously toward its source, and what is high at the present is high because of proximity to a transition that lies ahead, not because a transition has been crossed.

The consequence for periodization is that a boundary drawn in the past, an Anthropocene, on any of its proposed placements, marks a change in stakes rather than in kind, and continuity is what denies it the second reading. The argument should be run from the principle and not from the failure to locate a line, since inability to place a boundary is a sorites and not a proof. What the Route World adds is a positive alternative: the transition the periodization is reaching for lies *ahead*, at the Golden Point, which is the one place the framework does mark a change in kind — and which §6.8 forbids anyone from dating.

# Part IV: Scores, Meaning, and Conduct

Part IV is the normative and existential engine. The Score is derived and fenced, Golden ethics is developed with the Departric result at its center, the commitment-node crisis is diagnosed and given its criterion, and the third route, inferred meaning, is stated at its two tiers, placed against its literature, and defended against the reduction to religion (§9). The part yields conduct guidance under the computation ban and the meaning claims at their ladder of grades.

## 9 Golden scores: a decision rule and its limits

§2 adopted a grounding and left it uncomputable. This section computes it, and the computation is available in the model's own terms rather than by analogy: two options' survival tails carry the same power of $t$, the power cancels, and what remains is the ratio of their persistence-harmonic values — the Doob density that defines the Golden Biosphere. The decision rule proposed informally in the earlier material is therefore $\mathfrak{h}(A)/\mathfrak{h}(B)$, weighted by how attainable each option is.

Three things accompany it and are stated rather than deferred. The identity is exact for the hard-wall functional and a bounded-factor comparison for the soft-killing one, which is what the inputs' grades purchase. The map from a decision to a point in the model's state space does not exist, so the formula is exact and not yet usable, and Remark 5.34's reasoning says why supplying one in haste would be worse than waiting. And the rule licenses a comparison of conditional probabilities, never a duty: $\Phi2$ stands, and §9.5 restates it at full strength, including the consequence that no actor is scored by it.

### 9.1 The Initiative

§2 ended with an agent's grounding; this section asks what the grounding computes. Between the two sits one definition, and one sentence relating it to the mathematics.

**Definition 9.1** (The Golden Initiative; [Φ])**.** The Golden Initiative is a biosphere's organized effort toward aligning practical outcomes with the Golden Answer.

The relation to the Tendency is the subsection: *the Initiative is the conscious-agency component to what the Tendency describes mathematically.* The Tendency is an ensemble fact, futures reweighted, nothing striving (Remark 2.10), and the Initiative is not its cause but its agent-side face: what a biosphere's practice looks like when organized around the affirmative rather than the negative. The paper has already consumed this object once. The middle step of Proposition 2.13 quantifies over exactly it, *a biosphere whose practices are organized around the negative does less to persist*, so the Initiative is the affirmative case of the practice-organization that step needs, named rather than left implicit. And the scale discipline of Remark 2.14 carries over unchanged: the Initiative is a biosphere-scale object, and an individual participates in it without being a lever on it.

One expectation attaches, and its kind must be stated before its confidence. In ensemble form it is the Tendency at the level of practice, and needs nothing new: biospheres found at horizon $t$ have, increasingly in $t$, more of their practice so organized — Principle 1 read through the middle step. The trajectory form, one biosphere's Initiative deepening over time, in the far tail plausibly the analog of what self-preservation is to an organism now, is a different kind of claim, and Remark 2.11 locates rather than forbids it: it becomes available only where technoanatomy carries its own dynamics (§5.6.3, Corollary 5.43), and those dynamics are **[O]**. The forecast is filed where §2 says trajectory claims live, and nothing below consumes it.

## 9.2 The rule

Let an agent hold two mutually exclusive options in disagreeing superposition, $A$ with weight $P(A)$ and $B$ with weight $P(B)$, and let $G$ be the state of the biosphere surviving forever. The 2024 material proposed the comparison

$$P(A)\,P(G \mid A) \;\geq\; P(B)\,P(G \mid B), \tag{9.1}$$

equivalently

$$\frac{P(G \mid A)}{P(G \mid B)} \;\geq\; \frac{P(B)}{P(A)}. \tag{9.2}$$

The feature worth keeping is slightly counterintuitive and real: *the difficulty of adopting an option factors into its evaluation.* An option held at weight 0.2 against an alternative at 0.8 must carry a compensating factor of four in conditional survival before the comparison favors it — the rule prices not only where an option leads but how far from the agent's present superposition it sits.

As written, however, the rule is degenerate exactly where the paper lives. In the critical regime $P(G) = 0$ identically, so (9.2) is $0/0$ and the comparison says nothing. The degeneracy is not a defect of the idea but of the finite form, and the repair is the limit — which is the next subsection's derivation, not a further stipulation.

## 9.3 The limit form and the harmonic function

Replace $G$ by survival to a horizon, $S_t$, and take the horizon out:

$$\lim_{t\to\infty} \frac{P(S_t \mid A)}{P(S_t \mid B)} \;\geq\; \frac{P(B)}{P(A)}. \tag{9.3}$$

In the critical regime this limit is computable in the paper's own machinery — but the computation must be run at the grade its inputs carry, and they carry two. Claim 5.30 gives the survival tail as $\mathbb{P}(T > t) \asymp C\,t^{-1/4}$, and the paper's convention fixes $\asymp$ as bounded ratio: $0 < \liminf f/g \leq \limsup f/g < \infty$. Two quantities each $\asymp$ the same power have a ratio bounded between positive constants; nothing converges. What delivers convergence is the *refined* form, $\mathbb{P}(\tau > t) \sim c\,\mathfrak{h}\,t^{-1/4}$, which Theorem 5.29 with Proposition D.8 (general start, [58]) supplies for the hard-wall functional $\tau$, and which Theorem 5.75 already assumes. Ansatz 5.18 transfers $T$ to $\tau$ only at $\log \mathbb{P}(T > t) = \log \mathbb{P}(\tau > t) + O(1)$ — $\asymp$-grade, which does not upgrade.

**Proposition 9.2** (The Golden Score; **[P]** given **[A]** and **[C]** inputs)**.** *Identify the options $A$, $B$ with starting states (the identification map pending; see below). For the hard-wall functional $\tau$,*

$$\lim_{t\to\infty} \frac{P(\tau > t \mid A)}{P(\tau > t \mid B)} \;=\; \frac{\mathfrak{h}(A)}{\mathfrak{h}(B)}, \tag{9.4}$$

*the Doob transform density of* (5.47)*, and* (9.3) *is equivalent to the single computable functional*

$$P(A)\,\mathfrak{h}(A) \;\geq\; P(B)\,\mathfrak{h}(B). \tag{9.5}$$

*For the soft-killing functional $T$ the limit exists but is* not *taken in $\mathfrak{h}$. Splitting $\mathbb{P}_x(T > t) = \mathbb{E}_x[e^{-\Lambda_t}]$ on $\{\tau > t\}$, the surviving term is $\mathbb{P}_x(\tau > t) \cdot \mathbb{E}^{(t)}_x[e^{-\Lambda_t}]$, and the conditional factor converges to a* state-dependent *constant $\mathbb{E}^x_Q[e^{-\Lambda_\infty}]$; the sub-wall term contributes a renewal series over excursions. Hence*

$$\lim_{t\to\infty} \frac{P(T > t \mid A)}{P(T > t \mid B)} = \frac{\mathfrak{h}_T(A)}{\mathfrak{h}_T(B)}, \qquad \mathfrak{h}_T(x) = \mathfrak{h}(x)\,\mathbb{E}^x_Q\big[e^{-\Lambda_\infty}\big] + (\textit{excursion series}), \tag{9.6}$$

*which differs from $\mathfrak{h}(A)/\mathfrak{h}(B)$ in general, since $\mathbb{E}^x_Q[e^{-\Lambda_\infty}]$ increases with the start. Using $\mathfrak{h}$ in place of $\mathfrak{h}_T$ is the bounded-factor approximation, and $\mathfrak{h}_T = c\,\mathfrak{h}$ exactly in the hard-wall limit.*

**What the soft-killing statement costs, and what discharges part of it.** (9.6) is stronger than the bounded-factor language it replaces, the rule is *exact* for $T$ as well, in a different object, and it is owed three conditions. The first is that the $Q$-expected residual hazard be finite, $\sup_t \int_{t_0}^t \mathbb{E}^{(t)}[h^{\mathrm{tot}}]\,ds \to 0$; the other two govern the excursion series (finite crossing mass, and convergence of the renewal sum), and both remain open **[O]**.

The first is not open for this model. Under $Q$ the credit grows as $x_s \asymp s^{3/2}$, so a background decaying only polynomially, $\varsigma_b(x) \sim x^{-p}$, would give residual hazard $\int s^{-3p/2} ds$ and require $p > 2/3$. But (5.7)'s muffling factor is logistic and therefore decays *exponentially*: $\varsigma_b(x_s) \sim e^{-s^{3/2}}$, and the trough contributes $\Gamma e^{-\rho s^{3/2}}$. Both integrals converge absolutely whenever $\rho > 0$ — which is the regime the Score is stated for anyway, by Theorem 5.38. So the condition is met with room to spare, and the polynomial threshold is recorded because it is a real constraint on any *other* choice of soft wall.

The consequence for the identification map is worth naming: what a decision must be mapped to is a starting state for $\mathfrak{h}_T$, and $\mathfrak{h}_T$ carries the hazard the biosphere will actually accrue rather than only the geometry of the wall. The proof is the cancellation: both tails carry the same $t^{-1/4}$, under $\sim$ the prefactors are all that survive, and the prefactors are harmonic values. So the decision rule stated informally in 2024 is computable in the model's own terms, and the quantity it computes is the same object that defines the Golden Biosphere — a derivation, not an analogy. The qualification in the proposition is not a retreat: a comparison exact for $\tau$ and bounded-factor for $T$ is precisely what the inputs' grades purchase, and stating it so is what the grading discipline is for.

*Aurutance.* The natural linearization is the logarithm. Define the Aurutance of $A$ over $B$, under the same hypotheses, as

$$\mathrm{Aur}(A,B) \;=\; \lim_{t\to\infty} \ln \frac{P(S_t \mid A)\,P(A)}{P(S_t \mid B)\,P(B)} \;=\; \underbrace{\ln\frac{\mathfrak{h}(A)}{\mathfrak{h}(B)}}_{\text{log-Doob density}} \;+\; \underbrace{\ln\frac{P(A)}{P(B)}}_{\text{log-prior}}, \tag{9.7}$$

a portmanteau of *aurum* and *importance.* The sign convention falls out of (9.5): positive Aurutance is the comparison favoring $A$, in the exact sense of the limit and in no other sense — what further sense an agent attaches to it is §9.5's subject. One reading is fenced now rather than later: Aurutance is the logarithm of a probability ratio, and the additivity of (9.7) across independent comparisons is the additivity of logarithms — it licenses no accounting metaphor beyond that, and in particular no stock of scores to hold, spend, or owe.

*Two gaps, flagged rather than papered over. First, the identification map is missing.* $\mathfrak{h}$ is a function of the model's state $(\mu, D)$; $A$ and $B$ are Aethic states — additions to an agent's Aethus. Nothing in the paper supplies the map from a decision to a point in $(\mu, D)$, and without it (9.5) is exact and unusable. The gap is **[O]**, and it is the same shape as the reservation already recorded at Remark 5.34 about measuring $\rho$: a premature proxy for $\mathfrak{h}$ would be a scored index, inviting exactly the Goodhart failure that $\rho < 0$ names. The formula waits for the map; it does not license improvising one. Remark 5.65 names the one place a candidate might be found: the schedule channel delivers a distribution over completion states, and completion states are initial data. *Second, the computation is regime-specific.* The cancellation is a fact about the critical cell alone — the next subsection tables what happens elsewhere.

One qualitative consequence is available now, at the class level and no finer. Expenditure that raises technostrength while leaving technoanatomy untouched moves neither $\mathfrak{h}$ nor the comparison, it generates no Golden Score, and by Remark 5.41 may generate a negative one. That is a claim about a class of activity, it follows from results already proved, and it is as far as the rule reaches: Theorem 6.18 forbids certifying any system's position from finite observation, so no actor, named or unnamed, is scored by it — a bar §9.5 states in full.

## 9.4 Regime-dependence

What the rule computes depends on the cell, in the pattern of §6.2.1:

| *Regime* | *Survival tail* | *The rule computes* |
|---|---|---|
| Viable | $\mathbb{P}(T=\infty) > 0$ | an ordinary conditional ratio; no limit needed |
| Critical | $\sim c\,\mathfrak{h}\,t^{-1/4}$ (refined form) | the harmonic ratio, Proposition 9.2 |
| Trapped | $\le e^{-h_* t}$, frozen $B$ (Theorem 5.36(ii)) | a comparison of floors $h_*$, via (5.20) |

The trapped cell deserves finishing rather than dismissal. Across different anatomies the ratio diverges or vanishes, and what the rule collapses to is a comparison of the floors themselves: $h_*$, which (5.20) gives in closed form as a geometric interpolation between the external and internal scales, weighted by anatomy. The trapped verdict is therefore *lower the floor*, which is to say *move* $\rho$, exactly the intervention Remark 10.4 names as the one that changes the indicator rather than the multiplicand. Even where the harmonic structure is gone, the rule points at anatomy.

The asymmetry deserves its sentence: the rule is *most* informative exactly where survival is *not* achievable, the critical cell, where $P(G) = 0$ and the harmonic structure is richest, and least informative where survival is either assured in measure or foreclosed. §6.2.1 records the same inversion about the mutual-singularity claim; this is the second instance of that structural fact, and noticing the repetition costs one sentence and buys the pattern.

**The rule's neighbors: maxipok, ergodicity, ruin.** Three decision-flavored literatures sit near the Score, reached from premises this work does not share, and the pattern across all three is the same: convergent conclusion, divergent grounds, with the divergence carrying the content.

*Maxipok.* Bostrom's rule, maximize the probability of an "OK outcome", is the Score's nearest published neighbor [107], arrived at from astronomical-waste aggregation, exactly the expected-value machinery §2.7 declines. The deltas: the Score's ranking is by survival-harmonic weight rather than bare probability, so options are priced by what they make attainable and not only by what they avoid; the "OK outcome" is undefined where $\nu F$ is a fixed point; and where maxipok invites computation, the Score forbids it absent the identification map. Same direction of travel, different vehicle, and the vehicle is the claim.

*Ergodicity economics.* Peters's program replaces ensemble-expectation with time-average growth, on the ground that a multiplicative process's expectation is dominated by branches no history occupies [108]. The kinship is real and structural, Aurutance is a log-ratio, additive across independent comparisons, and the refusal to multiply small probabilities by large payoffs is common property, and the delta is the target: there the object whose growth is averaged is wealth, here the object is the survival-harmonic weight, and the family-versus-measure distinction (Remark 6.2) is what the time-average critique looks like when the multiplicative quantity is existence.

*Ruin problems.* The precautionary literature's structural core, ruin is an absorbing barrier, so cost–benefit reasoning that averages across it is invalid [109, 85], is the Base-level shadow of this section's machinery. The delta is the closure: their ruin is an event, $\mu G$ is the inevitability closure of the event, strictly larger (3.3), and the separations of Remark 2.19 live entirely in the difference. A precautionary rule keyed to observable ruin guards the wrong boundary: it fires late where departure is silent, and it would forbid, in the critical regime, exactly the exposure that constitutes staying alive.

## 9.5 What it does not license

The fence, at full strength, because this is where a reader most wants the rule to be more than it is. Φ2 stands: the rule licenses a comparison of conditional probabilities, and a comparison is not a duty. Calling one option *better* imports Φ6 *and* the Golden Answer, both of which are the agent's adopted grounding rather than the apparatus's output — the split of §2.2, consumed here exactly as stated there, and the reason the enumeration's fourth step said *comparisons follow*, not *duties do*. An agent who has adopted neither reads (9.5) as arithmetic about an ensemble, and the paper gives them no argument that they are wrong to.

Two further bars. Theorem 6.18 forbids certifying any system's position on the interpolation from finite observation, so the rule cannot be used to score an actor — not a person, not an institution, not a biosphere, one's own included; the class-level consequence of §9.3 is the most the machinery yields. And the missing identification map means no number anyone produces today *is* an Aurutance: a computed score requires a map the framework does not yet contain, and anything offered in its place is a proxy of exactly the kind Remark 5.34 warns against. Remark 10.7 fences the neighboring stakes argument at this same scope and in this same tone; the two fences are one commitment stated twice.

## 9.6 Golden ethics

§9.5 said what the rule does not license. This subsection says what it licenses *for an agent who has adopted the grounding of §2*, and that qualification is the whole of its content. No premise is added here: everything below is downstream of Φ6 and the affirmative answer to the Golden Question, and an agent declining either is committed to nothing in it.

### 9.6.1 The commitment, and why it is not an issued obligation

The tempting formulation is a duty — *one ought to prefer the option with the higher Golden Score*. Stated so it is a second premise, and Φ2 forbids it: §2.2 is explicit that Φ6 is the single doorway through which obligation enters this work, and a free-standing Golden Obligation would be a second doorway. The formulation that costs nothing reads the commitment off the stakes instead of issuing it. §10.1.4 establishes that forfeiting is not the same as never having had, and Remark 10.5 that the stakes so understood rise monotonically along a lineage. Those are results, not instructions. What Φ6 adds is an agent standing in a grounding under which those stakes are *theirs*. The commitment is then not an imperative anything issued but the shape those stakes take once occupied — and it is revocable in one specific respect and not in others. Three parts, easily run together and worth separating.

*That there is an obligation* is not conditional at all. Condition 1 of §2.1 places the agent under forced action, and Proposition 2.4 establishes that they hold a grounding whether or not anything is said. An agent already acts, and already on some footing — so the normative is in place before Φ6 speaks, which is exactly what §2.2 means in denying that any *ought* is derived here from an *is*. None of this issues from the apparatus, so Φ2 is untouched by it.

*What the obligation is about* is the conditional part, and the only one. Φ6 selects which frozen question a grounding turns on, and an agent declining it does not thereby escape obligation — they hold a different one, on a footing they have not chosen and cannot inspect. *And how much rides on it* the stakes supply. §10.1.4 and Remark 10.5 give the magnitude, and Principle 4 extends it backward: the stakes are continuous with those our ancestors faced without agency, so nothing about them begins with us. Stakes supply weight and Φ6 supplies direction, and neither does the other's work. *Golden virtue* names conduct answering to that shape and claims nothing about virtue in any prior sense.

### 9.6.2 Two moralities, and the case that separates them

Write $\mathcal{G}$ for the Golden Point, $\mathcal{N}_0$ for the Cosmic Nexus, and let $\varphi$ range over stated attributes — dispositions, policies, practices. Two quantities are available and they come apart.

**Definition 9.3** (Automorality)**.** For an Aethus $A$ carrying $\varphi$, with $A'_\varphi$ its complement under $\varphi$,

$$\mathrm{AM}(A,\varphi) = \ln \frac{\mathbb{P}(\mathcal{G} \mid A, \mathcal{N}_0)}{\mathbb{P}(\mathcal{G} \mid A'_\varphi, \mathcal{N}_0)}. \tag{9.8}$$

The gain from the attribute's having been carried — read modally, *if this agent does this, does the Golden Point become likelier?* It is act-indexed and indifferent to intent.

**Definition 9.4** (Relative morality)**.** For a reference Aethus $B$, with $B_{\varphi,A}$ replacing $B$'s attribute under $\varphi$ by $A$'s,

$$\mathrm{RM}_B(A,\varphi) = \ln \frac{\mathbb{P}(\mathcal{G} \mid B_{\varphi,A}, \mathcal{N}_0)}{\mathbb{P}(\mathcal{G} \mid B, \mathcal{N}_0)}. \tag{9.9}$$

The gain from the disposition's being *adopted* — read modally, *if an agent in general does this, does the Golden Point become likelier?* It is maxim-indexed rather than act-indexed, and asks what transplanting the disposition would do rather than what carrying it did.

They agree in the ordinary case and separate in the instructive one. Take an attribute exercised with intent to harm which, by the stepping-stone structure of §3, happens to open a route that mattered. Its automorality is positive: it helped. Its relative morality is negative: installed in a reference agent it would be expected to do the destructive thing, and the favorable outcome was the accident rather than the disposition. Table 1 places this at *Biotic evil*, evil by disposition, Biotic by outcome, one cell along from *Departric evil*, which is the same disposition where the accident did not occur, and diagonally opposite *Departric walk*, where the disposition was the good one and the accident went the other way. The pair therefore holds apart what an act *did* from what a disposition *is*, rather than reducing either to the other.

| | **Departric** (autoimmoral) | **Neffective** (autoneutral) | **Biotic** (automoral) |
|---|---|---|---|
| **relatively moral** | Departric walk | Neffective walk | *Golden good* |
| **relatively neutral** | Departric power | Neffective | Biotic power |
| **relatively immoral** | *Departric evil* | Neffective evil | *Biotic evil* |

Table 1: Dispositions classified by the two moralities. Each name is built from its coordinates, and the prefix is always a term of art, so no cell can be mistaken for ordinary usage. The *prefix* reports the outcome: *Departric* where automorality is negative, *Neffective* where it is zero, *Biotic* where it is positive — so *Bio-* occurs in the automoral column and nowhere else. The *suffix* reports the disposition: *walk*, *power*, *evil* as relative morality runs +1, 0, −1 — so *evil* occurs in the bottom row and nowhere else. *Walk* is chosen for the top row because English supplies no neutral term for a disposition worth adopting: *merit*, *virtue* and *goodness* all import an approving verdict on the outcome, which two of that row's three cells do not have. A walk is a manner of proceeding rather than an act or a result, so the prefix is left free to report how the proceeding turned out. *Good* is not a suffix in the scheme at all, since it asserts a beneficial outcome that only one column has; it is reserved for the corner where merit meets a Biotic outcome, which takes *Golden* in place of *Biotic* to mark the double alignment. The centre cell is bare because both coordinates vanish there and the ratio of Remark 9.5 is undefined rather than unbounded. So the bottom row's two evils are one disposition-class distinguished by what it happened to produce — the distinction the pair exists to draw and the one a single coordinate cannot — while *Departric walk* is its mirror, a disposition that would have helped meeting an outcome that harmed. *Neffective evil* imputes no malice, relative immorality covering negligence and folly and requiring no ill intent. The names are stipulative in the way *force* is stipulative in mechanics, and are not offered as analyses of ordinary usage.

Cell by cell. The first clause reports what the disposition does if adopted generally, the second what carrying it did here; the two are indexed differently, which is why they may disagree without contradiction.

- *Departric walk.* Generally beneficial; harmful here. A disposition one would want adopted, meeting a circumstance in which carrying it moved things toward Departure — misfortune rather than fault.

- *Neffective walk.* Generally beneficial; inert here. A disposition one would want adopted, which on this occasion found no purchase, with no verdict on its bearer implied either way.
- *Golden good.* Generally beneficial; beneficial here. The one cell with both coordinates positive, which is what earns *Golden* and what earns *good*.
- *Departric power.* Generally inert; harmful here. Nothing in the disposition to transplant, and an outcome owed entirely to where the attribute happened to fall.
- *Neffective.* Generally inert; inert here. Neither a disposition nor an outcome, and nothing for position to stand between — the one cell about which the pair has nothing to report.
- *Biotic power.* Generally inert; beneficial here. The attribute contributes nothing transplanted, yet in the circumstance it occupied it raised $\mathbb{P}(\mathcal{G})$: what would be nothing anywhere was decisive here.
- *Departric evil.* Generally harmful; harmful here. Disposition and outcome agree unfavorably — the cell ordinary usage means by *evil*, and the only one it fits without loss.
- *Neffective evil.* Generally harmful; inert here. A disposition that failed to find its occasion; since relative immorality covers negligence and folly, no ill intent is implied.
- *Biotic evil.* Generally harmful; beneficial here. The disposition is the destructive one and the favorable outcome was the accident — the case the two coordinates exist to separate.

The diagonal is agreement and the off-diagonal divergence, and the divergence is where a single moral verdict has to choose what to throw away. Ordinary usage collapses *Departric evil* together with *Biotic evil*, since both are dispositions one would not want adopted; and it has no settled name at all for *Departric walk*, which is why §9.6.2 needed two coordinates rather than one.

*Remark* 9.5 (Position against disposition). The ratio $|\mathrm{AM}/\mathrm{RM}|$ measures something neither term does alone: how far an outcome was owed to the *position* an attribute was exercised from rather than to the attribute. Large ratios mark dispositions whose effect exceeds what the same disposition would achieve elsewhere, which is what is ordinarily meant by leverage. This is the position-versus-purchase distinction of Remark 2.14 appearing on the normative side, and it inherits that remark's caution: occupying a position of leverage is not a causal claim about its occupant.

Three notes on the vocabulary this fixes. *Walk* carries its older moral sense, a manner of proceeding, as in a walk of life, and not the stochastic one the model's integrated random walk uses; the modifier disambiguates at every occurrence, and the two senses meet in no single passage. *Neffective* names the automorality-zero column — a disposition whose carrying did not move $\mathbb{P}(\mathcal{G})$ at all, which is the Golden analogue of an attribute of zero linkage in the companion's sense [5]. And *power* is the suffix for the relatively-neutral row, which repays being read literally rather than as a row of weak effects. $\mathrm{RM}_B(A, \varphi) = 0$ says the attribute transplanted into the reference circumstance does *nothing at all*; $\mathrm{AM}(A, \varphi) \neq 0$ says that in the circumstance it actually occupied it did something. The row is therefore an *exchange*: multiply the attribute into a standard Aethus and it is inert, multiply it into this one and it is consequential, and the whole of the outcome is owed to which Aethus received it. That the ratio above is unbounded there is not a pathology but the correct reading — all position, no disposition.

Which makes the row the disposition-side analogue of a structure the companion already names. A Miracle is an event of negligible generic weight whose occurrence on one particular line was decisive [5]; a power cell is an attribute of null generic effect whose carrying in one particular circumstance was. The parallel is structural and not an identity, those are events and these are dispositions, and the companion's linkage is indexed to observer existence where these are indexed to $\mathcal{G}$, but it is the same shape, and it is why *power* is the right word: the capacity to matter here that would matter nowhere.

The centre cell is the one place this fails, which is why Table 1 leaves it unsuffixed. Where automorality vanishes as well, the ratio is $0/0$ and undefined rather than unbounded: there is neither a disposition nor an outcome for position to stand between, so nothing is being exchanged.

*Remark* 9.6 (The act and the maxim, and what the pair is doing to a familiar dispute). The modal readings above have a shape worth naming. Automorality asks after an act, in a position, by an agent; relative morality asks what follows if the disposition is adopted generally. That is the distinction between an act and a maxim, and it is the distinction the categorical imperative's universalization test needs in order to be stated at all.

The resemblance should not be inflated into an encoding, and three differences say why. Kant's test is for *contradiction*, a maxim that cannot be conceived or cannot be willed as universal, whereas RM tests for a change in probability, which is consequentialist content wearing a universalizing form. Kant universalizes over all rational agents, whereas RM substitutes into a stated reference $B$ and carries it essentially, so what is recovered is a *localized* universalization with the full one as a limiting case, obtained by ranging $B$ over a population in the manner §9.6.3 describes. And Kant's test returns a verdict where RM returns a magnitude.

What the pair does supply is a measurement of a fault line rather than a position on it. Automorality is act-consequentialist; relative morality is the universalizing form with consequentialist content, which is to say rule-consequentialist. The cells where the two diverge, *Biotic evil* and its mirror in Table 1, are exactly the cases over which act- and rule-readings of consequence have always disagreed, and the two coordinates *quantify* the disagreement instead of adjudicating it. A framework that had to choose between them would lose the information the divergence carries.

One differentia remains, and it is the one §9.6.4 turns on. Every consequentialism needs a quantity to be a consequence *of*, and the usual choice is welfare aggregated over occupants. Here it is $\mathbb{P}(\mathcal{G})$ — probability of reaching the asymptote, not a sum over anyone. So this is a normative ethics fitted to infinite-horizon biosphere survivorship rather than to utility, which is why it ranks as it does and why the agreement with aggregative views is occasional rather than principled.

### 9.6.3 What this cannot do

Three limits, the first severe enough that the construction should be read as an exercise rather than a proposal. *It requires a vantage this work proves unavailable.* Both (9.8) and (9.9) consume $\mathbb{P}(\mathcal{G} \mid \cdot)$ for arbitrary counterfactual Aethae. Theorem 6.18 forbids certifying any system's position from finite observation, and no weaker vantage supplies these either. So Golden ethics is not awaiting data — it is a well-defined construction on a counterfactual the framework itself rules out. That is a stronger disclaimer than impracticality, and it is the correct one.

*It does not score persons.* Both quantities are defined over *attributes*, and §9.5's prohibition stands unamended, the rule cannot be used to score an actor, "not a person, not an institution, not a biosphere, one's own included", so the apparatus remains articulate about dispositions and classes of conduct and silent about their bearers. A table of dispositions is not a table of people, and reading Table 1 as the latter is the error that remark exists to prevent.

*And no score is absolute.* RM carries its reference $B$ essentially rather than incidentally; there is no view from nowhere against which a disposition scores. The nearest substitute is an extremal score across a stated population, which is a different object and should be labeled as one.

### 9.6.4 Why this is not longtermism, substantively

Remark 6.7 answers the objection structurally and concedes that in the viable regime it must be met by $\Phi 2$ alone. This closes that gap, because even granting a normative reading the two orderings differ. Golden reasoning grades by probability of reaching the boundary, not by aggregate life. A thousand persisting far outrank a million persisting less far, since what is conditioned on is survival to the asymptote and not a sum over occupants; and a population growing while decaying exponentially in prospect ranks poorly for the same reason, being no sustainability at all.

The standard test case separates them cleanly. Asked whether to found a settlement whose occupants will live and die young, an aggregative longtermism answers yes — existence is preferred to non-existence and the sum increases. Golden reasoning has no such argument available: the settlement's contribution to $\mathbb{P}(\mathcal{G})$ ranges from uncorrelated to adverse, and at no point does the framework return a positive verdict on that ground. Where the two agree, as over the fate of a biosphere, the agreement is a consequence and not a shared principle.

### 9.6.5 Past the Departure Point

A stronger version of the following is tempting and unsupportable, so the weaker one is stated exactly. The tempting version says that beyond a biosphere's Departure Point all virtue and meaning cease to exist. Nothing here supports that. What is supportable is that past $\mu G$ the viability event has probability zero, so $Q$ is not defined as a conditional measure and every quantity indexed to it, Golden Scores, both moralities above, the ordering of (9.5), is undefined rather than small. That is a claim about an index, not about the world.

The claim has a published cousin that locates it from the outside. Scheffler's collective-afterlife argument holds that the mattering of present activity depends, more than we notice, on humanity's continuing after us — his doomsday and infertility scenarios drain present projects of point [110]. That is this subsubsection's content met in the psychological register: his scenarios are conditionally Departric worlds, and what he argues we would *feel*, the deflation of mattering, is what the adopted ordering *computes*, indices undefined past $\mu G$. The deltas are two and they cut in opposite directions. His claim is empirical psychology and holds for whoever's motivations it describes; this one is $\Phi 6$-conditional and holds for whoever adopts the currency — narrower in reach, harder inside its reach. And his argument concerns valuers who precede the loss, while the striking here concerns the comparison itself; the two agree exactly where Remark 8.19's mixed-currency reader disagrees with both.

It has a consequence for forced choices, and the shape of the argument needs exact statement, because it resembles a wager and is not one. Suppose an agent under $\Phi 6$ faces two courses: one carrying a small probability of reaching the Golden Point, the other none. The obvious objection is that the second may be far safer in the ordinary sense — that the agent's own survival is likelier under it. The reply is that this survival lies entirely in the region past $\mu G$, across which the adopted ordering is indifferent by the paragraph above: every continuation there is equivalent under it. So the ordering does not fail to rank the safe course. It ranks it at *zero*, and the survival that recommended it falls inside an equivalence class the ordering reads as one point.

What remains is a comparison of two values of a single well-defined quantity, one strictly greater than the other — a percent against none. Nothing unbounded enters, and no small probability is multiplied by a large payoff, which is the structure this is most often mistaken for. It differs in a second way as well: a wager stakes something on a fact already settled, whereas here the probability is of the agent's own course succeeding, and which probability the agent faces is exactly what the choice determines. The quantity is contingent on the action rather than a credence about the world.

The conditionality is unchanged and does all the limiting work. An agent who has not adopted Φ6 holds no such ordering, faces no such comparison, and is owed no such conclusion — and nothing above converts the argument into one for them.

*Remark* 9.7 (What this is a formalization of). The structure above is not novel as an intuition, and saying what it formalizes is the quickest test of whether the formalization is any good. It is the familiar narrative case in which an agent accepts a course they are unlikely to survive because the alternative forecloses something they take to matter at large scale — "for the good of humanity," in the usual phrasing.

Ordinary decision theory reads such a choice as noble and irrational: near-certain survival traded for near-certain death. What the paragraphs above supply is the reading on which it is neither. The safe course is not being outweighed; it is being scored at zero, because the survival recommending it lies wholly inside a class the adopted ordering reads as one point. The agent is not overriding their interests but applying an ordering on which the safe course has no value to override with.

The formalization earns its keep by *discriminating*, since the trope has a familiar failure mode. The argument requires the risky course to carry nonzero probability of the thing valued. Where it does not, the comparison is zero against zero, the ordering is silent, and no gesture is licensed — which is Remark 10.4's structure exactly: faced with options of equal viability zero, the response is not to select among them but to find something that is not zero. Heroism that accomplishes nothing gets no support here, and that it gets none is a point in the construction's favor.

One stipulation should be visible rather than smuggled. The scenario assumes an agent with genuine leverage — no one better placed, and the act decisive. Remark 2.14 denies that in general: an individual is not a lever on their biosphere's hazard, and holds a position rather than a purchase. The argument concerns the shape of the comparison where such leverage obtains, and is not a claim that agents ordinarily face choices of this kind or should act as though they did.

## 9.7 Judging the nodes: commitment primitives and the inference-side crisis

The rule above evaluates *options*. This subsection concerns something the evaluation of options presupposes and moral philosophy mostly leaves in the dark: the undefended placements from which practical arguments are launched. It states a phenomenon, a diagnosis, and what the framework uniquely supplies — and it ends with the fences that keep the supply honest.

*The phenomenon, at its provenance grade.* Put a room of capable people to a live dilemma and ask for conduct. What returns is an array of stances that mutually render one another absurd — each participant certain the others are wrong, and each systematically unable to convey *why* on demand, however far the case is elaborated; the deeper the push, the more irresolvable the clash. The author witnessed this twice as an undergraduate, once in a philosophy-of-technology seminar, once in a philosophy-of-science seminar, both times over the ethics of de-extinction, and reports the structural content exactly: worldviews obligating vastly different conduct, not from anything intrinsic like religion or culture, but from a shallow mutual throwing-out of reasoned stances on which the whole room then stuck. Neither occasion was written down while fresh, so the exhibit is carried at recollection grade per the methodology companion's own practice [30], with the slot held open for a contemporaneous record when the dynamic is next observed. Two distinctions keep the phenomenon exact. It is not the wisdom-of-crowds setting: aggregation helps where many voices estimate one shared quantity with independent errors, and here there is no shared quantity — the divergence is not noise around a target but different targets, so no happy medium exists to converge on. And it is not the peer-disagreement problem of the epistemology literature [111], which asks how peers sharing evidence and methods should update on one another; the seminar deadlock fails the setup's antecedent, since what differs is neither evidence nor method.

*The diagnosis.* What differs is the *nodes*. Attention in argument goes to the propagation layer, the reasoning, and the equivalence classes of outcomes it defends, while beneath it sit commitment primitives: "it is paramount that we preserve this," placed undefended at some spot in the normative landscape, from which everything else propagates. Nobody in the room knows what anything really *is*, and it is precisely that non-understanding which frees each participant to place their primitive anywhere; the propagations then conflict because the placements do. And the crisis is one level deeper than the clash, because it attacks testimony itself: knowing that a speaker's node is sourced in their biographical and neurological contingency, knowing they do not know the context and are guessing at it, voids the evidential value of everything downstream, *however valid the propagation*, since there are no good guesses where guessing has no target, the source is not excusable from the conclusions, and the source is known to be no good. Call this the *inference-side crisis*: not that people disagree, but that each participant is right to discount every other's testimony at the root, and knows the others are right to discount theirs. Its first-person phenomenology has a canonical name in the literature, the anguish of the value-chooser [112], engaged with its tradition at §9.9.

*The nearest neighbors, and the delta that matters.* The diagnosis has a distinguished ancestry and the framework should say so before saying what it adds. MacIntyre opened *After Virtue* on exactly this phenomenology, interminable modern moral disagreement, rival premises incommensurable, premise-choice arbitrary, and Anscombe's diagnosis of a deontic vocabulary surviving the loss of its root is the same finding one register up [3, 113]. The delta is the remedy, and it is total. MacIntyre, holding a universe-sourced telos to be unavailable to post-Enlightenment thought, retreats to teleology internal to traditions and practices; the present framework's entire claim is that the unavailable object exists: a root sourced in the actual conditioning structure of the universe, the Golden Point, with §4.6 supplying its constitutive standing under Strong, rather than in any tradition, any practice, or any reasoner's contingency. Where the diagnosis is shared nearly verbatim, the framework answers the crisis with the object its sharpest diagnostician declared missing. One provenance fact belongs on the record here, because the methodology companion makes it evidential [30]: the diagnosis above was arrived at independently, from the seminar observations, dated in the collaboration record, and located in the literature only afterward, at placement. By the companion's own standard, independent convergence on a diagnosis is evidence *for* the diagnosis while moving none of the contribution's locus, which sits at the remedy; the author has confirmed the hinge accordingly, and the paper treats it as one: the crisis this subsection names is the crisis the framework's categorical and ontological completions (§4.6) exist to answer.

*What the framework supplies: a criterion for the nodes themselves.* Judge commitment primitives the way Remark 5.7 judges categories: by their tracing to the root. A node is non-arbitrary to the degree its content is individuated by contribution to the Golden Point — the task-object machinery of §4 run on normative primitives, with the contribution referent of Remark 4.2 pricing what the commitment's object actually carries. The paradigm case is already on the books: the frozen answer to the Golden Question is a commitment primitive whose holding is survivorship-forced rather than biographically sourced (Proposition 2.13) — the existence proof that a universe-sourced node is possible at all. And the de-extinction dilemma that occasioned the observation is, by no accident, a case the criterion can actually pose: "what would we be restoring, and what would its restoration contribute" is a task-object identification and a contribution question stated in the framework's own primitives — recorded here as a worked-case candidate **[O]**, not worked.

*Why it iterates where rule-axiomatizations dissipate.* The author's second advertised property deserves its structural ground. A rule-form axiomatization, Kantian universalization is the standing example, and Remark 9.6 locates this framework's relation to it, is a finite formula, and finite formulas jam on complexity: hard cases multiply conditions until the rule under-determines or self-conflicts. The Golden root is not a formula but a *conditioning*, and a conditioning is defined wherever its measure is: it refines under any partition of cases, composes across scales, and grades rather than jams (the $\varepsilon$-apparatus, the per-case individuation of task-objects, the Score's comparisons). That is the exact sense in which the Golden answer "operatively fits the shape of the situation rather than dissipating with complexity": the criterion's domain is the whole space of situations because it is carried by a measure on that space, not by a clause-list over it.

*Remark* 9.8 (The deleted middle term, re-derived). The engagement with the interminability diagnosis can be made exact, because the diagnosis has a precise anatomy [3]: classical ethics was a three-part scheme, the agent as they happen to be, the agent as they would be were their telos realized, and ethics as the discipline of the transition, and modernity deleted the middle term when its warrants (metaphysical biology, divine law) died, leaving the outer fragments with nothing to connect them. On that anatomy the Enlightenment justification project failed necessarily: it inherited the fragments' content while rejecting every framework that had made content intelligible. The framework's reply maps term by term, and the mapping belongs on the page rather than in the implication. The agent as they happen to be: a present state as a weighted superposition of the poles (Remark 3.15, Remark 3.12). The agent realized: $\nu F$ — with $\mu G$ as its privation, so the scheme's evaluative asymmetry arrives as the complementarity theorem rather than as an assumption. The discipline of the transition: the anatomy condition and the Score's pricing of movements between the poles (Proposition 9.2, §8.6). And the twist that makes the middle term re-derivable where its classical warrants died: $\nu F$ is *telos-shaped without final causation*, a fixed point of a modal operator over a conditioning, in which nothing acts, pulls, or intends (Remark 7.7, Remark 3.40), so the modern objection that killed the classical middle term has no purchase on its successor, which is not a cause at all.

Two identifications complete the anatomy. The verdict that emotivism, false as an account of meaning, became true as an account of use is, in this subsection's vocabulary, the testimony collapse observed from outside: once every node is known to be biographically sourced, moral utterance *can* function only as preference plus pressure, however valid the propagation — the collapse is the mechanism of the incommensurability, not a further symptom of it. And the historical argument that the Kantian project had to fail is answered at its joint rather than disputed: a finite rule *inherited* its content and jammed, as above, where the present framework inherits no content, it derives the middle term first and prices content against it, and the criterionless choice at which the interminability account locates the final breakdown survives here as *priced* choice: adoption remains free (Φ2), and what is deleted is only the criterionlessness. The claim all of this makes is calibrated at the meta-level, as the fences below require: the hole's damage was that no dimension of comparison existed among rival premises, so there was no fact of the matter to be wrong about; the criterion restores the dimension, commensurability with respect to the root, determinate though uncertifiable (Remark 8.20), and thereby re-poses a question the diagnosis showed to be ill-posed. Re-posed, not solved; the distance between those words is priced below. One resonance is recorded rather than argued: the interminability tradition models its remedy on the response to Rome's fall, local communities preserving practice through a dark age, while this framework's backward chain prices Rome's fall as a node in the conditioning itself (Remark 8.18). The same event serves one tradition as the exemplar of retreat and this one as a link in the route, which is as compact a statement of the delta as the record supplies.

*Remark* 9.9 (The crisis, classified: Departric from its vantage). The framework can now say what the post-Nietzschean condition *is*, in its own vocabulary, and the identification is exact rather than evocative. Past the Departure Point the adopted ordering's indices are undefined rather than small (§9.6.5); comparison does not fail by returning bad values, it fails by returning none, and from inside that failure every direction is as good as every other — the principle of explosion, lived. Nietzsche's report of the condition, up is down, the horizon wiped away, straying as through an infinite nothing [1], is the first-person phenomenology of exactly that invalidity, produced without the concept of the Golden Point, without even the concept of the Departure Point, and the implicative parallel is nonetheless exact: a meaning-system whose root is gone *is*, from its own vantage, in the Departric condition, and the metaphysics a vantage inhabits is, in the epistemic sense this framework has made precise (Remark 4.6, Remark 8.20), what the metaphysics from that vantage *is*. The dark age the interminability tradition forecast is the sociology of that condition run forward.

Which is why the reversal works at the level it works at, and why the author ranks this formulation as he does. Departric invalidity is *conditional*: the indices are undefined given departure, and what the mere *statement* of the Golden Point does, the concept, prior to any question of attainment, is restore the pole against which the indices are defined. A vantage that represents $\nu F$ is, by that fact alone, no longer in the epistemic Departric condition, whatever else remains uncertain; the explosion is not argued down but *deprived of its antecedent*. This is the precise sense of the author's claim, entered here in his voice at [Φ]: the concept prevents the forecast dark age by nipping it at the metaphysical root — not by winning the trajectory-argument on its own terms, but by changing what the metaphysics from the present vantage is. And it is why the fallback the interminability tradition lacked now exists: a centralization-allowing meaning and moral root that requires neither tradition, nor practice, nor retreat — so the response to the forecast need not be the Benedictine one, because the analogous fall of Rome is, on this showing, avoidable at the root rather than survivable in enclaves (Remark 9.8). The claim's grade is inherited from its parts: the identification is exact, the reversal is a statement about vantages and is available at every rung of the ladder, and what remains Strong-conditional is only the ontological backing, priced where it is claimed.

*The fences, which this subsection needs more than any other.* Nothing here solves the seminar. The identification map is missing and the computation ban of §9.5 governs every sentence above: the criterion supplies the *direction of judgment* on nodes and the standard of their non-arbitrariness, never verdicts on demand. The criterion inherits the ceiling — node-verdicts are disciplined, not certified, and anyone claiming certified ones has left the framework. The sourcing claim is graded by the Strong/Weak/ Void ladder of §4.6, degrading gracefully to the semantic premise's comparative defense under Void. And $\Phi 2$ stands untouched: a non-arbitrary node is not thereby an obligatory one — the framework prices roots, and the reader still chooses.

## 9.8 The third route: inferred meaning

The crisis of §9.7 has a cultural face, and the framework's answer to it deserves statement at the same register as its academic face — not least because the cultural face is the one nearly every reader arrives already inhabiting.

*The settlement.* For most of the modern era, educated reflection on cosmic questions has operated under what may be called the Nietzschean settlement — the name is convenient rather than exact, since Nietzsche marked an inflection in a longer arc rather than its origin [1]. Its content: traditional meaning-structures have been undermined by the cumulative successes of natural science in ways no honest agent can reverse by appeal to faith, and the only remaining honest response is to construct meaning oneself and commit to it as a deliberate act. Existentialism's authenticity, absurdism's defiant embrace, secular humanism's procedural commitments, and the sophisticated modern naturalisms are strategies *within* the settlement, not alternatives to it [21, 112, 114]. The settlement's evidential base is real: an arc running from Copernicus through Darwin to the present in which scientific accuracy and existential significance have run negatively correlated, each deepening of comprehension shrinking the apparent space for cosmic meaning, whose most honest statement is Weinberg's: the more the universe seems comprehensible, the more it also seems pointless [2]. The framework's reading of that sentence is charitable in exactly the sense its own machinery licenses: Weinberg was not committing a category error; he was accurately extrapolating four centuries of trajectory from within a vantage at which the resolving object did not exist. The strengthening the author intends should be stated exactly, because it is stronger than charity: from those vantages the pessimistic extrapolation was not merely reasonable but *uniquely* reasonable — the alternative, expecting the trajectory to reverse without evidence after the Enlightenment had spent centuries systematically dismantling every prior ground of reversal, would have been the deus-ex-machina hypothesis. Nietzsche sits near the descent's midpoint, barely an inflection, since that leg runs mostly linear, and his own positioning report says as much: *I have come too early; my time is not yet* [1]. Weinberg, nearer the present, extrapolates the leg he stands on, and the honest content of his sentence is that the descent is probably not bounded from below. Watching the prophecy arrive on schedule, only a fool would have extrapolated to anything but a dark age; the interminability tradition's forecast (§9.7) is that extrapolation done carefully. The same holds one register up for the diagnosis of §9.7: Nietzsche, Weinberg, and the MacIntyre–Anscombe finding are one trough with three faces, prophetic, physical, academic, and this framework engages all three with one object.

*Remark* 9.10 (The inference operator, and where it flips). "Uniquely reasonable" can be made mechanical, and the corpus contains the precedent that shows how. Treat best inference as an *operator*: for a fixed question, rationality's verdict is a function of the evidence corpus available at the vantage — indexed, per the author's own scoping and at his stated hedge **[S]**, by the epistemic state and not by anything as deep as an Aethus, since what varied across the generations was awareness, with meaning *uncertain* to the prior vantages rather than their ontology superposed. On this schema the trough is not a mood but an operator output: given the corpus running Copernicus through Weinberg, the downward extrapolation was the unique verdict, and holding today's conclusion *from that corpus* would have been inferentially out of line — which is the exact and final form of the charity this section has extended throughout.

*The precedent.* The companion's Tricky Principle is the same event one framework earlier [4]: whether the unobserved operates by the observed's rules is unfalsifiable — *until the discovery of quantum mechanics* — and the principle's own procedure records the flip in both directions: pre-quantum, the razor demanded the uniformity solution, and it was the post-quantum corpus — one demonstrated object beyond observation running by different rules — that made the razor *demand the reverse*, forcing a new object tailored to unobserved states, the Aethic blank. Two disciplines transfer from that instance. The flip is passive: the empiricism arrives and the razor turns, with the author imposing nothing — and the generalization is tailored rather than scaled, the companion's own caution, so that what the new corpus licenses is a fitted conclusion and never a blunt one.

*The present instance, and its flip condition.* The question is the trajectory of meaning, and the evidential arrival that flips the operator is named exactly: the *coupling of Miracles* [5]. While a record's improbabilities stand as a tally of independent surprises, each is severally dismissible, selection artifact, coincidence, a multiple-comparison debt incurred per item, and the downward verdict stands. The companion's coupling results change the object, not the mood: committing to one labeled improbability *forces* others, the record's label-set is the closure of a root commitment with the statistical debt paid once at the root, and disagreement between the cosmic and observer-conditioned weightings is a *detector* of coupling rather than a suspicion of it. A tally can be shrugged at item by item; a closure cannot, because there is only one commitment to audit. That is the speedbump. Past it, the operator's output inverts, upward becomes the razor's demand rather than anyone's hope, and the inversion matures into its end state when the coupled cascade is read as what it is: exotic structure proper, the record's improbable features as a categorical kind rather than a tally of surprises, at the interpretive grade the companion assigns that reading. The claim's pedigree is thereby auditable end to end: the flip inherits the coupling results at their stated grades, and nothing in it is asserted from taste.

*What the flip buys, in three wirings.* It dates the arc's inflection at an evidential event rather than a generational temperament — the trough's far side begins where the corpus includes the coupling, not where anyone decides to feel better. It converts the call to action into an obligation transfer: this paper imposes no conclusion; it delivers the empiricism, and the reader whose corpus now contains the coupling finds their own operator flipped — running the trough's extrapolation from the new corpus is what is now out of line. And it identifies the phase-transition prediction's mechanism (Remark 9.11): the transition is this flip at population scale, propagating exactly as far and as fast as the evidence does, with the concept-level reversal of Remark 9.9 as the flipped operator's output stated at the metaphysical root.

*The taxonomy, which is the load-bearing move.* The settlement's authority has always rested on an exhaustiveness claim: meaning is *constructed* (projected by the observer onto a world that contributes nothing) or *revealed* (delivered by the world through a non-natural channel, the observer merely receiving), and since revelation is closed to honest agents, construction is all that remains. The corpus's claim is that the dichotomy was false. There is a third route: *inferred* meaning — discovered by the observer through study of the natural record under the conditioning structure, the observer doing the inferring and the world supplying the structure inferred from, neither pole doing all the work. The Nexic companion opened the route in the backward direction: the record's measured improbability cascade as significance present in the rocks, quantifiable, and specific to the observer it produced [5]. The present work completes it in the forward and ontological directions: the Golden root as what things and categories arise from (§4.6), the vantage itself included (Remark 4.6), and a universe-sourced criterion for the commitment primitives on which judgment stands (§9.7). Backward significance and forward root: the two chambers of the hole the settlement's sharpest diagnosticians described, filled by one object at two tenses.

*The claim, at its two tiers.* Stated with the grading the framework requires of itself. The *unconditional* tier: breaking the dichotomy does not require the Strong reading, because "the universe is meaningful" becomes, on this framework, a graded, priced, falsifiable position for the first time — a reader who ends by assigning the Strong reading low credence has still lost the settlement's exhaustiveness claim, since a third route now exists to assign credence *to*. The *Strong-conditional* tier: under the Route World's Strong reading the historical correlation itself reverses, comprehension and significance become positively coupled, since every deepening of the record's analysis is a deepening of the measured structure the conditioning concentrates on, and Weinberg's sentence acquires its successor: the more the universe seems comprehensible, the more it also seems *significant*. The successor is not a denial of Weinberg's reading but the next stage of the same arc, exactly as his sentence was the next stage of Copernicus's.

*The phenomenological signature, recorded because it is a test.* Constructed meaning carries a tell: what one invented oneself, one can usually feel that one invented — it is the kind of thing one must keep talking oneself into. Inferred meaning carries the opposite tell, and the companion's register is its exhibit: a level of luck in the face of improbability *too vast to be conveyed and yet too personal to be named* [5] — the vertigo of the unchosen, arriving with the specific force of what one demonstrably did not author. The pair is offered as a matched diagnostic, not as proof: feelings certify nothing (the ceiling governs here as everywhere), but a route whose phenomenology is the structural opposite of construction's is at minimum not construction relabeled.

*Remark* 9.11 (The displaced-valor diagnosis, and the phase-transition prediction). One further observation of the author's belongs beside the taxonomy, because it converts the vacancy from an abstraction into a pressure reading. The exemplars that reliably produce the aesthetic charge of valor, martial music as the drums enter, anthems at their perilous-fight lines, the scored last stand, are, at their load points, *nationalistic*: humans rising to defeat other humans, with the pride requiring the put-down of the defeated as its complement. The diagnosis is charitable in the register this section has used throughout: *nobody is to blame for this.* The drive being scratched, the burning attachment of agency and valor to something that matters, is real and honorable, and if something externally meaningful were *known*, it would be loved; nothing is known, so nationalism is what the drive can reach. That is the node crisis (§9.7) read affectively: the valor has no non-arbitrary object, so it attaches to the available arbitrary ones.

From this the author enters a prediction, at **[S]** and falsifiable in the only way reception claims are: the vacancy is a *low-pressure region of vast extent*, and the introduction of a comprehensible external catalyst into it would kick off a phase transition rapid enough to surprise effectively every observer in some respect — and the mechanism is now named: the transition is the inference flip of Remark 9.10 at population scale, propagating as far and as fast as the coupling evidence does. The prediction is stated so it can be scored, and it is stated *with its danger attached*, because the two are one claim: a catalyst capable of a phase transition is capable of capture, and the anti-capture architecture of §9.10 is therefore not an appendix to this remark but its load-bearing condition — the six deltas are what permit a meaning-root to centralize *attachment* without centralizing enforcement. The same distinction prices the author's strategic observation: the standing attempts to centralize progressive coordination have run through the economic base [115], and their historical capture record is the dragon pattern in secular dress, centralized enforcement seizing centralized meaning, where a root carrying the six properties has, by construction, no certification to sell and no compliance lever to seize. What is centralized is the object of the drive; what is not created is any instrument of its policing. The fences hold as everywhere: the framework outputs no platform and no party, $\Phi 2$ stands, "progress" here denotes the Golden-directed conduct this work has already priced (§8.6) and nothing else — and the long game the seed is planted for is the one the whole paper is about: the deliberate run to the Golden Point itself.

*Remark* 9.12 (The arrival's timing, read under the framework's own guards). The author observes that, under the Route World, it is close to predetermined that this material be written, and written now of all times — at the trough, when morale is lowest and the vacancy widest. The observation is admissible, and it must be entered wearing every fence the framework owns, because it is the framework applied to its author. What the retrocausal reading licenses, at Strong-conditional grade, is a claim about the *role*: among $G$-reaching continuations, weight concentrates on those in which the concept arrives, and the invested reading prices the timing — the cheapest arrival is into maximal vacancy, where uptake pressure is greatest, which is the trough. What no reading licenses is the promotion of role to occupant: that *this* formulation by *this* author is the arrival is occupancy until audited (Remark 8.18's discipline, applied reflexively), necessity is never election (Remark 7.7, Remark 3.40), and the narrative-bias guards of Remark 8.17 apply with special force to an author reading his own moment as pivotal — the deflationary alternative, that every author at every trough feels sent, is recorded beside the claim as its standing rival. What survives the fences is exactly this much, and it is not nothing: if the framework is right, then the arrival of *some* such concept, at *some* such trough, is what the conditioning concentrates on — and the reader holding this page is, on that reading, not witnessing a coincidence. Whether they are witnessing this sentence's version of it, no one certifies.

**The meaning ledger: scaffolding for the Introduction.** The forthcoming Introduction carries this material in the author's own voice; this passage is its pointer map, so that every manifesto sentence has a paying result and the manifesto is never flat. (The master's own front door now exists at §1.4, in the paper's register; the manifesto remains the rhetorical instrument, and this map serves both.) "The meaning by which things take shape in the universe" is §4.6 (constitution) with Remark 4.6 (the standpoint). "Break free of the existential freefall" is §9.7 (the node criterion) with Proposition 2.13 (the one universe-sourced frozen commitment already on the books). "Not a return" is the taxonomy above plus the no-agent toggle (Remark 4.5; the conditioning does not act, Remark 7.7). "Weinberg inverted" is the Strong-conditional tier above, on the Accordance reweighting. "You cannot separate out the biosphere" is Remark 5.7 with Remark 4.6 — the severing is not merely unimaginable but ill-posed, since the individuation runs through the carrier. And the Renaissance the Introduction will call a Miracle is the same node the backward chain prices (Remark 8.18). Each pointer is a load-bearing wall behind a rhetorical façade; the Introduction may be read first, but it is paid here.

## 9.9 The meaning literature, placed

The settlement was engaged above as a frame; its interior contains distinct programs, and the discipline this paper applies to its rivals elsewhere, strongest form first, deltas named, what survives stated, is owed here most of all, since these are the positions nearly every serious reader arrives holding.

**The constructed route's interior: the strongest programs.** *Nietzsche's remedy.* The settlement's founder also wrote its most ambitious constructive response: revaluation — values legislated by the sovereign individual, eternal recurrence as the affirmation-test, amor fati as the achieved stance [1]. Three deltas. The node diagnosis of §9.7 applies to the remedy directly: self-legislated values are commitment primitives whose source is the legislator, which is exactly the arbitrariness the testimony collapse prices — so the remedy does not resolve the crisis but universalizes it, each sovereign legislator a new node, mutually void. The recurrence test is structurally a fixed-point condition run against the agent's own capacity to affirm; the framework's test runs against the world's conditioning — affirmation against inference, and the difference is the whole of the third route. And amor fati: where Nietzsche *commands* love of the realized route, the framework *derives* a portion of what that command asserts, the necessity-content of the realized route under the Golden target (Remark 8.18), which is the difference between an imperative and a theorem, with the dark-investment fence still governing: necessity is never endorsement, so what is derived is smaller than what was commanded, and honestly so.

*Camus.* The absurd is a two-premise clash: the human demand for meaning against the universe's silence, with suicide and the leap both refused and revolt held as the lucid posture [21]. The framework's engagement concedes the structure and contests the second premise as what it is — an *empirical* claim. Silence is a measurement, and the Accordance reading returns a record that is not silent but dense: the cascade is in the rocks, measured, specific to its reader. And the leap-charge is met on Camus's own integrity terms: inference from a public record with published falsifiers is not philosophical suicide — the third route passes his lucidity test while failing, deliberately, his inventory of options. Nothing here asks Sisyphus to smile; it disputes the mountain's blankness. Given the inventory Camus possessed, construct or leap, revolt was the honest posture, and the framework's quarrel is with the inventory, never the integrity.

*Sartre.* Abandonment, no signs, no given values, is the settlement's exhaustiveness claim stated as ontology; existence precedes essence; anguish marks the choosing; bad faith names the flight into pretended givens [112]. The framework contests the precedence claim at the scale where it operates: under the Strong reading the conditioning structure precedes and individuates (§4.6) — essence-like without an essence-giver, since nothing acts. Anguish it does not contest but *locates*: it is the first-person phenomenology of the node crisis, lived. And bad faith it turns: flight into pretended givens is bad faith, but givens *argued*, graded, fenced, falsifier published, are not fled to; they are audited, and the third route is the audit. What survives of Sartre is the word he actually needs: the choice at adoption is real and remains the reader's ($\Phi 2$), so the framework deletes his inventory and keeps his freedom.

*Kierkegaard.* The revealed route's most sophisticated modern defense: objective uncertainty embraced as the very medium of the leap, truth as subjectivity, the knight of faith [116]. The delta is clean — he makes the object's unreachability load-bearing, and the framework supplies the object — but the engagement ends on a consilience that should be recorded rather than hidden: the ceiling forbids certification (Theorem 6.18), so the third route, too, is *held rather than possessed* — commitment under proven uncertainty, passion with a referent. The difference is that his uncertainty was constitutive and welcomed, while ours is theorematic and priced; but a reader formed by Kierkegaard will recognize the posture, and is owed the acknowledgment.

*Heidegger, structurally.* Two joints map and are claimed, and only these. Thrownness, finding oneself already delivered into a situation not chosen, receives a formal referent: one is thrown into a *conditioned* record (Remark 8.4, Remark 4.6). And being-toward-death as the individuating horizon is engaged at biosphere scale by the hazard base itself: mortal moments forever, with the finitude tradition's descendants answered at Remark 2.17 [117]. The engagement is structural, not exegetical, and claims nothing about the corpus beyond the two mapped joints.

*Remark* 9.13 (Allies and the analytic wing). *Frankl is the century's nearest ally, and the nearest miss.* Logotherapy's founding claim is the third route's slogan stated clinically: meaning is *detected rather than invented*, discovered in the world [118]. The deltas locate the contribution. Frankl asserts the discovered character on phenomenological and clinical grounds and supplies no ontology on which detection could be veridical; the framework supplies the ontology. His scale is biographical where the root here is cosmic, with individual cases inheriting through task-object individuation (§4). And his attitudinal wing, meaning available in unavoidable suffering, resonates, at **[S]** and no stronger, with where this framework locates hope: the critical regime, written off by every event metric, is where the entire survivorship structure lives.

*Nagel, and which external view.* The absurd, on Nagel's sharper statement, is the clash between engaged seriousness and the external view from which nothing matters [45]. The framework's reply is a question: *which* external view? The one that deflates is the unconditioned one, the bare Cosmic Nexus, whose typical face is the void, and deflation is exactly what running that measure over a survivor's record produces. The honest external view of a conditioned record is the conditioned one, and run with the right measure the outside view returns significance rather than deflation: the deflation was an artifact of conditioning-blindness, not a discovery about the world. One rung deeper, the view from nowhere is engaged by Remark 4.6: a literal nowhere is the abyss, every determinate view is indexed, and the question was never whether to view from somewhere but which index the viewing consumes. Nagel's irony survives as a tone; its ground does not.

*Wolf's placeholder, filled.* The most influential contemporary account holds meaning to arise when subjective attraction meets objective attractiveness, fitting fulfillment, with its author explicit that no theory of the objective side is offered [119]. The framework slots in as exactly that missing half: contribution to the root (Remark 4.2) is a theory of objective attractiveness, graded and priced, and Wolf's bipolar structure survives intact — the subjective side remains genuinely necessary, since adoption is chosen (Φ2). The third route does not replace the fitting-fulfillment view; it completes it.

*Metz's taxonomy, and where this files.* The field's standing survey sorts theories into supernaturalist and naturalist, the latter into subjectivist and objectivist [120]. For reviewers who will want the filing: this framework is a naturalist objectivism of a type the taxonomy does not anticipate, *conditioning-based* rather than property-based, external in source, agent-free, graded by the Strong/Weak/Void ladder, with its objective side carried by survivorship structure rather than by metaphysical depth, adjacent to Metz's fundamentality intuition in direction, disjoint from it in mechanism. The density of near neighbors across this subsection is, per the methodology companion, the proximity corollary in action [30]; the deltas above are the scoring key.

## 9.10 The reduction to religion, refused

The third route will be met, predictably and from both directions, with one reduction: *this is religion again* — theism in survivorship clothing. The reduction deserves the same treatment the Precipice received: its strongest form stated first, then the deltas, each checkable on the page, because a defense that consists of checkable structure is reliable in a way a defense that consists of indignation is not.

*The objection at its strongest.* A serious critic will not say the framework mentions God; it does not. They will say the reduction is *functional*: an external source of meaning, a reverent register, a vocabulary of Miracles and divinity, a future condition shaping the present, a first-principles root for norms — the structure occupies religion's ecological niche, and whatever occupies that niche will be captured by the dynamics that captured its predecessors. The canonical statement of this style of objection is Midgley's, against evolutionary cosmologies that smuggle escalator myths and salvation narratives into scientific dress [121], and it is a good objection: frameworks *have* done this, and the burden of showing this one does not is properly on this work.

*Six deltas, each verifiable on the page.* First, *no agent, at any grade*: the conditioning is a measure operation — the Doob transform at biosphere scale, the Nexus conditioning at cosmic scale — and it does not act, choose, or design (Remark 7.7, Remark 3.40); adding an agent would add no explanatory power and would break the derivations, which is why the reduction is not merely uncharitable but *ontologically incoherent* — it attributes to the formalism the one object the formalism cannot host. Second, *the epistemic channel*: revelation, scripture, and authority against derivation from stated postulates with stated falsifiers — the Golden-categorical premise carries its falsifier (Remark 5.7), the fluctuation thesis its quantifier check (§4.6), the Ansatz its numerics that could have fired and did not; a religion does not publish the conditions of its own refutation. Third, and sharpest: *the ceiling is an anti-dogma theorem.* Dogma requires certifiable favor, assurance of standing on the saved side, and the framework's central epistemic result forbids exactly that certification, to every finite observer, forever (Theorem 6.18, Remark 6.19). A structure that *proves* no one can certify the favorable side cannot function as a salvation guarantee; the trough channel and the losing condition are the structural opposite of an escalator myth, which is precisely the Midgley checklist run and failed — in the framework's favor. Fourth, *the fear mechanism is absent by construction*: no damnation counterfactual (the Departric results concern undefinedness of comparison, never torment), no in-group salvation split, no compliance lever — $\Phi2$ stands, the rule outputs no duties, the node criterion disciplines without certifying, and there is no institution whose authority substitutes for the derivations, which anyone may rerun. Fifth, *severability*: the claims are graded and independently adoptable — a reader at Void keeps the entire mathematics and the Nexic backward route; no faith is demanded as an entry price, where the reduction's target is package-deal by nature. Sixth, *the genealogical delta*: on the record's own showing, dragon-mediated meaning sources in the fear-inheritance and is captured for control, while inferred meaning sources in measured improbability structure — different provenance, and checkable, which is what "inferred" means.

*The vocabulary, priced openly.* The critic will quote "computational divinity," "Miracle," and their kin, and the paper should pre-quote itself: the corpus borrows sacred register deliberately, to mark magnitudes for which no secular vocabulary exists — and defines every such term mechanically ("computational divinity" is a statement about computation time; a Miracle is an exotic-structure Prerequisite; nothing in either definition hosts a will) [5]. The borrowing is a cost, accepted with eyes open, for honesty about the phenomenology: the awe is real, is shared with religious experience as a matter of psychology, and is re-sourced — the delta lives at the source and the structure, never at the feeling, and a framework that denied the feeling would be falsifying its own exhibit.

*Why the refusal matters, in both directions.* The reduction is refused not to win a taxonomy dispute but because accepting it would be costly in exactly the two ways the record documents. Ontologically it is incoherent, as above. And practically it would re-license the capture channel: whatever is filed as religion inherits religion's captors, and the fear-for-control mechanism that seized the premodern meaning-structures, documented in this corpus without euphemism, would have precedent to seize this one. The six deltas are therefore also a *prophylaxis*: each is a structural property whose presence makes dragon-capture difficult in principle, no certification to sell, no duties to enforce, no agent to speak for, no package to hold hostage, and the paper commits to them in advance, in writing, where the commitment can be checked.

*What is not claimed.* That religious persons cannot read, adopt, or contribute to this framework — severability cuts that way too, and the mathematics is indifferent to its reader's metaphysics. That every religious phenomenon reduces to the capture mechanism — the corpus itself holds up universalist strands whose structure the mechanism displaced, and the critique's target is the mechanism wherever it occurs, secular instances included. And that refusing the reduction settles the third route's truth — it settles only its *category*, which is what a reduction disputes. The route's truth is priced where it is claimed, at the tiers and grades stated above, and nowhere else.

The objection also has a canonical *case study*, and running the checklist against it by name is cheaper than letting a reviewer do it first. Tipler's Omega Point [100, 99] is the future-conditioned cosmology that *was* captured by the salvation register, physical eschatology, resurrection mechanics, guaranteed convergence, and it fails this subsection's deltas exactly where the present framework passes them: an agentive terminus where this conditioning has no agent at any grade; physical commitments since falsified where this work carries published falsifiers; certifiable salvation where the ceiling proves certification unavailable; a package where this work is severable under Void. "FAP redux" is therefore the misfiling this subsection exists to preempt, and the six deltas are precisely the distance between the Golden Point and the Omega Point; Remark 8.8 states the placement in full.

*Remark* 9.14 (The tradition, named: Miracleism). The meaning tradition this part develops has a name in the author's usage, and the name is load-bearing rather than decorative: it is *Miracleism*, and pointedly not Goldenism. The Miracles themselves are the source of the meaning, the record's improbabilities read as the closure of one root commitment in the companion's technical sense [5], which is why the third route is *inferred* at all: the source already stands in the record as evidence rather than awaiting construction as a project. The Golden Point's role is different and narrower, and naming the tradition after it would misstate the theory: the Departure Point, in destroying the biosphere, is the one thing that can destroy the meaning, and the Golden Point therefore plays the role of actualization, preventing everything from being in vain. Source in the record, preservation in the future, vanity as the single failure mode; individual mortality is untouched, since the guarantee is against collective vanity and not against death, the moments remaining mortal exactly as §2.4 prices them.

The nearest published neighbor makes the division sharp. Scheffler's collective-afterlife conjecture holds that much of what matters to us now depends on humanity's survival after our deaths [110], and Miracleism agrees about the direction of the conditioning while relocating the source: on this account survival actualizes a meaning whose source already stands in the record, rather than survival being the thing on which mattering depends simpliciter. The delta is the difference between a future that funds the present and a future that keeps the present's funding from being voided. One vocabulary cost is priced where all such costs are priced: the term Miracle carries the companion's technical sense throughout, and the religious overtone the word invites is exactly the misfiling that §9.10 exists to refuse; the name is kept because the source-term should name the source.

**Routinized awe: the Archimedes point ([S], reception-grade).** The finest speeches in the record render their hearers noble bearers of a greater cause and land the note of awe, and the author's claim is that for Miracleism done well this is routine, because what once required the greatest rhetorical power in the species becomes deduction. The analogy is to the passage from Archimedes to the calculus: volumes that the method of exhaustion made available only to the finest intellect alive became routine for an advanced student once the general structure existed, genius having substituted for an absent method until the method arrived and routinized the result. The orator, likewise, rebuilt the greater-cause principle from scratch on each occasion and wired it by hand to the feeling it had to produce; with the structure handed over deductively by the framework, competent conveyance of that structure suffices, and the labor moves from persuasion to exposition.

The Gettysburg Address makes the point sharper than the analogy does, because it does not merely exhibit the effect, it performs the structure. Its central sentence, "that these dead shall not have died in vain," is the anti-vanity move this tradition formalizes: what was given is the source, and the cause's continuation is what keeps the giving from being voided. The Golden Point is the general form of "the cause" in that sentence, and the Miracles the general form of what was given, so the speech is an instance of the exact structure rather than of awe-rhetoric in general, constructed once by hand. Why deduction can now do what rhetoric alone increasingly cannot is the inference-side diagnosis of §9.7: root commitments admit no inference, so persuasion across them fails at the root, whereas derived structure admits inference, so exposition crosses where oratory could not, which is the obligation-transfer mechanism of Remark 9.10 restated for the speaker rather than the reader.

"Done well" carries the weight of a fence. Routinized awe is precisely the dragon's business model described in §1.4, and the difference is that the structure travels with its fences: the ceiling denies certified favor, no agent stands at any grade, and the whole is severable, so that awe with the fences intact is Miracleism and awe with the fences dropped is the dragon in new dress. A conveyance that lands the feeling by discarding the fences has not routinized the tradition but reinvented the thing it replaced. One refinement the author supplies keeps the claim honest about talent. Rhetorical power divides into formulation and execution: the search over a barren meaning landscape for a structure that can bear the weight, and the landing of that structure on an audience once it is in hand. Lincoln needed both, since he had to find the greater cause before he could deliver it. What the framework routinizes is formulation only. The landscape is no longer barren, the structure is supplied, and the search that was the greater part of the labor is gone, while execution still scales with talent exactly as skill in applying the calculus still does. The claim is therefore not that the effect becomes independent of the speaker but that the game becomes vastly easier, its floor rising to where competent execution suffices and its ceiling still belonging to the gifted. The scoreable form follows. With the structure in hand, competent expositors who are not orators should land the effect at all, where before they could not, and if landing it still requires the formulation genius of the finest speeches, the routinization claim fails; that the effect's magnitude continues to scale with execution talent is predicted, not counted against it. It is entered at **[S]**, alongside the other reception-scale predictions of this part.

## 9.11 The load path, stated once

Part IV's results are not a collection; they are one chain, and it should be written out once, link by link, with each link's site and grade, so that the whole can be audited the way its parts are. *The coupling of Miracles*, the record's improbabilities as the closure of one root commitment, not a tally ([5], at the companion's stated grades), forces *the inference flip*: the vantage-indexed operator's output inverts, passively, on the Tricky-Principle precedent (Remark 9.10). The flipped operator's licensed object is *the Golden Point stated as a concept* — the third route's referent, at the two-tier grade of §9.8. The stated concept *de-conditions the explosion*: Departric invalidity loses its antecedent, at every rung of the ladder (Remark 9.9). With the pole restored, *the middle term returns*, telos-shaped without final causation (Remark 9.8), and with the middle term, *the node criterion*: commitment primitives become judgeable, testimony recoverable in principle (§9.7). A non-arbitrary object for the valor drive permits *re-attachment* (Remark 9.11), whose population-scale form is *the phase transition*, the flip propagating as far and as fast as the evidence does, whose test is *the species-awareness criterion*, with its predicts and its forbids (Remark 10.1). And the chain's terminus is *owned conduct*: the anatomy work of §8.6, now representable as being *about* something, aimed at $\nu F$ under $\Phi 2$ — the reader still chooses.

The chain's grading is its honesty. Links two through six run at or near every rung of the ladder, the flip is evidential, the de-conditioning is logic, the criterion is semantics, and this near-unconditional prefix is where the paper's meaning-claims carry their weight. Links seven through nine are reception-grade, **[S]** and scoreable, entered with their dangers attached and the anti-capture architecture as their load-bearing condition (§9.10). Link ten is conduct and is nobody's but the reader's. Each link consumes only its predecessor, so a reader who stops anywhere keeps everything upstream — and a critic who breaks a link inherits the obligation the chain makes explicit: to say which one, since the sites and grades are now listed in a single place for exactly that purpose.

## 9.12 Ledger

**Costs.**

1. *The identification map* (decision $\to (\mu, D)$), **[O]** — the section's principal cost. Without it (9.5) is exact and unusable, and the section says so in the body rather than in a footnote.

2. *The Golden Score* (Proposition 9.2), **[P]** given **[A]** and **[C]** inputs: exact for the hard-wall functional; for soft killing, under Ansatz 5.18 strengthened to the refined form, else a comparison up to bounded factors. Regime-scoped, with the table of §9.4 recording the other cells.

3. *The Initiative's trajectory form*, routed to §5.6.3 at the **[O]** of those dynamics; the ensemble form free (the Tendency at the level of practice); neither consumed below.

4. *The class-level consequence* (technostrength without technoanatomy scores nothing, possibly negatively), [Φ] over Remark 5.41; bounded above by Theorem 6.18's bar on scoring actors.

**Buys.** A decision rule stated informally in 2024 turns out to be computable in the model's own terms, with the computed object the Doob density — a derivation, not an analogy, now stated as a tagged proposition with its hypotheses in the bracket. The $\asymp/\sim$ qualification, a shared defect of both drafts caught at referee, makes the result more creditable rather than less. The rule's counterintuitive feature (difficulty prices in) survives the formalization intact. The trapped cell, finished, connects the rule to the paper's one anatomy-moving intervention rather than dying in a table row. And the section closes §2's forward commitments: every `gs:` label pointed at from there now exists.

**Costs of §9.6.** *Golden ethics* — [Φ] throughout and Φ6-conditional at every step, adding no premise; *falsifier:* Φ6's and nothing further. Its two moralities consume $\mathbb{P}(\mathcal{G} \mid \cdot)$ for counterfactual Aethae, which Theorem 6.18 forbids anyone from supplying, so it is exhibited as an exercise on a counterfactual the framework rules out rather than offered for use.

**Not claimed.** That the rule outputs duties (Φ2 stands, and the fence restates it). That any actor's Golden Score can be computed, today or from any finite observation. That Aurutance can be measured absent the identification map — no proxy is licensed. That the soft-killing limit converges as the Ansatz stands — bounded factors are what it purchases. That Aurutance is a currency — the additivity is that of logarithms and licenses no stock to hold, spend, or owe; Remark 7.8 states what the economic vocabulary *does* license, which is an exchange rate within one comparison and not a stock. That the Initiative causes the Tendency, or that any individual's participation moves a biosphere's hazard. That the viable or trapped regimes support the harmonic form.

**Inherited dependency.** The section inherits §2's exposure whole — under fixed adverse $\rho$ there is nothing for the rule to compare, since $\mathfrak{h}$ collapses along the single surviving path — and adds one of its own: everything from Proposition 9.2 onward stands on the refined tail asymptotics — Theorem 5.29 for $\tau$, and for $T$ the strengthened Ansatz — so a repair there reprices this section before it reprices §2.

**Meaning-wing addendum (structural pass).** Entries added after this ledger's first writing, folded in at the move that restored the ledger to the section's end: the node criterion and the inference-side crisis, with the middle-term mapping and the Departric classification of the trough (§9.7); the third route with its two-tier claim and the inference flip that dates its licence (§9.8); the placement of the meaning literature, deltas as scoring keys (§9.9); and the reduction refused on six checkable deltas (§9.10). Dependencies: the wing consumes the ceiling, the conditioning ladder, and the ontological completion (§4.6); what it adds to the section's own exposure is confined to the reception-grade predictions, each entered scoreable. The load path of §9.11 states the wing's chain whole, with the grade of every link.

# Part V: The Human Position

Part V locates humanity in the structure the earlier parts built. Two standing errors about the human role are refuted (§10), the species-level self-awareness criterion is stated with its test and its falsifier, and adversivity supplies the microfoundation of technoanatomy at the scale where humans actually act (§11). What leaves this part is the reader's own position, which is the paper's point.

# 10 The human role: two errors about it

Two positions occupy the ends of a spectrum concerning what humans are for. At one end, techno-optimism: capability is the thing, and more of it is the answer. At the other, a strain of environmentalism holding that humanity is a net harm to the biosphere and that its voluntary disappearance would serve the living world. Golden reasoning finds both mistaken, and mistaken in ways that are structurally opposite — each neglects exactly the factor the other over-weights.

The argument against techno-optimism is already in hand and is consolidated here: it neglects technoanatomy, and by Remark 5.41 acceleration without anatomy is not merely insufficient but actively worse than stasis (§10.2). The argument against extinctionism is new and is the section's substance: on the extinctionist's *own* valuation, the biosphere's persistence, removing the biosphere's only hazard-lowering agent lowers the thing they mean to protect (§10.1). Both arguments are conditional, and the conditions are stated rather than assumed.

What emerges is a position on environmental value that neither end occupies: nature just *is* the biosphere, the Golden Point is the only route by which it acquires an unbounded expectancy, and human existence is on that accounting not an accident at odds with the living world but structurally implicated in its only route to persisting (§10.3). The threat is not thereby diminished — $\rho < 0$ remains the live danger, and this section is as much an argument for urgency about anatomy as against despair about presence.

*Remark* 10.1 (Species-level self-awareness: a criterion, and where it places the discontinuity). The section's two errors share a presupposition worth naming: both treat humanity as the kind of thing that already knows what it is doing. The author's thesis is that it is not, that *humanity, as a collective, is not yet self-aware*, and the framework can make the thesis exact instead of aphoristic. Individual self-awareness is, on the companion's account, not merely ancient but *basal*: the latch result reads the spark as forged in the shared Mesozoic crucible acting on the whole early-mammalian stock, plesiomorphic across the crown group, inherited by descent, with three consequences the received picture does not take: the mirror test [122] is category-error-shaped as an instrument, measuring a visual modality where the property is representational (the companion's blunt reductio: on the received reading dogs are not self-aware; on the latch's, that verdict is an artifact of testing a scent-first animal through glass); the latch *predicts* modality-appropriate self-recognition where the modality is corrected, with the olfactory-mirror result the worked instance of independent convergence [5, 123]; and the latch *forbids* findings that would count against it, stated in advance. The property itself is representational: an entity is self-aware when it can represent its own situation as its own, including what it would be for that situation to be won or lost. The species-level criterion below therefore does not introduce a new concept; it *lifts one that already carries evidential pedigree* — the same property, one scale up, with the proxy-critique transferring wholesale (species-scale behavioral proxies would be as category-error-shaped as the mirror) and with the latch's own discipline inherited: the criterion *predicts* that concept-reception produces a measurable coordination shift — which is the phase-transition prediction of Remark 9.11, now recognizable as this criterion's test — and it states what would count against it: a species that received the poles' concept and exhibited no ownership shift and no coordination consequence would leave the criterion idle, and the idleness would be evidence. Lift the property to species scale and the criterion writes itself: *a species is self-aware when it possesses the concept of its own modal poles* — when its losing and winning conditions, $\mu G$ and $\nu F$, are represented within it as such. By that criterion humanity has, to date, failed the test: it acts, at scale, with the executive competence of an agent — and has no representation of what its actions are *of*.

Two consequences follow, and the second dissolves an apparent paradox. First, the discontinuity is relocated. On the author's reading, the evolution of individual agency was smooth, exotic structure throughout, no single monolith moment, so the turning-point picture, if it applies anywhere, applies not to the lineage's past but to the species' *conceptual* history: the one genuine discontinuity available is the reception of the poles' concept, an event datable in principle and, as of this writing, ahead. Second, the record's shape before that event is not paradoxical. That humanity already exhibits agency-shaped conduct, coordinated, executive, consequential, without possessing the concept its conduct would be about is exactly what the conditioning predicts: under Past-Alignment the record is Golden-conditioned (§3.5, Remark 8.4), so $G$-consistent conduct precedes $G$-representation the way any conditioned trajectory precedes its description. The species could be augmented with the executive character of agency before knowing there was a "something" its agency was toward — and the concept's arrival, when it comes, changes not the conduct's existence but its *ownership*. The criterion is the claim entered here; whether any particular formulation is the one through which reception occurs is occupancy, priced at Remark 9.12, and the exhibits the author intends for the Introduction are reserved there.

**The evolutionary ladder, and the regime shift that communication caused ([S]).** The received picture of the evolutionary ladder runs by cognition. A human child produces art at the level of the most accomplished adult chimpanzee, and the picture extrapolates: whatever occupies the rung above us, its children would do with ease what our greatest individuals barely manage. The author's proposal is that the extrapolation fails at exactly one point, and that the failure is the species' own history. When our ancestors first developed communication robust enough to preserve an idea for arbitrarily long, the lineage left the cognition-ladder regime altogether and entered one in which further ranking by individual cognition ceased to be the relevant property, so that continuity with the ladder is preserved only by shifting what is valued on our side of the break. On our side the climb consists in shedding the metaphysical biases inherited from the ancestors, and the phenomenologically equivalent action to a rung is the training of humans to see the clearer way; the next rung is not another species but our own, once it clears the next hurdle and propagates the result to itself. Individual cognition was not surpassed but turned into gauge: whether one raises what a node can receive, process, and transmit, or rewires the connections between nodes to the same effect, ceases to matter once communication is the medium, because communication makes per-node cognition a degree of freedom that the machinery of transmission can shift around and replace. The concrete case is already in the companion. Human infants do not reason in counterfactuals and society trains them to, and the counterfactual method so instilled is itself, as [4] shows, a compromise partway toward a coherent Aethic ontology, a compromise no one knew had been struck; the cleanly stated next step is to propagate the more coherent metaphysics to those among us best fit to engage with it on the biosphere's behalf, often from a young age, which achieves what a rung achieves without any change in cognition at all. The routinization claim of §9.10 is an instance of the same mechanism applied to meaning, and the species criterion of Remark 10.1 is what the propagated result would have to contain for the rung to count: a species that has taught itself the concept of its own poles has climbed, whatever its individuals can or cannot compute.

**The alignment problem, from the vantage of species self-awareness ([S], the author's conjecture).** The author records a hope and a suspicion here, at the site where the species criterion is stated, because the two belong together. The hope is that awareness that the entire production of us and of our world is conditioned on the Golden Point, in enabling the species-level self-awareness the preceding remark defines, could supply the paradigm needed to address the alignment problem of artificial intelligence [124, 125, 126] without shooting in the dark in the way the field presently must. This is not to say that the problem does not require a great deal of work from there; it is to say where the work would begin. The suspicion is sharper: that a notable part of the current existential risk from artificial intelligence, as the field itself now assesses it [127, 128] and as warnings from inside the laboratories escalate at the time of writing [129], is tangled up with our sheer lack of awareness of the golden causality binding together the Aethic reality in which the decisions are being made. At the least, it is something a species would want to be aware of before making its next key moves with the technology, and the conjecture is entered at that minimal strength as well as at its full one.

What the framework supplies, stated so that the conjecture is more than a hope, comes to four things. The first is an answer to the question the alignment problem cannot escape and rarely states, aligned to *what*: every proposal that answers with human preferences inherits their arbitrariness and their rivalry, and the framework supplies a target that is neither, $\nu F$, structurally defined and non-rivalrous, the same object a self-aware species needs in order to own its own conduct, so that aligning the technology and aligning ourselves become one problem with one object rather than two problems with none. The second is technoanatomy. Artificial intelligence is capacity, and the two-condition theorem says the variable that decides survival is the sign of the hazard response to capacity and not its magnitude, so that alignment is a technoanatomy question at the species level, whether the capacity added feeds or starves the internal channel; on the design of §5.7.11 it is the largest increase in leverage yet delivered to the pushes on $\rho$ during the settlement window, which is why it matters more than any previous capacity and not because it is different in kind. The third is the ceiling: no alignment can be certified, only held at a grade, and a governance built on that fact is differently shaped from one that promises safety. The fourth is the reading the framework gives of "shooting in the dark": conduct that does not represent what it is conduct *of*, which is Departric conduct in this work's sense, the idleness the species criterion forbids, performed at the highest leverage in the record.

Two fences hold, and the second is the one this passage exists to state. The conjecture is the author's, at **[S]**, and what would count against it is alignment progress that proves independent of any representation of the conditioning target, while what would count for it is proposals that, on inspection, reduce to a target of the framework's shape. And the occupancy discipline applies to artificial agents exactly as to human ones: an artificial intelligence, or its makers, declaring itself the agent of the Golden Point is the occupancy claim in machine dress, and the target is an object at every grade, never a principal to be obeyed; a system aligned to the Golden Point would be aligned to what the conditioning is *on*, and to nothing that could speak for it.

## 10.1 Against extinctionism

### 10.1.1 The position, in its own terms

The target is a specific and explicitly biosphere-valuing position, and it must be stated in its own terms or the argument does not engage it. The Voluntary Human Extinction Movement holds that humanity should abstain from reproduction so as to bring about its own gradual disappearance, on the ground that this would prevent environmental degradation [130]. The reasoning proceeds *from* the value of the living world: the movement's own framing weighs the greater value of Earth's entire biosphere against the lesser value of one species, and holds that once one envisages the biosphere entire and grants the intrinsic value of every life form, voluntary extinction begins to make sense.

That framing is what makes the position addressable here. Its premise, that the biosphere's persistence and flourishing is what matters, is one this framework can evaluate, because persistence of a biosphere is the quantity §5 models.

**What this argument does not touch.** It is essential to distinguish the target from a neighboring position it is often assimilated to. Benatar's antinatalism holds that coming into existence is always a serious harm, and concludes that human extinction would be preferable, on an asymmetry between the badness of pain and the goodness of pleasure [131]. That argument does not proceed from biosphere value at all; its currency is suffering, and extinction is a consequence rather than the objective.

*Nothing below engages it.* An argument that removing humans lowers the biosphere's expectancy is no reply to someone who never valued the expectancy. The suffering-focused position must be met on its own ground and this section does not attempt it. What follows applies to extinctionism *in its environmental form* — and that restriction is not a weakening, since the environmental form is the one that appeals to a premise the framework shares.

### 10.1.2 The argument

**Proposition 10.2** (Removing the muffling agent forfeits viability; **[P]** given the model)**.** *In the model of §5, a biosphere with no hazard-lowering agent is the case $x \equiv 0$: no muffling is accumulated, and the total hazard reduces to the background, $h_t = B_t$. Then $\Lambda_t = \int_0^t B_u\, du \to \infty$ (Lemma 5.27 with $S = [0, \infty)$), so*

$$\mathbb{P}(\Lambda_\infty < \infty) = 0$$

*identically — not small, but zero, and for every parameter value.*

*Compare the alternative. By Theorem 5.38, a biosphere that does accumulate muffling has $\mathbb{P}(\Lambda_\infty < \infty) = \mathbf{1}\{\rho > 0\} \cdot \big(s(\mu_0) - s(-\infty)\big) / \big(s(+\infty) - s(-\infty)\big)$, which is strictly positive whenever anatomy is positive and the driver escapes — transient toward $+\infty$, or (by the theorem's repaired recurrent clause) positive recurrent with positive stationary mean, which* widens *the class of muffling dynamics this comparison can invoke. The extinctionist proposal is therefore a choice of an option with viability probability exactly zero over one whose viability probability may be positive.*

*Two scopings keep the zero honest at the site. The **[P]** is model-internal: it prices the fixed-configuration model, and the model-to-world step remains this section's ledger cost at [$\Phi$]. And the zero is configuration-conditional: the recovery branch, a successor lineage re-evolving the muffling capacity, is precisely the branch a fixed-biosphere model does not represent, and the* *corpus prices rather than ignores it: that branch is the $\varepsilon$ of $\varepsilon$-Departure (Definition 3.14), where the pricing sentence lives.*

The magnitude of what is forfeited is not left abstract. The companion treats the external hazard as roughly constant over long timescales, since the cosmic background does not change appreciably from era to era, giving a survival probability $e^{-\lambda t}$ and an external life expectancy of $1/\lambda$ [5]. The framework's own reading of the Rare Earth and High-Contrast Earth hypotheses is that $\lambda t_0 > 1$, where $t_0$ is the length natural history actually took — which is to say that *Earth's remaining expectancy under the external hazard alone is shorter than the span already elapsed.*

*Remark* 10.3 (Undisturbed nature is not preserved nature)*.* This is the point on which the position turns, and it is worth stating without ornament. The extinctionist picture is of a living world continuing undisturbed once the disturbance is removed. But "undisturbed" is not a stable state; it is the case $x \equiv 0$, in which the biosphere has no means of lowering its own hazard and the background runs unopposed.

The reframing of Definition 3.5 sharpens what is being chosen, and sharpens it against the position. Under a reading on which the relevant event is biological death, "leave nature alone" sounds like survival, and the argument must show that death follows anyway. Under the reading on which the Departure Point is the closure of possibility, no such demonstration is needed. At $x \equiv 0$, Proposition 10.2 gives $\mathbb{P}(\Lambda_\infty < \infty) = 0$ identically: *no* continuation available to that biosphere advances. The modal quantifier of Definition 3.5 is therefore satisfied at once, and the Departure Point is not a fate awaiting the undisturbed world but its present condition. In the terms of §3.4.1: $x \equiv 0$ lies in $\mu G \setminus \mathrm{Base}$ — it is a departure at which no hazard has yet fired, entered by configuration rather than by event. The extinctionist proposal is a proposal to enter the cascade deliberately. A biosphere left alone has not been preserved; it has been placed in the one configuration from which nothing leads anywhere. The extinctionist is not choosing a world that survives without us. They are choosing a world that cannot go anywhere. Destabilization of the star, a sufficiently close hypernova, orbital perturbation by a passing mass — these do not become less likely because humanity has withdrawn. They become the only things that happen. Remark 5.53 reaches the same configuration from the other side: a biosphere that has not completed its schedule has not yet acquired what the extinctionist proposes to give up, and the two sit at the same bare background.

And the choice is not between a damaged biosphere and an undamaged one. It is between a biosphere with a positive probability of persisting indefinitely and one with none. On the extinctionist's own valuation, the second is worse.

### 10.1.3 The concession, and why it does not rescue the position

The argument has a real limit, and stating it is what keeps it honest.

*Remark* 10.4 (Adverse anatomy does not vindicate the position)*.* The natural concession at this point is that under $\rho < 0$ the extinctionist is closer to right, since $x \equiv 0$ leaves hazard merely at background while adverse anatomy drives it upward. The concession should not be made, and seeing why changes which options the argument is really between.

*First, the local comparison goes the other way.* A biosphere with adverse anatomy that *halts* at the optimum $x_*$ of Proposition 5.35(iii) sits at hazard $h_*$, and $h_* < h(0)$ for every $\rho < 0$. Numerically, at $B = 1$, $b = 3$, $k = 6$:

| $\rho$ | $h(0)$ | $x_*$ | $h(x_*)$ | expectancy gain |
|---:|---:|---:|---:|---:|
| $-0.05$ | 1.00 | 11.47 | 0.0046 | 217× |
| $-0.20$ | 1.00 | 8.88 | 0.0176 | 57× |
| $-1.00$ | 1.00 | 4.28 | 0.407 | 2.5× |
| $-2.00$ | 1.00 | 1.82 | 0.898 | 1.1× |

Even at $\rho = -2$ — anatomy adverse enough that capability doubles internal hazard twice over — halting still beats having no muffling at all. So $x \equiv 0$ is strictly dominated for every $\rho < 0$. Adverse anatomy is worse than absence only if the biosphere *fails to halt*, which is a failure to stop rather than a property of the anatomy.

*Second, and this is the point the local comparison obscures, both options forfeit.* By Theorem 5.38, $\mathbb{P}(\text{viable}) = \mathbf{1}\{\rho > 0\} \cdot (\text{escape factor})$. At $x \equiv 0$ there is no muffling agent, hazard sits at background, and $\Lambda_\infty = \infty$. At $\rho < 0$ halted at $x_*$, hazard sits at $h_* > 0$, and $\Lambda_\infty = \infty$. *Both are zero.* The extinctionist and the biosphere that settles at $x_*$ are not choosing between a worse and a better outcome; they are choosing between two ways of dying, and the model does not distinguish them in the only respect that admits an answer other than death.

The response to adverse anatomy is therefore not to select among the zeros. It is to move $\rho$, which is the one intervention that changes the indicator rather than the multiplicand — and which §11 exists to model. A reader persuaded of the danger and unpersuaded that $\rho$ can be moved still has a coherent position, and it remains the strongest form of the view opposed here; but it is a claim about the tractability of anatomy, not a defense of extinction.

### 10.1.4 Why forfeiting is not the same as never having had

Purpose and its grounding are treated once, at §2.4; this subsection confines itself to stakes. The preceding leaves a gap that arithmetic alone will not close. If both options carry $\mathbb{P}(\text{viable}) = 0$, why prefer either to the other, and why regard the choice as weighty rather than indifferent? The answer is that the two are not symmetric with respect to what has already been spent — and note that under Definition 3.5 what is at stake is the trajectory rather than the lives on it, which is why the accumulation is the right object to weigh. A biosphere at $x \equiv 0$ has not merely a low viability probability; it has *surrendered* an accumulation, and the accumulation is the subject of the companion's Miracle sequence [5]. Whatever its correct magnitude, the record is a chain of events each of which had to succeed for the next to be attempted: an origination, a transfer if panspermia obtained, the eukaryotic transition, the Dark Era, the end-Cretaceous impact, the $\alpha$-gal epidemic. The companion's estimate for a single such step, the interstellar transfer, is of order $10^{-21}$ [5].

*Remark* 10.5 (The stakes rise monotonically)*.* One structural feature of that chain is worth extracting, because it is what makes the argument an argument rather than an appeal to sunk cost. Each Miracle in the sequence is *more* consequential than the last — in the companion's formulation, because the gap to the Golden Point was by then that much closer [5]. The reason is not sentimental but combinatorial: a step late in the chain carries all the improbability of every prior step behind it, so the expected cost of failing at step $n$ is the whole product $\prod_{i<n} p_i$ rather than $p_n$ alone.

Two consequences follow. The stakes attaching to a decision rise monotonically along the trajectory, so the present moment carries the highest stakes the lineage has yet faced — not by self-importance but by construction. And sunk cost is not the structure here: the classical sunk-cost fallacy consists in letting past expenditure govern a choice whose future payoffs are unchanged by it, whereas here the past expenditure has *purchased* the option that remains, and forfeiting is forfeiting that option rather than honoring a debt. The distinction is that between a bad reason to persist and a live asset one may discard.

*Remark* 10.6 (Forfeiture is not observable). The cascade of §3.4.1 adds a feature to the stakes argument that deserves separate statement, because it is the feature that makes the argument urgent rather than merely large. A biosphere enters $\mu G$ at the moment the last advancing route closes, and by (3.3) that moment precedes the firing of any hazard — possibly by a great deal. Nothing marks it. There is no event to observe, no loss to register, and by §6.8 no finite observation that would distinguish a biosphere which has just crossed from one which has not. The accumulation is not forfeited in some future catastrophe one might see coming and reconsider; it is forfeited silently, at whatever point the routes ran out, and the catastrophe is only the execution.

This does not make the stakes argument more permissive, see the limits below, but it does relocate what the argument is about. It is not a claim that a visible disaster would be very bad. It is a claim that the relevant transition is invisible, which is why deferring attention to anatomy until the danger is legible is deferring it past the point where deferral is possible.

*Remark* 10.7 (What this does and does not license). The stakes argument establishes an asymmetry between forfeiting and never having had, and nothing beyond it. In particular it does not license: that any present cost is acceptable, since the argument is silent on rates of exchange; that the Golden Point will be reached, which §6.8 forbids anyone from knowing; or that persons alive now bear a duty derived from the accumulation, which $\Phi 2$ excludes and which the framework has no machinery to generate.

What it does license is narrow and, we think, sufficient: that the two options which forfeit viability are not the neutral defaults they are often taken for, and that between an option which changes the indicator and options which do not, the model has a determinate preference.

### 10.2 Against techno-optimism

The opposite error requires less new argument, since the result it founders on is already proved. Techno-optimism holds that continued growth in capability is the answer to the risks capability creates. In the model's terms it supplies the escape factor of Theorem 5.38 and is silent on the anatomy factor — $\rho$ does not appear in any accounting of energy capture, capability, or rate of progress, and cannot be inferred from them.

The verdict is not that this leaves the conclusion unsupported but that it inverts it. With $\rho < 0$, feedback does not fail to secure survival; it *accelerates the failure*, since $\mu_t \to +\infty$ drives $x_t \to +\infty$ and hence, by Proposition 5.35(iii), $h \to \infty$ superexponentially. Supplying capability alone is strictly worse than supplying neither: a poorly-constituted biosphere that stagnates sits near its hazard floor $h_*$, while one that accelerates runs at its own cliff.

*Remark* 10.8 (The cascade forces anatomy to be a process). Both arguments of this section, taken with §3.4.1, deliver a verdict on the fixed-$\rho$ model that is worth confronting rather than absorbing. If $\rho < 0$ and $\rho$ is a *constant*, then Theorem 5.38 gives $\mathbb{P}(\Lambda_\infty < \infty) = 0$ identically — at every time, including $t = 0$. By the modal definition the biosphere is then in $\mu G$ from the outset: no continuation available to it ever advances, and it departed before anything happened. That is a degenerate verdict. It says the question of what such a biosphere should do was settled before it was asked, which is not a claim about biospheres but a symptom of the model.

The degeneracy has exactly one exit, and it is not a matter of taste. It is avoided if and only if $\rho$ can change — if adverse anatomy now leaves open a continuation in which anatomy is not adverse later. That is the fluctuating-anatomy setting of §5.6.3, where Corollary 5.43 and Proposition 5.42 take over, and where the question becomes whether the escape is completed in time rather than whether it was ever available.

So the cascade converts emergent anatomy from a desirable extension into a *requirement of coherence*. A framework that treats technoanatomy as a parameter and departure as modal closure will pronounce every adverse-anatomy biosphere already lost; only a framework in which anatomy has its own dynamics can say the thing this section is trying to say, which is that the matter is open and the window is closing. The arguments above should be read as holding under the fluctuating reading, and the fixed-$\rho$ statements as the limiting case in which the window has already shut.

**The two errors are the same error.** Placed side by side, the positions are not opposites but a single mistake taken in two directions. Both treat capability as the only variable. The techno-optimist maximizes it; the extinctionist minimizes it; neither carries a parameter for whether capability, once acquired, is turned inward. Once $\rho$ is in the model the spectrum they occupy is revealed as one axis of a two-dimensional space, and the interesting region, positive anatomy with accumulating capability, lies on neither end of it. That is why the disagreement between them has proved unresolvable by argument along their shared axis: the quantity that settles it is not on that axis.

The refutation's arsenal has grown since this subsection was written, and the additions belong on display here. The invariant of §6.2 says the operative threshold does not live in the space capability measures at all; the grabby engagement says visibility-weighted selection is not persistence-weighted selection, so the loud are not thereby the lasting (Remark 6.10); and $\beta < \alpha$ is the one inequality on which capability-first hope depends and which its proponents never argue (Remark 6.12). One further point is sharper than an arsenal item and is entered with its edge showing. The escalator-myth checklist that this framework was run against and passed (§9.10) *fires* when run against techno-optimism: guaranteed ascent, salvation through capability, the future as reward — the structure Midgley named is present here in secular dress [121], with the two-condition theorem as the specific fact the myth requires readers not to check. The reduction-to-religion charge, wherever it is finally owed, is owed to the position this subsection refutes.

## 10.3 What this implies for environmental value

Three consequences follow, of decreasing formality and increasing interest. They are stated as consequences of the foregoing and inherit its conditions. *Nature is the biosphere.* The unit §3.1 adopts for methodological reasons, the totality of living and technical activity associated with a world, taken as one system, coincides with what environmental thought means by nature when it speaks of intrinsic value in the whole rather than in particular organisms. The framework does not need to argue for that identification; it arrives at the same unit from the requirement that a conditioning target apply across the whole record (§3.4).

*The Golden Point is the only route to an unbounded expectancy.* Every alternative has a finite one. A biosphere without a hazard-lowering agent has expectancy set by the external background; a biosphere with adverse anatomy has a shorter one still. Only the conjunction of positive anatomy and accumulating capability yields $\mathbb{P}(\Lambda_\infty < \infty) > 0$. Whatever else environmental value recommends, if it recommends the living world's persistence then it recommends passage through that conjunction, since no other route is available. The modal form of the same claim, that passage through the Golden Point is the only route by which the Departure Point is avoided, is §2.6's, where it rests on the complementarity of (3.3) rather than on the expectancy comparison made here.

*Human existence is not ontologically at odds with the living world.* This is the claim most likely to be over-read, so its content and its limits are given precisely.

*Remark* 10.9 (What is and is not established). *Established, conditional on the foregoing:* the outcome in which the biosphere persists indefinitely runs through the existence of an agent capable of lowering its hazard. Humanity is, so far as the record shows, that agent's occurrence in this biosphere. So on this accounting human existence is not contingent noise afflicting an otherwise self-sufficient nature; it is structurally implicated in the only trajectory on which nature persists. Under Golden-conditioning (§3) the implication is stronger still, since the record is then read as produced under a conditioning that the passage occur.

*Not established:* that the harm is small, that any particular human activity is vindicated, or that the present trajectory is good. The threat is real, $\rho < 0$ is the live danger, and Remark 10.4 declines the concession that a biosphere with adverse anatomy is worse than none, and the refusal is an argument rather than an optimism. The claim concerns humanity's *structural position*, not its conduct, and the two are independent: an agent can be necessary to an outcome and be failing at it.

*Also not established, and worth disclaiming explicitly:* any conclusion about the standing of human meaning-making at large. It is tempting to read the above as supplying, from a source outside human self-interpretation, a verdict that humanity is not a mistake — and as answering a broader cultural pessimism thereby. The temptation should be resisted at this grade. What the framework delivers is a conditional structural claim about one biosphere's route to persistence, resting on Φ3 and on the Golden-conditioning argument, both of which are argued rather than proved. That is a substantial thing to have. It is not a philosophy of history, and presenting it as one would trade a defensible result for an indefensible one. §2.4.3 makes a claim in this neighborhood and does not disturb the refusal: what it offers is conditional on an adopted premise (Φ6) rather than issued by the apparatus, and the difference between a conditional grounding and a verdict from outside is the distinction this remark is drawing.

## 10.4 Ledger

**Costs.** *The valuation premise* — the argument against extinctionism is conditional on valuing the biosphere's persistence, and is no argument at all against suffering-focused antinatalism [131]; *falsifier:* none required, the restriction is stated. *The identification of humanity as the hazard-lowering agent* — [Φ]; *falsifier:* a mechanism by which a biosphere lowers its own hazard without anything playing that role. Φ3 *and Golden-conditioning*, inherited by §10.3 and carrying their grades unchanged.

**Buys.** An argument against environmental extinctionism that proceeds *on its own valuation* rather than by denying it. A diagnosis on which the two ends of the spectrum share one error rather than opposing each other. And a position on environmental value that is neither anthropocentric nor misanthropic, obtained without adding a premise for the purpose.

**Not claimed.** That human conduct is vindicated; that the danger is overstated; that suffering-focused antinatalism is answered; or that anything here bears on the standing of human meaning-making beyond the structural claim stated.

# 11 Adversivity: institutional balance and the microfoundation of technoanatomy

Technoanatomy entered §5 as a parameter: the exponent $\rho$ governing whether capacity acquired is capacity turned inward. Nothing there says where it comes from. This section builds the apparatus intended to supply it, a model of institutions that inflict hazard on one another and reorganize under a single principle, and reports how far that apparatus gets on its own. It does not yet reach $\rho$; §11.6 states precisely what remains.

The principle is that every institution seeks *net adversivity balance*: its weighted alliances should cancel its weighted enmities. Its two motivating cases are historical, and the section's numerical work reproduces both. Two literatures border it closely and are placed in §11.2; the principle is not a refinement of either, and against structural balance in particular it has *disjoint* ground states.

## 11.1 The principle

*Premise* Φ7 (Net adversivity balance; *governing principle*)*.* Each institution assigns every other a signed weight, positive for alliance and negative for enmity, and seeks to make the sum of those weights, taken against the others' sizes, vanish. It is pressed away from having only allies exactly as it is pressed away from having only enemies.[24]

Two cases motivate it, and both are recovered numerically in §11.5. *Removal of a common enemy.* When the Welsh drove back the Anglo-Saxons, the alliance among Welsh tribes did not survive the threat that had produced it. On the principle, the alliances were never about one another: they were the positive weight required to offset a large negative one, and their justification vanished with it.

*Arrival of a common enemy.* Conversely, the only circumstance under which Earth's states would plausibly unify is the arrival of something opposed to all of them — whereupon, on the principle, they would unify not by persuasion but by the same arithmetic running the other way.

**The load-bearing departure.** The principle's distinctive content is that *all-allied is unstable*. That is what the removal case requires, and it is precisely where the principle parts from structural balance, in which all-allied is a ground state. Everything in §11.2 turns on this one disagreement.

## 11.2 Placement: two literatures, and a disjointness

**Structural balance.** Signed-graph balance theory begins with Heider [132]. It was formalized for graphs by Cartwright and Harary [133]. A triad is *balanced* when the product of its three

[24] *Provenance.* The principle is recorded in our journal of August 2025, with both cases below and with the accompanying diagnosis that political theory has attended to its various doctrinal categories where the underlying process is one of stable and unstable equilibria: "if there are all alliances, the weights shift and institutions start targeting each other, and if there are more adversaries than not, alliances form simply to counter that weight and for no particular other reason." The formalization, the placement against the two literatures of §11.2, and everything numerical below are later and are from the drafting collaboration.

signs is positive, so that $(+,+,+)$ and $(+,-,-)$ are stable while $(+,+,-)$ and $(-,-,-)$ are not. The structure theorem states that a complete signed graph is balanced if and only if it splits into at most two camps, positive within and negative between. Continuous-time dynamics were given by Marvel, Kleinberg, Kleinberg and Strogatz [134] and a statistical-physics treatment by Antal, Krapivsky and Redner [135].

The Welsh case is instructive in that vocabulary. While the Anglo-Saxons are present, every pair of tribes sits in a triad with two negative edges, which forces the third positive: the tribes are allied by triad closure, and nothing in their own relations holds them together. Remove the shared enemy and those props vanish, leaving unbalanced triads that must repartition.

**Disjoint ground states ([P]).** The two principles are not variants. Writing $\hat{E}_{\mathrm{H}}$ for the normalized Heider energy and $\hat{E}_{\mathrm{A}}$ for mean-square net adversivity, at $n = 5$ with unit weights:

| configuration | net adversivity | Heider energy |
|---|---|---|
| all-allied (Heider ground state) | $(4, 4, 4, 4, 4)$ | $-10$ |
| two blocs 3/2 (Heider ground state) | $(0, 0, 0, -2, -2)$ | $-10$ |
| pentagon$^+$/pentagram$^-$ | $(0, 0, 0, 0, 0)$ | 0 |

The adversivity optimum is Heider's *worst* case — maximally frustrated — and both Heider optima fail adversivity balance. Nor is two-bloc structure available even approximately: with unit weights a majority bloc of size $p$ and a minority of size $q$ would need $p-1 = q$ and $q-1 = p$ simultaneously. The principle's stable states are frustrated ones, which fits the observation that real international systems do not settle into two clean camps.

**Balance of power.** The second neighbor is a different tradition, with a name confusingly close to the first. Realist theory in international relations holds that states balance against concentrations of power [136], refined by Walt into balancing against perceived *threat* rather than raw capability [137]. Both motivating cases above are textbook for that tradition, and our stipulation that perceived threat tracks an adversary's present unity rather than its capacity to recombine is close to Walt's central move.

The present model sits between the two: the signed-network formalism of the first, the node-local balancing condition of the second. The nearest existing constructions we know of are the energy-based structural-balance dynamics of Marvel, Strogatz and Kleinberg and the continuous-time balance models descending from Kułakowski, both of which are edge-local; a node-local quadratic penalty over signed degree does not appear in either, and we have not found it elsewhere. That is a negative search and not a demonstration of novelty, and the claim is held at **[O]** accordingly.

## 11.3 The energy

Institutions are nodes; $w_{ij} \in [-1, 1]$ is $i$'s weight on $j$; $s_j > 0$ is $j$'s size.

### 11.3.1 The model

The model is a two-term energy on a constrained set:

$$\min_{W} \quad a\, E_1(W) + b\, E_2(W) \qquad \text{subject to} \qquad W = W^{\top}, \quad \sum_{j \neq i} |w_{ij}| = \beta, \tag{11.1}$$

with

$$E_1 = \frac{1}{n}\sum_i \Big(\sum_{j\neq i} w_{ij}s_j\Big)^2 \qquad \text{adversivity balance; node-local, quadratic} \tag{11.2}$$

$$E_2 = \frac{-1}{2\binom{n}{3}}\sum_{i<j<k}\big(w_{ij}w_{jk}w_{ki} + w_{ji}w_{kj}w_{ik}\big) \qquad \text{Heider frustration; triad-local, cubic} \tag{11.3}$$

and $\beta = n - 1$. Equation (11.3) is computed from $\mathrm{tr}(W^3)$, which equals three times the triad sum when the diagonal vanishes.

There is *one* free parameter, the ratio $b/a$, and the two constraints are substantive rather than technical: relations are *mutual*, and attention is *finite*. Both are claims about what an institution is, and §11.3.2 records what goes wrong if either is dropped.

### 11.3.2 The relaxation used for computation

Hard constraints are awkward to anneal, so the numerical work below minimizes a four-channel relaxation in which each constraint is replaced by a penalty:

$$\hat{E}_3 = \frac{1}{8n(n-1)}\sum_{i\neq j}(w_{ij} - w_{ji})^2 \qquad \text{reciprocity; relaxes } W = W^\top \tag{11.4}$$

$$\hat{E}_\mathrm{b} = \frac{1}{n}\sum_i \Big(\sum_{j\neq i}|w_{ij}| - \beta\Big)^2 \qquad \text{engagement budget; relaxes the norm} \tag{11.5}$$

with $E = a\hat{E}_1 + b\hat{E}_2 + c\hat{E}_3 + d\hat{E}_\mathrm{b}$. The two forms agree on everything reported here. At $n = 9$, $b = 4$, over independent seeds:

| | four-term relaxation | two-term constrained |
|---|---|---|
| $a = 0$ | lopsided 6/3, 7/2 | lopsided 6/3, 6/3 |
| $a$ small | even 4/5, $\hat{E}_1 = 1.78$ | even 4/5, $E_1 = 1.78$ |
| $a = 1$ | frustrated web, $\hat{E}_1 \approx 0$ | frustrated web, $E_1 = 0.03$ |

The relaxation is therefore a computational convenience and (11.1) is the model. Three facts about the relaxation are nonetheless worth recording, because each explains why the corresponding constraint is in (11.1) at all.

**The budget is not optional.** Drop the norm constraint and the principle has a trivial optimum: *be indifferent to everyone*. Numerically, pure $\hat{E}_1$ reaches exact balance at $n = 6$ by leaving eight of fifteen ties weak. Balance by apathy is not what the principle intends, and finite attention, a fixed total spent across one's ties, is the substantive claim that rules it out. This is why the norm appears in (11.1) as a constraint rather than as a preference to be traded against others.

**Reciprocity is not emergent.** $E_1$ depends only on row sums and is therefore blind to the antisymmetric part of $W$; $E_2$ constrains cyclic products and does not force symmetry either. Removing $\hat{E}_3$ from the relaxation and re-optimizing produces asymmetries at the maximum, $|w_{ij} - w_{ji}| = 2$: total one-sided devotion. Symmetry must therefore be *imposed*, which is why it is a constraint in (11.1). Its near-zero value at relaxed optima indicates that it is cheap to *satisfy*, not that it is redundant — an inference the low value invites and which the test above refuses.

*Remark* 11.1 (Normalize by the mean, not the maximum). This concerns the relaxation, and is the one point at which a natural choice is badly wrong. Dividing $\hat{E}_1$ by its maximum, attained at the all-allied state, crushes it, since near any configuration of interest it then varies over only $O(n^{-2})$, and the Heider term dominates at any comparable weight. The mean-square form (11.2) instead assigns the balanced two-bloc state a cost of exactly 1 *for every* $n$, since there $\sum_j w_{ij} = (p-1) - q = -1$ regardless of size, matching the Heider floor of $-1$. Only under this normalization is $b/a$ an interpretable weight, and only then is the model's behavior independent of $n$. No re-powering of $\hat{E}_2$ achieves the same effect, and taking roots of it would destroy the sign structure that makes it frustration in the first place.

## 11.4 Ground states

**Proposition 11.2** (Perfect balance requires $n \equiv 1 \pmod 4$; **[P]**)**.** *Among symmetric configurations with all ties saturated at $\pm 1$ and equal sizes, exact adversivity balance is attainable if and only if $n \equiv 1 \pmod 4$. Each node requires $(n-1)/2$ positive ties, forcing $n$ odd; and the total number of positive edges, $n(n-1)/4$, must be an integer, forcing $n \equiv 1 \pmod 4$. The admissible sizes are $n = 5, 9, 13, 17, \ldots$*

*Proof.* Both conditions are immediate; their conjunction holds exactly on $n \equiv 1 \pmod 4$. Exhaustive search confirms the minimum of $\sum_i (\mathrm{net}_i)^2$ is zero at $n = 5$ and positive at $n = 3, 4, 6$, the first case being realized by the pentagon of positive ties with the complementary pentagram of negative ones. □

**Proposition 11.3** (Two failure modes, by parity; **[P]**)**.** *Write $P$ and $M$ for the numbers of alliances and enmities, so that $P + M = \binom{n}{2}$. Summing every node's ledger counts each tie twice:*

$$\sum_i \mathrm{net}_i \;=\; 2\,(P - M). \tag{11.6}$$

*Since $P - M$ and $P + M$ share a parity, the total imbalance is* forced *away from zero whenever $\binom{n}{2}$ is odd: $\sum_i \mathrm{net}_i \equiv 2 \pmod 4$. Two regimes follow.*

Even $n$: the imbalance spreads. *Here $n-1$ is odd, so every* $\mathrm{net}_i$ *is a sum of an odd number of $\pm 1$ terms and is therefore odd;* no *node can balance, and the cheapest option distributes the residual — at $n = 4$ the optimum is $(-1, -1, -1, -1)$.* $n \equiv 3 \pmod 4$: the imbalance concentrates. *Here individual balance is possible but the total is not, and because the penalty is* quadratic *it is cheaper to load a mandatory total of* 2 *onto one node ($\sum \mathrm{net}_i^2 = 4$) than to spread it over three ($-2, -2, +2$ sums correctly but costs* 12*). Squares reward concentration.*

**The scapegoat, and what it does not license.** Proposition 11.3 accounts for every computed residual:

| $n$ | $\binom{n}{2}$ | forced $\sum \mathrm{net}_i$ | predicted min $\sum \mathrm{net}_i^2$ | computed |
|---|---|---|---|---|
| 3 | 3 (odd) | $\equiv 2$ | 4 | 4 |
| 4 | 6 (even) | $\equiv 0$ | 4 | 4 |
| 5 | 10 (even) | $\equiv 0$ | **0** | 0 |
| 6 | 15 (odd) | $\equiv 2$ | 6 | 6 |
| 7 | 21 (odd) | $\equiv 2$ | 4 | 4 |

At $n = 7$ the optimum leaves six nodes exactly balanced and one carrying the whole residual; annealing under continuous weights reproduces it, six nodes within 0.17 of zero and one at $-1.64$. So where the arithmetic forbids universal balance, the cheapest arrangement is not shared discomfort but a single universally-disliked member — and this is forced by counting rather than emergent from dynamics.

Two limits on the reading. The result assumes unit weights and equal sizes; with continuous weights every $n$ reaches balance by weakening ties, so the congruence marks where balance is *free* rather than where it is possible. And the scapegoat mechanism, unlike the congruence, survives that relaxation: the quadratic penalty rewards concentrating an unavoidable residual on one member whatever the weights, which is the part worth carrying forward.

**Where factions appear.** Under the normalization of Remark 11.1, the two channels are commensurate at $b \approx a$, and at $n = 9$ the positive-tie components remain a single connected web up to $b \approx 2$, splitting into two factions between $b = 2$ and $b = 4$. The interesting regime is $b \approx 4$, where the system exhibits *both* a 5/4 split *and* near-perfect balance, $\hat{E}_1 \approx 0.005$; only above $b \approx 8$ is balance abandoned, $\hat{E}_1 \to 1.6$, in favor of saturated bipolarity.

At large $b$ the adversivity term takes on a further role, and it is more than tie-breaking. Heider admits two ground states, one bloc and two blocs, both at $\hat{E}_2 = -1$; since all-allied costs $\hat{E}_1 = (n-1)^2$ while two-bloc costs 1, any positive $a$ breaks that degeneracy in favor of two blocs. But Heider is also indifferent among two-bloc *partitions*: every bipartition sits at $\hat{E}_2 = -1$ whatever the bloc sizes. Numerically at $n = 9$, $b = 4$, the effect of $a$ across three seeds:

| $a$ | faction sizes found | $\hat{E}_1$ | $\max_i \lvert\text{net}_i\rvert$ |
|---:|---|---:|---:|
| 0 | 6/3, 3/6, 7/2 — lopsided | 12.1 | 4.67 |
| 0.001–0.1 | 4/5, 5/4 — even, every seed | 1.78 | 2.00 |
| 1 | no clean bipartition | 0.003 | 0.10 |

Heider alone produces *lopsided* blocs, since it cannot see their sizes. A thousandth of $\hat{E}_1$ makes them *even*, because near-equal blocs balance far better — which is a structural prediction (bipolar systems should have near-equal camps) rather than a degeneracy-breaking convenience. At $a \sim 1$ the bipartition dissolves into a frustrated balanced web. The adversivity term therefore selects among Heider's ground states on a criterion Heider does not possess, at every weight at which it is present.

## 11.5 Two experiments

Both motivating cases are simulated by representing the common enemy as an exogenous field: each institution carries a fixed tie $-h$ to it, so its ledger reads $\sum_j w_{ij} s_j - h$. A first attempt that instead *initialized* an enemy node and let it optimize failed, the node simply joining a faction; a threat must be exogenous rather than a participant balancing its own books.

### 11.5.1 Removal: the Welsh

Seven institutions equilibrated at $h = 1$, then relaxed at $h = 0$ from that state.

| | mean tie | allied | opposed | $\hat{E}_1$ |
|---|---|---|---|---|
| threat present | +0.146 | 6 | 3 | 0.728 |
| threat removed, before relaxation | +0.146 | 6 | 3 | 1.476 |
| after relaxation | +0.004 | 4 | 4 | 0.004 |

The middle row is the mechanism displayed: nothing has moved, and the same configuration has doubled in cost. The alliances were balancing the field; without it they are surplus. Relaxation then sheds them, and the two factions present under threat become *three plus an isolate*.

**Fragmentation, not repolarization — and no flips.** Two features distinguish this from what either neighboring theory would predict, and the second is a limitation. Balance theory predicts repolarization into two fresh camps, since two camps is what a balanced complete graph must be. The present model predicts *fragmentation*: with no external field, the cheapest balance is many small mutual enmities rather than two large ones. The historical record on the Welsh after the Anglo-Saxon reverse, increased factional subdivision without descent into general war, is consistent with the second and not the first. We record this as a retrodiction and not a test: the simulation was run before the record was consulted, but the parameters were not fixed in advance and a single case cannot discriminate.

A third feature was first recorded here as a limitation and is better read as a match. *No alliance inverted*: no pair went from ally to enemy. What happened instead is that two weak alliances lapsed to neutrality while one previously neutral pair turned hostile, so alliances fell from six to four and enmities rose from three to four. That is coalition dissolution accompanied by fresh hostility among parties who were already distinct — not the betrayal of close allies. The historical pattern is the former: the Welsh tribes were separate before the reverse and became mutually hostile after it, rather than turning on partners with whom they had been closely bound. A model producing ally-to-enemy inversions would in fact overshoot the record.

The distinction is worth keeping because the two are easy to conflate and make different predictions. Betrayal requires a commitment term penalizing intermediate $|w|$, so that ties cannot fade and must flip; dissolution-plus-new-hostility requires no such term and is what the present energy produces. Which of the two a given historical episode exhibits is checkable, and the model is committed to the second.

### 11.5.2 Arrival: the aliens

Nine institutions equilibrated alone, then a tenth appended with size exceeding all nine combined and with its own weights *frozen* at $-1$ toward each — it takes no view among them.

| | mean tie among the nine | allied pairs | $\hat{E}_1$ |
|---|---|---|---|
| alone | +0.000 | 10 of 36 | 0.001 |
| threat present | +0.583 | **28 of 36** | 8.1 |

Eight previously opposed pairs became allied — inverted, not merely softened. The reinforcement sustaining those enmities loses its force once every ledger is dominated by a single enormous term.

**Incomplete unification, and an asymmetry.** Two results here were not designed for.

*Unification saturates short of completion.* Only 28 of 36 pairs allied, with $\hat{E}_1$ blown up to 8.1: even universal alliance supplies at most +8 against a threat term of −18, so the system cannot balance and carries permanent strain. Overwhelming threat produces near-total but incomplete unification, which is a more qualified prediction than the verbal argument yields. *Arrival and removal are not symmetric.* Adding a threat produced eight sign inversions; removing one produced none. The asymmetry follows from size-weighting, an adversary larger than everyone combined dominates every ledger at once, where a unit-sized field leaves only a small surplus to shed, and yields the prediction that *threat arrival is a sharp transition while threat removal is a slow relaxation.* Nothing was inserted to produce this.

## 11.6 Toward technoanatomy

The purpose of this apparatus is to supply what §5 assumes. Internal hazard there is $\Gamma e^{-\rho x}$ with both constants primitive; here internal hazard would be what institutions inflict on one another, and $\rho$ would be how that scales with capability. In the notation of §6.6.1, where $\rho = -(\alpha - \beta)$ with $\alpha$ the growth of reach and $\beta$ of enforcement, the adversivity model supplies referents: $\alpha$ is how fast the damage one institution can do another grows with technology, and $\beta$ how fast joint-agreement structure grows.

**What is missing, stated exactly.** The section does not deliver $\rho$, and the obstruction is specific rather than a matter of unfinished labor. This model contains conflict dynamics and no coordination scaling: $\hat{E}_1$ through $\hat{E}_\mathrm{b}$ describe how institutions arrange themselves given their capacities, and nothing in them says how the *cost of maintaining joint agreement* grows as those capacities grow. That quantity is $\beta$, and technoanatomy is entirely a claim about it. Balance dynamics of any kind will supply $\alpha$ and be silent on $\beta$.

The route from here is therefore twofold, and both parts are open **[O]**. A coarse-graining must be specified, merging positively-tied clusters into single nodes at the next scope, with sizes adding and weights averaging, and the energy must be shown to keep its form under it, so that every scope obeys the same dynamics and the biosphere's $\rho$ is the top of a flow rather than a separate posit. And a coordination cost must be added, whose scaling with capability is the anatomy the whole construction is for.

## 11.7 Ledger

**Costs.** *The principle* (Φ7) — **governing premise**; *falsifier:* an institutional system observed to rest stably in universal alliance without an external threat, or to be indifferent to the balance of its commitments. *The engagement budget* — **structural premise**, without which the principle is trivially satisfied by indifference; *falsifier:* institutions observed to hold arbitrarily many ties at full strength. *The weighting $b/a$* — the model's **sole free parameter**, controlling the multipolar-to-bipolar transition; *falsifier:* it is fitted rather than derived, and a system whose faction structure is inconsistent with any single value across eras would embarrass the model. *Mutuality and finite attention* — **constraints**, not preferences; *falsifiers:* stable one-sided regard between institutions, and institutions holding arbitrarily many ties at full strength, respectively. *Completeness* — the two-faction structure theorem assumes a complete graph, and real institutional networks are sparse; the incomplete case admits more than two factions and the clean result is unavailable.

**Buys.** A single principle reproducing both motivating cases, with the removal case matching the historical pattern more closely than either neighboring theory. A counting constraint that produces scapegoat structure without it being posited. An arrival/removal asymmetry that was not designed for. A construction with one free parameter and two interpretable constraints, the four-channel form used for computation being a relaxation that agrees with it. And a stated, checkable route to $\rho$, together with an exact account of why this apparatus cannot reach it alone.

# A Horizon theory

## A.1 The driftless horizon is a timescale, and it is maximally soft

In the driftless model the naive horizon quantity $\Pi(\mu, D) := \mathbb{P}_{(\mu,D)}(D \text{ never returns to } 0)$ vanishes identically (Theorem 5.26): null recurrence leaves no escape probability to be gradual about, and the horizon must be recast in the only currency the model possesses — time.

**Proposition A.1** (Recovery cost is sublinear in credit; **[P]**)**.** *Let $\varrho(d)$ denote the median of the return time* $\inf\{t : D_t \le 0\}$ *from $(\mu, D) = (0, d)$, $d > 0$. Then exactly*

$$\varrho(d) = \varrho(1)\left(\frac{d}{\sigma}\right)^{2/3}\sigma^{2/3}\cdot\sigma^{-2/3} = \varrho_1\left(\frac{d}{\sigma}\right)^{2/3}, \qquad \varrho_1 := \varrho(1)\big|_{\sigma=1} \in (0,\infty), \tag{A.1}$$

*and the return time is finite a.s. with tail index $\frac{1}{4}$. Doubling accumulated credit multiplies the recovery timescale by only $2^{2/3} \approx 1.59$: credit is far easier to undo than a linear intuition suggests.*

*Proof.* Finiteness a.s. is contained in the proof of Theorem 5.26 (the set $A$ there is unbounded). The scaling identity is Proposition 5.22 with $c = d^{-1/3}$ applied to the median, a scale-equivariant functional; positivity and finiteness of $\varrho_1$ follow since the return time is a nondegenerate positive random variable. The tail index is Theorem 5.29 read for the returning coordinate. □

**Proposition A.2** (Horizon level sets are $\kappa$-rays; **[P]**)**.** *Every scale-invariant function on the open quadrant is constant exactly on the curves $D \propto \mu^3/\sigma^2$; if a horizon exists in any scale-invariant sense, it is a ray of constant $\kappa$, and $\kappa$ is its natural transverse coordinate.*

*Proof.* Proposition 5.23(iii). □

**Definition A.3** (Deadline-relative horizon and its width)**.** Fix a time budget $\mathcal{T}$. The *deadline horizon* is the level set $\{t_*(\mu, D) \asymp \mathcal{T}\}$, located at $\mu^* \asymp \sigma\mathcal{T}^{1/2}$, $D^* \asymp \sigma\mathcal{T}^{3/2}$. Its *softness* is the logarithmic sensitivity of survival to state.

**Corollary A.4** (Maximal softness; **[A]**)**.** *Under Claim 5.30, $\dfrac{\partial \log \mathbb{P}(T > \mathcal{T})}{\partial \log t_*} = \dfrac{1}{4} + o(1)$. Changing survival probability by a factor $e$ therefore requires changing $t_*$ by $e^4 \approx 55$ and, since $t_* \propto D^{2/3}$ in the credit-dominated regime, changing $D$ by $e^6 \approx 403$. The driftless horizon is smeared over roughly two orders of magnitude in timescale and nearly three in credit: there is no regime in which resistance has arbitrarily little effect, only a $\frac{1}{4}$-power degradation — about as gentle a degradation as a power law permits.*

## A.2 The scale-function criterion: Pólya's theorem, correctly translated

Let the driver be a regular diffusion on $\mathbb{R}$,

$$d\mu_t = b(\mu_t)\,dt + \varsigma(\mu_t)\,dW_t, \qquad s(\mu) := \int_0^{\mu} \exp\Big(-\int_0^{v} \frac{2b(w)}{\varsigma^2(w)}\,dw\Big)dv, \tag{A.2}$$

with $\varsigma > 0$ locally bounded away from 0 and $\infty$; $s$ is the scale function and Feller's test (see [71, Ch. 15] or [72, Ch. VII]) classifies the boundary behavior.

**Theorem A.5** (Trichotomy; **[P]**). *For the model* (5.1) *with driver* (A.2)*:*

(i) *If $s(+\infty) < \infty$ and $s(-\infty) = -\infty$, then $\mu_t \to +\infty$ a.s., $\Lambda_\infty < \infty$ a.s., $\mathbb{P}(\mathcal{I}) = 1$, and $\mathbb{P}(T = \infty) = \mathbb{E}[e^{-\Lambda_\infty}] \in (0, 1)$.*

(ii) *If $s(-\infty) > -\infty$ and $s(+\infty) = \infty$, then $\mu_t \to -\infty$ a.s., $\Lambda_\infty = \infty$ a.s., and $T < \infty$ a.s.*

(iii) *If both limits are finite, then*

$$\mathbb{P}(\mathcal{I}) \;=\; \frac{s(\mu_0) - s(-\infty)}{s(+\infty) - s(-\infty)}, \tag{A.3}$$

*with $\Lambda_\infty < \infty$ on $\{\mu_t \to +\infty\}$ and $\Lambda_\infty = \infty$ on the complementary event $\{\mu_t \to -\infty\}$.*

(iv) *If both limits are infinite, the driver is recurrent and the outcome is* not *determined by $s$; see Proposition A.6.*

*Proof.* (i) $\mathbb{P}(\lim_t \mu_t = +\infty) = 1$ is Feller's test [71]. On this event choose $t_0(\omega)$ with $\mu_t \geq 1$ for $t \geq t_0$; then $D_t \geq D_{t_0} + (t - t_0)$, so $\Lambda_\infty \leq \Lambda_{t_0} + h_0 e^{-(D_{t_0} - D_0)} \int_0^\infty e^{-v} dv < \infty$ a.s. Positivity of $\mathbb{P}(T = \infty)$ follows since $e^{-\Lambda_\infty} > 0$ a.s.; it is $< 1$ since $\Lambda_\infty \geq \Lambda_1 > 0$ a.s. (ii) is the mirror image: eventually $\mu_t \leq -1$, so $e^{-(D_t - D_0)} \to \infty$ and $\Lambda_\infty = \infty$; death then occurs in finite time a.s. by Definition 5.4. (iii) The hitting probabilities of $\pm\infty$ are the classical scale-function formula, and the two events partition the space up to null sets; apply the pathwise arguments of (i) and (ii) on each. (iv) Recurrence under unbounded scale is Feller's test again; Example A.7 shows $s$ alone cannot decide. ◻

**Proposition A.6** (Recurrent drivers; **[P]**). *Suppose the driver* (A.2) *is positive recurrent with stationary law $\pi$ and $\int |\mu|\,\pi(d\mu) < \infty$; write $\bar{m} := \int \mu\,\pi(d\mu)$.*

(a) *If $\bar{m} > 0$ then $D_t/t \to \bar{m}$ a.s., $\Lambda_\infty < \infty$ a.s., and $\mathbb{P}(\mathcal{I}) = 1$.*

(b) *If $\bar{m} < 0$ then $\Lambda_\infty = \infty$ a.s. and $T < \infty$ a.s.*

(c) *If $\bar{m} = 0$ and the cycle integral defined in the proof is nondegenerate (automatic for non-degenerate diffusions), then $\Lambda_\infty = \infty$ a.s.:* the critical case falls on the mortal side.

*For null-recurrent drivers no general statement is offered **[O]**; the Brownian driver, the fundamental null-recurrent case, is settled unconditionally by Theorem 5.26.*

*Proof.* (a), (b): the ergodic theorem for positive recurrent one-dimensional diffusions gives $D_t/t \to \bar{m}$ a.s. (see [72, Ch. X]); if $\bar{m} > 0$ then eventually $D_t \geq D_0 + \frac{\bar{m}}{2}t$ and $\Lambda_\infty < \infty$; if $\bar{m} < 0$ the integrand $e^{-(D_t - D_0)} \to \infty$.

(c) Fix interior levels $a < b$ and regenerate at up-crossings: let $R_0 := \inf\{t : \mu_t = b\}$ and inductively $S_k := \inf\{t > R_{k-1} : \mu_t = a\}$, $R_k := \inf\{t > S_k : \mu_t = b\}$. By the strong

Markov property the cycles over $[R_{k-1}, R_k)$ are i.i.d.; positive recurrence gives $\mathbb{E}[L] < \infty$ for the cycle length $L := R_1 - R_0$, and the cycle occupation identity yields, for the cycle integral $\xi_k := \int_{R_{k-1}}^{R_k} \mu_s \, ds$,

$$\mathbb{E}[\xi_1] = \mathbb{E}[L] \int \mu \, d\pi = 0, \qquad \mathbb{E}|\xi_1| \leq \mathbb{E}\Big[\int_{\text{cycle}} |\mu_s| ds\Big] = \mathbb{E}[L] \int |\mu| \, d\pi < \infty.$$

Thus $S_n := D_{R_n} - D_{R_0} = \sum_{k \leq n} \xi_k$ is a nondegenerate mean-zero i.i.d. walk, so $\liminf_n S_n = -\infty$ a.s. by the classical oscillation dichotomy for random walks (see, e.g., [138]); in particular the events $B_n := \{D_{R_n} \leq -1\}$ occur infinitely often a.s. Choose $\delta > 0$ and $M < \infty$ with

$$q := \mathbb{P}_b\Big(L \geq \delta, \sup_{0 \leq u \leq \delta} \Big| \int_0^u \mu_s \, ds \Big| \leq M\Big) > 0,$$

possible since $L > 0$ a.s. and the supremum is finite a.s. Let $G_n$ be the corresponding event for cycle $n+1$; by the strong Markov property $\mathbb{P}(G_n \mid \mathcal{F}_{R_n}) = q$, so $\sum_n \mathbb{P}(B_n \cap G_n \mid \mathcal{F}_{R_n}) = q \sum_n \mathbf{1}_{B_n} = \infty$ a.s., and Lévy's extension of the Borel–Cantelli lemma (see, e.g., [138]) gives $B_n \cap G_n$ infinitely often a.s. On $B_n \cap G_n$, for $u \in [0, \delta]$ one has $D_{R_n+u} \leq -1 + M$, hence

$$\int_{R_n}^{R_n+\delta} e^{-(D_s - D_0)} \, ds \;\geq\; \delta \, e^{\,1-M+D_0} \;>\; 0,$$

and these windows are disjoint. Summing over infinitely many of them, $\Lambda_\infty = \infty$ a.s. □

*Example* A.7 (Recurrence alone does not decide)*.* The mean-reverting Ornstein–Uhlenbeck driver $d\mu = \theta(\bar{\mu} - \mu)dt + \sigma dW$ with $\theta > 0$ has Gaussian-growing $s'$, hence $s(\pm\infty) = \pm\infty$: it is (positive) recurrent for every $\bar{\mu}$. Yet Proposition A.6 gives $\mathbb{P}(\mathcal{I}) = 1$ for $\bar{\mu} > 0$ and certain death for $\bar{\mu} \leq 0$. A horizon exists here, sharp, at $\bar{\mu} = 0$, but it lives in *parameter* space, not in the initial condition, which is again pure gauge.

**Two warnings.** Two traps in this circle of ideas deserve explicit flags. *First*, it is tempting to summarize (A.3) as "recurrent driver ⇒ certain death"; Example A.7 shows this is false — in the recurrent class the order parameter is the stationary mean $\bar{m}$, with the critical case $\bar{m} = 0$ mortal, and only in the null-recurrent Brownian case is symmetry itself the whole story. *Second*, under adverse drift it is tempting to read the survival tail off the typical path, which yields doubly-exponential decay; in fact the tail is only *exponential*, dominated by an optimal "fighting-back" fluctuation, Proposition A.10 below, a Freidlin–Wentzell phenomenon in miniature.

| **Random walk (Pólya)** | **Hazard dynamics** |
|---|---|
| dimension $d$ | scale function $s$ of the driver |
| $d \leq 2$: recurrent, no escape | $s(\pm\infty) = \pm\infty$: no viability from data |
| $d \geq 3$: escape probability $> 0$ | $s(\pm\infty)$ finite: viability probability in $(0, 1)$ |
| $d = 2$: null recurrent | Brownian driver: certain death, infinite mean life |
| escape probability from $x$ | normalized scale function at $\mu_0$ |

**Slogan.** *Existence* of a horizon in the initial condition = boundedness of the scale function; *sharpness* of the horizon = steepness of the scale function. The Brownian driver has $s(\mu) = \mu$, unbounded and of constant slope: its horizon degenerates not by becoming infinitely sharp but by becoming infinitely smeared — a linear scale function distinguishes no location. Finally, recall from (5.2) that (A.3) computes *viability*; immortality proper is $\mathbb{E}[e^{-\Lambda_\infty}\mathbf{1}_{\mathcal{I}}]$, strictly smaller, and this is the unique juncture at which the gauge parameter $h_0$ re-enters (Lemma 5.10) — affecting values, never dichotomies.

## A.3 An exactly solvable gradual horizon

**Theorem A.8** (Unstable Ornstein–Uhlenbeck driver; **[P]**)**.** *Let the driver be*

$$d\mu_t = \nu\,\mu_t\,dt + \sigma\,dW_t, \qquad \nu > 0. \tag{A.4}$$

*Then $e^{-\nu t}\mu_t \to Z$ a.s. and in $L^2$, where $Z \sim N\big(\mu_0,\ \sigma^2/2\nu\big)$, and*

$$\mathbb{P}(\mathcal{I}) \;=\; \mathbb{P}(Z > 0) \;=\; \Phi\Big(\frac{\mu_0\sqrt{2\nu}}{\sigma}\Big). \tag{A.5}$$

*On $\{Z > 0\}$ the hazard decays doubly exponentially, $\log h_t \sim -(Z/\nu)e^{\nu t}$, and $\Lambda_\infty < \infty$; on $\{Z < 0\}$ the hazard explodes doubly exponentially and $T < \infty$ a.s.*

*Proof.* Variation of constants gives $\mu_t = e^{\nu t}\big(\mu_0 + \sigma\int_0^t e^{-\nu s}dW_s\big) =: e^{\nu t}M_t$. The martingale $M$ has $\mathbb{E}\langle M\rangle_\infty = \sigma^2\int_0^\infty e^{-2\nu s}ds = \sigma^2/2\nu < \infty$, hence converges a.s. and in $L^2$ to a Gaussian limit $Z$ with the stated mean and variance. On $\{Z > 0\}$ pick $t_0(\omega)$ with $\mu_t \ge \frac{Z}{2}e^{\nu t}$ for $t \ge t_0$; then $D_t \ge D_{t_0} + \frac{Z}{2\nu}(e^{\nu t} - e^{\nu t_0})$ and the integrand $e^{-(D_t - D_0)}$ is dominated by a doubly-exponentially decaying function, so $\Lambda_\infty < \infty$. On $\{Z < 0\}$ symmetrically $\mu_t \le \frac{Z}{2}e^{\nu t}$ eventually, $D_t \to -\infty$ doubly exponentially, the integrand diverges, $\Lambda_\infty = \infty$, and $T < \infty$ a.s. Finally $\mathbb{P}(Z = 0) = 0$, and $\mathbb{P}(Z > 0) = \Phi(\mu_0/\sqrt{\sigma^2/2\nu})$. ☐

**Consistency with the scale-function formula.** For (A.4), $s'(v) = \exp(-\nu v^2/\sigma^2)$, so $s(\pm\infty)$ are finite and the normalized scale function at $\mu_0$ is the mass below $\mu_0$ of the centered Gaussian with variance $\sigma^2/2\nu$ — precisely (A.5). The exactly solvable case is thus the generic case of Theorem A.5(iii) in its purest form.

**Definition A.9** (Horizon profile, location, width)**.** For a driver with bounded scale, the *horizon profile* is $\Pi(\mu_0) := \mathbb{P}(\mathcal{I} \mid \mu_0)$, the *location* its median point $\Pi = \frac{1}{2}$, and the *width* the natural spread of the transition, e.g. the interquartile range of the measure $d\Pi$. For (A.4): location $\mu_0 = 0$; width

$$w = \frac{\sigma}{\sqrt{2\nu}}, \qquad \text{sharpness} \quad \Sigma := w^{-1} = \frac{\sqrt{2\nu}}{\sigma}. \tag{A.6}$$

As $\Sigma \to \infty$ (small noise or strong instability), $\Pi \to \mathbf{1}\{\mu_0 > 0\}$: a hard event horizon. As $\Sigma \to 0$, $\Pi \to \frac{1}{2}$ everywhere: no horizon, a fair coin.

**Sharpness is a small-noise limit; general principle.** The width (A.6) is exactly the scale of fluctuation of the terminal charge $Z$ capable of overturning the deterministic verdict sign$(\mu_0)$: the sharp horizon is the Freidlin–Wentzell limit [139] of the soft family. We record the principle in the form answering the question that motivates this part: *a horizon is never intrinsically sharp;*

*its width is the fluctuation scale of the driver, and it appears sharp exactly when that scale is small against the spread of initial conditions.* Operationally, comparing $w$ to the population spread of $\mu_0$ yields three regimes: $w \ll$ spread — sharp horizon, the population splits into the fated; $w \sim$ spread — genuinely gradual horizon, initial conditions matter but do not determine; $w \gg$ spread — no horizon, the outcome is noise.

### A.4 Model atlas and the role of driver smoothness

**Proposition A.10** (Adverse linear drift: the price of survival; **P/S**). *Let $\mu_t = \mu_0 + \nu t + \sigma W_t$ with $\nu < 0$. Then:*

(i) ***[P]*** $\liminf_{t\to\infty} \dfrac{1}{t} \log \mathbb{P}(T > t) \;\geq\; -\dfrac{\nu^2}{2\sigma^2}$*;*

(ii) ***[P]*** *there exists $c > 0$ with $\mathbb{P}(T > t) \leq e^{-ct}$ for all large $t$;*

(iii) ***[S]*** *the exact rate is*

$$\lim_{t\to\infty} \frac{1}{t} \log \mathbb{P}(T > t) \;=\; -\frac{3\nu^2}{8\sigma^2}, \tag{A.7}$$

*attained by the optimal survival profile $\mu_{tv} \approx \frac{|\nu|t}{4}\, v\,(2-3v)$, $D_{tv} \approx \frac{|\nu|t^2}{4}\, v^2(1-v)$, $v \in [0,1]$: the conditioned survivor front-loads credit, lets the rate go negative at $v = \frac{2}{3}$, and coasts so as to exhaust its credit exactly at the deadline.*

*Proof.* (i) By the Markov property and a finite-time event of positive probability we may assume $\mu_0 \geq 1$; constants below depend on the data. Let $a := |\nu|/\sigma$, $b_0 := (1-\mu_0)/\sigma \leq 0$, and

$$A_t := \Big\{ W_u \geq b_0 + au \;\; \forall u \in [0,t], \quad W_t \leq b_0 + at + 1 \Big\}.$$

On $A_t$, $\mu_u \geq 1$ for all $u \leq t$, hence $D_u \geq D_0 + u$ and $\Lambda_t \leq h_0$, so $\mathbb{P}(T > t) \geq e^{-h_0}\mathbb{P}(A_t)$. Girsanov with drift $a$ (density $\exp(-a\widetilde{W}_t - a^2 t/2)$, $\widetilde{W} := W - a\,\cdot$ a Brownian motion under the tilted measure $Q$) gives

$$\mathbb{P}(A_t) \;\geq\; e^{-a^2 t/2}\, e^{-a(b_0+1)}\, Q\big(\widetilde{W}_u \geq b_0 \;\forall u \leq t,\; \widetilde{W}_t \in [b_0, b_0+1]\big),$$

and the $Q$-probability is of order $t^{-3/2}$ by the reflection principle: polynomial corrections do not affect the rate. (ii) With $K := \frac{h_0}{4}t$, $\mathbb{P}(T > t) \leq e^{-K} + \mathbb{P}(\Lambda_t \leq K)$. On $\{\Lambda_t \leq K\}$, $\mathrm{Leb}\{u \leq t : D_u \leq D_0\} \leq K/(h_0 e^{-D_0} \wedge h_0) \lesssim t/4$ (adjust constants by $D_0$), so there exists $u_* \in [t/2,\, 3t/4+1]$ with $D_{u_*} > D_0$, forcing $\sigma I_{u_*} \geq |\nu| u_*^2/2 - \mu_0 u_* \geq |\nu| t^2/16$ for large $t$. By the Borell–TIS inequality for the Gaussian process $I$ on $[0,t]$ (whose maximal variance is $t^3/3$ and whose expected supremum is $O(t^{3/2}) = o(t^2)$), $\mathbb{P}(\sup_{u\leq t} I_u \geq |\nu|t^2/16\sigma) \leq \exp(-c' t^4/t^3) = e^{-c't}$; combine. (iii) Rescale $W_{tv} = \sqrt{t}\, B_v$, so $I_{tv} = t^{3/2}\int_0^v B$ and $D_{tv} = \mu_0 tv + \frac{\nu t^2 v^2}{2} + \sigma t^{3/2}\int_0^v B$. Keeping $\Lambda_t = O(1)$ requires, at leading order, $D_{tv} \geq 0$ for all $v \in [0,1]$; with $B = \sqrt{t}\,\gamma$ this reads $\int_0^v \gamma \geq \beta v^2$, $\beta := |\nu|/2\sigma$, at Cameron–Martin cost $\exp(-t \cdot \frac{1}{2}\int_0^1 \dot\gamma^2)$. The variational problem $\min \frac{1}{2}\int_0^1 \dot\gamma^2$ subject to $\int_0^v \gamma \geq \beta v^2$ $\forall v$, $\gamma(0) = 0$, is solved by relaxing to the endpoint constraint $\int_0^1 \gamma = \int_0^1 (1-s)\dot\gamma(s)\, ds \geq \beta$: Cauchy–Schwarz gives $\int \dot\gamma^2 \geq 3\beta^2$ with equality for $\dot\gamma(s) = 3\beta(1-s)$, i.e. $\gamma(v) = 3\beta(v - \frac{v^2}{2})$, which satisfies the *full* constraint, $\int_0^v \gamma = \frac{\beta v^2}{2}(3-v) \geq \beta v^2$ on $[0,1]$ with equality only at $v = 1$. Hence the minimum is $\frac{3}{2}\beta^2 = 3\nu^2/8\sigma^2$, and unwinding the substitutions yields the profiles stated: $\mu_{tv} = \mu_0 + t(\nu v + \sigma\gamma(v)) \approx \frac{|\nu|t}{4} v(2-3v)$ and

$D_{tv} \approx t^2\big(\sigma \int_0^v \gamma - \frac{|\nu| v^2}{2}\big) = \frac{|\nu| t^2}{4} v^2(1-v)$. Elevating this computation to a full large-deviation proof (upper bound with matching constant, treatment of the soft constraint) is routine but lengthy and is not carried out here. □

*Remark* A.11. Three comments. *(1)* The naive lower-bound strategy "hold $\mu \geq 1$ throughout," used in (i), costs $\nu^2/2\sigma^2$ per unit time and is strictly suboptimal: the true optimizer spends the final third of the budget with $\mu < 0$, spending credit rather than defending the rate — survival to a deadline is cheaper than survival in perpetuity, and the discount is exactly the factor $\frac{3}{4}$. *(2)* Conditioned on the rare event $\{T > t\}$, the path concentrates on the stated profile: once again the appearance of purposeful resistance is manufactured by conditioning. *(3)* The analog for a mean-reverting driver with $\bar{\mu} < 0$ is an exponential rate given by the corresponding Freidlin–Wentzell action; we do not compute it here.

| **Driver** $\mu_t$ | $\mathbb{P}(\mathcal{I})$ | **tail of** $\mathbb{P}(T > t)$ | $\mathbb{E}[T]$ | **horizon** |
|---|---|---|---|---|
| constant $\mu_0 > 0$ | 1 | $\to e^{-h_0/\mu_0} > 0$ | $\infty$ (atom) | trivial step |
| constant $\mu_0 < 0$ | 0 | $\exp\big(-\frac{h_0}{\lvert\mu_0\rvert} e^{\lvert\mu_0\rvert t}\big)$ | finite | — |
| $\mu_0 + \sigma W_t$ | 0 | $t^{-1/4}$ **[A]** | $\infty$ | none (null rec.) |
| $\mu_0 + \nu t + \sigma W_t$, $\nu > 0$ | 1 | $\to$ const $> 0$ | $\infty$ | dichotomy in $\nu$ |
| $\mu_0 + \nu t + \sigma W_t$, $\nu < 0$ | 0 | $e^{-\frac{3\nu^2}{8\sigma^2} t(1+o(1))}$ **[S]** | finite | — |
| OU, mean $\bar{\mu} > 0$ | 1 | $\to$ const $> 0$ | $\infty$ | sharp in $\bar{\mu}$ |
| OU, mean $\bar{\mu} = 0$ | 0 | $t^{-1/2}$ **[S]** | $\infty$ | null rec., new exponent |
| OU, mean $\bar{\mu} < 0$ | 0 | $e^{-ct}$ | finite | — |
| unstable OU (A.4) | $\Phi\big(\mu_0\sqrt{2\nu}/\sigma\big)$ | mixture | $\infty$ | **gradual, width** $\sigma/\sqrt{2\nu}$ |
| general bounded scale | (A.3) | mixture | $\infty$ | shape = scale fn. |

**Exponents measure smoothness, not recurrence class.** The mean-zero mean-reverting driver is mortal (Proposition A.6(c)) with expected tail $t^{-1/2}$, not $t^{-1/4}$: at large scales its credit is Brownian-like (an additive functional obeying an invariance principle [140]), so the relevant persistence exponent is that of Brownian motion, $\frac{1}{2}$, not of its integral, $\frac{1}{4}$. The survival exponent is a *persistence exponent of the credit process* and is set by the smoothness of the driver rather than by its recurrence class: smoother driver, smaller exponent, heavier survival tail. For a fractional Brownian driver of Hurst index $H$ the persistence exponent of the driver itself is $1 - H$ [141]; the exponent for its integral, the relevant one here, is not known rigorously **[O]**: the numerical evidence of Molchan–Khokhlov [142] supports the conjecture $\theta(H) = H(1 - H)$ for persistence from the origin, which correctly returns $\frac{1}{4}$ at $H = \frac{1}{2}$; see also [56].

# B The background as a free energy

The background must be positive, variable, and never near zero. Construction (5.6) achieves all three: a sum of $n$ positive terms is small only if *all* are small, an event of probability $\Phi(-a/\epsilon\sqrt{t})^n$, exponentially small in $n$. This is the mechanism we identified. What was not anticipated is that the construction is a familiar object.

## B.1 Freezing and Gompertz

**The background is a partition function.** Writing $\epsilon W_t^{(i)} = \epsilon\sqrt{t}\, g_i$ with $g_i$ i.i.d. standard normal,

$$B_t = \sum_{i=1}^{n} e^{\beta_t g_i}, \qquad \beta_t := \epsilon\sqrt{t},$$

the partition function of $n$ i.i.d. Gaussian energy levels at inverse temperature $\beta_t$. Since $\beta_t$ *increases* with $t$, the background is a disordered system that *cools as it ages*: it is Derrida's random energy model [143, 144] with time as inverse temperature. The corresponding limit theory for sums of random exponentials is [145]; the mathematical treatment of the REM is [146].

**Theorem B.1** (Freezing; **[C]**). *With $\theta = \epsilon\sqrt{t/\log n}$, as $n \to \infty$ at fixed $\theta$,*

$$\frac{\log B_t}{\log n} \longrightarrow \begin{cases} 1 + \theta^2/2, & \theta < \sqrt{2} \quad (\textit{delocalized}), \\ \theta\sqrt{2}, & \theta \geq \sqrt{2} \quad (\textit{frozen}), \end{cases} \tag{B.1}$$

*in probability, with freezing time $t_{\mathrm{fr}} = 2\log n/\epsilon^2$.*

*Proof and status.* The two candidates are the annealed free energy $\log \mathbb{E}B_t = \log n + \epsilon^2 t/2$ and the extremal contribution $\max_i \epsilon W_t^{(i)} \approx \epsilon\sqrt{2t\log n}$; normalized these are $1 + \theta^2/2$ and $\theta\sqrt{2}$, and $(1+\theta^2/2) - \theta\sqrt{2} = \frac{1}{2}(\theta - \sqrt{2})^2 \geq 0$ with equality only at $\theta = \sqrt{2}$ — the annealed value dominates everywhere and the curves meet tangentially, the REM's characteristic contact. The second moment gives $\mathbb{E}B_t^2/(\mathbb{E}B_t)^2 = e^{\epsilon^2 t}/n + (n-1)/n \to 1$ iff $\epsilon^2 t < \log n$, so concentration is immediate for $t < t_{\mathrm{fr}}/2$; the full delocalized range needs the truncated second moment, and the frozen phase is the standard extreme-value computation. Both are classical [143, 146, 145]. □

**Our two mechanisms are the two phases.** The brief describes the background as held up both by "the high ones dominating" and by "the fuzz of all the intermediate-sized ones adding together." These are not two descriptions of one regime but exact descriptions of the two: the frozen phase is carried by the single largest term, the delocalized phase by the bulk.

**Corollary B.2** (Gompertz; **[P]**). *For $t < t_{\mathrm{fr}}$, $B_t = n\, e^{\epsilon^2 t/2}(1 + o(1))$ in probability: the background grows exponentially in age at rate $\epsilon^2/2$. This is the Gompertz law [147], arriving unbidden — the construction was chosen for positivity, not to reproduce it.*

## B.2 The background is gauge

**Lemma B.3** (Channels define moving walls; **[P]**). *Set $D_t := x_t - x_0$. Each channel becomes $O(1)$, i.e. contributes appreciably to $\Lambda$, at a* killing wall*:*

$$x_t^{\mathrm{crest}} = \log B_t + b, \qquad x^{\mathrm{trough}} = \frac{\log \Gamma}{\rho}, \tag{B.2}$$

*the first random and slowly moving, the second a fixed constant. the driftless case's soft-wall reduction applies with its fixed wall replaced by* max *of these.*

*Proof.* Solve $B_t e^{b-x} = 1$ and $\Gamma e^{-\rho x} = 1$. □

**Theorem B.4** (Exact absorption of a linear background; **[P]**)**.** *Suppose* $\log B_u = a + cu$ *exactly — the Gompertz phase,* $a = \log n$*,* $c = \epsilon^2/2$*. Then persistence above the crest wall* (B.2) *is* identical *to the driftless case's fixed-wall problem with initial data*

$$\mu_0 \longmapsto \mu_0 - c, \qquad D_0 \longmapsto D_0 - a - b. \tag{B.3}$$

*By that section's gauge proposition the background therefore moves prefactors and timescales alone, and no asymptotic exponent.*

*Proof.* With $D_u = D_0 + \mu_0 u + \sigma I_u$, $I_u = \int_0^u W_s\,ds$, the constraint $D_u \geq a + b + cu - x_0$ reads $\sigma\big(I_u - \frac{c-\mu_0}{\sigma}u\big) \geq a + b - x_0 - D_0$, and $I_u - \kappa u = \int_0^u (W_s - \kappa)ds$ is the integral of a Brownian motion started at $-\kappa$: the same Kolmogorov diffusion from shifted data. $\square$

**A headwind.** An exponentially growing background is a *constant headwind* of size $\epsilon^2/2$ on the improvement rate: the system must improve at that rate merely to hold its hazard fixed, and its initial credit is debited by $\log n + b$. Aging, here, is not a new mechanism but an offset.

**Theorem B.5** (Sublinear walls are invisible; **[A]**)**.** *If both walls are* $o(u^{3/2})$ *almost surely, which holds here, the crest wall being* $O(u)$ *in the delocalized phase and* $O(\sqrt{u})$ *in the frozen phase, and the trough wall constant, then, granting the moving-boundary robustness of the driftless case's Ansatz class, the persistence exponent is* $\frac{1}{4}$*, unchanged.*

*Sketch and status.* After rescaling by $t^{3/2}$ the boundary converges to zero uniformly on compacts, so the constraint converges to the fixed-wall constraint. Converting this to an exponent identity is the moving-boundary robustness discussed in the driftless case and surveyed in [56]. $\square$

# C The waiting room and the gain profile

## C.1 Two kinds of wall

A distinction must be drawn that the informal account elides. There are two different senses in which a channel "has a wall", and both are needed.

**Definition C.1** (Shape and killing walls)**.** The *shape walls* are where a channel rises above the background floor, i.e. where it becomes comparable to $B_t$:

$$x_{\text{shape}}^{\text{crest}} \approx b, \qquad x_{\text{shape}}^{\text{trough}} \approx -\frac{k_t}{\rho}, \qquad k_t := \log\frac{\mathbb{E}[B_t]}{\Gamma}, \tag{C.1}$$

so that the room is the *asymmetric* interval $[-k_t/\rho,\, b]$, containing the origin in its interior. The *killing walls* are where a channel makes $\Lambda$ accumulate, i.e. where it becomes $O(1)$ in absolute terms; these are (B.2). The shape walls govern what the hazard *looks like* as a function of $x$; the killing walls govern the driftless case's persistence geometry. They are different objects and are not interchangeable.

*Remark* C.2 ($\Gamma$ primitive, $k$ derived)*.* $\Gamma$ is a primitive of Definition 5.11; the width $k_t$ is *not* a parameter but a bookkeeping quantity recording where $\Gamma$ sits relative to the background. Two consequences follow immediately and are worth separating, since they run in opposite directions:

- *The room widens.* In the delocalized phase $\mathbb{E}[B_t] = ne^{\epsilon^2 t/2}$, so $k_t = \log(n/\Gamma) + \epsilon^2 t/2$ grows linearly and the shape-wall separation $k_t/\rho$ widens at rate $\epsilon^2/2\rho$: *grace grows as the background rises.*
- *The killing walls separate faster.* The crest killing wall $\log(2B_t)$ rises at $\epsilon^2/2$ while the trough killing wall $\rho^{-1}\log\Gamma$ is constant, so their separation grows at $\epsilon^2/2$.

Both are harmless asymptotically, being $O(t) = o(t^{3/2})$.

## C.2 The void, and when it is flat

**Proposition C.3** (The room, and when it is flat; **[P]**)**.** *The crest is a sigmoid of transition width $O(1)$ centered at $x = b$: $\varsigma_b \approx 1$ for $x \lesssim b-2$ and $\varsigma_b \approx e^{b-x}$ for $x \gtrsim b+2$. Hence the total hazard is flat, at $\approx B_t$, on*

$$-\frac{k_t}{\rho} \ll x \lesssim b-2, \tag{C.2}$$

*a genuine plateau when $k_t/\rho + b \gg 1$. It contains $x = 0$ in its interior precisely when $b \gtrsim 2$, which is the structural role of the offset.*

*Proof.* $\varsigma_b(x) \in (0.88, 1.05)\cdot$ its plateau value for $x \le b-2$, and the trough exceeds the crest below $-k_t/\rho$ by Proposition 5.13(i). □

**Constraint 5** (Internal onsets before external)**.** As the amplitude grows, $x$ reaches the trough wall before the crest wall if and only if

$$\boxed{b > \frac{k_t}{\rho}} \tag{C.3}$$

— the offset must exceed the trough's half-width. Under (C.3) a system of unfavorable anatomy suffers internal damage first, and only later does its external hazard begin to fall.

*Remark* C.4 (Why the offset is needed). At $b = 0$ the origin sits at the crest's inflection, where the local gain is $\frac{1}{2}$: a system with no muffling at all would already be half effective, contradicting the reading of the void as the regime in which the system contributes nothing. The offset moves the origin into the plateau, where by Theorem C.7(iii) the gain is $O(e^{-b})$. The void then contains its own center, and (C.3) orders the two onsets.

**Void as mutedness.** The void is where the system contributes nothing. That is stated qualitatively by Remark 5.12 and quantitatively by Proposition C.8: the gain $\Psi'$, which measures how much the driver moves the hazard, falls to $\kappa(\rho)e^{-(\rho b+k)/(1+\rho)}$ on the void, exponentially small in the room's width. A system in the waiting room is not merely at low hazard; its efforts have almost no purchase, and the wider the room the less.

## C.3 Activation: time supplies the interpolation

**Proposition C.5** (Activation time; **[S]**)**.** *In log-time $s = \log t$,*

$$\log \mathcal{A}_{e^s} = \log\sigma + \tfrac{3}{2}s + \log R_{e^s}, \qquad R \textit{ stationary}, \tag{C.4}$$

*so the amplitude is exponentially small at early log-times and grows exponentially, with a stationary fluctuation riding on a deterministic ramp of slope $\frac{3}{2}$. The void therefore ends, and the channels engage, at*

$$t_{\mathrm{act}}^{\mathrm{int}} \asymp \left(\frac{k}{\rho\,\sigma}\right)^{2/3}, \qquad t_{\mathrm{act}}^{\mathrm{ext}} \asymp \left(\frac{b}{\sigma}\right)^{2/3}, \tag{C.5}$$

*the times at which the typical amplitude reaches the trough and crest walls respectively. Under Constraint 5 these are ordered, $t_{\mathrm{act}}^{\mathrm{int}} < t_{\mathrm{act}}^{\mathrm{ext}}$: the internal channel engages first and the external one only later.*

*Scaling argument.* $\mathcal{A}_t = \sigma t^{3/2} R_t$ with $R_t = O_P(1)$ stationary; setting $\sigma t^{3/2} \asymp k/\rho$ and $\sigma t^{3/2} \asymp b$ give (C.5). Dimensions: $[(k/\rho\sigma)^{2/3}] = (\mathsf{T}^{3/2})^{2/3} = \mathsf{T}$. A sharp constant would require the first-passage law of the amplitude, which we do not compute. □

**Why no capacity parameter is needed.** Revision 1 installed a capacity (strength) parameter with a leak, whose purpose was to produce a default regime in which the system contributes nothing and $\mu$ matters only rarely. Equation (C.4) supplies that regime for free: the amplitude of $x$ is exponentially small in log-time at early times, so the void is not an assumption to be installed but *what the model already does*, and it is left at (C.5), in two stages, internal then external, without any additional primitive. The interpolation parameter is $k$, large $k$, long void, background-dominated; small $k$, immediate engagement, and $k$ is itself derived (Remark C.2).

One distinction is worth drawing here rather than left for a reader to trip over, since §5.7 reinstates a leak. The parameter dropped above was a *standing* primitive: a permanent damping of $x$, installed to manufacture a regime the model turned out to supply unaided. Definition 5.49's leak is a *transient* and carries no new primitive of that kind — it is indexed by the schedule's stage, it vanishes identically at completion, and the model above is its $k = n$ boundary case. What was dropped was a permanent assumption; what is added is a temporary one that removes itself.

*Remark* C.6 (The cost of dropping capacity, recorded)*.* This is a real loss and should not be glossed.

| | capacity (revision 1, dropped) | $k$ (retained) |
|---|---|---|
| what it moves | an *exponent*, $\frac{1}{4} \to \frac{1}{2}$ | a *timescale* |
| sharpness | genuine threshold at $S_c$ | crossover at $t_{\mathrm{act}}$ |

Revision 1's sharp-threshold theorem is therefore gone, and its "sharp in parameter space" register weakens to a crossover. What replaces it is the threshold at $\rho = 1$ (Proposition 5.13(ii)), beyond which the background still reaches the survivor's hazard and below which the system closes off from the external world. That is a threshold in a structural parameter with a mechanical meaning rather than one installed to produce a threshold — but, as Remark 5.14(c) records, it moves no exponent. The exchange is favorable and honest, not free.

## C.4 The exact gain

**Theorem C.7** (Soft-kink gain; **[P]**)**.** *Fix $B_t \equiv B$ constant and define the* gain

$$\Psi(x) := -\log \frac{h^{\mathrm{tot}}(x)}{h^{\mathrm{tot}}(0)}, \qquad \textit{so that} \qquad h^{\mathrm{tot}}(x) = h^{\mathrm{tot}}(0)\, e^{-\Psi(x)} \tag{C.6}$$

*identically, with $h^{\mathrm{tot}}(0) = B + \Gamma$. Then:*

(i) $\Psi(0) = 0$ *and* $\Psi$ *is smooth and strictly increasing;*

(ii) $\Psi(x) = x + O(e^{-x})$ *as* $x \to +\infty$ *and* $\Psi(x) = \rho x + O(1)$ *as* $x \to -\infty$*: a* soft kink *of slopes* $1$ *and* $\rho$*;*

(iii) *the local gain is a hazard-weighted average of the two channels' slopes,*

$$\Psi'(x) = \frac{h^{\text{crest}}(x)\,\frac{e^x}{1+e^x} + h^{\text{trough}}(x)\,\rho}{h^{\text{crest}}(x) + h^{\text{trough}}(x)} \;\in\; (0, \rho), \tag{C.7}$$

*with* $\Psi'(0) \approx e^{-b} + \rho e^{-k}$*,* $\Psi' \to 1$ *as* $x \to +\infty$*,* $\Psi' \to \rho$ *as* $x \to -\infty$*, and an interior minimum on the void quantified in Proposition C.8.*

*Consequently, with constant background, the model is exactly the driftless case's with the credit passed through* $\Psi$*: Constraint 4 is met.*

*Proof.* (i) is immediate from (C.6) and positivity of $h^{\text{tot}}$; monotonicity from (iii). (ii): as $x \to +\infty$, $h^{\text{tot}} = 2Be^{-x}(1+O(e^{-(\rho-1)x}))$, so $\Psi = x + O(\cdot)$; as $x \to -\infty$, $h^{\text{tot}} = \Gamma e^{-\rho x}(1+O(e^{(\rho-1)x}))$, so $\Psi = \rho x + O(1)$. (iii): for a sum $h = h_1 + h_2$ one has $-h'/h = (h_1 s_1 + h_2 s_2)/(h_1 + h_2)$ with $s_i = -h_i'/h_i$; here $s_2 = \rho$ and, differentiating $2B/(1+e^x)$, $s_1 = e^{x-b}/(1+e^{x-b}) \in (0,1)$. The stated limits follow by evaluating the weights, using $\Gamma/B = e^{-k}$ at $x = 0$. □

**Proposition C.8** (The void is muted, exponentially in $k$; **[P]**)**.** *On the void the gain* (C.7) *attains an interior minimum at*

$$x^* = \frac{b - k + 2\log\rho}{1+\rho}, \qquad \min\Psi' = \kappa(\rho)\, e^{-\frac{\rho b + k}{1+\rho}}, \qquad \kappa(\rho) := \rho^{\frac{2}{1+\rho}} + \rho^{\frac{1-\rho}{1+\rho}}. \tag{C.8}$$

*The gain therefore does not merely dip but is* exponentially small in $(\rho b + k)/(1+\rho)$*: the wider the room, on either side, the more completely the driver is muted within it. For* $\rho = \frac{3}{2}$*,* $\kappa = 2.045$*; for* $\rho = \frac{5}{2}$*,* $\kappa = 2.443$*.*

*Proof.* On the void the crest is saturated, so its weight is $\approx 1$ and its slope is $s_1 \approx e^{x-b}$, while the trough weight is $\approx (\Gamma/B)e^{-\rho x} = e^{-k}e^{-\rho x}$. Hence $\Psi' \approx f(x) := e^{x-b} + \rho e^{-k}e^{-\rho x}$, and $f'(x) = 0$ gives $e^{(1+\rho)x} = \rho^2 e^{b-k}$, which is (C.8). Substituting back, both terms carry the common factor $e^{-(\rho b+k)/(1+\rho)}$ with coefficients $\rho^{2/(1+\rho)}$ and $\rho^{(1-\rho)/(1+\rho)}$, summing to the stated value. (The approximation is verified numerically to four significant figures over $\rho \in \{1.5, 2.5\}$, $b \in [6, 14]$, $k \in [8, 20]$.) □

*Remark* C.9 (Correction to the reference specification)*.* Two points where the informal specification requires amendment, both recorded in the ledger.

*(a)* The closed form $\psi_\rho(x) = -\log(e^{-x} + e^{-\rho x})$ is *not* an identity for (5.7): at $x = 0$ the model gives $B + \Gamma$ while $\psi_\rho$ gives $2h_0$. It is the clean representative of the soft-kink class, not the model's own gain; (C.6) is exact by construction and has the same asymptotic slopes, so nothing structural is lost. *(b)* The gain range is $(0, \rho)$ and not $[1, \rho]$. The crest channel's local slope is $e^x/(1+e^x)$, which tends to 0, not 1, as $x \to -\infty$. This is not a blemish but the sharpest available statement of what the void *is*: on the plateau the gain falls to $\kappa(\rho)e^{-(\rho b+k)/(1+\rho)}$ (Proposition C.8), so the driver is muted and its efforts scarcely move the hazard. The gain profile thus encodes all three regimes — exponentially small in the void, $\to 1$ under shielding, $\to \rho$ under damage. The mutedness is a property of a *wide* room: if $\rho b + k$ is $O(1)$ the minimum is $O(1)$ and there is no muted regime, consistently with Proposition C.3.

### C.5 What the original driver was concealing

**Corollary C.10** (State-dependent gain; [**P**]). *Along a path, the effective negative logarithmic derivative of the total hazard is*

$$\tilde{\mu}_t = -\frac{d}{dt}\log h_t^{\mathrm{tot}} = \Psi'(x_t)\,\mu_t, \qquad \Psi' \in (0,\rho). \tag{C.9}$$

*That is:* the driftless case's driver was the present driver times a state-dependent gain*, and the old Brownian motion was absorbing that gain as extra noise. The revision therefore extracts structure from the original noise term rather than adding a primitive to it.*

*Proof.* Chain rule on (C.6) with $\dot{x} = \mu$. ☐

**Reading of the gain.** The new $\mu$ is correspondingly narrower than the old, and the difference is now explicit rather than hidden. Where the old model attributed all variation in the log-hazard's slope to the driver, the present one attributes part of it to *where the system is*: the same effort buys a full unit of log-hazard reduction under shielding, $\rho$ units of damage in a trough, and nothing at all in the void.

## D Transfer, in detail

### D.1 Reduction to the driftless case

**Proposition D.1** (Reduction; [**P**]). *On the slice $\epsilon = 0$ (so $B \equiv n$), $\Gamma \to 0$, and $x \gg 0$, the model is the driftless case's with $h_0 = 2n$ and credit $x - x_0$. More usefully: for $\epsilon = 0$ and any $\Gamma$, the total hazard is a deterministic function of $x$ alone, and Theorem C.7 identifies it as the driftless case's hazard composed with a fixed gain.*

*Proof.* At $\epsilon = 0$, $B \equiv n$; for $x \gg 0$, $h^{\mathrm{tot}} \approx 2ne^{-x}$ since the trough is $O(e^{-\rho x}) = o(e^{-x})$. ☐

### D.2 Asymptotics: the driftless case survives intact

**Theorem D.2** (the driftless case is undisturbed; [**A**]). *In the two-channel model, for every $\rho > 0$, $\Gamma > 0$ and $b \geq 0$:*

(i) *the persistence exponent is $\frac{1}{4}$, exactly;*

(ii) $\mathbb{P}(T > t) \asymp Ct^{-1/4}$*;*

(iii) $\mathbb{E}[T] = \infty$ *while* $T < \infty$ *a.s. (null recurrence);*

(iv) *the survivorship interpolation of the driftless case, its pure Doob limit, its $s/4t$ linear response and its critical mode, are unchanged.*

No truncation window and no $\gamma = 0$ caveat*: Constraint 3 holds as a theorem.*

*Proof.* By Lemma B.3 the killing walls are $\log(2B_t)$ and $\rho^{-1}\log\Gamma$. The latter is constant, hence an $O(1)$ shift, which we proved to be gauge. The former is $O(t)$ in the delocalized phase and $O(\sqrt{t})$ in the frozen phase, hence $o(t^{3/2})$, and is absorbed exactly in the linear case by Theorem B.4 and invisibly in general by Theorem B.5; the offset $b$ enters (B.3) as a further $O(1)$ shift and is likewise gauge. Survivors live at large positive $x$, where by Proposition 5.13(ii) the dominant channel is the crest if $\rho > 1$ and the trough if $\rho < 1$; in either case the dominant channel is an exponential in $x$, so the killing wall is a level of the credit and the persistence problem is the driftless case's up to a constant rescaling. The conclusions are then the driftless case's, transported — and, notably, for every $\rho > 0$ rather than only $\rho > 1$. □

**This repairs revision 1's defect.** Revision 1's additive internal hazard contributed $\gamma t$ to $\Lambda$, destroying the $t^{-1/4}$ tail and forcing a truncation window with finite life expectancy $\asymp g^{-3/4}$. The trough channel does not, and the reason is structural rather than fortunate: $\Gamma e^{-\rho x}$ is small exactly where survivors live, whereas a constant is not small anywhere. Both walls are $O(1)$-or-gauge shifts, and we proved such shifts to be gauge.

**What was given up.** Both channels are decreasing in $x$, so the total hazard is monotone and *more strength is always better*. Revision 1's U-shaped hazard, and with it the optimal finite strength and the $(S, A)$ optimization, are gone. This is the price of Constraint 3: an interior optimum requires a term that grows with $x$, and any such term destroys the tail. An optimum and an undisturbed the driftless case were never going to coexist. What survives of the earlier optimization story is the threshold at $\rho = 1$ and the trade recorded in Remark C.6.

Theorem D.2 asserted that the driftless case's conclusions carry over. This subsection makes the transfer quantitative: it enumerates the four ways the two-channel hazard differs from $h_0 e^{-x}$, shows that each lands in gauge or in the prefactor, and isolates the one place where the two-channel model is genuinely *better* conditioned than the driftless case.

## D.3 The four differences

| difference | size | where it goes | reference |
|---|---|---|---|
| wall location $\log B_u + b$ | $\Theta(u)$ | gauge (exact absorption) | Thm. B.4 |
| graded boundary of width $W_u$ | $\Theta(u)$ | gauge (collapses under rescaling) | Thm. D.4 |
| waiting-room passage | $O(1)$ | prefactor | Prop. D.6 |
| background fluctuation | $O(\sqrt{u})$ | gauge | Thm. B.5 |

Nothing reaches an exponent. The transfer is therefore exact at the level at which the driftless case states its results.

**Lemma D.3** (Above the room the model is the driftless case's; **[P]**)**.** *For $x \geq b$,*

$$\tfrac{1}{2}\big(1 + e^{-b}\big)e^{\,b-x} \;\leq\; \varsigma_b(x) \;\leq\; \big(1 + e^{-b}\big)e^{\,b-x}, \tag{D.1}$$

*so $h^{\text{crest}} \asymp B_u e^{\,b-x}$ with constants in $[\frac{1}{2}, 1] \cdot (1 + e^{-b})$. If moreover $\rho > 1$ then $h^{\text{trough}}/h^{\text{crest}} \to 0$ as $x \to \infty$, so $h^{\text{tot}} \asymp B_u e^{\,b-x}$: above the room the hazard is the driftless case's $h_0 e^{-x}$ with $h_0 = B_u e^{b}$, up to bounded multiplicative constants.*

*Proof.* For $x \geq b$ one has $1 \leq e^{x-b}$, hence $e^{x-b} \leq 1 + e^{x-b} \leq 2e^{x-b}$; invert. The trough comparison is Proposition 5.13(ii). □

## D.4 The graded boundary collapses

The substantive structural difference is that the killing boundary is neither hard nor soft but *graded*. Above $\log B_u + b$ the hazard is negligible; across a band below it the hazard is $\asymp B_u$, bounded; below $(\log \Gamma)/\rho$ it diverges.

**Theorem D.4** (The band is asymptotically invisible; [**A**]). *Let*

$$W_u := \underbrace{\log B_u + b}_{\textit{crest wall}} - \underbrace{\frac{\log \Gamma}{\rho}}_{\textit{trough wall}} = \Big(1 - \frac{1}{\rho}\Big) \log B_u + b + \frac{k}{\rho} \tag{D.2}$$

*be the width of the graded band. In the delocalized phase* $W_u = \Theta(u)$ *and in the frozen phase* $W_u = \Theta(\sqrt{u})$*; in both,*

$$\frac{W_u}{u^{3/2}} \longrightarrow 0.$$

*Hence under the driftless case's rescaling the whole band contracts to a single point, and the graded boundary is asymptotically the hard wall of that section. Granting the moving-boundary robustness of its Ansatz class, the persistence exponent is* $\frac{1}{4}$.

*Proof.* Insert $\log B_u = \log n + \epsilon^2 u/2$ (delocalized, Corollary B.2) or $\log B_u = \epsilon\sqrt{2u \log n}$ (frozen, Theorem B.1) into (D.2); both are $o(u^{3/2})$. The rescaling and the robustness step are as in Theorem B.5. □

**Softening and re-hardening do not cancel each other — they both cancel against self-similarity.** It is tempting to read the situation as a competition: the logistic bounds the hazard from above and would soften the wall, while the trough is unbounded and re-hardens it, the two effects offsetting. That is not what happens. The two effects act at *different places*, the logistic at the top of the band, the trough at its bottom, and neither undoes the other. What renders both irrelevant is the third thing: the band they enclose has width $o(u^{3/2})$, while the credit they constrain fluctuates on scale $u^{3/2}$. A boundary structure of any shape whatsoever, confined to a band of width $o(u^{3/2})$, is invisible to the exponent. The net effect of the logistic and the trough is therefore exactly nil asymptotically — not by mutual cancellation but because self-similarity outruns them both.

**Proposition D.5** (The tolerance shrinks: this model is better conditioned than the driftless case's; [**P**]). *Survival to time* $t$ *requires*

$$\mathrm{Leb}\{u \le t : x_u \lesssim b\} \;=\; O\big(B_t^{-1}\big), \tag{D.3}$$

*since the hazard on the band is* $\asymp B_u$ *and* $\Lambda_t = O(1)$ *is required. In the delocalized phase* $B_t^{-1} = n^{-1} e^{-\epsilon^2 t/2} \to 0$*, so the admissible occupation below the wall tends to zero: the problem converges to* strict *persistence.*

*Proof.* $\Lambda_t \ge \big(\inf_{u \le t} B_u\big) \cdot c \cdot \mathrm{Leb}\{u \le t : x_u \lesssim b\}$ with $c = \varsigma_b(b) > 0$; survival to $t$ with probability bounded below forces $\Lambda_t = O(1)$. □

**The Ansatz gap narrows.** the driftless case's sole load-bearing heuristic was that persistence with *bounded occupation tolerance* shares the exponent of strict persistence. The two-channel model does not evade that assumption, but it weakens what is being assumed: by Proposition D.5 the tolerance here is not $O(1)$ but $O(B_t^{-1}) \to 0$. The richer, more ontologically committed model is thus *closer* to the rigorous case than the simpler one — an unusual direction for added structure to move a proof, and worth recording as such.

## D.5 The waiting room is a prefactor, and it is not a shelter

**Proposition D.6** (Cost of the room passage; **[S]**). *The room* $[-k/\rho, b]$ *lies entirely below the crest killing wall* $\log B_u + b$*, since* $\log B_u > 0$*. A trajectory therefore traverses it only before escaping upward, at time* $t_{\mathrm{act}}^{\mathrm{ext}} \asymp (b/\sigma)^{2/3}$ *(Proposition C.5), accumulating*

$$\Lambda_{\mathrm{room}} \;\asymp\; B_0\, t_{\mathrm{act}}^{\mathrm{ext}} \;\asymp\; n\Big(\frac{b}{\sigma}\Big)^{2/3}, \tag{D.4}$$

*an* $O(1)$ *quantity in* $t$*. Its entire effect is to multiply the prefactor in* $\mathbb{P}(T > t) \asymp Ct^{-1/4}$ *by* $e^{-\Theta\left(n(b/\sigma)^{2/3}\right)}$.

**Grace, but not shelter.** The room deserves a more careful reading than "grace interval" suggests. Inside it the hazard is *exactly the background*: neither raised by damage nor lowered by shielding. It is therefore grace relative to the trough, unfavorable anatomy costs nothing while the amplitude is small enough to keep $x$ above $-k/\rho$, but it is a *cost* relative to the crest, because survivors must be above the wall and time spent in the room is time not spent shielding. Formula (D.4) prices that cost. The wider the room and the larger the background, the more expensive the wait: the offset $b$ buys the ordering of the two onsets (Constraint 5) and pays for it in the prefactor, at rate $n(b/\sigma)^{2/3}$ in the exponent.

## D.6 The conditioning ladder

We quantify how each conditioning enhances survival.

**Proposition D.7** (Degrees of the initial data; **C/A**). (i) Initial rate. *From* $(m, 0)$ *with* $m > 0$*: the rate stays positive until* $W$ *reaches* $-m/\sigma$*, a time of order* $t_\mu = m^2/\sigma^2$ *during which* $D$ *increases; thereafter, exactly by scaling,* $\mathbb{P}_{(m,0)}(\tau > t) \sim c_1(t_\mu/t)^{1/4} \propto m^{1/2}$ (degree $\frac{1}{2}$: a tenfold gain in survival odds costs a hundredfold increase in $m$). (ii) Initial credit. *From* $(0, d)$ *with* $d > 0$*: by scaling with* $c = d^{-1/3}$, $\mathbb{P}_{(0,d)}(\tau > t) = \mathbb{P}_{(0,1)}(\sigma^{2/3}\tau > t\, d^{-2/3}\sigma^{2/3}) \asymp (t_D/t)^{1/4} \propto d^{1/6}$ (degree $\frac{1}{6}$). (iii) Unification. *All initial data act only through the equivalent timescale they generate:*

$$\mathbb{P}(T > t) \;\asymp\; \Big(\frac{t_*}{t}\Big)^{1/4}, \qquad t_* \;\asymp\; \sigma^{-2/3} \max\big(1,\; \hat\mu_0^2,\; \hat D_0^{2/3}\big), \tag{D.5}$$

*and which term dominates is decided by the horizon coordinate: by* (5.15), $t_D/t_\mu = \kappa_0^{2/3}$*, so rate dominates for* $\kappa_0 \ll 1$ *and credit for* $\kappa_0 \gg 1$.

*Proof.* The scaling identities are exact (Proposition 5.22); transfer to $\mathbb{P}(T > t)$ is Ansatz 5.18. That the exponent from interior starting states is unchanged is part of Theorem 5.29. □

**Proposition D.8** (Homogeneity of the harmonic function; **P/C**). *On the open quadrant define*

$$\mathfrak{h}(\mu, d) \;:=\; \lim_{t\to\infty} t^{1/4}\, \mathbb{P}_{(\mu,d)}(\tau > t);$$

*the limit exists, is finite and positive [54, 58]. Then* $\mathfrak{h}$ *is invariant-harmonic for the killed Kolmogorov diffusion, and homogeneous of degree* $\frac{1}{2}$ *under the scaling of Proposition 5.22:* $\mathfrak{h}(c\mu, c^3 d) = c^{1/2}\mathfrak{h}(\mu, d)$*. Consequently*

$$\mathfrak{h}(\mu, d) = \mu^{1/2} F\big(\kappa\big) = d^{1/6} \tilde F(\kappa), \qquad \kappa = \sigma^2 d/\mu^3, \tag{D.6}$$

*for profiles $F, \tilde{F}$ determined by the explicit confluent-hypergeometric formulas constructed in [58]. (We flag a provenance point: the special functions arising here are confluent hypergeometric, not Airy; Airy functions belong to the neighboring parabolic-drift literature.)*

*Proof.* Given existence of the limit, homogeneity is immediate from Proposition 5.22:

$$t^{1/4}\,\mathbb{P}_{(c\mu,c^3 d)}(\tau > t) = c^{1/2}\left(t/c^2\right)^{1/4}\mathbb{P}_{(\mu,d)}\left(\tau > t/c^2\right) \;\longrightarrow\; c^{1/2}\,\mathfrak{h}(\mu,d).$$

Harmonicity follows from the Markov property and the same limit taken one step later. Representation (D.6) is homogeneity restricted to the orbit coordinates of Proposition 5.23(iii). □

*Remark* D.9 (Conditioning on survival forever: the Doob transform; **[C]**). Conditioning on $\{T = \infty\}$ is vacuous (Theorem 5.26); its canonical regularization is the $\mathfrak{h}$-transform, the $t \to \infty$ limit of conditioning on $\{\tau > t\}$. Under the transformed measure the process never returns to zero credit and, by self-similarity,

$$\mu_s \;\asymp\; \sigma s^{1/2}, \qquad D_s \;\asymp\; \sigma s^{3/2}, \qquad \text{hence} \qquad \log h_s \;\sim\; -\sigma s^{3/2}:$$

conditioned survivors exhibit superexponentially decaying hazard and appear, from the inside, unambiguously immortal — although the unconditioned process is certainly mortal and the model contains no drift. The drift is manufactured by selection; this is the strongest true form of the event-horizon intuition, and it is a statement about ensembles, not about state space. The accompanying Yaglom-type statement, convergence in law of $(\mu_t/\sigma t^{1/2},\, D_t/\sigma t^{3/2})$ conditionally on $\{\tau > t\}$ to a nondegenerate limit, is the correct "stationary profile of a long-lived system," stationary only in self-similar coordinates; see [58, 56], and [60] for the general theory of quasi-stationary limits against which this self-similar variant should be read. Section 5.8 promotes this remark to the section's central result: the conditioning that manufactures the drift becomes itself the object of study, and the transform described here is identified as the exact limit of the survivorship interpolation.

# E Reference sheet: the filter at a glance

This subsection collects, without proof, the quantitative content of the section in the form most useful for reference. Everything here is established above; cross-references point to the source.

## E.1 The conditioned dynamics in closed form

Under the survivorship limit $Q$ of Theorem 5.75 the system is

$$d\mu_t = \frac{\sigma^2}{\mu_t}\,G(\kappa_t)\,dt + \sigma\,dW_t^Q, \qquad dD_t = \mu_t\,dt, \qquad \kappa = \frac{\sigma^2 D}{\mu^3}, \tag{E.1}$$

with $G(\kappa) = \frac{1}{2} - 3\kappa F'(\kappa)/F(\kappa)$ read off $\mathfrak{h} = \mu^{1/2}F(\kappa)$ as in (D.6)–(5.50). Dimensionally $[\sigma^2/\mu] = \mathsf{T}^{-2} = [\mu]/[t]$, as required. The shape function interpolates between two endpoints (Theorem 5.78):

$$G(0^+) = \tfrac{1}{2}\ \ (\text{drift } \sigma^2/2\mu,\ \text{maximal}), \qquad G(\infty) = 0\ \ (\text{drift } 0,\ \text{none}),$$

so that *survivorship pushes hardest near the wall and not at all once credit is abundant*, the strength along any ray of fixed $\kappa$ scaling as $1/\mu$.

**The endpoints are ray limits, not coordinate limits.** One may not simply let $\mu \to 0$ in $\sigma^2/2\mu$: at fixed $D > 0$ that sends $\kappa \to \infty$ and hence $G \to 0$, and the two factors compete. Only the statement along rays of constant $\kappa$ is unambiguous, which is precisely why $\kappa$ and not $(\mu, D)$ separately is the coordinate in which the conditioned dynamics should be read (Proposition A.2).

## E.2 The three levels, before and after

| | **Before** ($\mathbb{P}$) | **After** ($Q$) |
|---|---|---|
| *Level 1: the driver $\mu$* | | |
| law | Gaussian, driftless | non-Gaussian, drift $(\sigma^2/\mu)G(\kappa) \geq 0$ |
| autonomy | $\mu$ Markov by itself | only the pair $(\mu, D)$ is Markov |
| quadratic variation | $\sigma^2\,dt$ | $\sigma^2\,dt$ *(unchanged)* |
| increments | stationary, independent | neither |
| self-similarity | index $3/2$ | index $3/2$ *(preserved)* |
| *Level 2: the hazard $h$* | | |
| path regularity | $C^1$ | $C^1$ *(unchanged)* |
| law at time $s$ | lognormal, log-variance $\sigma^2 s^3/3$ | non-Gaussian, one-sided |
| range | exceeds $h_0$ infinitely often, unbounded above | $h_s < h_0$ for *all* $s$ (if $D_0 = 0$) |
| credit $D$ | null recurrent, $\liminf_s D_s = -\infty$ | transient, $D_s \to \infty$, $D_s \asymp \sigma s^{3/2}$ |
| *Level 3: survival* | | |
| $\Lambda_\infty$ | $= \infty$ a.s. | $< \infty$ a.s., $\asymp \hat{h}_0 = h_0\sigma^{-2/3}$ |
| quenched curve $e^{-\Lambda_t}$ | $\to 0$; collapse $\exp(-e^{\sigma t^{3/2}})$ | plateau at $e^{-\Lambda_\infty} > 0$ |
| population curve | $\asymp Ct^{-1/4}$ | $\equiv 1$ |
| lifetime | $T < \infty$ a.s., $\mathbb{E}[T] = \infty$ | $T = \infty$ |

Sources: Level 1, Theorems 5.75 and 5.78; Level 2, Definition 5.2 and Theorem 5.80; Level 3, Theorem 5.26, Claim 5.30, Corollary 5.31.

**Notable structures at Level 2.** Three features are worth isolating. *(i)* The hazard is one derivative *smoother* than the noise driving it, a direct consequence of the log-derivative placement (Remark 5.9); conditioning does not change this, since Girsanov alters drift and never path regularity. *(ii)* Before conditioning $\log h_s$ is exactly Gaussian with variance $\sigma^2 s^3/3$, *cubic* in time, the integrated-Brownian fingerprint, so $h_s$ is lognormal. *(iii)* After conditioning, the survivor's hazard never once exceeds its initial value, whereas before it does so infinitely often and by unbounded amounts: this is the wall of §5.2.6 in its most concrete form. Finer still, Groeneboom, Jongbloed and Wellner [58] identify $t \mapsto t^{9/10}$ as a critical curve for the conditioned process, the expected time spent below $t^\alpha$ being finite for $\alpha < 9/10$ and infinite for $\alpha \geq 9/10$: survivors do lag their typical $s^{3/2}$ growth, and 9/10 is exactly how much.

**Proposition E.1** (The quenched–annealed gap; **[P]**)**.** *In the driftless model, almost surely*

$$\log \Lambda_t = \sup_{u \leq t}\bigl(D_0 - D_u\bigr) + O(\log t), \tag{E.2}$$

*and $t^{-3/2}\sup_{u\leq t}(D_0 - D_u) \to \sigma \sup_{v\leq 1}(-I_v)$ in law, a nondegenerate positive limit. Hence the quenched survival curve collapses* doubly exponentially*: writing $c_t := t^{-3/2}\log\Lambda_t$,*

$$e^{-\Lambda_t} = \exp\bigl(-e^{\,c_t\, t^{3/2}}\bigr), \qquad c_t \xrightarrow{d} \sigma \sup_{v\leq 1}(-I_v) \;>\; 0,$$

*while by Claim 5.30 the population curve decays only as $Ct^{-1/4}$. The entire population tail is therefore carried by a set of paths of vanishing probability.*

*Proof.* Write $M_t = \sup_{u \le t}(D_0 - D_u)$ and $\Lambda_t = h_0 \int_0^t e^{D_0 - D_u} du$. The upper bound $\Lambda_t \le h_0 t e^{M_t}$ is immediate. For the lower bound, $D$ is $C^1$ with $\dot{D} = \mu$ and $\sup_{u \le t} |\mu_u| = O(t^{1/2})$ a.s. up to logarithmic factors, so on an interval of length $\asymp t^{-1/2}$ around a near-maximizer the integrand exceeds $e^{M_t - 1}$, giving $\Lambda_t \gtrsim h_0 t^{-1/2} e^{M_t}$. Taking logarithms yields (E.2). The scaling limit is Proposition 5.22 applied to $I$, with $\sup_{v \le 1}(-I_v) > 0$ a.s. $\square$

**The interpolation as a one-parameter family.** Between the two columns, the survival curve of a $t$-survivor is

$$\mathbb{P}(T > t + s \mid T > t) \;\approx\; \left(1 + \frac{s}{t}\right)^{-1/4},$$

a scale-free family flattening from a power law at $t = 0$ to the constant 1 as $t \to \infty$; equivalently the $t$-survivor carries effective hazard $1/4t$ (Corollaries 5.31 and 5.31). The filter does not deform the survival curve arbitrarily: it slides it along a single one-parameter orbit.

## E.3 What conditioning can and cannot do

**Proposition E.2** (Rigidity of the survivorship transform; **[P]**). *Let $\widetilde{\mathbb{P}}$ be any measure locally absolutely continuous with respect to $\mathbb{P}$ on each $\mathcal{F}_s$ — in particular $Q$. Then:*

(i) The kinematics survive exactly. *$dD_t = \mu_t\, dt$ holds $\widetilde{\mathbb{P}}$-a.s. unchanged: credit remains literally the integral of rate. Conditioning cannot alter it.*

(ii) Only the driver may be tilted. *Since the noise enters solely the $\mu$ coordinate in* (5.3)*, Girsanov can add drift only to $\mu$; the transform has exactly one functional degree of freedom.*

(iii) Self-similarity fixes its form. *If in addition $\widetilde{\mathbb{P}}$ respects the scaling of Proposition 5.22, that degree of freedom is forced into the shape $(\sigma^2/\mu)G(\kappa)$ of* (E.1)*, a single function of one dimensionless variable.*

(iv) The cascade is destroyed. *Under $\mathbb{P}$ the system is triangular: $\mu$ is autonomous and $D$ is slaved to it. Under $Q$ the credit re-enters the drift of $\mu$ through $\kappa$, so $\mu$ alone is no longer Markov although the pair remains so.*

*Proof.* (i) and (ii) are Girsanov's theorem for (5.3), whose diffusion matrix is degenerate with range spanned by $\partial_\mu$; the finite-variation coordinate admits no equivalent change. (iii) is Proposition D.8 together with Theorem 5.78. (iv) is the $\kappa$-dependence in (5.50), non-constant by Theorem 5.78(ii)–(iii). $\square$

**Survivorship coupling is collider bias in continuous time.** The feedback of Proposition E.2(iv) is informational, not mechanical: nothing pushes $\mu$. Under $\mathbb{P}$ the credit accumulated by time $s$ and the future increments of the driver are independent, but both feed a common outcome, survival; conditioning on that outcome renders them dependent. This is exactly collider bias, Berkson's paradox, or "explaining away", realized in continuous time, and (E.1) is its quantitative form: little credit banked implies the remaining path must work harder, and the drift says by how much. The same object appears in large-deviation theory as the driven or effective process associated with a rare event. The moral of Proposition E.2 is then sharp: survivorship cannot bend the kinematics of a system, only the statistics of what drives it — and, when the system is self-similar, only along a one-parameter family of shapes.

# F Specification of numerical tests

The one load-bearing heuristic, Ansatz 5.18, admits a direct test, which we specify for completeness; this subsection may be read as an appendix.

**Lemma F.1** (Exact simulation of the pair; **[P]**)**.** *For a step* $\Delta > 0$ *let* $\xi := W_{t+\Delta} - W_t$ *and* $\eta := \int_t^{t+\Delta}(W_s - W_t)\,ds$*. Then* $(\xi, \eta)$ *is independent of* $\mathcal{F}_t$*, centered Gaussian with covariance* $\left(\begin{smallmatrix} \Delta & \Delta^2/2 \\ \Delta^2/2 & \Delta^3/3 \end{smallmatrix}\right)$*, and the update*

$$\mu_{t+\Delta} = \mu_t + \sigma\xi, \qquad D_{t+\Delta} = D_t + \mu_t\Delta + \sigma\eta$$

*reproduces the law of* (5.3) *at the grid points with zero discretization error.*

*Proof.* $(\xi, \eta)$ is a measurable functional of the post-$t$ increments and is a copy of $(W_\Delta, I_\Delta)$; apply Lemma 5.21. The update identity is $I_{t+\Delta} = I_t + W_t\Delta + \eta$ multiplied through by $\sigma$. □

The remaining error is the quadrature of $\Lambda_t = h_0 e^{-D_0}\int_0^t e^{-(D_u - D_0)}du$, whose integrand varies on the scale $|\Delta D| \lesssim 1$, i.e. $\Delta \lesssim 1/|\mu| \sim t^{-1/2}$: the constraint tightens with time. Accumulate in logarithmic space; the integrand ranges over hundreds of orders of magnitude. To reach the tail, probabilities of $10^{-3}$ to $10^{-4}$ require $t/t_*$ up to $10^{12}$ and beyond, use multilevel splitting along the thresholds $D_k = \sigma(\theta^k t_*)^{3/2}$, or importance sampling tilted by the harmonic function of Proposition D.8, the asymptotically optimal tilt. Validate the code first on the hard-wall functional, where (5.18) is a theorem; then run the soft-killing functional and fit $\log \mathbb{P}$ against $\log t$ over at least three decades, discarding $t \lesssim 10\,t_*$. *Falsifiers:* a fitted slope robustly different from $-\frac{1}{4}$ (Ansatz fails: the killing term carries its own scale); any dependence of the slope on $h_0$ (gauge violation, contradicting Proposition 5.23); non-convergence of the conditioned rescaled credit $D_s/\sigma s^{3/2}$ (failure of the self-similar Yaglom picture of Remark D.9).

*Results (executed August 2026; [148], §6).* The falsifiers did not fire. With exact pair simulation, $N = 1.5 \times 10^5$ paths, and a hybrid uniform–geometric grid to $t = 3 \times 10^4$ (control run at half the step sizes agreeing within intervals): hard-wall slope $-0.2499$ $[-0.254, -0.243]$ against the theorem's $-\frac{1}{4}$; soft-killing slopes $-0.2487$, $-0.2480$, $-0.2483$ at $h_0 \in \{0.3, 1, 3\}$, with the $h_0 = 30$ transient relaxing toward $-\frac{1}{4}$ over late windows exactly as Lemma 5.10 predicts; the ratio $\mathbb{P}(T > t)/\mathbb{P}(\tau > t)$ constant to three decimals over two fitted decades ($[0.1768, 0.1788]$ at $h_0 = 1$) — sharper than the Ansatz asserts; the kinematic factor matching $(1 - s/t)^{-1/4}$ at grid spacing ($0.2874 \pm 0.0133$ measured, $0.2859$ exact); Yaglom rescaled-credit distributions coinciding to KS 0.010 between $t = 2 \times 10^3$ and $3 \times 10^4$; and the survivorship drift positive at the predicted order for small $\kappa$ and vanishing at large $\kappa$. The Ansatz is accordingly **[A]** with direct numerical support; the deep-tail computation at $t/t_* \gtrsim 10^{12}$ via splitting or harmonic-function importance sampling remains open. The survivorship results of §5.8 add three sharper tests: estimate the window bias $\beta_t(s)$ under splitting and regress it against $\mathfrak{h}(X_s)$, whose proportionality tests Theorem 5.75; regress conditioned increments $\Delta\mu$ on $1/\mu$ at small $\kappa$, whose slope $\sigma^2/2$ tests the wall asymptotics of Corollary 5.79; and verify the linear response $\beta_t \approx 1 + s/4t$ at large $t/s$, whose coefficient $\frac{1}{4}$ tests Proposition 5.77.

Four tests specific to this part, additional to the driftless case's. *(i) Freezing.* Simulate $B_t$ for $n \in \{10^3, 10^4, 10^5\}$ and plot $\log B_t/\log n$ against $\theta$; expect collapse onto (B.1) with tangential contact at $\theta = \sqrt{2}$, and sample-to-sample fluctuations $o(1)$ before the transition and $O(1)$ after, the frozen phase being non-self-averaging. *(ii) The two coordinates under conditioning.* Verify the sign truncation and the anatomy selection separately, since they are the section's two measured claims and they differ in kind. For the first, estimate $\mathbb{P}(x_T < 0 \mid T_{\text{death}} > T)$ by the particle filter and compare with the unconditioned $\frac{1}{2}$ of Theorem 5.24; the prediction is an abrupt collapse, to order $10^{-3}$ within a few multiples of the intrinsic timescale. For the second, place a prior on the phase, $\rho \sim \text{Unif}(-1, 3)$ serves, and track the conditioned $\mathbb{P}(\rho > 0)$ against horizon; the prediction is a *gradual* rise from the prior together with steady truncation of the negative tail. A selection on $\rho$ as abrupt as the selection on $\operatorname{sign} x$, or an absence of selection on $\rho$ altogether, would each contradict §5.5.

*(iii) Exponent robustness.* Estimate the survival tail for a grid of $(\epsilon, n, \Gamma, \rho)$ and fit the local slope. Prediction: $-\frac{1}{4}$ throughout, with no dependence on any of the four — this is Theorem D.2 and the section's central falsifiable claim. *(iv) The gain profile.* Compute $\Psi'$ from (C.7) and check the limiting values of Theorem C.7(iii); then test Proposition C.8 by regressing $\log \min \Psi'$ on $\rho b + k$ across a grid of $\rho$. The predicted slope is $-1/(1+\rho)$, the predicted intercept $\log \kappa(\rho)$, and the minimizer is at (C.8); all three are sharp, making this the most stringent test in the section.

*Falsifiers.* A fitted exponent differing from $-\frac{1}{4}$, or depending on $\Gamma$ or $\rho$ (contradicting Theorem D.2); a gain reaching a floor of 1 rather than 0 on the plateau (contradicting Remark C.9(b)); an absence of anatomy selection under conditioning (contradicting Theorem 5.36, which makes adverse anatomy eventually fatal and hence selectable).

## A biographical note (not formal)

The dated record cited at the Golden Point's definition fixes the concept's provenance; one further item from the same period is recorded here for completeness, and it is biographical rather than formal: nothing in the argument depends on it, and nothing in it is graded.[25]

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[25] In the author's words. "As a natural fantasy worldbuilder, I also daydreamed in 2021 about a potential future in which 'Tragedism' exists, who are basically cultists that worship the Departure Point and want my head on a stake. The weird thing about them though is that they are almost like the Sith in Star Wars in that they fully accept Nexic reasoning as truth, (ie like force-sensitivity), and so branch from Miracleists later than classical religious fundamentalists and evolution-deniers, but branch just short of interpreting the Golden Point favorably. Hence it ends up being the case that future me has to fight the fundamentalist Level 1 goblins but also the elite Level 10 Tragedists, who actually have the power to harness Nexic reasoning but use it for evil. They would hate each other's guts, yet both would view me as the embodiment of the enemy for in their eyes threatening their way of life. And also would they team up, or would they hate each other too much for that, etcetera for the worldbuilding questions, (although my alliance model suggests they very much would team up if I was 'winning'). So of course I don't imagine that fictional account to actually hold up literally, but a main inspiration of mine with defining the Golden question and Golden answer was to ask myself exactly what I would have wanted to propose in the present day to get the upper hand on the Tragedists of the future, since I do after all have the first move. A very sci-fi digression indeed! (And yes, the weird recursion of it, opened up by Aethic future-conditioning, is how eerily much Tragedists serve as an allegory for a story that hasn't even happened yet.)"

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